FiF-avhandling, No 141

Argument fronting in English and Swedish written texts

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LINKÖPING UNIVERSITY

Filosofiska fakulteten – FiF avhandling 141
Licentiate Thesis No. 141

Argument fronting in English and Swedish written texts

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Linköping 2026

This is a Swedish Licentiate's Thesis

Swedish postgraduate education leads to a doctor's degree and/or a licentiate's degree.
A doctor's degree comprises 240 ECTS credits (4 years of full-time studies).
A licentiate's degree comprises 120 ECTS credits.

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Typeset using XeTeX

Printed by LiU-Tryck, Linköping 2026

Edition 1:1

© Mikael Svensson, 2026
ISBN 978–91–8118–764–9 (print)
ISBN 978–91–8118–765–6 (PDF)
https://doi.org/10.3384/9789181187656
ISSN 1401-4637
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ABSTRACT

This thesis investigates argument fronting in English and Swedish written texts, focusing on the relationship between syntactic structure, prosodic representation, information structure, and discourse context. The main object of study is Information-structural fronting (IS-fronting), in which a non-subject element of the argument structure, such as an object or predicative complement, occurs clause-initially. Related constructions, including VP-proform fronting and inversion fronting, are also considered.

The empirical investigation is based on a subset of the English-Swedish Parallel Corpus (ESPC), consisting of original English and Swedish texts and their translations into the other language. The parallel-corpus design enables comparison of both formally corresponding constructions and expressions serving equivalent discourse functions. The corpus analysis is complemented by theoretical analysis and introspection, particularly because written data cannot directly reveal prosodic realisation. The analysis therefore considers the interaction between syntax, phonology, information structure, and co-text.

The results show that argument fronting is substantially more frequent and more widely distributed in Swedish than in English. Swedish material contains numerous fronted noun- and adjective-headed phrases functioning as direct objects or predicative complements, as well as VP-proforms and clausal complements. In English, by contrast, the most frequent type is fronting of obligatory adverbials accompanied by subject–verb inversion, while fronted objects and predicative complements are comparatively rare. The translation data reinforce this difference: only 8% of Swedish fronted constructions are translated into English using argument fronting, whereas 73% of English fronted constructions are rendered as argument fronting in Swedish.

The thesis argues that these differences are not primarily due to fundamentally different discourse functions or restrictions on which constituents may be fronted. Rather, they result from the interaction between language-specific morphosyntactic properties and the mapping between syntax and prosodic phonology. In particular, the verb-second structure of Swedish permits a different relationship between the fronted constituent and the intonational phrase than English does, affecting the prosodic prominence and discourse interpretation of the fronted element. Information structure, including Topic–Focus relations, Given–New status, and the salience of discourse referents, further contributes to the contextual distribution of fronting.

Overall, the study demonstrates that argument fronting is shaped by the interaction of syntax, phonology, and discourse. It also shows the value of parallel-corpus data for identifying discourse-functional correspondences across languages. The thesis concludes that further research based on spoken data and additional languages is needed to develop a broader account of the syntax–phonology–discourse interface in argument fronting.

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Chapter 1

Introduction

1.1 Sentence and discourse

This thesis is about the relation between sentence and discourse, or more accurately about the relation between syntactic structure, phonological representation, and co-text; a type of phenomena that Neeleman and Vermeulen (2013, p. 1) refer to as "discourse-related word order variation". The thesis investigates this relation by studying sentences structures that place non-subject elements of the argument structure sentence initially. The main focus will be on a structure where an object or predicative complement is placed before the finite verb in declarative clauses, which we will refer to as Information-structural fronting or IS-fronting for short, illustrated in (1).

(1) The Rolls-Royce I leave to my favourite nephew.

However, we will also have something to say about two other structures that prepose elements of the argument structure, such as *verb phrase anaphora fronting* (VPA-fronting), shown in (2) and *inversion fronting* demonstrated in (3)

(2) That I will.

(3) In the middle of the room stood a small table.

Collectively, we will refer to any structure that places an element of the argument structure before the finite verb as *fronted structures* and the process of placing them there as *fronting*.

The reasons for distinguishing these structures from IS-fronting are in the case of VPA-fronting that there is a difference in terms of how much of the canonical structure is fronted, whereas in Inversion fronting, fronting is combined with other structural changes of the canonical structure. In VPA-fronting, the VP-pro element includes both the verb and the argument (Mikkelsen, 2015) and in Inversion fronting, the preposing of obligatory adverbials or predicative complements are combined with the subject being placed after the finite verb, traditionally called subject-verb inversion (Quirk et al., 1985, p. 1379) or subject postposing (Huddleston & Pullum, 2002, p. 97). The equivalent structures in Swedish are somewhat different, primarily because of the word order difference between English and Swedish. English is a strict subject-verb-object (SVO) language, unlike all other Germanic languages, including Swedish, which are *verb-second* languages (V2), i.e., the finite verb is always the second constituent in declarative clauses (Holmberg, 2015). It is therefore important when comparing the two languages to describe these differences and how

they contribute to the discourse functions of fronted structures in the two languages. A short descriptive overview of English and Swedish declarative sentences will be given in section 1.4.2.

In this study, we will make a contrastive comparison of the discourse functional characteristics of fronted structures in English and Swedish, arguing for an analysis which connects these structures possible discourse functions to the way morphosyntactic structure interacts with phonology and discourse. The fact that Swedish fronted constructions are somewhat less constrained in terms of the co-texts in which they can be used is claimed to be a result of how the different syntactic structures in English and Swedish are matched up with the prosodic representation of each language.

The argument that prosodic phonology can explain restrictions in the use of written constructions is the most unorthodox element in this thesis, as the main empirical data do not include any phonological information. However, it is claimed that implicit prosody Fodor (2002a) restricts the use of IS-fronting in English to a greater extent than it does in Swedish. The argument is that IS-fronting in English will be perceived as giving prominence to the fronted element in a way that Swedish, and other verb-second languages, do not, allowing the latter languages to use IS-fronting in more contexts.

The relation between sentence and discourse can be approached in different ways. We can take a top-down approach by starting with the notion of discourse and focus on its properties. By looking at discourse both as a dynamic process and as a structured linguistic entity, we can study how it is built up by sequential contributions from one or several participants, as well as how it is organised, and what distinguishes discourse from a random list of sentences.

However, we can also take a more bottom-up approach, taking our main object of investigation to be the sentence, and the way discourse enables and restricts its form – morphosyntactic and phonological – when producing some linguistic expression to perform some communicative function. These different approaches are in no way incompatible but will naturally have different primary areas of interest. This thesis takes the latter approach, focussing primarily on the morphosyntactic, semantic, and phonological properties of individual sentences, and how these properties are affected by, and interact with the context in which the sentences are produced.

To achieve a relevant description and explanation of any aspect of reality in a single study, it is necessary to delimit the aim and scope to some specific phenomenon. Still, even when we focus our attention on some clearly demarcated object of study, we are in most cases trying to incrementally work our way towards an understanding of some more general problem. For this study, the bigger picture is trying to get to grips with the fact that simply producing grammatical sentences expressing the correct propositions, in the right order, with the right illocutionary force is not enough to create a well-formed text. From the perspective of the sentence, the principles that apply in its production within the parallel domains of morphosyntax, semantics, and prosodic phonology are inescapably linked with the context that each new sentence is simultaneously interacting with and being a part of, so much so that even when we try to abstract away the context, we cannot escape it.

At the same time, the need for scientific rigour requires that the description and the theoretical model are both adequate and specific to make precise predictions that can be easily refuted or confirmed. It is argued in this thesis that applying a generative syntactic theory with autosegmental-metrical phonology through the use of information structure is the best way to achieve this.

1.2 The phenomenon of fronting

The main phenomenon investigated in this study is the sentence structure that in the literature is commonly referred to as *topicalization*, but in this thesis is called IS-fronting. The reason for going against the terminological grain is that the term topicalization relies on the specific information-structural notion of topic that has proven notoriously difficult to pin down and thus offers poor guidance on the nature of the syntactic structure. Moreover, there are cases of topicalization/IS-fronting, which are generally agreed to *not* involve topical elements, in which cases the term is simply a misnomer. Using the term topicalization would therefore follow a naming convention whose logic is like the one illustrated by the 90s advertising campaign for Red Rock Cider: "It is not red and there's no rocks in it!".

Although the primary focus will be on IS-fronting, VPA-fronting and Inversion also exhibit displacement of the internal argument to a pre-verbal position. Thus, any claims regarding the relation between argument fronting and discourse function of IS-fronting need to be coherent with the observations of the discourse functional properties of VPA-fronting and Inversion.

The clearest way to define the syntactic structure of argument fronting is to first establish the notion of *canonical clause structure*. The purpose is to use this notion as the baseline and then define other syntactic structures in terms of how they differ from the canonical one. A canonical clause is not claimed to be more common, in the sense of being more frequent, nor is it intended to indicate that it is more fundamental and from which non-canonical structures are derived. It is primarily a descriptive convenience that simplifies the comparisons of different sentence structures within a particular language, and also helps to establish what are equivalent structures across languages.[1]

One of the most detailed accounts of the notion of canonical clause structure is found in Huddleston and Pullum (2002) which describes a canonical clause as having a set of properties which they argue are "more basic or elementary" than non-canonical ones. According to Huddleston and Pullum, a canonical clause in English is a positive, declarative, active, non-coordinated, main clause. Each of these properties is the canonical value of a specific grammatical category or dimension, e.g. the positive value *positive* is the canonical value of the category *polarity*. The categories which the different values represent are fairly straightforward, except for the value of *active*, which is not restricted to representing the category of *voice* but rather includes different ways to syntactically realise the same propositional assertion. In that sense, it is most similar to the clause type dimension (declarative, interrogative, imperative, exclamative), but representing a less established grammatical category. In addition to active and passive clauses, it includes clefts, right/left-dislocation, *there*-existential construction and the one that will be investigating in this study: argument fronting. The common dimension is thus that they represent different ways to syntactically "package" an assertion (Chafe, 1976). Essentially, this dimension deals with differences in the way information in a sentence is presented in relation to discourse.

Using the notion of canonical structure for Swedish adds some complexity because of the verb-second phenomenon. As is the case for all Germanic languages, with the exception of English, the finite verb must be the second constituent in main clauses (Holmberg, 2015, p. 343). Thus, both an adjunct and an argument in initial position will push the subject to a position after the finite verb. However, maintaining the idea that the notion of canonical structure is a way to establish some sentence structural baseline to which other structures can be compared, we will argue that the canonical structure in Swedish is also SVO. Although both declarative clauses without a fronted adjunct, shown in (4a) and clauses with, illustrated in (4b) fulfil the criteria for a canonical clause, i.e. a positive, declarative, active, non-coordinated main clause. However, placing an adjunct in the first position in a Swedish main clause will restrict the discourse context in which the clause can be used felicitously. Consequently, as the goal is to establish a

descriptive base structure to which other structures can be compared and whose felicitous use is not dependent on any particular discourse configuration, SVO will also be the canonical structure in Swedish.

(4) a. Jag har sett huggormar i den här skogen
I have seen adders in this here forrest.DET
'I've seen adders in this forrest'

b. I den här skogen har jag sett huggormar
in this here forrest.DET have I seen adders.
'In this forrest I've seen adders '

Having thus established the notion of canonical clause, we can now define the structure of argument fronting by comparing it to a canonical one. Example (5) illustrates the canonical structure and (6) shows argument fronting.

(5) John parked *the car* across the street.

(6) *The car*$_{\text{i}}$ John parked _____ $_{\text{i}}$ across the street.

In the canonical clause, the internal argument, which in this case is a direct object *the car*, is positioned after the main verb, whereas in the argument fronted construction the object is placed clause initially, before both the subject and the verb. The position of the object in the canonical clause is indicated in (6) with an underlined gap which is co-indexed with the fronted object to illustrate the relation between the object and the canonical position. In the literature, it is common to express this relation as *movement* or *displacement* of the internal argument to a clause initial position. However, this does not necessarily indicate a theoretical stance on whether the object actually originates in the canonical position and is moved to its fronted position, and is often just a way to follow a conventionalised way of describing this structure as a relative notion with respect to the canonical structure.

In addition to this type of argument fronting, we will also include a number of other constructions that are syntactically distinct but which also displace the internal argument clause initially. VPA-fronting moves the argument to the front of the sentence together with the entire VP, as shown in (7).

(7) He said I would like her, and *like her*$_{\text{i}}$ I do _____ $_{\text{i}}$

As this example illustrates, the use of this structure when the VP is realised as a full phrase is limited to context where a previously mentioned VP is repeated in a fronted construction confirming the truth of the initial proposition. However, when the VP is realised as a proform as in (8), it offers more contextual flexibility.

(8) He said I would like her, and *that*$_{\text{i}}$ I do _____ $_{\text{i}}$

Moreover, in Swedish the equivalent proform fronting is a common structure for promoting the internal verb inside the VP (9).

(9) Han sa att jag skulle gilla henne och *det*$_{\text{i}}$ gör jag.
He said that I would like her and it do I
He said that I would like her, and that I do

The final structure with which we will be concerned is inversion fronting, which shares with IS-fronting the movement of the internal argument to the front of the sentence, but differs by the fact that the subject is placed after the verb, resulting in an inversion of the internal and the external argument, as shown in (10).

(10) On the TV_i $stood_j$ a figurine _____ $_j$ _____ $_i$

Characteristic of these structures, such as (5) and (6) is that despite the differences in object placement, semantically the two sentences assert the same propositional content and are thus truth-conditionally equivalent. Yet, most would agree that the two sentences differ in interpretation. The interpretative difference can thus not be explained only by employing standard truth-conditional semantics. The general description is that the word order in (6) has more specific constraints in terms of the types of contexts in which it can be felicitously used (Huddleston & Pullum, 2002, p. 1372).

In order to make this more explicit, it is common to use a question to represent the co-text of the declarative sentences under analysis. By using different types of questions we can get an understanding of what information is already established or presupposed and what information is being presented as new. We can also establish what questions the sentence is a possible answer to, something which is traditionally referred to as *question-answer congruence*.

The most general and least restrictive co-text is often referred to as an *out-of-the-blue context*, and commonly represented by a question like "What happened?" or "What is going on". By manipulating the information the question establishes as presupposed, we can test the co-textual constraints of a particular declarative sentence. If we apply this diagnostic on our canonical and fronted sentences in (5) and (6), we recognize that the canonical sentence is coherent with both an out-of-the-blue question like "What happened?" as well as the slightly more restricted "What is John doing?" which prompts for what is predicated about John. In comparison, the argument-fronted sentence in (6) is not congruent with these contexts. Instead, we need to establish the notion of *the car* in the question for (6) to be possible, as in "What did John do with the car?". This type of context would be possible also with the canonical version, but we would have to indicate the established status of *the car* by using a pronoun or attenuate the noun phrase prosodically. Our diagnostic thus indicates that the argument fronted construction is more contextually constrained than the canonical version; simply put, (5) can be used in more types of context than (6).

In English, the constraints for using argument fronting appear to be fairly restrictive, making the distribution and use of these structures rather limited, despite them expressing the same proposition as the canonical structures. We can tentatively describe this constraint as the requirement that the initially placed object must have contextual properties that allow it to be placed in a particular structural relation to the rest of the sentence as in (5), i.e. the well-formedness of the sentence is not solely dependent on the relations among the elements within the syntactic domain but also on the relation between elements in the sentences and their co-text. A more precise description and theoretical model of the way formal and contextual properties interact is one of the objectives of this study.

The methodological approach taken in this study is to compare the properties of argument fronted constructions in English, with the equivalent structure in Swedish. The motivation for comparing fronting in English and Swedish is based on anecdotal observations and some exploratory findings in previous studies (M. Johansson, 2002; Rahkonen, 2000) which indicate that there is a noticeable difference in the frequency of fronting in English and Swedish. Although the fronted construction in both languages is more contextually constrained than its canonical equivalent, fronting in Swedish seems relatively less restricted than the equivalent structure in English, i.e. there appears to be more types of context where it will be felicitous to use a

Swedish fronted construction and where the English fronted construction is either not possible or dispreferred, than the other way around. This perceived difference between the languages can be illustrated by observations such as those in (11) and (12) where fronting is possible in Swedish but not in English (The fronted element is in italics).

(11) a. (Vill inte du ha kaffe?)
Want not you have coffee?
'Don't you want coffee?'

b. Nej, *kaffe* dricker inte jag
No, coffee drink not I.
'No, I don't drink coffee'

(12) a. (Hur tog han avslaget?)
How took he the rejection?
'How did he handle the rejection?'

b. Ganska bra, men *lite* *sårad* blev han nog.
Fairly well, but a little hurt became he probably.
'Fairly well, but he was probably a little bit hurt.'

The examples in (11) and (12) show that, for reasons that need to made explicit, in these contexts the English equivalent cannot be a fronted construction, although argument fronting is syntactically possible given different contextual conditions, such as in (13), in which there is an explicit contrast set to the fronted element.

(13) I like Kombucha and Chai but coffee I don't drink.

However, it is less clear what context, if any, is required for (12) to be felicitously fronted in English. We can also note that in both Swedish examples where fronting has been used, it is also possible to use a canonical structure, i.e., *jag dricker inte kaffe* in (11) and *...men han blev nog lite sårad* in (12) are both felicitous in the same context. This indicates that there are points in the text where it is possible to use both a canonical and a fronted construction. Some of these may be realised as fronted constructions, and others with canonical ones. We therefore need to establish not only which contextual properties license the use of fronting constructions but also whether there are factors that actively *trigger* their use.

Finally, and less predictably, we need to include the way these constructions can be phonologically realised. Even though we are dealing with written data, an important aspect in the way a particular syntactic structure interrelates with its co-text relies on the way its prosodic phonology is interpreted. We will argue that, although written text is not a record of a particular instance of phonological realisation by a speaker, any attempt at explaining the difference in the use of fronting in Swedish and English must include the way prosody influences syntactic structures in a particular context. The current project is thus aligned with the view expressed by Féry (2006, p. 69) that including prosody in the analysis of grammatical structures will be able to solve problems that have previously appeared puzzling. In order to begin to give a systematic account of these facts, we need to get a broader empirical overview of the type of fronted constructions that can be found in English and Swedish, and some way to compare the two languages where the contextual parameters are kept as constant as possible.

The research presented in this thesis is partly an effort to investigate how the intuition about fronting in Swedish and English can be verified or refuted, but more importantly it is an attempt to explore how the description and theoretical generalisations of fronted structures can be incorporated in some current model of language competence and communicative usage in order to explain any differences and similarities between the English and Swedish fronted constructions.

1.3 Why study fronting?

The structures of argument fronting, VP-fronting and argument inversion have all been studied fairly extensively both from a discourse functional and formal-syntactic perspective in English, e.g. (Birner & Ward, 1998; Chomsky, 1977; Netz & Kuzar, 2007; E. F. Prince, 1984; Rizzi, 1997), and in Swedish (Engdahl & Lindahl, 2014; M. Johansson, 1996; Lehti-Eklund, 1984; Lindahl & Engdahl, 2022; Rahkonen, 2000, 2006). However, although it has always been recognised that these constructions cannot be satisfactorily described and explained without taking both of these aspects into consideration, the focus has in the majority of cases been restricted to one of them. Thus, in more functionally-orientated studies the aim has been to look at the semantic and pragmatic aspects of these fronting structures, analysing them in terms of their communicative function within theories of information structure, discourse-status, and prosodic phonology, whereas in more syntactically focused studies, the chief interest has been on the formal mechanism of phrase structure and constituency order, trying to find ways to account for the constraints delimiting the general principles to those that are well-formed in individual languages.

Moreover, in cross-linguistic studies, there has been a tendency to restrict attention to syntactically equivalent structures in the language under study. Thus, an argument fronted structure in English is compared to an argument fronted structure in, say German, in order to bring out aspect of the structure in either language not obvious by only studying it within a single language.

It is argued in this thesis that such approaches will fail to sufficiently address the fact that it is only by combining a more formal theoretical model of the architecture of grammar – separating linguistic competence into distinct modules that operate on module specific principles – with a discourse functional theory of the interaction between syntax, semantics, prosodic phonology, and co-text/common ground that we can hope to arrive at an understanding of the different properties of fronting structures.

Using corpus data ensures that we will be able to study these structures within a natural co-text, and by using a parallel corpus, we will be able to find discourse functionally equivalent structures, i.e. the comparison will not be between syntactically defined structures in different languages, but between fronting structures in one language with its discourse functional equivalent in the other language, which may not have any syntactic similarity to a fronting structure.

This methodology has its origin in contrastive linguistics (Chesterman, 1998; James, 1980; Krzeszowski, 1990), an approach to comparative language study that has traditionally been used for applied investigations such as translation studies and various language learning applications (Meir, 2023). It focuses on contextualised language usage, such as the investigation on how specific constructions or patterns in a source language are idiomatically rendered into a target language, but a more general applicability for the use of contrastive linguistics has been argued for by S. Johansson (2012), who promotes its potential to explore communication-focused phenomena more widely; especially the use of parallel corpora to explore discourse semantically/pragmatically equivalent expressions rather than only formally comparable expressions.

However, although studies within contrastive linguistics have made considerable contributions to the study of language structure and use, the emphasis on empirical results and the lack of a coherent theoretical framework have tended to restrict the results to primarily descriptive generalisations. In order to achieve more explanatory developments, a better integration of the contrastive methodology and the theoretical models is necessary. This study is an attempt to contribute to achieving this.

Table 1.1: Distribution of fronted constituents by text type

Text type	Fronted objects	Subjects	Fronted adverbials
Interviews	14.0	61.9	18.5
Radio broadcasts	1.6	73.0	22.0
Conversations/debates	9.4	59.8	23.1

1.4 Morphosyntax and Distribution of Fronted Construction

In order to get an understanding of fronted structures in English and Swedish, we will look at what is already known about different aspects of fronting in the two languages. This section will look at what has been reported about the frequency and distribution of fronting in different types of corpora in English and Swedish, as well as the morphosyntactic and discourse functional characteristics of fronted structures.

1.4.1 Frequency and distribution of fronted constructions

There are a fairly limited number of corpus investigations that have studied the type of more general fronted constructions that we focus on in this study. One reason might be that fronting often requires corpora that can be searched for patterns of clause elements, which requires a parsed corpus, also called a treebanks (Marcus et al., 1993). However, Biber et al. (1999) provides some statistics on the fronting of what they call core elements, (Biber et al., 1999, pp. 120—122) which essentially aligns with the definition of arguments used in the current study.

Biber et al. (1999, pp. 909–910) report approximately 50–100 fronting constructions per million words in conversation and news, and 200–300 per million in fiction and academic prose; fronted objects and other nominals are particularly frequent in conversation and fiction.

The low frequency of fronting in English written text is given some support from Oostdijk and de Haan (1994) who investigate the Nijmegen Corpus of 130,000 words of syntactically annotated English text. They show that object fronting in English written texts are very scarce with only 9 cases out of 1814 monotransitive main clauses. In ditransitive clauses the frequency is higher with 3 instances out of 90. Unfortunately, the description of what type of objects that are fronted is not detailed.

For Swedish, there are no studies looking at argument fronting in general. However, Hörberg (2018) investigated the occurrence of object fronting and found 869 OVS sentences in a corpus of approximately 1.27 million words of written Swedish, corresponding to approximately 685 OVS sentences per million words; OVS sentences constituted 5.6% of the transitive sentences included in the analysis.

While a direct quantitative comparison between English and Swedish fronting frequencies is difficult due to differing analytical criteria, the corpus data indicates a markedly higher baseline frequency in Swedish. Because Hörberg's figures are restricted specifically to OVS transitive structures—excluding other types of fronted arguments—the true frequency gap between English and Swedish non-subject argument fronting is likely even wider than these raw figures suggest.

Although Jörgensen (1976) studies spoken Swedish discourse of three different types: interviews, radio broadcasts and conversations/debates, the investigation provides some indication of the structural and prosodic possibilities for Swedish fronting. The distribution of constituents in pre-finite position of declarative clauses shows large variation between the different discourse types, as seen in Table (1.1).

Looking at the types of constructions that are included in Jörgensen's study, the data reveal

some interesting facts about fronted objects in Swedish spoken discourse. A great majority of the examples of object fronting presented have the third person singular pronoun *det* (transl. 'it/that'), as the fronted constituent, as illustrated below in (14). (Jörgensen consistently transcribes "det" as "de").

(14) a. de tycker hon
that thinks she
'she thinks so/that'

Taken together, the available corpus evidence suggests that argument fronting is substantially more frequent in Swedish than in English, although differences in corpus composition and analytical criteria make direct quantitative comparison difficult. The existing English studies indicate that fronting is relatively rare, particularly in written texts, whereas the higher frequency of OVS constructions reported for Swedish points to a greater possibility for non-subject argument fronting. The Swedish data further suggest that fronting is not restricted to a single discourse type or structural configuration, but varies considerably across spoken genres and may involve a range of prosodic and discourse functions. Overall, the existing literature provides some evidence for substantial cross-linguistic and discourse-related variation, while leaving the structural, discourse-functional, and prosodic factors governing argument fronting in need of further study.

1.4.2 Morphosyntactic constraints on fronted elements

Another port of call on the journey towards a fuller understanding of the form and function of argument-fronting in English and Swedish is to establish the basic structural possibilities of placing verb-licenced elements, including the verb phrase/VP-pro form, in sentence initial position. This requires establishing the structural relations between a fronted element's original and fronted position, as well as identifying which types of phrasal elements can be fronted.

Moreover, the fact that fronted constructions are often sensitive to the co-text for their acceptability makes it particularly important to look at both syntactic possibility and discourse constraints when studying fronting of different types of elements. Also, since the main focus of this study is IS-fronting as defined in section 1.2, we will mainly describe the characteristics of this structure. Finally, we will focus on fronting in non-embedded contexts.

A notable syntactic difference between the two languages is the verb-second constraint of Swedish. Thus, English is able to combine several constituents clause initially, leading to fronted constructions in the two languages having different syntactic characteristics.

Looking at the word order across main clause types in English, leaving left-dislocated elements aside, we can observe that in English imperatives (15), and polarity interrogative clauses (16) cannot place any arguments before the finite verb, whereas constituent interrogatives are limited to placing a single *wh*-element in this position, regardless of it being an argument (17) or an adjunct (18). In effect, constituent questions exhibit verb-second word order, characterised by Rizzi (1990, 1996) as *residual verb-second*, since English has historically been a verb-second language. Declarative clauses (19a) are the only clause type that can combine the external argument with an internal argument (19b) or an obligatory adverbial ahead of the finite verb.

(15) a. Buy him the paper!
b. * **The paper** buy him!
c. * **Him** buy the paper!

(16) a. Has he bought the paper?
b. * **The paper** has he bought?

(17) a. **Who** has bought the paper?
b. * **The paper who** has bought?

(18) a. **Why** has he bought the paper?
b. * **The paper** why has he bought?

(19) a. **He** has bought the paper.
b. **The paper he** has bought.

Comparing the main clause pattern in English and Swedish, we note that although the verb-second constraint in Swedish is a fundamental syntactic difference, its effect on the possible types of pre-finite structure is limited to declarative main clauses. In imperatives, polarity interrogatives, and constituent interrogatives, the possibilities are the same, no obligatory element can precede the finite verb in imperative or polarity questions in both English and Swedish or the *wh*-element in constituent questions.

In Scandinavian syntactic tradition, the restrictions on placing elements before the finite verb are formulated in terms of sequential areas, not hierarchical structure. The clause is described as consisting of three areas: the one before the finite verb is called the *initial field* (*Sw. initialfält*) (Teleman et al., 1999, Vol. 4, p. 5ff), with the two following being the *middle field* (*Sw. mittfält*) and the *end field* (*Sw. slutfält*)(Teleman et al., 1999, Vol. 4, p. 5ff). The latter two are collectively set apart from the pre-finite area by being referred to as the *clause core* (*Sw. satskärna*). The *initial field* can be filled or empty, and the elements placed in this field are either subordinating conjunctions or an element functioning as a clause element. When the pre-finite position consists of a clause element, it is called a *clause base* (*Sw. satsbas*).

Imperatives (20b, 20c) and polarity interrogatives (21b) cannot have any clause bases (Teleman et al., 1999, vol. 4, p. 11), and constituent interrogatives and expressives are restricted to a *wh*-interrogative element (22b) and an expressive element (23b, respectively (Teleman et al., 1999, vol. 4, p. 411). However, in declarative clauses, where the ability to combine elements in English makes this sentence type distinct in the two languages.

(20) a. Köp honom tidningen!
Buy him paper.DET
Buy him the paper!
b. * **Tidningen** köp honom!
Paper.DET buy him
c. * **Honom** köp tidningen!
Him buy Paper.DET

(21) a. Har han köpt tidningen?
Has he buy.SUP paper.DET
Has he bought the paper?
b. * **Tidningen** har han köpt?
Paper.DET has he buy.SUP

(22) a. **Vem** har köpt tidningen?
who has buy.SUP paper.DET
What paper has he bought.
b. * **Tidningen** har vem köpt?
the paper has who buy.SUP

(23) a. **Varför** har han köpt tidningen?
Why has he buy.SUP paper.DET
Why has he bought the paper?
b. ? **Tidningen** har han köpt varför?
Paper.DET has he buy.SUP why

(24) a. **Han** har köpt tidningen.
He has buy.SUP paper.DET.
He has bought the paper.
b. **Tidningen** har han köpt.
Paper.DET has he buy.SUP
The paper he has bought.

It should be pointed out that the ungrammaticality of (21b) rests on its interpretation as an interrogative; as a declarative, this structure is fine. Also, the structure in (23b) is marginally possible as an in-situ constituent question.

The data in (16) – (24) show that in both English and Swedish, it is primarily in declarative clauses where there is a possibility to front an element of the argument structure in main clauses, and it is unclear from only looking at the fronted construction if there is any systematic difference in their communicative interpretation and discourse function.

However, a well-known characteristic of Germanic verb-second languages is the fact that the word order is different in subordinate and root clauses. By comparing the order of the finite verb and a sentence adverbial in main and subordinate clauses in the two languages, a key difference between a V2 and a non-V2 language is revealed. In English, the order in both main and subordinate clauses is to have the auxiliary before the adverbial (25a) and (25b), whereas in Swedish, the auxiliary is before the adverbial in the main clauses (26a) but after in the subordinate clauses (26b).

(25) a. He **has not** bought the paper.
b. (I think) that he **has not** bought the paper.

(26) a. Han **har inte** köpt tidningen
He has not buy.SUP paper.DET
b. (Jag tror) att han **inte har** köpt tidningen
(I believe) that he not has buy.SUP paper.DET
(I think) that he has not bought the paper.

Although the order of auxiliary and adverbial does not appear to have a major impact on the communicative interpretation of declarative sentences, this difference has been a key piece of data in the syntactic theorising of the structural properties of clauses in general and will be a central part in our analysis of the differences in communicative function of English and Swedish IS-construction. The theoretical implications will be discussed in more detail in chapter 3.

To sum up, at the surface level, it seems that the main difference in English and Swedish fronting is the order of the finite verb when a non-subject argument is fronted, resulting in the difference shown in (27) and (28).

(27) The newspaper he has bought.

(28) Tidningen har han köpt.
Newspaper.DET has he bought.
'The newspaper, he has bought'

Part of the central argumentation in this thesis is that the underlying structural difference between these clauses, which is revealed by the difference in the order of finite verb and sentence adverb in subordinate clauses, affects the discourse functions of fronting in the two languages. Moreover, we cannot consider the meaning of these sentences and their degree of well-formedness, or felicity, without assigning them an implicit prosodic representation and some co-text, e.g., placing the nuclear accent on the fronted object or on the final verb. The differences in the way syntactic structure is mapped onto the phonological representation in the two languages will have somewhat different interpretative effects and restrictions, one of which is the possibility of fronting objects in Swedish in contexts that do not require the fronted object to be interpreted as contrastive (Hörberg, 2018; Lindahl & Engdahl, 2022).

1.4.3 Types of fronted elements and relation between fronted and in-situ position

Having established that in English and Swedish, fronting in main clauses is limited to declaratives, our next step is to give an account of the type of phrases and clause elements that are able to front. In addition, the relation between the fronted element and its canonical or in situ position should also be determined.

Although fronting appears to be restricted to a fairly limited range of discourse contexts in both languages, particularly in English, there are, somewhat paradoxically, relatively few morphosyntactic restrictions on what can be fronted (for English, see Biber et al., 1999; Huddleston and Pullum, 2002; Quirk et al., 1985, and for Swedish, Teleman et al., 1999, vol. 4, p 413ff). This suggests that the constraints are not primarily syntactic, or at least not only syntactic, as has been observed for both English (Birner & Ward, 1998) and Swedish (Brandtler & Håkansson, 2012, p.75). In both languages, it is possible to front practically all types of clause elements and phrasal categories. According to Birner and Ward (1998, p. 45ff), fronting is not limited to any particular phrasal category, which they illustrate with the following examples of fronted NP (29), PP (30), VP (31), and AP (32), Birner and Ward's examples (40a – 40d) respectively:

(29) *One of these rugs* Chambers delivered to Harry Dexter White.

(30) *For that last bold assertion* there are no statistics.

(31) ... and *pass* I did

(32) *Humble* they may be. But *daft* they ain 't.

Moreover, fronting is possible not only from constituents of the main clause but also from elements of embedded subordinate clauses, such as demonstrated by Jackendoff (1972, p. 128) in example (33).

(33) Beans$_i$, you'll never persuade me to force my kids to eat _____ $_i$.

Here, the fronted DP *beans* is the argument of the verb *eat*, which is embedded as an argument of the verb *force*, which in turn is embedded in the root clause. An example from Quirk et al. (1985, p. 1378n.) illustrates the same point in (34.

(34) *Everything – or nearly everything – that the Labour movement exists to stop the Tories from doing*$_i$, Labour will be asked to support the Cabinet in doing _____ $_i$.

In (34) the fronted element is again twice embedded; moreover, the fronted element is a quantified DP, not a referential expression (see Lasnik and Stowell (1991) for the claim that fronted quantified DPs are ungrammatical and the counterarguments by Postal (1993) and Ruys (2004)). These examples show that embedding as such does not seem to be a constraining aspect of fronting in English, and the same situation is also the case for Swedish, as shown in example (35), taken from Teleman et al. (1999, vol. 4, p. 421)

(35) Den boken vet jag att hon har läst
That book.DET know I that she has read
'This book I know that she has read.'

The possibility in Swedish of fronting clause elements from subordinate clauses in Swedish was in the 1980s (L.-G. Andersson, 1982; Engdahl, 1982), as it contradicted the claim that such displacement should be universally prohibited (Ross, 1967). As this change in analysis illustrates, the theoretical analysis of fronted constructions within a generative framework has developed over time (Chomsky, 1977; Baltin, 1982, among others) arriving at an extended CP analysis following Rizzi (1997) in the late 90s, which has been influential and generated solutions based on the ideas of Rizzi, as in Haegeman (2004, 2012). The current study will be fairly agnostic as regards the formal characterisation of a fronted structure, apart from deviating from Rizzi (1997) in not regarding movement to the left-periphery of the clause to be driven by specific information-structural features. However, this will be discussed in more detail in 3.

At the same time, it is recognised that fronted structures are pragmatically marked by requiring specific discourse contexts to be felicitous (Birner & Ward, 2009, p. 1167).

1.5 Discourse function of argument fronting in English and Swedish

The phenomenon of placing some verb-licensed constituent in sentence initial position for various discourse functional reasons is well established in languages with more flexible word order (Firbas, 1971) and has been observed in the literature for English at least since Hockett (1958) and given both formal (Ross, 1967) and functional treatments (Halliday, 1967) since the late 60s. In Mainland Scandinavian, its discourse function has been studied by among others L.-G. Andersson (1982), Lehti-Eklund (1984), M. Johansson (2002), Rahkonen (2000), Hörberg (2018), and Lindahl and Engdahl (2022).

The current section attempts to give a short overview of the different treatments of this structure in terms of its discourse function in English and Swedish, with some comparison to German, to place the current investigation in a research context that can give an indication of what are foundational and what are issues of contention.

1.5.1 Discourse function of argument fronting in English

The construction of argument fronting is often referred to as topicalisation, which was also the term used by Ross (1967) in his seminal thesis, which was highly influential in establishing the terminology for different types of syntactic structures. However, the name suggests that a topic construction's main function is to introduce a non-subject constituent as the current topic, i.e. the clause element about which the rest of the sentence makes some assertion (see section 2.6 for a discussion on the efforts to define the information structural function of topic in a more comprehensive way). This was recognised to be too simplistic by the existence of object-fronted structure where the object was focal (Gundel, 1977; E. F. Prince, 1981b) or contrastive (Chafe,

1976) or both. A key object therefore became to account for the different discourse functions of this construction and the contextual conditions required to use this construction with different functions.

In their corpus study of spoken English,Gregory and Michaelis (2001) classify fronted constituents using Gundel et al.'s (1993) Givenness Hierarchy. Their comparison is mainly between topicalisation and left-dislocation, and their dataset consists of both full NPs and pronominal elements. Their empirical findings show that unstressed object pronouns (e.g., *it*, *them*) are strictly excluded from fronting. When pronominal elements such as demonstratives (*this*, *that*) are fronted, they systematically carry prosodic stress. From a theoretical perspective (cf. E. F. Prince, 1981b, 1986), this structural organisation provides a dual-focus template, allowing prosodic prominence to fall simultaneously on both the preposed element and the clause-final focus. However, the acceptability for using these constructions varies considerably depending on the sentential and discoursal co-text in which they are used.

1.5.1.1 Birner and Ward 1998

Among the more comprehensive investigations of complement fronting in English are Ward (1988) and Birner and Ward (1998). The former is exclusively focused on fronted elements, or preposing, whereas the latter investigates various information structural constructions in English, including complement fronting. Birner and Ward develops the ideas in Ward (1988) and attempts to arrive at a general explanation of the correspondence between various word order phenomena in English and their discourse functions.

The data in the two studies comprise recorded observations of oral and written discourse, and thus include naturally occurring language usage and descriptions of the discourse context. The data are described as consisting of several thousand tokens of non-canonical structures collected over a period of approximately ten years, and are characterized as standard American English. It represents both written and spoken data from a wide range of genres and includes random samples as well as data from compiled corpora like the Brown Corpus. Birner and Ward point out that since the data collection is a database of particular constructions, they can make no claims of representativeness, frequency, or distribution in natural language. They claim, however, to be able to say something about what occurs and does not occur in English based on their data.

Birner and Ward (1998) is an important study that provides one of the most comprehensive investigations of information structure on naturally occurring data in English that has been undertaken. However, it is a rather bold claim to be able to draw conclusions about what structures do not occur in English from a database of occurrences of fronting. Even if they can demonstrate that alternative constructions would have been infelicitous in their recorded tokens, it is impossible to make general claims about their impossibility only on the basis of the non-occurrence of certain structures in their data.

Ward (1988) provide a great deal of statistics on the frequency of various constructions, but Birner and Ward (1998) only occasionally present any numbers on their data. Although it is not explicitly stated that the two studies use the same database, the figures that are included in the later work all seem to correspond to statistics in the earlier one, and it is therefore assumed that the two studies are using the same database. The database contains a number of different structures, but those that include preposed constituents are reported to be 915 tokens.

In addition to investigating the collected examples of constituent displacement, Ward (1988) conducted a small corpus study to test the hypothesis that the lack of prosody would make preposing more frequent in writing as a way to signal relations that are prosodically marked in speech. Ward looked at a written text by Richard Nixon and compared it with transcripts of his speeches, but found that the situation was the opposite with only .3% of the written sentences

having preposed constituents whereas the prepared speech had .7%. It is also worth noting that both had a frequency below 1%.

Ward (1988) and Birner and Ward (1998) show that any type of constituent can be preposed, although the majority are noun phrases. From an information structural point-of-view complement fronting in English can be categorised into two general types:

- A preposed element functions as topic (topicalization)
- A preposed element functions as focus (preposed focus)

The distinction between these types derives from the analysis made in E. F. Prince (1981b) where preposed focus is characterised as the fronting of an element that is "new" in the discourse and given nuclear stress, whereas topicalization is the preposing of any non-focal element which is often de-accented but could receive secondary stress, whereas the nuclear accent is on the non-preposed focal element. In the data, topicalizations account for approximately 64% of the tokens and preposed focus slightly over 9%.

In addition, the database also includes structures with non-subject preposed elements and an inversion of the subject and finite verb resulting in "full inversion" structures, such as "In the middle was a tent", which is of interest from a contrastive perspective as it mirrors the typical structure found in V2-languages.

According to Birner and Ward both topicalization and preposed focus share the general constraint of requiring that the denotation of the preposed element should be "anaphorically linked to the preceding discourse" (Birner & Ward, 1998, p. 32). They also conclude that to be felicitous, a preposing construction requires an open proposition to be salient or inferrable from the context. The open proposition is defined as a proposition containing one (or possibly several) variable, which is being instantiated by the focus of the utterance. The open proposition is consequently the presupposed information in the sentence. Following E. F. Prince (1986), Birner and Ward (1998) argue that the preposed structure is used to make the open proposition salient.

A preposed structure can thus be analysed into minimally the following elements: an open proposition, a focus element, and a partially ordered set (poset) which links the preposed element with the previous discourse. An utterance like (36a) will then be analysed as in (36b)(Birner and Ward's (11))

(36) a. I think she was Japanese. No – *Korean she was.*
b. OP: She was X, where X is a member of the set {nationalities}

The connection between the preposed element and the previous discourse can be of many different kinds, but they can all be described as a poset relation. The relation is defined as being formed between a denotation in the previous context (trigger) and the preposed element (link) by both being members of a partially ordered set. In Birner and Ward (1998) this is exemplified by the following example (their (22)):

(37) Customer: Can I get a bagel?
Waitress: No, sorry. We're out of bagels? *A bran muffin I can give you*

In (37) the preposed "A bran muffin" is the link that inferentially evokes a poset by standing in a relation with the trigger "bagel" which Birner and Ward suggest is {breakfast baked goods}. However, the relation between the link and the preposed element is sometimes less direct, and more inferential steps are required to associate the two parts. An example of this is shown in (38) (their 27)).

(38) A: Where can I get the reading packet?
B: In Steinberg. [Gives directions] *Six dollars it costs.*

Here the link "six dollars" requires a way to connect to the common ground despite the fact that the previous utterance does not include a referent that could be inferred to belong to the same poset. This can be contrasted with (37) where the link "bran muffin" and the trigger "bagel" could be construed as belonging to the poset {breakfast baked goods}. Inferring a poset in (38) thus requires more inferential processing. A's question indicates A's interest and assumed intention of buying the reading packet. B will therefore assume that in addition to knowing where it can be acquired, it is also relevant for A to know its cost.

Birner and Ward correctly observe that A's question makes the price of the book salient, but claiming that pairing it with the preposed "six dollars" evokes a poset of {prices} requires a more detailed explanation of how links and triggers can be related. Some aspects of the topic "reading packet" seem to be available to act as implied triggers, such as weight (39a), and length (39b), but not other information related to the contents of the reading packet (39c), judging from their felicity.

(39) A: Where can I get the reading packet?
B: In Steinberg.
a. 2 pounds it weighs.
b. 600 pages it contains.
c. # All the texts it includes

It is also worth noting that none of the fronted responses will be felicitous if there is an explicit question from A, asking for the information in the fronted element.

(40) A: Where can I get the reading packet?
B: In Steinberg.
a. A: How much does it cost?
B: #Six dollars it costs.
b. A: How much does it weigh?
B: #2 pounds it weighs.
c. A: How many pages does it contain?
B: #600 pages it contains.

Although part of the definition is said to rely on the givenness of the discourse referent, the difference between topicalization and preposed focus is often more based on prosodic marking and internal sentence relation than on referential givenness.

According to Birner and Ward, example (41) (Birner and Ward's (29)) shows a topicalization, where "baseball" is the topic and "better" is the focus.

(41) G: Do you watch football?
E: Yeah. *Baseball I like a lot better*

The basis for this analysis is that E's answer prosodically marks both the preposed element "baseball" and the final word "better" which indicates that "better" is focus, and in cases where an element other than the subject is fronted, it will form a separate tone unit that is given nuclear accent. The topic status of "baseball" is achieved through being a link to the previous context by forming the poset {sports}, whereas the piece of information being added to the common ground is 'degree of liking'. This analysis is somewhat problematic. Had the initial question been more explicit, and if B's qualification had not been preposed as in (42), it is difficult to see why not both the specific sport and the degree of liking would be focus.

(42) Do you like football?
Yeah. (But) I like baseball a lot better.

In their data of topicalized sentences, Birner and Ward find five types of linking relations between the preposed element and the previous context: set/subset, part/whole, type/subtype, greater-than/less-than, and identity (Birner & Ward, 1998, p. 45). The great majority consists of fronted NPs, which appear to have fewer constraints on their usage than non-NP fronted constructions. Preposed Verb phrase (VP) and Adjective phrase (AP) primarily consist of structures that assert or negate the truth of the proposition of which the preposed element is part, placing the end focus on an auxiliary verb, as in (43) and (44).

(43) ...it was necessary to pass, if I was to stay at Oxford, *and pass I did* (VP)

(44) *Humble they may be, but daft they ain't* (AP)

These structures are consequently fairly restricted in the way they can be used. In the vast majority of the examples, AP fronting requires a combination of two preposed constructions that are contrasted, as the example in (44). In contrast, the majority of non-AP do not occur in environments with contrasting preposed structures.

Birner and Ward (1998) give a great deal of attention to describing these types of preposed VP and AP topics, which they call "proposition assessment". They identify three different types, "proposition affirmation", "proposition suspension" and "proposition denial", shown in (45.

(45) a. ...it was necessary to pass, if I was to stay at Oxford, and *pass I did.* (PROPOSITION AFFIRMATION)
b. Mark submitted his report late, *if submit it he did.* (PROPOSITION SUSPENSION)
c. *Light as a feather this bird is not.* (PROPOSITION DENIAL)

The two first types "proposition affirmation" and "proposition denial" requires that the trigger is linguistically realized in the previous discourse and that it is morphologically similar to the preposed element, e.g. "...necessary to pass...pass I did" in (45a) and "..submitted..if submit it he did" in (45b). Birner and Ward show that it is possible to make an even finer categorization by dividing the "affirmation"-constructions into three sub-categories.

(46) a. We went to Canada to learn, and *learn we did.* (LOGICALLY INDEPENDENT PROPOSITION AFFIRMATION)
b. Waiting in long lines to pay someone else more money than they seem to be entitled to is lunacy. But *wait in line they did.* (CONCESSIVE PROPOSITION AFFIRMATION)
c. They stopped when they hit the mud. And *hit the mud they did.* (SCALAR AFFIRMATION)

These structures all share the characteristic of repeating a previously expressed predication to emphatically affirm or refute it. It is clear that a non-preposed structure would not be felicitous, unless it includes some type of focus adverb such as "indeed" or "certainly", as in (47).

(47) They stopped when they hit the mud. And they DID/did INDEED/CERTAINLY hit the mud.

The "proposition denial" differ in this respect since it cannot have an explicit phrase that expresses the link, but instead the link has to be inferred, as is shown in (48).

(48) The NBA's new collective-bargain agreement sounds as though it was written by the same people who put together the Internal Revenue Service's long form. *Simple it is not*

This construction generally receives an ironical interpretation caused by the use of understatement. This type of "affirmation"-construction is, however, similar to the other two categories by placing the focus on the negation or affirmation of the predication.

Birner and Ward (1998) argue that all preposing constructions, regardless of whether they front a focus or a topic element, share the property of being "discourse-old", i.e. part of the "common ground", either by explicit mention in the previous discourse or by inference. In this way Birner and Ward are able to give a more general explanation for the properties of preposed elements. However, it requires that a fairly wide range of relations between the preposed element and the previous discourse can be accommodated within the same concept of "discourse-old". This requires a more elaborate theory of both "discourse-statues" and inference making. In terms of developing the idea of "discourse-old", Birner and Ward argue for the distinction between being "discourse-old" and properties such as specificity and definiteness.

Although no clear consensus has been reached on the discourse-functional possibilities and distribution of object fronting, one of the more detailed analyses for object fronting in English can be traced back to E. F. Prince (1981b) and has become part of the reference literature of the English language (Huddleston and Pullum, 2002, ch16). It was developed in the works of E. F. Prince (1981a, 1984, 1992, 1999), Birner (1996), Ward (1988), Ward and Prince (1991) and Birner and Ward (1998)). It attempts to explain the discourse functions of fronting and the conditions that make this construction felicitous. The most developed account is the one in Birner and Ward. In the more descriptive Huddleston and Pullum (2002, p. 1372), the constraints are reduced to two: (a) the fronted complement must be previously established in discourse, and (b) act as a link to some other entities previously evoked. However, Birner and Ward (1998) develop these descriptive features into a theoretical analysis that attempts to explain this in terms of information status.

1.5.2 Discourse function of argument fronting in Swedish

Corpus-based studies of argument fronting in Swedish is relatively rare and generally focuses on statistical and distributional data, rather than aspects of discourse status or information-structural functions, which often have methodological reasons.

Westman (1974) conducts a corpus study of different genres of written Swedish where she presents the frequency of clause-element types preceding the finite verb in Swedish main clauses. Although Westman, in contrast to the current study, include reported clauses that precede the reporting clause as object, the percentage of fronted objects in most genres is between 1.3 and 2.3, with only newspaper text including 4.2; however, half of those are reported speech clauses.

Among the examples of object-fronting presented by Westman, we only find a minority that include any anaphoric elements. Since no contexts are provided, it is hard to estimate the way the fronted elements are linked to the previous discourse. However, most of the examples do not have any obvious requirements of contrastive contexts, although a common characteristic of many of these examples is that they include a sentence final adjunct which expresses high communicative dynamism and often a salient or highly generic subject, as illustrated in the examples in (49.

(49) a. Krav på disciplin möter du på de flesta andra områden i samhället.
Demands for discipline encounter you on the most other areas in society
'You will encounter demands for discipline in most other areas of society'

b. Folkpensionen får man oberoende av tidigare arbetsinkomst.
National pension recieve one independent of previous work.income
You will receive national pension independently of previous earned income

c. Sådana reformer har Kina och Tanzania just genomfört.
Such reforms have China and Tanzania just carried out
China and Tanzania have recently implemented such reforms.

Intuitively, both (49a) and (49b) would be possible also with canonical word order even if the link between the fronted element and the previous context might be somewhat weaker. However, since the focal elements will be sentence final regardless of the position of the object and the fronted element is a full DP and the subject is a generic reference, the difference is slight.

The example in (49c) is different in that the fronted element is anaphoric and is promoted from a canonical sentence-final position. Prosodically, the object is thus moved from a position where is would be post-focal and instead get left-edge prominence and a clearer discourse link.

Rahkonen (2000, 2006) report on a study done on the one million word corpus the Stockholm-Umeå Corpus (SUC) together with the corpus Gymnasistsvenska (GSV) containing 88 757 words. The main aim of the study is to investigate the requirements for topicalization in active declarative clauses in Swedish, and what factors allow or trigger topic-fronting.

Since the surface word order in Swedish is indistinguishable between canonical and topic-fronted sentences when the arguments do not show case, see (50), Rahkonen is particularly interested in the way the ability to distinguish between the sentence types influences the use of topicalization in Swedish.

(50) Kain dödade Abel.
Cain killed Abel.
'Cain killed Abel/Abel Cain killed'

In contrast, using pronominal elements, which have distinct subject and object forms, would remove any clause type ambiguity. Rahkonen reports that the majority of subjects in topicalized structures show case, but recognises that this may be a consequence of a greater use of pronouns as subjects in declarative clauses overall.

One of the most significant results, according to Rahkonen is the distributional properties of the propositional pronoun *det(ta)* which make up a large part of the fronted non-subject in both corpora (55.6% in GSV/38.5% in SUC) and among all instances of the pronouns *(det(ta)* a majority are fronted. A typical example is shown in (51).

(51) Äktenskapet av idag har stora svårigheter, beroende på jäkt, ekonomisk press,
marriage.DET of today has large difficulties, depending on stress, economic pressure,
statusjakt och mycket annat. Detta påpekar både Stig Hellsten och Barbora
pursuit of status and much other. This point out both Stig Hellsten and Barbora
Backberger.
Backberger.
'Today's marriage has great difficulties, caused by stress, economic pressure, pursuit of status and much more. This is pointed out by both Stig Hellsten and Barbora Backberger'

The use of propositional *det(ta)* could be seen as a result of a general use of fronted anaphoric pronouns, but other types such as the personal pronouns *honom* 'him', *henne* 'her', *dom, dem* 'them' show very low numbers of fronting compared to propositional *det(ta)*

Since Rahkonen is limited to searching for formally defined types, he has to approach the question of information-structure as a matter of the inclusion of anaphoric elements in the canonical and fronted objects. If they include anaphoric elements, they are classified as continuous, and discontinuous if they don't. Rahkonen both looks at the number of continuous NPs of all fronted elements and the number of continuous NPs of canonical structures. In the first category, he

finds that approximately 70% of all fronted elements are continuous, whereas only about 30% of canonically placed objects are continuous. Based on these numbers Rahkonen (2006) draws the conclusion that fronted objects are associated with connecting to some antecedent in the previous discourse making them topical, and the motivation for placing them at the left-edge is a way to align the constituents according to communicative dynamism or in a topic-comment progression.

However, some caution should be taken in drawing any firm conclusions from these statistics, since we need to have at least some idea of the anaphoricity/continuity of the other constituents in the sentence first. Looking at the distribution of pronominal subjects in the SUC corpus in fronted and canonical structures, we find that the degree of personal pronouns in the fronted constructions are higher (61.5%) than in the canonical structures (33%), which suggests that in the fronted constructions both the fronted element *and* the subject both in a large number of cases include anaphoric elements. This makes the argument that fronting is a way to place less informative elements before more informative ones questionable. Admittedly in his conclusion Rahkonen formulate a weaker proposition that the primary reason to front non-subjects is "place a continuous, activated, non-subject in sentence-initial position". That leaves the primary question of why this placement is needed unanswered. Rahkonen (2000, 2006) are of interest since they are among the relatively few corpus studies that focus on IS-fronting, and they show both that there is a dominance of propositional proforms among the fronted constituents and give an indication in the example used that Swedish has the ability to front full noun phrases without any obvious contrastive interpretation.

Although it only briefly touches on fronting in Swedish, Vallduví and Engdahl (1996) includes one of the fuller descriptions in terms of observations specific to the discourse-functional properties of these structures in Swedish. Vallduví and Engdahl's use of the term 'topic fronting' is done in a theoretical context where most types of argument fronting are seen as 'links fronting', which is most similar to what Büring (1997, 2016) calls 'contrastive topics', i.e. constituents defined as having a specific L*+H accent. Contrastive topic fronting, which includes all types of fronting in English, is contrasted with Swedish topic fronting which involves fronting a weak, unstressed pronoun, as in (53a), example 94 and 95 in Vallduví and Engdahl).

(52) Igår fick jag en bok.
yesterday 1s-pst-get I a book
'Yesterday I got a book...'

(53) a. Den ska jag läsa ikväll.
it fut I read tonight

b. Jag ska läsa den ikväll.
I fut read it tonight
'I will read it tonight.'

The discourse function is described by Vallduví and Engdahl as fronting the referent that is "most topical or salient" and explicitly distinguished from 'link-fronting' or 'topicalisation' primarily because these latter constructions must have full DPs or strong pronouns, and that in English they will have a pitch accent. Also, the function of 'link-fronting' is to establish some central point of reference for the updating of the discourse model or common ground, which means in terms of thematic progression there is a shift from the reference centre in the previous sentence to a new one. A phenomenon that in other descriptions is referred to as 'shift-topics'. Swedish topic fronting, on the other hand, is primarily a way to continue with the already established reference centre, or a continuation topic. In addition Vallduví and Engdahl associate the use of Swedish topic-fronting with temporal sequencing.

One of the most thorough studies on object-fronting in Swedish spoken discourse is Lindahl and Engdahl (2022), which provides insight into the different discourse functions of object-fronting. It also points out what is probably the most significant difference between Scandinavian languages and other Germanic languages, including Dutch, German, and English, which explains the discourse functions of fronted structures in English and Swedish: in the mainland Scandinavian languages non-subject topics can be fronted without resulting in a contrastive interpretation. However, they do not attempt to explain why this should be the case.

1.5.3 Discourse function of argument fronting in other Germanic languages

Research relevant to IS-fronting in the other Mainland Scandinavian languages Norwegian and Danish has included the phenomenon both in terms of its communicative functions (Norwegian) and how fronting can be analysed theoretically (Danish). In his 2007 book S. Johansson devotes a chapter to corpus research on sentence openings, which is essentially the pre-verbal constituents of a sentence, and highlights a number of studies by Hasselgård (Hasselgård (1997), Hasselgård (1998), Hasselgård (2004b), Hasselgård (2004a)) that have collected data on Norwegian pre-verbal structures and the way they have been rendered in English. A number of observations indicate that there are similarities between Norwegian and Swedish fronting such that a greater frequency of initial non-subject elements in Norwegian and that these elements are less marked than is the case with fronted non-subject elements in English.

As the current study bases its argumentation on the different discourse-functional properties on the syntactic difference between Swedish as a verb-second language and English as a predominant subject-first language, it is relevant to look at what has been found when studying other verb-second languages. German is fairly well-investigated from this perspective among the Germanic language.

In Fanselow et al. (2008, p. 414), they state that the possibility to place any element clause initially in German is "textbook knowledge" but that subject and object initial sentences differ in terms of sentence processing, with the latter being harder to comprehend. The reason has been suggested to be a greater number of constraint violations (Keller, 2000). However, Fanselow et al. conclude that the low frequency of object-initial constructions in German corpora may also be the effect of the restricted context in which these structures can be used felicitously. In order to gain greater insight into the pragmatic conditions that allow object-initial statements, they conduct a number of controlled acceptability rating experiments. The conclusion is that object fronting in German is possible not only when the fronted object carries narrow focus, but also in context where focus is broader, such as on the entire VP.

1.6 Aims

One of the aims of this investigation is to find out whether the intuition that Swedish fronting is less contextually restricted than its English equivalent has any empirical grounding in the corpus material used in this study. However, more importantly, the aim is to establish the syntactic, phonological and semantic/pragmatic properties of these structures in English and Swedish in order to arrive at the way fronted structures interact with their contexts to create coherent, felicitous discourse, how their properties differ from non-fronted declarative sentence in English and Swedish respectively, and what the similarities and differences are between the equivalent fronted structures in the two languages. More specifically the following research questions will be investigated:

1. What types of fronting construction do we find in Swedish and English written texts?
2. How are fronted constructions represented in the other language, and what does that tell us about the meaning and use of these structures in English and Swedish?
3. What restricts the use of fronted constructions in English and Swedish?
4. How can the restrictions in using fronted constructions be explained in terms of general linguistics principles in a theoretical model?

1.7 Scope

The empirical scope of this study focuses on the syntactic, phonological and discourse-pragmatic properties of non-subject fronting in English and Swedish. However, because fronting in initial position encompasses a heterogeneous set of syntactic constituents, establishing clear empirical boundaries requires a systematic distinction between different types of preposed elements. In particular, a fundamental structural and functional divide exists between the preposing of core clause participants, arguments or complement, and that of peripheral modification or adjuncts. The following subsection details the theoretical and diagnostic criteria used to distinguish argument fronting from adjunct fronting, delineating the exact constituent types that fall within the scope of this investigation. It also briefly discuss the way indirect speech has been handled.

1.7.1 Argument vs. Adjunct fronting

The current study delimits the investigation to constructions, where the fronted item is either the main verb and/or a structurally obligatory element licensed by the verb. In terms of clause elements, this includes: verb, subject, any objects or predicative complements, and verb-licensed adverbials. The terms generally used to refer to these elements are either predicate - argument or verb - complement. These concepts are related, but have different theoretical foundations. Still, they capture the same fundamental distinction between linguistic elements that are dependent on the verb, *arguments/complements*, and those elements that are structurally freer and generally optional, which are normally termed *adjuncts* (note, however, the different use of the term in Quirk et al. 1985).

The main interest is on the constituents of the clause structure that express the representation of the state-of-affairs being presented in the clause, i.e. the who-does-what-to-whom. We have different conceptualisations of this core structure of the clause, Systemic-Functional Linguistics calls this 'transitivity' or the ideational component, whereas within formal linguistic theories it is generally covered by the predicate and its internal argument or the verb and its complement.

However, although the principle idea of how to distinguish complements from adjuncts is reasonably robust, it is not always straight-forward to draw clear categorical boundaries between the two. Methodologically, we need a way to operationally select which constituents to include and which to exclude when collecting our corpus data. The standard approach has historically been to identify properties that can be tested diagnostically and on the basis of which the separation of complements or adjuncts has been made. The three main reference grammars of the English Language (Biber et al., 1999; Huddleston & Pullum, 2002; Quirk et al., 1985) present a set of distinguishing criteria. Section 1.7.2 will give a brief overview of these and present a number of cases where it has been necessary to consider different possible analyses, and brief discussions about the arguments for the analyses used.

1.7.2 Complement criteria

The motivation for making a categorical distinction between elements that are licensed by the verb, as well as the verb itself, from other constituents when it comes to clause initial placement is based on the assumption that the close relationship that exists between the verb and its complements has a fundamental impact on the ability to place complements in other positions than in its canonical position. Adjuncts, on the other hand, can more freely be placed in different positions in the clause; they are often optional, and can occur together in various structural positions. The difference in constraints on word-order variation has also been observed by Birner and Ward (1998).

A central distinction has to be made between argument/complements on the one hand and adjuncts on the other. Following Huddleston and Pullum (2002, pp. 219), the following criteria are taken as distinguishing between the two: licensing, obligatoriness, anaphora, and position.

1.7.2.1 Licensing

The type and number of other necessary constituents in the clause are decided by the properties of the verb. As can be seen in (54) the verb *mention* permits a nominal object, whereas the verb *allude* does not. The verb *mention* is said to license the complement/arguments of the clause (Huddleston and Pullum (2002, examples 11 i-ii).

(54) She mentioned the letter.

(55) * She alluded the letter.

A complicating factor in establishing the number and types of complements for a particular verb is the fact that individual verbs can often be constructed with different patterns. Some verbs that normally take a nominal can also be used without any complements/arguments at all. Thus, a range of diagnostics is needed to arrive at a classification.

(56) He is eating.

(57) He is eating a sandwich.

1.7.2.2 Obligatoriness

This criterion is closely related to the notion of licensing but Huddleston and Pullum express the difference as a matter of *allowing* or *requiring*. Licensing captures the property of the verb, *allowing* certain elements in the clause, whereas obligatoriness *requires* certain elements to form a grammatical sentence. The difference between obligatory complement, optional complement and optional adjunct is illustrated in (58a – 60b).

(58) a. She perused the report. (obligatory complement)
b. * She perused.

(59) a. She read the report. (optional complement)
b. She read.

(60) a. She left because she was ill. (optional adjunct)
b. She left.

1.7.2.3 Anaphora

The possibility of using the anaphoric expression *do so* to refer to a verb and additional constituents indicates that the constituents are complements/argument rather than adjuncts.

(61) *Jill keeps her car in the garage but Pam does so in the road.

(62) Jill washes her car in the garage but Pam does so in the road.

Constituents of a clause that can be combined with the *"do so"*-expression can be identified as adjuncts, but those that fail cannot be conclusively categorised as complements, as some adjunct will also fail the *"do so"*-test.

1.7.2.4 Position

A significant criterion for the current study is the relative positional fixedness of complements compared to adjuncts. Normally the complements will receive some additional implicational meaning when not placed in its canonical position and would have more restricted application in terms of contexts. Consequently, (63a) will be more restricted than (63b).

(63) a. To Kim he gave the beer. (complement)
b. In the afternoon, we played tennis. (adjunct)

An important point in this context, and one that is reiterated throughout this study, is that most positional changes will result in differences in discourse congruence. Thus, it is often claimed that as a response to a question of 'What did you do in the afternoon?', the sentence in (63b) is a more felicitous answer than having the time adverbial in final position, i.e., 'We played tennis in the afternoon'. However, this also depends on the phonological realisation. Since the default placement of nuclear accent is on the final content constituent, we tend to apply that also in silent reading as argued by Fodor (2002a), which creates the impression of incongruity. However, if the nuclear accent is placed on 'tennis' rather than 'afternoon' as indicated in (64), and 'in the afternoon' de-accented, this improves the acceptability.

(64) Q: What did you in the afternoon?
A: We played TENNIS in the afternoon.

Consequently, this diagnostic should be treated as all those listed here, as an indication of a constituent's complement status, which, in combination with other diagnostics, can provide the basis for a classification. Not as a criterion that on its own can identify a constituent being a complement or adjunct.

1.7.2.5 Arguments

The difference between the notions of complement and argument is often described as the former being a structural concept and the latter a semantic one. Thus, the structure of the lexical verb and its complement is represented semantically as a predicate and its arguments. Although this distinction is often, but not always easy to maintain, there is generally a correspondence between the structural and semantic relation between verb/predicate and complements/arguments. Huddleston and Pullum (2002) take this as another way to distinguish complements from adjuncts. By identifying the semantic relation expressed by the verb, it is possible to identify the arguments involved in that relation. Thus in a sentence like (65) *read* is the relation and the arguments are *he* (*agent*) and *the paper* (*patient*).

(65) He always reads the paper before breakfast.

This is in contrast to *before breakfast* that is not involved or assigned a role in the 'reading' relation.

Still, the correspondence between complement and argument sometimes breaks down as with complements that lack semantic content such as the subject of weather verbs like *it is raining* where there is an explicit complement *it* but the predicate does not have any corresponding argument.

Moreover, in structures with raising verbs (Postal, 1974), such as *seem*, as illustrated in (66), the superordinate verb assigns no argument role to the subject, which instead receives an agent role from the subordinate verb *mislead*.

(66) Pat seems to have misled them.

These limitations in complement/argument correspondence and the general issue of unambiguously establishing semantic relations make this a criterion with some restrictions, but it contributes as one more element in the palette of diagnostics to tease apart adjuncts and complements.

1.7.2.6 Selection and Role

As in the case of the criterion of argument, the selection and role aspects are semantic in character and provide some additional indication of a constituents structural and semantic status. The idea is that a verb will delimit the arguments based on its semantic properties. Thus, it will require or select arguments that have certain features, e.g. Huddleston and Pullum (2002) points out that verbs like *frighten* and *enjoy* need an animate subject to be felicitous.

Although this restriction can be circumvented with the right context, there needs to be some constituent that can be identified as the one experiencing the fear or the enjoyment, which connects to the previous argument criterion. Moreover, the verb will assign thematic roles to its arguments and will consequently select a number of arguments with compatible roles, as well as other properties compatible with a particular verb, such as having the ability to be frightened or to enjoy something.

These criteria will be used to distinguish between complements and adjuncts, and in the corpus study restrict the collection of data to those where there is a fronted argument/complement. In a small number of cases, additional considerations have to be taken, as will be discussed in the next section.

1.7.3 Analysis of indirect speech

The diagnostic tests in section 1.7.1 generally provide reliable criteria for operationally distinguishing between adjuncts and arguments, but in some instances the categorisation does not conclusively establish the role of the fronted element. The analysis and motivations to include or exclude these structures in the current study are presented in this section.

1.7.3.1 Indirect speech

One issue with interpreting the frequency and distribution of data in some of the corpus investigations that have been conducted on English and Swedish word order is the way sentences of direct speech have been analysed, especially where the subject and the reporting verb is placed after the speech clause, as in (67).

(67) Tom is an idiot, he said.

One approach is to analyse the syntactic structure of direct speech as analogous to indirect speech with the subordinator left out, as shown in (68).

(68) a. He said that Tom is an idiot.
b. He said ~~that~~ Tom is an idiot.

Following such an analysis, the structure in (67) would qualify as an example of argument fronting. The fact that this type of structure is quite frequent in fiction texts, such an analysis would significantly increase the number of argument fronted tokens, and require some explanation for the readiness for this particular argument to front. However, Huddleston and Pullum, 2002, p. 1024, in a discussion of *Indirect Reported Speech*, convincingly argue against regarding the speech clause in direct speech as subordinated. The argumentation is based on the data in (69) and (70).

(69) a. She said that she lived alone.
b. She lived alone, she said.

(70) a. Did she say if I'll be invited?
b. Will I be invited, did she say?

In the (a) sentences the reported speech exhibits three grammatical properties not found in the (b) sentences. They are syntactically subordinated with an initial subordinator (*that/if*); the subordinate clause is obligatory (unless the sentence occurs in a context where the complement clause can be elided), and it is licensed by the main verb *say*. In comparison, the (b) sentences show no syntactic subordination; they do not require the reporting frame, and the reporting clauses are not licensing the reporting clause. In the (a) cases the reporting frame does not form a constituent in the sense normally assumed of constituency in formal syntactic theories in that the external argument *she* combines with only the verb, not the entire VP. In the (b) sentences these incomplete structures *she said* and *did she say* are instead regarded as parentheticals.

Huddleston and Pullum reject structures as those in (69) and (70) as preposed on the basis of these formal differences, but also on the difference in meaning in the case of the indirect interrogative in (70). In (a) the question concerns whether a particular statement has been made (did she or did she not say X), in (b) it is about whether the speaker receives an invitation (will I or will I not be invited). An answer of 'no' would therefore have different interpretations in the two cases (No, she didn't say vs. No, you will not be invited). Moreover, parentheticals can follow the reported speech but may also be inserted in the middle as in (71):

(71) When she got home, I told her, we would have to have the locks changed.

A similar observation is made by Quirk et al. (1985, p. 1378) which they suggest indicates that the exceptional readiness of indirect speech clauses to front is because it is, in fact, a dependent comment clause.

The discussion in Huddleston and Pullum (2002) is based on English, but the reasoning is also applicable to Swedish, as is demonstrated by the fact that a similar type of argumentation for Swedish is found in Petersson (2008). Consequently, indirect speech clauses have not been considered to be argument fronted and therefore excluded in the corpus investigation.

1.8 Method

To investigate the syntactic, prosodic, and discourse-pragmatic properties of non-subject fronting in English and Swedish, this study adopts a multi-methodological approach. While empirical

corpus data provides authentic examples of how fronted constructions are distributed in natural discourse, corpus data alone cannot account for the full range of syntactic possibilities, nor can it directly capture prosodic effects on discourse functional constraints. Therefore, the corpus investigation is complemented by formal theoretical analyses and phonological considerations. The following subsections outline the methodological framework of the study, detailing the interplay between corpus data and introspection (1.8.1), the criteria used to evaluate degrees of well-formedness (1.8.2), and the strategies for analysing phonological and prosodic properties within written text (1.8.3).

1.8.1 Corpus data and introspection

The empirical part of the present investigation is collected from the English-Swedish Parallel Corpus (ESPC) and the methodology of comparing how a formally defined structure such as argument fronting is used in discourse in a source language (English and Swedish original written text) with how it is rendered in a target language (Swedish translation of English originals and English translation of Swedish originals) in the equivalent context requires a parallel corpus. However, given that the ESPC at the time when the examples were collected was not tagged or parsed, and the fact that it is not trivial to construct a word pattern that would be able to find all the relevant structures, the corpus had to be searched manually. Paired with the fact that argument fronting as reported by larger corpus studies and corpus-based reference grammars occurs in low numbers, especially in written text, the corpus material is therefore small, and no significant statistical result can be presented.

However, the primary purpose was never to give a complete overview of the range of the types of argument fronted structures in the two languages or to achieve frequency results. Instead, the main motivation for the corpus study was to discover patterns of equivalence between argument fronted constructions in the two languages and in both originals and translations. Thus, in some ways, the use of corpus data in the current study does not comply with the traditional approach to corpus studies. At the same time, the analysis and argumentation for the differences that we observe in English and Swedish data are based on formal linguistic theorising, a line of research that is based on introspection and not reliant on natural language data, including corpus data. The choice of methodology and theoretical model may, therefore, seem unconventional, but is intended to avoid some of the problems found in investigations that rely too heavily on traditional pairing of theory and methodology. Much of the research on the interaction between prosody and morphosyntax that is relevant to this study makes use of some form of generative linguistic theory, and at the same time we have to limit conclusions based on questionable data. There are several examples in the literature where data are classified both as acceptable and unacceptable on less than solid grounds. An example of the former from Bayer and Dasgupta (2014, p. 311), which claims that (72) is unproblematic.

(72) Every dog, I did not want to talk about. Only JOHN's dog I was talking about.

This seems at best to be marginally possible and is not particularly compelling support for the general acceptability of fronted quantified DPs in English. The fact that it cannot be found in corpus data does not improve its status. Conversely, Holmberg (2015) claims that fronted objects cannot be the focus and illustrates it with (73)

(73) *? Allt åt Johan.
Everything ate Johan.
Johan ate everything.

In contrast to English, these types of structures, although rare, are well-formed and occur in Swedish monolingual corpora e.g. "Allt följer du via SVT:s direktsändningar" (M. Andersson, 2014). However, by using a proper noun, *Johan* in this example, the acceptability is affected negatively as this suggests that the subject is a switch topic, which normally should get prosodic prominence. If instead a pronominal element had been used *Allt åt han*, the acceptability would significantly improve.

1.8.2 Types of well-formedness

Well-formedness is a central notion for describing and explaining why linguistic expressions are acceptable to native speakers.At different levels of the linguistic model, we want to be able to categorise its form and usage as either well- or ill-formed. In practice, such judgements are not always categorical, and we end up with both different assessments of acceptability from different informants, and different degrees of well-formedness, e.g. we end up with some examples being marginally acceptable rather than unconditionally acceptable. Given that the criteria for well-formedness will vary depending on the grammatical categories we assess, we use differently defined and termed dimensions of well-formedness.

In the current study, we will distinguish between the standard concept of syntactic well-formedness, *grammaticality* from discourse semantic/pragmatic well-formedness, which is termed *felicity*. The claim that an expression is ungrammatical indicates that this structure is not generated by the morphosyntacic module of grammar, and notationally this is indicated with the standard notion of an asterisk '*', as in (74).

(74) * He are nice.

The notion of felicity concerns the well-formedness of the interaction between sentence and co-text, such that when the use of a particular structure in a specific context does not perform the intended discourse function or is not congruent with the co-text, the sentence is taken to be infelicitous. The term is derived from Austin (1962) where it is used for a range of pragmatic cases where a speech act is not successfully performed but in the literature it is conventionally applied to the use of a grammatically well-formed sentence that is not contextually congruent.

(75) A: Have you cleaned you room?
B: #I do.

Pragmatically, any context-sentence pairing where the communicative function is not successfully performed, or contextually incongruent is deemed infelicitous and marked with a hash sign('#'). In addition to these well-established notions, we also find it useful to characterise differences in felicitous uses along a scale of pragmatic markedness. A pragmatically marked use of a sentence is considered felicitous, but in some way deviating from pragmatic principles. A practical illustration of the differences between felicity and markedness is shown in (76) illustrating infelicity and (77) demonstrating pragmatic markedness.

(76) A: Where is Peter?
B: #I saw HIM in the staff room.
B′: I saw him in the STAFF room.

(77) A: Where is Peter?
B: HIM I saw in the STAFF room.

In (76), the placement of nuclear focus on the pronoun *him* will be interpreted as identification or corrective focus in a context of a question that requires focus on the part of the assertion providing the instantiation of the *wh*-element which is breaking principles of information structure and thus failing to provide a felicitous answer.

In contrast to the answer in (77) the nuclear accent is placed on the locative information and is thus a felicitous answer but pragmatically marked since a fronted element is a salience promoting process which is redundant as the discourse referent is already salient. It is pragmatically marked in comparison with (76 B′).

1.8.3 Phonology and written data

A central part in the analysis of the differences in communicative function of fronting construction in Swedish and English written texts is based on differences in the interaction between the two languages morphosyntactic structure and prosodic phonology. The fact that phonology is employed to carry a substantial explanatory burden for written data, which generally does not include any explicit information about its phonological realisation, needs to be argued for.

An important part of realising information structure in languages such as English and Swedish is prosodic phonology (Büring, 2016, p. 446; Cole, 2025; Myrberg and Riad, 2016, p. 449). Although there are important differences—such as Swedish having pitch accents, both at the level of the prosodic word and intonational phrase—the languages share fundamental properties in marking focus and givenness with intonation (Myrberg & Riad, 2015, p. 141).

Our understanding of what is the focus of a sentence is based on identifying the prosodic structure of a syntactic main clause, in the sense of the division into prosodic and intonational phrases and the position of nuclear accents. Moreover, the deaccenting of given information will affect the contextual constraints and prosodic well-formedness of some syntactic structures. A central research project within prosodic phonology is to establish the way syntactic structure restricts and controls the phonological realisation (Selkirk, 2005).

However, an important part of the syntax-phonology interface is that this interaction is necessary also for the interpretation of written language. This importance of phonology for written language is often assumed but not always explicitly stated or discussed, especially in formal linguistics, and is therefore not always given the descriptive and explanatory significance it should have. However, reference grammars like Quirk et al. (1985, p. 1355) often recognise this and emphasise the importance of phonology in writing, arguing that "intonation and other prosodic features are therefore essential in the processing and receiving of information, whether written or spoken".

The notion of felicity and congruence in both speech and writing is dependent on the discourse recipient being able to assign a prosodic structure to the syntactic representation, e.g. it is necessary to evaluate the position of the nuclear accent in a sentence like (78) to be able to incorporate the information in the sentence into the current discourse.

(78) John ate the pizza.

Placing the nuclear accent (indicated by capitals) on one of the four words in (78) as demonstrated in (79 - 82) will decide the interpretation and felicity of the use of the sentence.

(79) Q: Who ate the pizza?
A: JOHN ate the pizza.

(80) Q: What did John do with the pizza?
A: John ATE the pizza.

(81) Q: Which pizza did John eat?
A: John ate THE pizza.

(82) Q: What did John eat?/What did John do?
A: John ate the PIZZA.

In speech, the prosodic phrasing and accent placement of a sentence will be the result of a combination of its syntactic structure, prosodic phonology, and information structural properties together with the discourse context in which it is uttered to express the intended communicative message of the speaker. The listener will then interpret the meaning of the utterance from the interaction of morphosyntax, phonology, and context. However, in writing there is no explicit phonological information and the reader will have to work out the phonological structure of a sentence only based on morphosyntax and context. Thus, in the absence of any typographical indications of prosodic phrasing or accent placement, the writer has to predict how the reader will render a written sentence prosodically. This places a greater burden on the syntactic structure to guide the reader to the correct phonological form, possibly restricting the use of expressions that are dependent on a non-default prosodic structure in writing.

In psycholinguistics, the role of phonology for the interpretation of written text is well established, as in Fodor (2002a, 2002b) which describes the way phonology influences silent reading in terms of *implicit prosody*, and Bader (1998) outlines the process of mapping a phonological structure to written text as *phonological coding*. Fodor (2002b) argues for a view in which implicit phonology is a central part of the sentence processing of written language.

Furthermore, (McCurdy et al., 2013) provides evidence that readers use implicit metric information to resolve syntactic ambiguities in the parsing of written data; thus, indicating that prosodic information is involved in processing syntactic structure of written texts.

An consequence of introducing implicit prosody in the processing of written data is that it casts doubts on the traditional view of the way syntax and phonology interact. The standard view for some time has been that phonology only realises the linguistic information already included in the syntactic structure, rather than enhancing or influencing it; thus in written media where the language does not have to be audibly externalised, phonology appears to be redundant. However, it is clear that when an orthographic sentence is compatible with several possible syntactic structures, implicit prosody will resolve this ambiguity by assuming the "the most natural (default) prosodic contour" (Fodor, 2002a, p. 112). A sentence such as (78) will be interpreted to have the implicit phonology of (82). This will be of importance when we look at the structures of information packaging, such as argument fronting since these have a prosodic structure that is contextually restricted, i.e. they are not semantically incompatible, but they are infelicitous for the pragmatic effects of their implicit phonology.

Consequently, a substantial psycholinguistic literature suggests that prosodic structure is internally simulated during silent reading. Information-structural distinctions such as focus and givenness are mentally represented, and readers appear to project implicit prosodic contours onto written text (Fodor, 2002a; Gross et al., 2014; McCurdy et al., 2013).

Prosody is not only a physical phenomenon but also a cognitive one. Information structure — topic, focus, given/new distinctions — is represented mentally. Prosodic features such as nuclear stress and pitch accent are the phonological realisations of these underlying cognitive categories (Féry, 2013; Ladd, 2008; Lambrecht, 1994), but the categories themselves are not modality-bound. When a reader processes a written cleft construction like *It was Maria who solved it*, they mentally project a prosodic contour onto it. Psycholinguistic evidence suggests that readers internally simulate prosody (Breen, 2014; Fodor, 2002a).

Thus, the central argument for explaining the differences in the discourse-functional possibilities of argument-fronted structures in Swedish and English rests on findings concerning the relation

between morphosyntax and prosodic phonology. The analysis assumes that aspects of prosodic phonology are mentally represented during the processing of written language. On this basis, it argues that discourse-functional restrictions can only be explained by considering the interaction between morphosyntax and phonology in a particular discourse context.

1.9 Materials

The primary data in the current study are taken from the English-Swedish parallel corpus (ESPC) (Altenberg & Aijmer, 2000). The corpus is compiled with a number of different design goals. In terms of representativness it is a so called balanced corpus (Biber, 1993) which means that the texts included in the corpus have been selected to represent a number of different genres and communicative purposes in order to provide a sample of general written language. It thus differs from text archives, where the text types and the number of each type are of less importance, often in an effort to prioritise the size of the corpus rather than its representativeness of the language use. The ESPC also differs from specialised corpora (McEnery & Hardie, 2012) in which the texts that make up the corpus represent a particular type of language usage, such as a specific genre or language produced by a particular user, i.e. newspaper corpora, or learner corpora.

To be able to represent a fairly large number of different text in a corpus of a comparatively small size of less than 3 million words, the individual texts are not included in their entirety but are excerpts of approximately 14,000 words from the beginning of each text. There is a hypothetical possibility that different parts of a text may have different characteristics in terms of the type of language that are used, and that some structures may be over-represented for discourse structural or content-related reasons. This could then bias the corpus towards a certain type of language and make it less representative of language in general. However, no compelling evidence has been presented for this being the case, and the advantages of including a greater variety of text types seem to be a preferable prioritisation.

In terms of medium, the ESPC is restricted to written data, but it includes a wide range of genres, including fictional texts, which means that it contains emulated dialogue, and also, as we will see further on, the style of certain authors in the material includes language features which are often characterised as being typical of spoken data.

The ESPC is a contrastive corpus which includes English and Swedish written texts. However, it is very hard to ensure that the type of language that is being represented in similar genres in the two languages is comparable in terms of language use, i.e. it is not reliable to assume that categorisations based on content or cultural traditions will reflect the linguistic properties of the language used. A writer of crime fiction in English cannot be assumed to employ a language that corresponds to the language used by someone writing a crime novel in Swedish. Indeed, in fictional texts, the often idiosyncratic style of authors makes it hard to assume that two writers of the same genre in the same language will be similar as regards type and frequency of language features. Thus, in an effort to control for differences between individual writers, the ESPC is designed as a parallel corpus, which means that the corpus in addition to consisting of a balanced set of original texts in the two languages, also contains translations of the original text into the language being contrasted. In the ESPC, this means that the original English texts also contain their translations into Swedish, and the same original-translation relation exists with the Swedish original texts. This type of design permits a number of different comparisons between the languages in the corpus. These different possibilities can be represented as in Figure (1.1).

The boxes illustrate the different parts of the corpus and the arrows represent the different comparisons that this design makes possible. It is thus possible to compare the differences that are found between the original texts with the way an original text has been translated. If the

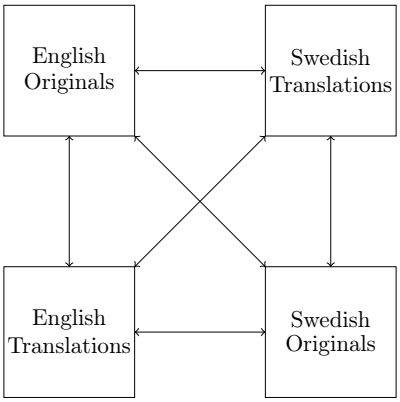


Figure 1.1: ESPC types of comparisons

same phenomena occur both between originals and between originals and translation, it goes some way towards eliminating the risk that the differences between originals are dependent on style or the content of the text. Moreover, the comparison between original texts enables a way to control for possible effects of a source text influencing the translation and creating non-native like or unidiomatic language so called 'translationese' Gellerstam (1996).

The ESPC is currently maintained by the Department of Languages and Literatures at Gothenburg university and is available through Språkbanken's search engine KORP. The size and composition of the ESPC can be seen in Table 1.2

Table 1.2: Number of texts and word counts in the English-Swedish Parallel Corpus

	Fiction		**Non-Fiction**		**Total**	
Category	**Texts**	**Words**	**Texts**	**Words**	**Texts**	**Words**
EO	25	340,745	39	364,448	64	705,393
ST	25	346,649	39	344,131	64	690,780
SO	25	308,160	47	353,303	72	661,663
ET	25	333,375	47	413,500	72	746,875
Total	**100**	**1,328,929**	**172**	**1,475,582**	**272**	**2,804,511**

The current study is based on a subset of the ESPC consisting of ten original texts from each language and their corresponding translations. Attempts have been made to maintain some balance as regards the number of fiction and non-fiction texts, and also the type of texts from each sub-genre.

The texts in the ESPC are divided into ortographic units, S-units, which generally consist of a stretch of discourse that starts with a capital letter and ends with a full stop, exclamation point, question mark or a line break. S-units may consequently consist of individual words or combinations of words that do not contain a complete clause, as illustrated with the italicised S-units in (83) – (85):

(83) Perhaps in a worn coat and a knitted hat like a tea cosy, sitting on a curb, with two plastic bags filled with her only possessions, muttering to herself. (MA1.1.2.s86)
Cordelia! (MA1.1.2.s87)

(84) I want some friends, friends who will be girls. (MA1.2.3.s35)
Girl friends.(MA1.2.3.s36)

(85) There would be champagne too – the boss's own.(JC1.1.1.s10)
And fruit, of course.(JC1.1.1.s11)
All this for just five birthday guests.(JC1.1.1.s12)

The accompanying code shown in these examples, e.g. MA1.1.2.s86, identifies text (MA1), hierarchical organisation such as chapter and section (1.2.) and the individual sentence (s86) that the example represents. In1.3 the design of the subcorpus is presented, giving an overview of the number of words and S-units included separated by language. Because the English and Swedish materials include both original texts and their translations, they allow for the various comparisons illustrated in Figure 1.1. As indicated by the number of S-units, the Non-fiction texts in both languages, and in both originals and translations, have fewer S-units, something that suggests longer clauses. This is of significance from the perspective of the way fronted structures were identified.

The ESPC is neither annotated for word classes (tagged) or syntactic constituents (parsed) which means that it is not possible to computationally search for phrasal categories (NP, PP, etc.) or syntactic patterns (OSV, SVO etc.). Moreover, the type of constituent that can be fronted can potentially include all major phrase types, and there is no consistent way of identifying fronted elements based only on the form of the initial elements. As a consequence, identification of fronted elements had to be done by manually scanning the text and annotating those containing a fronted element in a database.

To make the process of identifying the fronted constructions as efficient as possible, a decision was made to limit the study to the initial part of the S-unit. In the first step, each S-unit was scanned from the beginning of the unit up to the constituent that revealed the sentence type, and argument structure of the first main clause, normally the first finite or lexical verb, as illustrated by the italicised part in (86) and (87).

In most cases this made it possible to establish if the clause was a declarative clause and whether non-subject initial elements were arguments or adjuncts.

(86) *With a logic more poetic than functional, the market which engulfed the garden was known*, too, as the Soap Market, though soap was not on sale.

(87) *In 1999, ABB completed* 20 new bond issues for a total of $1,584 million, with the majority under ABB's Euro Medium Term Note (EMTN) program and a maturity of less than one year.

A drawback of this scanning method is that it omits structures beyond the point of scanning. This approach affects coordinated main clauses, main clauses concatenated with semicolons, and subordinate clauses in the later part of the sentence. Although it would have been preferable to catch all instances of argument fronting, this was considered to be the best way to be able to cover a fairly large number of sentences manually.

The selected texts and their translations are listed in 1.4 together with the codes for the different texts. The translations are identified by having a T after the code for the original, e.g. Margaret Atwood's novel *Cat's Eye* has the code MA1, and its Swedish translation has the code MA1T.

Table 1.3: Subcorpus design of texts taken from the English-Swedish Parallel Corpus

Sub-corpus	Genre	Texts (#)	Total Words	Total S-Units
English Sub-corpus (Total English Material)				
English Originals	Fiction	5	62,406	4,717
	Non-Fiction	5	69,966	2,977
English Translations	Fiction	5	66,726	4,851
	Non-Fiction	5	68,341	3,065
Subtotal English		**20**	**267,439**	**15,610**
Swedish Sub-corpus (Total Swedish Material)				
Swedish Originals	Fiction	5	62,002	4,923
	Non-Fiction	5	55,304	3,179
Swedish Translations	Fiction	5	66,291	4,655
	Non-Fiction	5	62,264	3,277
Subtotal Swedish		**20**	**245,861**	**16,034**

Table 1.4: Overview of ESPC Primary Sources in the study

Code	Author / Source	Original Title	Target Title
English Fiction			
MA1	Atwood, M. (1988)	Cat's Eye	Kattöga
JC1	Crace, J. (1992)	Arcadia	Arkadien
RD1	Dahl, R. (1988)	Matilda	Matilda
AT1	Tyler, A. (1985)	The Accidental Tourist	Den tillfällige turisten
RR1	Rendell, R. (1992)	Kissing the Gunner's...	Dödligt sällskap
Swedish Fiction			
KOB1	Bornemark, K.-O. (1986)	Handgången man	The Henchman
AL1	Lindgren, A. (1981)	Ronja rövardotter	Ronja, the Robber's...
AP1	Pleijel, A. (1989)	Hundstjärnan	The Dog Star
MS1	Scherer, M. (1980)	Kejsarvalsen	The Emperor Waltz
GT1	Tunström, G. (1983)	Juloratoriet	The Christmas Oratorio
English Non-fiction			
ABB1	ABB Ltd. (2000)	Group Annual Report '99	Årsredovisning 1999
JL1	Lovelock, J. (1988)	The Ages of Gaia	Gaia
DM1	Morris, D. (1990)	Animalwatching	Se på djuren...
CS1	Sanger, C. (1982)	Safe and Sound	Nedrustning, utveckling...
LT1	Timberlake, L. (1987)	Only One Earth	En enda jord
Swedish Non-fiction			
ASSI1	AssiDomän (1999)	Årsredovisning 1998	Annual Report 1998
CO1	Cedergren & Odén (1991)	I krisens spår	In the Wake of Crisis
HG1	Gustafsson, H. (1994)	Politisk interaktion...	Political Interaction...
BJ1	Johnson et al. (1990)	Hus och hälsa	Buildings and Health
CP1	Petersson, C. (1994)	Förskolebarn	Preschool Children

1.10 Organization of the thesis

The thesis will be organised as follows. Chapter 2 introduces the theoretical notions necessary to discuss the discourse functional properties of the different types of argument fronting investigated in this study, with a focus on IS-fronting. Discourse is developed by adding sentences that are incorporated into the discourse model constructed by each discourse participant. In order for this to be successfully achieved, the participants assume that they share a set of propositions which can be developed and updated, and that the information statues of discourse referents are communicated.

The theoretical concepts of information structure will be discussed and evaluated in order to arrive at a analytical model that can be used to explain the discourse function of argument fronting in the two languages and what enables and restricts the different functions of these constructions in the two languages. Given that prosodic phonology is central for information structure in both English and Swedish, and is made to provide a key part of the explanation for the different properties of argument fronting in the two language, the chapter also gives a short overview of auto-segmental phonology and its most basic notions, particularly how they relate to information structure.

In chapter 3, the general theoretical model of the language competence is presented in order to establish the main ideas concerning modularity, the centrality of morphosyntax, the semantic and phonological interfaces and where information-structure and contextual information are situated in such a model. It will build the foundation for what type of information is specific to each module, in what ways they interact, and how this can be theorised. Although these matters tend to be expressed in fairly broad terms, it anchors the theoretical discussion by making commitments to a number of fundamental principles grounded in a particular take on the nature of language and linguistic research.

Although the syntactic analysis is derivative of previous studies, it is necessary to establish in chapter four what type of syntactic description our subsequent syntax-phonology interface application assumes. Also, the influential theory of cartography within generative minimalism requires some discussion of the syntactization of information structure, and how that influences how to implement the way syntax and phonology interact.

The main argument in this thesis is that the differences in the way argument fronting works in English and Swedish can be explained by a number of language specific differences in morphosyntax and prosodic phonology, especially in the way syntactic structure maps on to phonology.

The fourth and fifth chapters contain the main empirical findings. Chapter 4 presents the different elements that have been fronted in the original text, and a comparison is made between the different types of fronting that were found in the original material. Chapter 5 looks at the way the fronted constructions were translated into English and Swedish. The context in which the fronted structures are used is compared with the type of construction that is used in the translation. The empirical results form the basis for formulating hypotheses about the principled ways morphology, syntax, prosody, and context interact to restrict or promote the use of particular information structural configurations.

The central chapter is chapter 6 which synthesises the theoretical model and corpus data of the previous chapter into an analysis that goes some way towards explaining the differences in discourse function that exist in English and Swedish fronted constructions.

In chapter 7, the contributions the thesis makes to the study of morphosyntax and discourse interactions are discussed.

Finally, chapter 8 summarises the conclusions and reflects on the way the findings in the current study can form the basis for further development of the theory and description of discourse functions from a contrastive perspective.

Notes

[1]Canonical clause structure should not be confused with the typological notion of *basic word order*, the latter term is based on a number of word order properties of a particular language and generally categorised as the relative order of Subject (S), main verb (V) and direct object (O). This may coincide with the constituency order of the canonical clause, as in English where a canonical clause like (88) has the basic word order of English, i.e. SVO,

(88) Cain killed Abel.

However, it may also differ as in German where the canonical clause would be SVO, but the basic word order is considered to be SOV (Vikner, 1990). Newmeyer (2003) argues for the usefulness of the notion of Basic Word Order from a theoretical and typological perspective, primarily as a reply to Lambrecht (1994), which uses the term Canonical Word Order. Lambrecht (1994) criticism of using Canonical Word Order as a notion is based on the often low frequency of this order in language use. By distinguishing the technical term Basic Word Order from the purely descriptive term Canonical Word Order some of the disagreements can be resolved.

Chapter 2

Information structure and prosody

This chapter gives an overview of the discourse functional notions that are used to describe the communicative function of fronted constructions and the theoretical model that is used to predict how individual sentences interact with their co-text to create and develop discourse. It also reviews previous discourse-functional studies of fronting.

2.1 Introduction

One of the central aims of the current study is to give an account of the fronted constructions in English and Swedish that can go some way towards explaining the differences in communicative function between the two languages observed in the corpus data from the English-Swedish Parallel Corpus, i.e. we want to both describe the discourse functions of the Swedish and English fronted structures in the corpus material and explain any communicative differences.

Several steps are necessary to achieve such an account. Firstly, we need to describe the general differences in discourse function between a canonical structure and a fronted structure, then to give some theoretical motivation for this difference. In a subsequent step, we then have to demonstrate the way that language specific properties in morphosyntax and phonology result in differences in discourse function between fronting in English and Swedish when realising general principles of discourse functions in each language.

The following chapter turns to how these functional properties are realised morphosyntactically and phonetically in English and Swedish, which will be argued to show the different properties and functional possibilities that fronted constructions have in the two languages.

Generally, the aspects of linguistic description and theory that concern the communicative function of sentences and the way syntax and context interact are categorised as *information structure* (Halliday, 1967) or *packaging* (Chafe, 1976). Within this general classification, we find a range of concepts that deal with various discourse-semantic and pragmatic phenomena and their linguistic realisations. In the present study, the central explanatory burden is carried by the way basic information-structural functions and properties are realised through the mapping between syntactic structure and prosodic phonology.

A complicating factor in establishing a coherent model of discourse functions and discourse creation is the lack of consensus surrounding the terminology and definitions used in information-structure research. This has led scholars such as Levinson (1983) to throw their hands up in despair, doubting the viability of arriving at a consistent model. However, although no completely coherent model of discourse functions exists, a number of notions can be discerned as making up the core set of any information structural theory. The basic concepts that will be discussed are to a

great extent consistent with the concepts presented in Krifka (2008), although the definitions used by Krifka do not fully align with those used in the current study. The current study ascribes to a separation between linguistic phenomena across sentence boundaries, which includes information structural principles, and rules which apply to relations within the boundaries of the sentence, i.e. morphosyntax. However, this separation is not uncontested and there is some disagreement about the type of phenomena that are governed by discourse/pragmatic principles and those that should be included within morphosyntax. In the next chapter, some of the suggestions that propose expanding the explanatory ability of morphosyntax over discourse semantics/pragmatics will be discussed, such as Rizzi (1997) and Kratzer and Selkirk (2020).

An important aspect of information structure is that it is not something that is completely determined by the pragmatic properties of the propositions and discourse entities of the text as presented at a certain point in discourse. The realisation of information structure is also influenced by the discourse participants' communicative intention of presenting information as having particular properties.

Naturally, the possible options that a discourse participant has are restricted by the discourse model that is assumed to be shared by all discourse participants, but within those limitations, it is possible for participants to present information as being, for instance, given or topical, although this information has previously not been explicitly expressed.

The options available to a discourse participant are constrained by the discourse model assumed to be shared by all participants. However, within these limits, speakers can still present information in different ways. For example, they may treat information as given or topical even if it has not been previously expressed explicitly.

This relation between the way the properties of discourse delimit the possible expression and the communicative choices made by discourse participants to change or explicitly manifest pragmatic status of information is generally implicitly assumed, but not always clearly expressed in the literature, which may occasionally results in some confusion about what these notions represent, e.g. whether the notions relate to mental states or instructions for relating sentences to common ground. However, the current study takes the position, expressed, among others, by Halliday (1967), Birner (1996, p. 140), and Mikkelsen (2004), that formal properties of linguistic elements have no direct link to any actual mental state of the listener only as instruction on how to relate new utterances to established information.

2.2 Central Concepts of Information Structure

The following sections will discuss the different discourse functions that are involved in building coherent discourse by updating the assumption of shared knowledge among discourse participants. This dynamic progression is achieved by anchoring new information to previously established knowledge that continuously updates the shared knowledge. Discourse referents are introduced and selected in order to connect propositional information to these referents. By employing ways to indicate what part of a sentence is the anchor and what is the information that updates the shared knowledge, it is possible to build complex discourse that in its central parts are shared by the discourse participant.

This general outline for discourse building is fairly established, as are the central notions (Féry & Ishihara, 2016; Krifka & Musan, 2012; Reich et al., 2024), there are often conflicting views on the exact definitions of these notions. The shared knowledge is generally formalised as *common ground*. Updates to common ground make use of already established information, to which either new information or already established information is added in a non-predictable way, indicating that a choice among possible options has been made (Rooth, 1992). The two types of *discourse*

status are termed *given* and *new*, the discourse referent to which new information is added is the *topic* and the part of the clause that indicates the a choice of alternatives *focus*.

Given that in languages such as English and Swedish both givenness and focus are signalled prosodically, the final part of the chapter will give an account of the area of prosodic phonology within the framework of autosegmental metrical phonology Ladd (2008) and the way information structure is expressed through the interaction between syntax and phonology.

The aim is to establish some general principles for the way we adapt the formal expression of propositional content to dynamically develop coherent discourse. The principles are argued to be language independent, but the implementation of these principles is based on language dependent principles of morphosyntax and phonology (Zimmermann & Féry, 2009).

The key feature of information structure is that it is an attempt to capture the way in which the formal properties of individual sentences indicate relations between the assumed shared knowledge of the discourse context and the information expressed in the sentence.

Our starting point is to establish a formalisation of shared discourse knowledge, or the common ground. We then explore formal representations for the discourse status of referents and propositional content. In doing so, we account for the linguistic mechanisms used to mark information as either given (already shared) or new (being added).

The next step is to show how the notions of topic and focus are used in selecting given information (topic) and assigning new information to that topic (focus) in order to update the common ground. By employing these theoretical concepts, it will be possible to discuss the communicative functional differences between fronted and non-fronted structures and how these functions are manifested in English and Swedish.

2.3 Common Ground and Shared knowledge

Fronted constructions involve the way discourse context restricts the felicitous use of sentences with particular formal properties (morphosyntactic and prosodic); consequently, an account of these expressions requires a way to describe and model contextual information. A key task is to capture contextual aspects in a way that is precise enough to explain the data at hand and allows us to predict the way context and sentence form interact to produce communicatively well-formed expressions.

When we communicate linguistically, we are aware of and make use of a great range of information that we have about the world in general and about the specific discourse situation, e.g. what we know about the other participants, the purpose of the communication, and the surroundings. We also assume that much of this knowledge is shared by the other participants taking part in the discourse. Linguistic interaction is therefore grounded in information that speech participants believe they all share, including the information exchanged in the ongoing discourse. This is what we can refer to as the *discourse context*. Although this is taken by most to be trivially true, it is less clear to what extent and in what way this shared information should be represented and formalised in linguistic theory.

A common approach in linguistic studies is to apply the notion of *common ground* (Stalnaker, 1978) to represent the knowledge that is assumed to be shared between speaker and recipient(s) in discourse. In many studies, common ground is simply treated as an abstract representation of the intersection of all interlocutors' knowledge. However, in order to be able to employ common ground theory to explain the use and meaning of particular language construction, it is necessary to have a more developed model of contextual information.

The notion of common ground is central for the theorising of the interaction between linguistic representations of sentences and their discourse context. In particular, it is a key component in

many developments of information structure (topic/focus) and information status (given/new). This section outlines the main features of Stalnaker's common ground theory and discusses some of the criticism that has been directed against it and some of its limitations.

The idea of shared knowledge among discourse participants is historically well-established in the linguistic literature, but the way to formalise this notion is generally regarded as originating with Stalnaker in the 1970s (Stalnaker, 1974a, 1974b, 1978). According to Stalnaker, interlocutors assume that they share a set of presuppositions, i.e. propositions that each individual engaged in discourse believes that all other participants take to be true. Individual participants do not necessarily believe these presuppositions *to* be true but only that all participants *take* them to be true. The source of these presuppositions is varied and includes the observable facts of the discourse situation, knowledge of the other discourse participant, and general knowledge of the world. The meaning of this set of presupposed propositions, the common ground, is formalised as a set of possible worlds, which Stalnaker calls the *context set*.

Essentially this discourse model is compatible with the way sentence meaning is defined in formal semantics i.e. the meaning of an assertion is the truth conditions of its proposition or more precisely the possible worlds in which the proposition is true. To carry this over into a dynamic model of discourse, the common ground is seen as a set of propositions that defines the context as the set of possible worlds in which all the propositions being presupposed are true. In the development of discourse, information is added or altered, and, if accepted, the set of possible worlds (i.e. the context set) is reduced to only those worlds in which the new set of propositions constituting the common ground are true. In this way, the discourse context is modelled to reflect what Stalnaker argues is the main purpose of communication, i.e. "distinguishing between the alternative ways that things may be".

Since assumptions are individual, the common ground may differ to some extent across speakers; Stalnaker refers to such a state as a *defective context*. In most cases, these differences will not become apparent and discourse will progress unimpeded. However, when it becomes clear that the common ground is defective, participants will adjust their presuppositions to achieve a shared common ground, since the aim of discourse is to communicate.

Stalnaker's model of discourse context goes some way towards formalising contextual information, but it is still a rather abstract representation and has been criticised for not being able to capture some basic ways in which utterances and context interact (Abbott, 2000, 2008; Burton-Roberts, 1989; Gauker, 1998).

One issue concerns the way in which the presuppositions of the common ground restrict what counts as a felicitous contribution to the development of discourse. It has been argued that Stalnaker's model predicts that a sentence containing a presupposition that is not already part of the common ground cannot be added to it. Consider the following utterance:

(89) I am sorry I am late. I had to pick up my daughter.

In Stalnaker's model, the presupposition that A has a daughter would have to be part of the common ground for the proposition expressed by (89) to be added to it. This appears to be at odds with the way such information is handled in actual discourse, where its introduction need not be disruptive or prevent the discourse from developing.

Both Stalnaker (1998) and von Fintel (2008) have refuted this criticism as failing to recognise the way the common ground interacts with new information. The central issues for Stalnaker and von Fintel's argumentation revolve around the type of information that is included in the common ground and the time when the proposition in the new information is compared to the common ground. Regarding the first issue, Stalnaker points out that it is important to recognise that common ground includes not only information exchanged through linguistic expressions, but also the events taking place during discourse, including the fact that a speaker is uttering

a particular expression. The simple fact that we need to recognise who is speaking when the pronoun 'I' is used shows that this needs to be represented in the common ground (Note, however, Abbott's (2008) objection to comparing presuppositions and deixis). Secondly, the timing of the introduction of new propositions in the common ground is central. Any presuppositions included in new information are checked against the common ground *before* the propositional content is added to the common ground, but *after* the utterance has been made. The ability to accept sentences containing presuppositions not already included in the common ground is an effect of proposition accommodation, which is based on the idea of *accommodation* as suggested by Lewis (1979).

Returning to the example in (89), accommodation allows the utterance to be interpreted even if the common ground does not contain the presupposition that the speaker has a daughter. The listener recognises that the utterance presupposes this information and presents it as already part of the common ground, and consequently adjusts the common ground to accommodate it. This is possible, provided that no proposition in the common ground directly contradicts the newly introduced presupposition.

Consequently, following Stalnaker and von Fintel, the propositional content of a sentence normally comprises both presupposed and asserted content. Presupposed content is assumed to be taken for granted by all discourse participants and is therefore already part of the common ground. Asserted content, by contrast, introduces new information and needs to be added to the common ground, provided that it is accepted by the recipients.

Abbott (2008) does not find Stalnaker's and von Fintel's defence satisfactory, and her two main issues are that the distinction between presupposing and asserting information is not clear, and that the definition of what it means to assert information is mistaken. She claims that Stalnaker's use of accommodation to clarify how presuppositions in a new utterance can be accepted despite not being in the common ground effectively equates presuppositions and accommodations. Essentially, there would be no difference between saying (89) and (90).

(90) I am sorry I am late. I have a daughter and I had to pick her up.

This may well be true, in the sense that in both cases the common ground is updated with the same propositional content from the linguistic expressions. However, this does not address the second issue discussed by Stalnaker: the common ground is updated not only with propositions corresponding to linguistic expressions, but also with information about the events in which those expressions are produced. This includes information about how propositional content is introduced into the common ground. For example, in (89) the proposition that the speaker has a daughter is recognised as presupposed rather than explicitly asserted, whereas in (90), it is explicitly asserted. Although the propositional content concerning the speaker and the speaker's daughter is the same in both cases, the way in which that information enters the common ground differs.

Abbott's second main objection to Stalnaker's common ground theory is his view on assertions. Stalnaker claim that the main function of assertions is to introduce new information in order to reduce the common ground to the set of possible worlds that accord with the actual world. Although this may be the case with many assertions, it is not always true. In a fair amount of assertions, no new information is being added, and therefore no reduction of the set of possible worlds takes place. Essentially, Abbott argues that Stalnaker (1998)'s common ground theory is too general and does not sufficiently take into account the linguistic form of an utterance. Often, she argues, assertion simply provides information that is already part of the common ground, such as:

(91) a. The next agenda item is our annual picnic.

b. It's been ages since I saw you

c. It is what it is.

Although Abbott is correct that Stalnaker's theory of common ground cannot capture more detailed distinctions between different contributions to discourse and therefore needs to be complemented by other principled accounts of discourse development, she overlooks an important aspect of Stalnaker's proposal. His model is intended to incorporate not only the propositions expressed by an assertion but also the fact that a speaker has made that assertion. The sentences in (91a)–(91c) add only information that is already presupposed, but they nevertheless change the common ground by adding the fact that they have been uttered in the discourse. Although this distinction may seem trivial in terms of the context set, it is nevertheless significant for the representation of the discourse: the context records not only what propositions are accepted, but also how they have been introduced into the common ground.

However, the issue then becomes whether we need a richer more elaborate theory of common ground, or if we need to take care of that by developing other theoretical frameworks, such as theories about information structure and information status. Stalnaker's view is that his theory is not intended to deal with these more specific issues. He explicitly addresses this matter in response to additional critique of the inadequacies of the notion of common ground from Kamp (1988), which regards the inability of common ground to predict the difference pronominal-antecedent development in different context with the same common ground presuppositions.

Kamp bases his criticism on a comparison of two short discourses.

(92) a. Exactly one of the ten balls is not in the bag.

b. It is under the sofa.

(93) a. Exactly nine of the ten balls are in the bag.

b. It is under the sofa.

The point Kamp (1988) makes regarding Stalnaker's theory of discourse context is that the different interpretations of the pronoun *it* in the two (b) sentences are unexpected if the two sentence pairs have the same common ground. Kamp (1988) argues that, if (92) and (93) have the same preceding context, and if each (a) statement provides a truth-conditionally identical context for the corresponding (b) statement, the common ground must be the same. Consequently, this representation of discourse context cannot explain why the pronoun *it* receives different interpretations. In (92a) the most natural interpretation would be to view the antecedent of *it* in *it is under the sofa* to be *one of the ten balls*, whereas in (b) there is no antecedent for *it* in (93a). Clearly, there is something missing in the common ground model of discourse context.

Stalnaker's response is that his theory of common ground is not intended to cover any language specific aspects of pronoun resolution. However, it is intended to predict that there should be a difference since the theory makes a distinction between the information deriving from the linguistic utterance, on the one hand, and the information deriving from the fact that the utterance is made, on the other. Thus, there will be a difference between two discourse contexts, such as (92) and (93) regardless of the identical truth conditional propositions, since the propositions were added to the common ground through different linguistically formed utterances. Consequently, Stalnaker claims that his theory has nothing to say about the specifics of the differences or how these differences are related to any specific language properties, only that it predicts that in addition to the changes to the discourse context being effectuated by the propositional content of the utterances, there will also be changes in discourse context by the act of producing these utterances.

From the perspective of studying specific language phenomena—such as fronted constructions and their contextual constraints—the principal value of common ground theory lies in its formalisation of two key concepts. It sharply distinguishes presupposed information (assumed to be already known) from asserted information (propositions that, if uncontested, reduce the context set). It also highlights the need to address phenomena beyond common ground, such as how referents become salient for pronoun resolution (Kamp, 1981). Furthermore, Stalnaker's common ground notion draws attention to the difference between changes produced by an utterance's proposition and those produced by the act of uttering itself. The latter can be interpreted to encompass meanings beyond truth-conditional semantics, including the types of phenomena covered by models of information structure.

It is clear that some language phenomena will require a more developed theory of the interaction between specific language expressions and common ground. Some information in the common ground will be more available as antecedents than others, which cannot be explained by a fairly abstract idea of an unordered set of propositions. Also, any model of discourse that has anything to say about natural discourse data will have to be able to handle other speech acts than assertions, and in what way this information will update the common ground. Moreover, the dynamic nature of discourse is basically handled by the process of updating common ground and reducing the context set, but little is said about how this is handled as a collaborative effort in dialogue or as a progressive development in monologue. Different suggestions on how to deal with these problems will be explored in the section 2.6.4 on discourse development.

2.4 Discourse status – Given and New

The notion of common ground is based on the view that discourse develops dynamically by updating already established information with new information. However, although intuitively appealing, it has turned out to be less than trivial to formalise this in some satisfactory way. In modern linguistics, the earliest attempt is generally considered to be undertaken by the linguists associated with the Prague School of Linguistics. Based on the ideas of Mathesius (1975), they develop the theoretical notion of *communicative dynamism (CD)*, a view of sentence structure as a way to develop information, prototypically going from the least to most informative language elements. Within the framework of the Functional Sentence Perspective (FSP) Firbas (1971) introduced the notions of *theme* and *rheme* that are regarded as two poles on a scale of informativeness, where *theme* is the least informative and *rheme* the most. This means that there is no categorical division between thematic or rhematic elements, and that all sentences will exhibit a value of informativeness on a scale going from thematic to rhematic, resulting in no absolute properties for the extreme poles in different sentences. The default principle for information development is taken to be thematic before rhematic, with informativeness thus increasing in a left-to-right order. Although the theme-rheme structure is considered the default order, it is also recognised that in some contexts it is possible to have a rheme-theme structure with a distinct discourse functional effect.

These ideas have been conceptually influential and form the basis for much of the subsequent description and theorising which take the communicative function of linguistic expressions as a central part of language research and where the dynamic development of discourse is of prime importance to a theory of language. However, the notion of relative informativeness has been difficult to operationalise, and a categorial division into information being already part of the common ground, *given*, and information altering or updating the common ground, *new*, has emerged as the standard dichotomy (Krifka, 2008; Lambrecht, 1994).

Still, the wide variety of different contextual relations that can render a language item given

or new has resulted in the development of a taxonomy of information status subcategories. In this area, influential work has been done by E. F. Prince (1981a, 1981b, 1984, 1999). In E. F. Prince (1981b), she identifies three uses for the term givenness in the literature: (a) shared knowledge, (b) cognitive salience, and (c) predicatability/recoverability, and suggest that they are best captured by being divided into two subcategories of givenness: inferrable and evoked.

E. F. Prince (1981b) discussion highlights the complexities in what has previously been categorised as simply given. Shared knowledge is essentially the same as information being part of the common ground, whereas salience is more dependent on the immediate co-text and captures information that is easily evoked. With the theoretical notion of common ground, there is no difference among information in terms of availability, whereas salience harks back to Chafe (1974) notion of cognitive activation, and indicates something already in the consciousness of the listener/reader. Thus, a salient referent can be referenced using a pronominal element. E. F. Prince (1981b) use of predictable/recoverable is rather specific as it is limited to cased where the linguistic element can be deleted. This introduces syntactic constraints in the definition of givenness, which is not normally part of the characterisation of types of information status.

Geluykens (1991) suggests a more flexible notion of recoverability, which does not require any particular syntactic constraints, only that the referential expression can be incorporated into the current discourse model. Consequently, signalling that a discourse referent is given in combination with it being made part of the discourse model is sufficient for a felicitous use. The key difference is that there is no requirement for a given element to be, in some sense, currently in the mind of the listener, only that it is possible to assimilate the referent into the discourse model. Thus, in a situation as in example (90) no previous mention of 'daughter' is needed or activation of this discourse referent in the listener. As long as the referent can be integrated in the model, with little cognitive effort, it is used felicitously.

E. F. Prince (1981b) original subcategorisation of givenness into inferrable and evoked is along the same line, but with more specific requirements of the relation that makes the discourse referent given. Evoked indicates some referent that has previously been introduced in the discourse and been reactivated or alternatively a referent from the communicative context, like *me* and *you*. In E. F. Prince (1981a), she presents a different taxonomy based on whether a referent is given or new based on its statues in the discourse or the speakers consciousness, i.e. discourse-old, discourse-new and speaker-old, speaker new.

In addition to providing a systematic description of different types of familiarity, E. F. Prince (1981b) argues that a theory of information status also requires a taxonomy of linguistic forms and an account of how these forms relate to the notion of givenness. Gundel et al. (1993) presents such a taxonomy for English and attempts to correlate linguistic form with information status.

Although much of the discussion aiming to define the different concepts of familiarity has centred on English data, the general assumption has been that these notions are language independent (Zimmermann & Féry, 2009, p. 1). The forms signalling the different familiarity statuses, however, are language dependent. In English and Swedish, these formal expressions involve morphosyntactic distinctions—such as different types of nominal expressions and information-packaging constructions (Huddleston & Pullum, 2002, p. 1365)—as well as phonological ones, including prosodic phrasing and pitch accent placement.

The close connection between information status and phonological realisation in English has resulted in much research being directed toward establishing the relation between the semantic/pragmatic notions of topic and focus and the phonological signalling of given and new elements, i.e., deaccenting and pitch accent, respectively. Although the general association is to correlate topic with deaccenting, or even ellipsis, and nuclear pitch accent with focus, the relation is more complex and garnered much attention in the literature, especially the definition, interpretation and principles where the nuclear pitch accent will be placed to express different

focal constituents.

2.5 Focus

Focus is a central notion in a theory of information structure, regulating how information in an utterance is interpreted relative to the discourse context. Giving an account of the theoretical foundations of focus includes examining its communicative functions, its distinct semantic and pragmatic interpretations, and the formal mechanisms used to mark focus across languages.

2.5.1 Dimensions of Focus

In analysing the notion of focus, Gussenhoven (2008) identifies three central dimensions: the focus constituent, the focus meaning, and the formal expression of focus.

First, the *focus constituent* refers to the specific part of a linguistic representation that carries focus meaning, typically classified along a continuum from broad to narrow focus. Second, the *focus meaning* captures the pragmatic function of the focalised element. This is often categorised into distinct types, such as identificational focus versus information focus (Kiss, 1998), or information versus contrastive focus. Finally, the *formal realization* concerns how focus is explicitly marked in a given language, primarily through morphosyntactic or phonological mechanisms.

However, the fact that Focus is a discourse semantic/pragmatic notion with only an indirect relation between the form and function of the Focus constituent makes it hard to deal with each aspect separately and requires a way to regard this as different aspects of a single concept.

The functional basis for the notion of focus is the use of given information to incorporate new information into the discourse model, with focus being prototypically associated with new information. The distinction can be illustrated with a question-and-answer pair, such as (94).

(94) A: What is the colour of your car?
B: My car is [BLUE.]$_{\text{Focus}}$

Speaker A's question introduces a proposition with an uninstantiated variable into the common ground: [the colour of your car is X], and the information in B's answer that provides the propositional value for the variable is the focus of the sentence. The part of B's answer that has already been introduced by A's question is given, and in a question-and-answer pair it can be, and often is left out, making a possible answer in (94) be only the focal information "Blue".

Informally, we can describe this as a way to communicate by providing enough established information to make it clear how the focal information should update the discourse participants' knowledge.

The pragmatic concept of the 'point of departure' in a sentence rests on the assumption that the information structure serves as an instruction to the addressee on how information added to discourse relates to information already in the discourse model The point of departure-element, the topic, indicates what discourse referent new information should be related to. Expressed in the metaphor of file cards Heim (1982), it tells the addressee on which file card to record new information. The way topic is signalled varies both within and across languages, but it is prototypically associated with the left edge of the sentence and phonological attenuation

(Gundel, 1988, p. 215; Lambrecht, 1994, pp. 199–201; Vallduví, 1993, p. 46). The indication of what is new information and how it relates to the topic is another central notion in a theory of information structure. Just as topic is often associated with prosodic deaccenting, focus is generally considered to be strongly associated with prosodic prominence (Rooth, 1992).

Using the traditional question-answer diagnostic previously demonstrated, we can identify focus as the part that provides the answer to the *wh*-question, i.e. in (95).

(95) A: What about Victor?
B: Victor [likes FISH]$_{\text{F}}$.

The pragmatic function of focus marking is to indicate the part of the sentence that updates the discourse model. The part of the sentence that lies outside the focus scope is regarded as presupposed or given, in the sense of information assumed to be shared by the interlocutors. As a common characteristic of topic is that it is given, it follows that it cannot be part of the focus. Topic and the presupposed parts of a sentence are, however, not always coextensive. Topic has a number of criteria in addition to being given, and will in some cases pick out a more restricted part of the sentence, as can be seen if we reduce the scope of focus in (95), as in (96).

(96) A: What does Victor like?
B: [Victor]$_{\text{T}}$likes [FISH]$_{\text{F}}$.

The part that is not in focus is thus presupposed and within the presupposed part we find the topic. The fact that the information that is not focused is presupposed does not mean that focused information cannot be presupposed. The focus of a sentence is "new" information, not in the sense of not having been previously introduced in the discourse model, but by providing the value for variables in an open propositions (E. F. Prince, 1981b). The open proposition is formed by the interaction between the information already established in the context, the topic and information relations that the addresser wants to establish. In (95) the open proposition is Victor X and the focus is then the instantiation of X, as in X = [likes fish]. The equivalent of 96 is Viktor likes X, with the focus being X = fish.

(97) A: What happened to the fish?
B: [Victor ate]$_{\text{F}}$ it.

In (97) both "the fish" and "Victor" are already in the discourse model. Since B uses the proper name "Victor", it is assumed that the discourse referent is in the common ground.

Examples like (97) provide a challenge for any theory that assumes that focus is assigned to syntactic constituents, since "Victor ate" is clearly not a syntactic constituent.

A development of the notion of focus is made in Lambrecht (1994) in which Lambrecht develops a theory of information structure that concerns "the morphosyntactic and prosodic encoding of the identifiability and activation statuses of the denotata of sentence constituents and of the topic and focus relations these denotata have with propositions".

2.5.2 Focus marking

Studies of how focus is realised show that there are differences both across and within languages. Although prosodic prominence has been given a great deal of attention in the literature, it is clear that focus marking can be realised by different linguistic systems, such as dedicated syntactic positions in Hungarian (Kiss, 1998), focus morphemes in Wambon, and morphology and word order in Aghem (Dik, 1997b, pp. 335–337).

The standard analysis of focus marking in English originates from Chomsky (1972) and Jackendoff (1972). It suggests a relation between focus constituents and prosody in which the highest stress in a sentence is placed on the syllable that is given the highest stress by regular stress rules. As a consequence, there is ambiguity in the scope of focus when focus falls within the verb phrase, as shown in (98a–98c).

(98) a. John gave Mary [the BOOK.]
b. John [gave Mary the BOOK.]

c. [John gave Mary the BOOK]

In these examples, it is assumed that the set of referents that can instantiate the variable in the open proposition is unrestricted. A distinction is therefore made between "new information" focus or *information focus*, where the set is unrestricted and "contrastive focus" or *identificational focus* where the set is contextually restricted (Kiss, 1998), as in (99).

(99) A: Who gave Mary the book, John or Paul?
B: JOHN gave her the book.

The correlation between specific phonetic features and the perception of focus marking has been difficult to identify and define with rigour. Although we can intuitively pick out elements as being prosodically prominent, it has been difficult to define this instrumentally.

One solution is to identify prosodic prominence on the basis of what is perceived to be prominent, which is a relative notion. That is, an element is perceived as prominent when it is more prominent than the surrounding elements or when it receives prominence that it would not receive in a default or canonical configuration.

An interesting observation is presented in Quirk et al. (1985, p. 1377n) in relation to a discussion of marked theme. Quirk et al. point out that prosodic prominence must be evaluated in a context of morphosyntactic structure, which they exemplify with (100).

(100) (Have you considered a holiday in Miami?)
a. |Miami BÒRES me.|
b. |Miami I've BÈEN to.|

They claim that *Miami* would probably have the same prosodic weight in both (100a) and (100b) if measured instrumentally. However, the latter would be perceived to have greater prominence due to its morphosyntactic structure. As they do not provide any evidence that there is no objective difference in prosodic weight, it is difficult to evaluate this suggestion. However, it highlights the question whether there is such a difference and what the implications are if there is only a perceived prominence of the fronted element. A study by Bishop et al. (2020) shows the complexity of the factors involved in the perception of prosodic prominence, but establishes that accent status, i.e., being accented or unaccented plays an important role. These results give some support to the theoretical argument, e.g. Féry (2006), that topicalisation has a prosodic reflex.

2.5.3 Functions of focus

In the literature (e.g., Gussenhoven, 1994; Halliday, 1967; Kiss, 1998; Rochemont, 1986) different types of foci have been identified. The fundamental distinctions that will be made here are between different discourse functions: *information focus*, and *identificational focus*. Other terms for this central distinction is *presentational focus* and *constrastive focus* respectively. In addition, the distinction between types of focus domains: *broad* and *narrow* is sometimes taken to cover the same notions as information focus and identificational focus.

2.5.3.1 Information focus

Information focus serves to introduce previously unmentioned entities into the discourse or to provide the missing value in an open proposition (e.g., Gussenhoven, 1994; Halliday, 1967). It directly responds to explicit or implicit *wh*-questions, such as *What happened?* or *Who arrived?*. Unlike contrastive or identificational focus, information focus carries no presupposition

of exclusivity, exhaustive identification, or correction against a closed set of discourse alternatives (Kiss, 1998).

A standard diagnostic for information focus is its felicity in response to non-restrictive information-seeking questions, as illustrated in (101):

(101) A: What did John buy at the market?
B: He bought [apples]F.

Cross-linguistically, information focus is prototypically marked by default prosodic prominence (such as nuclear accent placement in English) without requiring special syntactic movement or dedicated focus morphemes (Lambrecht, 1994). Within the domain of information focus, languages may distinguish between *broad focus* (where the focus domain spans the entire sentence or verb phrase) and *narrow focus* (where focus is restricted to a single constituent).

2.5.3.2 Identificational focus

In contrast to information focus, identificational focus identifies one or more members from a contextually restricted set of alternatives for which the predicate holds, while explicitly or implicitly excluding all other members from that set (Kiss, 1998; Rochemont, 1986). Prototypically, it expresses two core semantic/pragmatic features: *exhaustivity* (the assertion holds *only* for the specified entity) and *contrast* (evaluating the focalised entity against alternative candidates).

The canonical diagnostic for identificational focus involves testing for exhaustive exclusion or corrective responses, such as cleft structures in English or syntactic fronting in languages like Hungarian (Kiss, 1998). An example of identificational focus using a cleft in English is shown in (102):

(102) A: Did John buy apples?
B: No, it was [PEARS]F that he bought.

In terms of structural realisation, identificational focus may trigger non-default morphosyntactic operations, such as movement to a left-peripheral structural positions, insertion of focus particles/cleft construction templates, or special pitch accent tunes distinct from standard neutral nuclear stress.

The properties of exhaustivity and contrast are also part of Féry (2017, p. 142) description of what she terms *selection focus*. The distinctive characteristics of this type of focus is the explicit inclusion of a limited set in the context from which the focal information should be selected, as illustrated in (103).

(103) {Was it Mary or Peter who ate the lemon pie?}
[MARY]$_{\text{F}}$ ate it

This is an example that is traditionally included in the class of identificational focus, i.e. a focal element that selects a referent to the exclusion of all other possible referents. Féry (2017) also the type of focus illustrated in (102) where the contextual information indicates the assumption of the truth of a particular proposition, but where the added sentence corrects this assumption, as in (104). This type of contrastive focus Féry calls *correction focus*.

(104) {Unfortunately Peter ate the lemon pie}
[MARY]$_{F}$ ate it. {not Peter}

Féry's terms introduce a further subcategorisation compared to indentificational focus. Although the type of focus that Féry (2017) refers to as selection focus and correction focus share the property of exhaustively identifying the referent in contrast to informational focus, they are distinct in the respect that it is only selectional focus which explicitly "selects" one referent from a specified set. Correctional focus adds new information in much the same way as informational focus.

As will be discussed in section 2.6.1 on contrastive topic, the main distinction between information and identificational focus is also possible with the information structural category of topic, where we find a difference between aboutness topic as in (105) and a contrastive topic in (106).

(105) A: What are people doing?
B: [John]$_T$ is [washing the dishes.]$_F$

(106) A: Is Peter washing the dishes?
B: (No,) [John]$_T$ is washing the dishes.

Büring (2016) discusses different types of contrastive topic and includes a type he terms "purely implicational topics". The characteristic of this particular type is that, in contrast to other types of contrastive topic, its structure resolves the questions completely. However, it is different from clauses with non-contrastive topics in that it implies the existence of other questions, as illustrated in (109), originally Büring's example (16).

(107) (Where was the gardener at the time of the murder? –)
The GARDENER$_{CT}$ was in the HOUSE$_F$.

According to Büring, the effect of emphasising the topic prosodically through pitch accent and specific tonal contour is to indicate that there are other questions that could be asked, such as 'Where was [other referent] at the time of the murder?'. The prosodic emphasis on the topic thus implies "But other people could be in other places" which could generate answers like "The maid was in the library; the driver was in the outhouse. . . ".

In Büring's characterisation, an implicational topic generates the implicature by picking out a referent from a suggested set, which is generally not explicitly expressed. If such a set only consists of one element, i.e. the implicational topic is a singleton, the interpretation of the construction becomes infelicitous, as shown in (107), Büring's example (52).

(108) (Where did Mann write *The Beloved Returns?* –)
#MANN$_{CT}$ wrote it in [LOS ANGELES]$_F$.

Consequently, placing prosodic prominence on a non-focal element will result in the referent being interpreted as a contrastive topic. Thus, in an exchange like (109) the implication would be that there are some other referents, which could be evoked by A.

(109) A: Do you know George Wilson?
B1: I know him very well.
B2: ?I know HIM very well.

However, by fronting the object, it is possible to give added emphasis prosodically without resulting in a contrastive topic reading. Essentially, we have "topicalized" the referent by selecting it as the referent on which something is predicated.

(110) Do you know George Wilson?
Him I know very well.

This provides an account of the possible combinations of discourse context and sentence structure that motivate the fronting of an object in English without a clear contrast to a set of alternative referents and without the type of contrastive pitch accent, or B accent, (Büring, 1997, 2003; Jackendoff, 1972) often associated with contrast, which we will look at in section (2.6.1). It is worth pointing out, however, that in English, the fronted element requires prosodic prominence. This makes it possible to front in contexts like (110), but not in contexts where the topic is the default or continuing topic (Frascarelli & Hinterhölzl, 2007), since in such contexts the topic is expected to be deaccented or even elided. This contrasts with Swedish, where fronting is possible with continuing topics, as shown by Mörnsjö (2002). To give a fuller account of the way discourse functions and structural and prosodic possibilities constrain the use of fronted elements in English and Swedish, we need an operational definition of the information-structural notion of *topic*.

2.6 Topic

When we use a declarative sentence to assert a proposition, the sentence performs two basic communicative functions: (a) it activates a referent or a piece of propositional information, and (b) it provides some information 'about' this referent/piece of information. Traditionally, the former is called *topic* and the latter *comment*. Its basic discourse-functional account of the topic has remained essentially the same over time; however, because of this fairly vague description, numerous suggestions have been presented to improve it in terms of specificity and theoretical motivation, such as Chafe (1976), Gundel (1977), Krifka (2008), Lambrecht (1994), Reinhart (1981), and Vallduví (1993) among many others. The many different approaches with somewhat different views of the functional properties have led to a number of different definitions that only partially overlap or contain contradictory features.

In languages like English, the correlation between subject and topic is fairly high, but Hockett (1958, p. 201) provides an example where the topic and subject are represented by two different sentence constituents in (111).

(111) That new book by Thomas Guernsey, I haven't read yet.

This is intuitively appealing in that a fronted object performs some communicative function similar to what the subject generally does, i.e. draws the listener's attention to some referent and then provides information related to this referent. This functional description of the clause is also given some typological support, where we can find various constructions which in different ways separate the syntactic function of subject from the communicative function of presenting something that we intend to say something about. Hockett (1958, pp. 201–2) presents examples from Menomini which are translated into English as a left-dislocated structure, and Chinese examples where the topic is not even part of the argument structure of the sentence, but rather some general frame, as covered in more detail by Li and Thompson (1976), generally referred to as Chinese style topics. It is worth noting that, in contrast to a formal analysis of sentence elements into clause constituents, the notion of topic is defined by its discourse function, not by any particular morphosyntactic properties.

Yet, in many instances, the sentence's discourse functional division into topic and comment mirrors the structural separation into subject and predication. However, since information structure is regarded as independent of the way the different parts are realised structurally, using the term subject risks conflating the information-structural function with the sentence function. In this bi-partite division, much of the attention is given to the definition of the topic whereas the comment is regarded as the complement of the topic, that which is not the topic. The comment can then, as in (111), be a part of the sentence that does not constitute a syntactic constituent. The idea is that if we can find a definition of topic, the rest of the sentence will be the comment, regardless of the syntactic function of these elements or the syntactic relation between them.

This model of the general communicative functions of uttering an assertion appears intuitively sound, but it has turned out to be rather difficult to formulate robust definitions for these notions that can be applied across languages and which can be used operationally to identify how these functions are realised in different sentences. Especially, the notion of topic has remained rather vague and hard to pin down to something precise enough to have explanatory power. This has been pointed out so frequently in the literature (Büring, 2016; Erteschik-Shir, 2007; Jacobs, 2001; Roberts, 2011, among others) that it is a sign of the attractiveness of this original notion that despite the many objections, it is still considered a core element in information structure. In the following section, it will be argued that topic, in its original broader conceptualisation, is less central than the notions of focus and contrast, and that we should find more narrowly defined concepts for capturing the various, rather disparate discourse functions that are conflated in the original idea of topic. We should therefore qualify the general notion of topic and end up with a number of topic subcategories, as has been suggested by Jacobs (2001).

Lyons (1966, pp. 212–213) makes the observation that the division of the sentence into topic and comment is closely connected to the discourse context and often the topic is elided or pronominalised, as shown in (112a).

(112) Q: What did John do?

a. A: Run away.

b. A: He ran away.

It is not mainly a description of morphosyntactic structure but of the possibility to elide or pronominalise linguistic elements in some discourse context. Thus, Lyons regards the information structural notion of topic as primarily concerning, what in this study is called discourse status, i.e. the value of givenness and salience that referents and other propositional items have. This dual nature of topic is also reflected in the distinction made by Gundel and Hedberg (2016) into *relational newness/ givenness* and *referential newness/ givenness*, where the first covers the traditional discourse functions of topic and focus, while the latter is more akin to the traditional notions of given and new, and other related concepts such as salience, activation, and retrievability.

As argued by Rochemont (2019), the notion of topic is not simply identical to all given elements; however, the functional characteristic of topics, which is often described in terms of presenting a message as being about some referent paired with information predicated on this topical element, is better portrayed as a question about the activation and retrievability of discourse referents and the linguistic realisations of signalling this property in relation to the state of the discourse model at a particular point in the discourse.

Similarly to the way focus is analysed as a way to mark something as the selected value among a set of alternatives (Rooth, 1992), the traditional notion of topic is essentially a way to indicate that an element has the accessibility of being a salient or activated element in discourse. However, this is not a uniform function in all sentence structures and will have different functional characteristics depending on the type of relation it is involved in, resulting in the problem of finding a single definition.

Consequently, although the information-structural function of topic is generally regarded as an information-structural primitive, it can be argued that this notion is a descriptive convenience rather than a consistent and well-defined theoretical notion. Instead, a range of different discourse semantic and pragmatic phenomena will have to be decomposed into more specific notions.

To account for argument fronting in English and Swedish, this study adopts a tripartite interface model of syntax, prosody, and information structure. Because "topic" lacks a single, coherent definition in the literature, we adopt Frascarelli and Hinterhölzl's (2007) taxonomy of Aboutness and Given topics. We depart from their framework, however, by rejecting the category of Familiar Topic; following Vallduví (1993), default or familiar information is excluded from serving as a distinct topic type.

Moreover, the ideas promoted by Büring (1997, 2016) and Büring and Gutiérrez-Bravo (2001) introduce the notion of contrastive topic. As suggested by Büring (2016) contrastive topic is distinct and independent of any definition of aboutness topic which is generally what is being referred to when the unqualified term topic is used in discussions about information structure. Also, contrastive topic, unlike aboutness topic, is empirically and theoretically more robust, although it is unclear to what extent this notion can be argued to be universal or language independent. However, for the purposes of our comparison of English and Swedish, it will be argued be explanatory.

Almost all models of information structure contain some equivalent of this discourse function (although the terminology may shift, such as *theme* (Sgall et al., 1986; Steedman, 2000), or *link* (Vallduví, 1993; Vallduví & Engdahl, 1996)). Its basic discourse-functional role has remained essentially the same over time, although numerous suggestions have been presented to improve definition in terms of specificity and theoretical motivation (Chafe, 1976; Gundel, 1977; Lambrecht, 1994; Reinhart, 1981; Vallduví, 1993, among others), which has led to a number of different definitions that only partly overlap or contain contradictory features.

The problem of applying this concept in a language like English is that there are no clear morphosyntactic realisations of the topic function and no reliable diagnostics. Apart from the sequential aspect, i.e. you should start by presenting the topic, and then provide the comment, it is hard to claim that the sentence in (113) with a canonical structure is *not* as much about 'the new book by Thomas Guernsey' as (111).

(113) I haven't read that new book by Thomas Guernsey yet.

On the other hand, there is clearly some interpretive difference between (111) and (113), but the question is whether we can account for this difference by appealing to the functional role of topic and if so, there is some way of identifying this functional element. The examples in Hockett (1958) suggest that sentence elements placed at the left-edge of the sentence have this function, which is essentially the way topic is defined by Halliday (1967), although it is termed theme. However, a definition in terms of structural position turns out to be problematic in English, even if some languages are reported to have syntactically specific topic positions at the surface level (Kiss, 2008). Question-answer interactions such as (114) make the positional argument weaker in English.

(114) Q: Where is the laptop?
A: John took it.

Since the question includes the referent of the laptop and requests information about its whereabouts, we would expect the laptop to be the topic and the information about it to be the comment. With a structural definition of topic, we would expect to find the topic at the left-edge, but the answer in (114) has the pronoun referring to the laptop in the final position. Conversely,

structures with expletive subjects, such as *it's raining; there's food in the fridge* show that any structural definition at least requires a more elaborate formulation than simply the left-most sentence element is the topic.

This lack of clear structural position as a way to identify topics in English has resulted in a greater focus on the functional properties as the basis for picking out the topic of a declarative sentence. Two of the most cited efforts to come to grips with the notion of topic in English are Reinhart (1981) and Gundel (1988). Both are based on Hockett's characterisation of 'aboutness' but elaborate with more detailed descriptions and more developed theoretical argumentation.

Gundel's (1985, p. 86) approach is to elaborate on Hockett's description, making it more formalised and precise in order to improve the identification of topics in a sentence, ending up with the formulation in (115).

(115) *Topic definition*
An entity, E, is the topic of a sentence, S, iff in using S the speaker intends to increase the addressee's knowledge about, request information about, or otherwise get the addressee to act with respect to E.

The problems of finding a definition that enables us to identify topics and makes it possible to explain how the interaction of morphosyntactic structure and context enables us to build discourse is well-known in the literature, and it is therefore often suggested that an alternative is to find some good diagnostics to pick out the correct constituent and start generalising from the empirical data.

Both Gundel (1977) and Reinhart (1981) discuss ways to test the topic-status of various constituents in a sentence, Gundel (following Lakoff, 1970), proposes that sentence topics can be diagnosed by the felicity of frame setters such as *as for X, concerning X* or *about X*, as illustrated in (116).

(116) About Archie, Archie rejected the proposal.

However, these diagnostics only show that it is generally not possible to pick out any element that is given nuclear accent, i.e. these types of phrase are congruent with any argument as long as they do not carry nuclear pitch accent, e.g. the reason 'the proposal' is not felicitous with 116 is that we assume a default placement on the last content word. If we add a final temporal adverbial, the aboutness test works fine with "the proposal", as shown in (117) and (118).

(117) # About the proposal, Archie rejected the PROPOSAL.

(118) About the proposal, Archie rejected the proposal at the BOARD meeting.

The only thing these phrases test is non-focus which is not the same as topic. Thus, the inability to find any morphosyntactic characteristics of topics, as well as no conceptual definitions, no empirical descriptive characteristics, and no diagnostic makes 'aboutness' topic invalid as a notion, at least in languages like English and Swedish.

Although topic has been a central notion in information structure for some time in the effort to theorise the adaptation of sentence form to the relation between contextual information and propositional information, the problems with topic are well-known, at least among linguistic specifically focused on these phenomena (Büring, 2016; Roberts, 2011; van Bergen & de Hoop, 2009). In fact, the very premise of any declarative sentence being divided into some 'aboutness', and the predication of this aboutness have been criticised as having no obvious foundational basis. Krifka (2008) points out that a division into some aboutness element and some predicative element related to this aboutness does not follow from any theory of information, still he includes topic as a central information-structural category.

A distinct aspect of the analysis of information structure in this study is using a model with no unified notion of topic. The problematic nature of topic as a fundamental notion in information structure both in terms of its discourse functional properties and its phonological reflection. Jacobs (2001) breaks down the different functions which he perceives as being attributed to topical elements in the literature and concludes that it is hard to argue for a coherent and well-delineated definition of a notion of topic. Similarly, Rochemont (2019) suggests that topic should be replaced by salience since this latter concepts would enable clearer definitions and the possibility to operationally identify constituents possessing these properties.

What will be argued in this thesis is that the phenomena that have been claimed to involve a unified notion of topic should be looked at in terms of ways to achieve activation and re-activation of discourse referents, together with linguistic indications of the givenness or salience of discourse referents and other propositional information.

The traditional notion of topicalisation is not a structure that promotes an internal argument to the status of topic by placing it sentence initially. Instead, there are specific restrictions and properties with these structures that make them language specific. The subject in a fronted structure does not lose its relation to the predication in terms of being what the predication is about. However, a fronted pronominal element will receive added prominence, by not being deaccented, as is the default case if given and pronominal; at the same time it places the nuclear focus on the right-most element creating a contrastive interpretation, especially if the nuclear accent falls on the main verb, but may also fall on an adverbial which would make it less marked, as in (119).

(119) The legal documents they sent by mail.

However, in many cases, the fronted element has to select a discourse referent from a complement set, either one that is explicitly mentioned in the previous discourse or simply implied from the prominence given by position.

2.6.1 Contrastive topic

One of the problematic aspects of the effort to identify and define 'aboutness topic' is that it has often been regarded as a function within a sentence domain rather than some discourse property, i.e. as the property of being an activated referent. This is one of the messier areas of information structure as it involves the intermingling of the notion of function and the notion of property, as well as no clear separation of the domains within which an element can bear a particular function.

To disentangle the different definitions of topic, some scholars have distinguished between two types of topic that operate in different domains: pragmatic topic and sentence topic. Gundel (1985) suggests that a pragmatic topic is defined as in (115) and described as a "relation between a speaker and a sentence relative to a context", whereas a sentence topic is a relation between a sentence and a topic. She also concludes that a sentence topic is always a pragmatic topic, but the converse relation is not true. Moreover, a sentence topic may not necessarily be realised in a particular topic position or be identified by having other specific morphosyntactic characteristics. In fact, it may not be manifested in the sentence at all. Whereas Gundel's definition of semantic topic has had little impact, her definition of pragmatic topic has lived on and been referred to a great number of times, despite its rather vague nature. However, the way this pragmatic relation is realised has been more controversial and depends on language specific resources.

Büring (1999) also recognises two types of topic within two different domains: a sentence version, S-topic, and a discourse version, D-topic. In contrast to Gundel (1985), he elaborates at great length on the definition of S-topic, whereas the D-topic takes the form of the potential continuation at a certain point in the discourse, which means that it is formalised as a constituent

question which is answered by the following sentence. The D-topic is thus the sentence excluding the focus, e.g.

(120) a. John and Paul are starting their holiday tomorrow.
b. John is going to Ibiza.

If we analyse (120b) according to Büring's notions of S- and D-topic, a possible question for continuing (120a) may be "Where is John going" in which case the D-topic would be "John is going X" and the focus "to Ibiza", whereas the S-topic would be "John". Büring's primary interest is the S-topic which he relate to the aboutness-criteria that have been discussed previously, but also include phonological criteria for the languages he is discussing, English and German. Following Jackendoff (1972) he identifies topic with a rising pitch accent, L*H, that is contrasted with the focus accent which is falling, H*L.

The relation between Aboutness Topic and Contrastive Topic is not completely clear. In essence the main concern for theories of information structure was for a long time the definition of Topic with no obvious need to break this into subcategories. Thus, when discussing the intonational characteristics of Topic and Focus, Jackendoff (1972) associate Topic with a L*H pitch accent, called B accent following Bolinger (1972) and Focus with a H*L, called A accent.

To help the reader produce the pitch accents of structures with pitch accent on both the Topic and Focus elements, he suggests a diagnostic consisting of two questions, one general to pick out the Topic element, *What about...?* and a second constituent question where the focal element is the *wh*-constituent, and expressing the question with nuclear focus on the element prompted for in the first question.

According to Jackendoff, all preposed constituents will receive the B accent and normal Foci will receive an A accent. Resulting in examples (121a – 121d).

(121) a. BAGEL(B) I don't like to EAT(A).
b. At six o'CLOCK(B), FRED(A) walked in.
c. Of all the people I've ever MET(B), you are undoubtedly the UGLIEST(A).
d. As for FRED(B), I don't think HE(B) can MAKE(A) it.

The main exceptions, according to Jackendoff is preposed elements that are negative, *wh*-constituents and *only*-phrases

2.6.2 Discourse functions traditionally described as topic

Any definition of topic has to be explicit about whether all sentences have a topic, and whether it is possible to have more than one topic in a main clause. Many studies have claimed that not all sentences include an element that it is about. In a context such as (122).

(122) A: What happened?
B: The staff ate a cake.

The information in B's answer is not considered to be linked to an established referent, i.e. there is no obvious file card on which to add the information 'ate a cake'. Instead, it is assumed that the sentence result in a completely new file card being opened, on which the information in (122) is recorded.

Such topicless sentences are termed *all-focus* or *thetic* sentences and are contrasted with a sentence in which information is predicated of a topic, a so called categorical sentence. The difference between a thetic and a categorical sentence cannot be seen in the syntactic structure in

a language like English. However, it is manifested in intonation. In (123a) (from Vallduví and Engdahl, 1996) the categorical sentence has a canonical stress pattern with the main stress on the final constituent.

(123) a. The screen *died*.
b. The SCREEN died.

This signals that there is a topic-comment relation between the two constituents. In (123b) the main stress is on the subject, which indicates that the entire proposition is in some sense new to the discourse, i.e. a response to a more general knowledge gap, as in (122). As a consequence, (123b) is considered to lack a topic. The information is not linked to any established referent in discourse. Although it is clear that the referent of "the screen" is assumed to be known to the addressee, (123b) is not uttered to provide information about "the screen", it is more generally about what happened. See Vallduví (1993, p. 28) for discussion of Swedish topic fronting and the argument for not considering weak pronouns as part of the information structure Vallduví (1993, pp. 16).

The fronting of a non-subject complement is the resolution of the conflict between expressing a particular discourse referent as the topic and the need to express a particular predication where the topical element is not the highest thematic role among the arguments. As a consequence, it is necessary to indicate both its semantic relation to the predicate and its discourse functional role as topic. The different equivalences between languages reflect possible solutions to this conflict. Passives promotes a lower thematic argument to subject, fronting of a given element signals that it is the topical element where the rest of the clause is the information predicated about that element.

In terms of the subcategorisation of topics following Frascarelli and Hinterhölzl (2007), it is important to recognise the distinction between a familiar topic and a continuing topic. This distinction is discussed by Frascarelli and Hinterhölzl but in the current study we will argue for different definitions of familiar and continuing topic. The familiar topic is a topical element of a sentence that is assumed to be salient for the discourse participants, i.e. it is both active and given. In contrast to Frascarelli and Hinterhölzl (2007) and Vallduví (1993), this study will consider familiar topics as similar to shift/switch topics in having the functional property of being 'about' what is being predicated by the sentence. However, among familiar topics, we can define a subcategory of continuing topic, which has a particular relation to the previous context in having been the topic of the immediately preceding sentence. Continuing topics are thus examples of what Daneš (1974) captures with the notion of "continuous theme" whereas familiar topic also includes "simple linear progression". Although all familiar topics tend to be realised as proforms and often is prosodically attenuated, continuing topics have a greater ability to be elided. This can be illustrated by the following exchange.

(124) You know HA/RRY.

(125) a. He bought a new C/AR, yesterday/# ø bought a new car yesterday.
b. ø clearly can't stop BUY/ing stuff.

(126) a. He bought a new C/AR, yesterday.
b. It is the latest model FO\RD/# ø is the latest model FO\RD.
c. ø runs like a charm.

2.6.3 Stage topic

An alternative to claiming that sentences like (123b) are topicless, which paradoxically would make them "about" nothing, is to say that it is the temporal and spatial reference frame in which the sentence is used that is the topic (Gundel, 1977), so called "stage"-topics (Erteschik-Shir, 1997).

Another example of a sentence that can be interpreted as lacking an explicit referent as topic is (127).

(127) As regards the World Cup, all the group matches have now been played.

If we accept the "as regards"-diagnostic as a means of identifying the topic, *the World Cup* is not part of the *all the group matches have now been played*. It can be argued that there are reasons to distinguish this type of sentence from a thetic structure that is assumed to have little restriction in terms of expected information, i.e. prompted by a "what happened?" contextual frame. In (127) the topic as indicated by the "as regards"-phrase, is not part of the following clause, but there is an inferred link between "World cup" and "the group matches" (cf. (Jacobs, 2001, p. 642) on *free topic*).

Krifka (2008) discusses the role of initial elements in the clause which often pass the "as regards"-diagnostic but which Krifka is reluctant to accept as topic in the same sense as verb-controlled arguments, and discourse referents.

(128) A: How is John?
B: {Healthwise/As regards his health}, he is [FINE]$_{\text{Focus}}$

(129) A: How is business going for Daimler-Chrysler?
B: [In GERmany]$_{\text{Frame}}$ the prospects are [good]$_{\text{Focus}}$, but [in AMErica]$_{\text{Frame}}$ they are [losing MOney]$_{\text{Focus}}$.

Krifka argues that this clause-initial information or frame-setter, restricts the applicability of the information to a specific subdomain, a notion that is discussed in Chafe (1976) as *Chinese topics*. These types of constituents share similarities with the treatment of initial elements in Systemic Functional Grammar (Halliday, 1967), where the term *theme* is used for all elements that are placed at the left edge of a clause. Krifka points out that frame-setters are used in contexts where there are competing frames, and there is therefore a need to establish which frame is currently relevant. If no such competition exists, the frame information is redundant.

Krifka (2006b) claims that frame-setting is not a device that is used to indicate a way to link the current utterance with the context ("Common Ground" according to Krifka), i.e. it is not allocating the main information to a particular file card. Instead, it is a way to qualify how information is recorded on the file card. "is fine" will be registered on the card of "John", but the information is restricted by the frame *as regards health*.

The intonational pattern demonstrated by (123a) with the main pitch accent on the subject has, in certain contexts, been argued to constitute a third type of information structure, in addition to categorical and thetic structures. When (123a) is used to correct a previous assumption of what the current topic is, as in (130).

(130) A: I heard you got a new computer because the hard drive died.
B: The SCREEN died.

The analysis of B's utterance in (130) has been argued to be an example of argument focus (Lambrecht, 1994, pp. 228ff) which lacks a topic. Alternatively, the entire NP "the *screen*", is seen as a contrastive topic (Erteschik-Shir, 2007). Although "the screen" is not mentioned in

the context, it could be argued that it is anchored in the context by being part of "computer". Prototypical examples of contrastive topics are (131) and (132) (from Jackendoff, 1972, p. 261) where the topic is explicitly mentioned in the question and there is a presupposed restricted set of individuals who have been eating, and to which Fred is contrasted. Similarly, the stress and pitch accent on "beans" in (132) evoke a contrast set.

(131) A: What about FRED? What did HE eat?
B: [FRED]$_{CT}$ ate the BEANS.

(132) A: What about the BEANS. Who ate THEM?
B: FRED ate the [BEANS]$_{CT}$.

These types of prosodically signalled information structures have been given alternative analysis in Wagner (2012) and Titov (2013).

Lambrecht argues that there is no topic in the answer to the following exchange:

(133) A: Who went to school?
B: The CHILDREN went to school.

By saying that the answer in (133) does not predicate something about "the children" instead it identifies "the children" as the missing referent in the open proposition which is the question, i.e. [X went to school]. The information structure of the answer can thus be expressed as [X went to school: X = the children] whereas a categorical sentence answering a question like "What did the children do?" can be expressed as [The children did X: X = went to school].

Lambrecht distinguishes between pronouns whose relation to their antecedent is one of topic and those whose relation is one of focus. In (134), the pronouns "her" and "she" are related to "Pat" as topics.

(134) A: Has Pat been called yet?
B: a) Pat said they called her TWICE.
b) Pat said she was called TWICE.

(135) A: Who did they call?
B: a) Pat said SHE was called.
b) Pat said they called HER.

This distinction is based on the intonational prominence that is given to "she" and "her" in (135) and the corresponding lack of accent on the pronouns in (134). In addition, Lambrecht make use of cross-linguistic evidence by referring to Italian and French data, where topic and focus relations are signalled both prosodically and syntactically.

(136) a. IO PAGO (It) – MOI je PAYE (Fr) "I'll [PAY]"
b. Pago IO (It) – C'est MOI qui paye (Fr)"[I]'ll pay"

It is thus claimed that the topic of a sentence is a referent that has been previously established in discourse, but not all such referents are necessarily topics. The information in a sentence that is added to discourse has to be introduced as being about the referent. In a language like English, this makes the subject the default topic. However, if the context establishes a knowledge gap (The piece of information asked for in a constituent question, e.g. "What happened to the last biscuit?"), an utterance providing that piece of information will make the referent that fills that information gap the topic even though it is not the subject ("I ate it"), and even though the subject is also established.

Somewhat controversially, Krifka (2008), Lambrecht (1994, p. 147ff), and Erteschik-Shir (1997) argue that a sentence can have multiple topics. Lambrecht makes the point by saying that in a sequence such as in (137)

(137) I am proud of the time I spent writing this thesis.
The product I feel less good about.

The second sentence contains two topics: "The product" and "I". They are both marked as being topics, "the product" by being a lexical NP in non-canonical position, and "I" by being a non-accented pronoun. Lambrecht categorises "I" as the primary topic, based on the fact that the referent of "I" is what the previous context has as topic, whereas "the product" is a secondary topic, since the writer expresses information "about" the product. In addition to being "about" the two referents, a sentence with one or more topics also contains information about the relation between the different topics.

Erteschik-Shir (2007) makes a similar point with the exchange

(138) Q: Did you see Peter yesterday?
A: Yes, I saw him at school.

Her arguments are that both "I" and "him" are established and therefore topics, but for the answer to have any truth value, one of the topics has to be the main topic, the topic is the referent in relation to which the sentence is assessed in terms of truth value.

Lambrecht recognises the risk of making the notion of topic vacuous by permitting all given/old referents to be topics in a sentence. He claims, however, that there is a clear difference between "He married Rosa" in a context where Rosa is mentioned for the first time, and "He married her". The first sentence contains a single topic "he" whereas the second one contains two, "he" and "her". The crucial difference is the previous mention of the referent of "her" in the second sentence. The key point is therefore the relation that given referents have to the immediate context.

Lambrecht's analysis suggests that encoding a discourse referent as the subject gives that referent prominence in comparison to other discourse referents in the same sentence that have "aboutness" properties.

To sum up this rather lengthy discussion about the information-structural function of topic, it has been argued that there is no single coherent and clearly delineated notion of sentence topic, but rather a set of different discourse functions that traditionally have been combined into some composite concept, and that these should better be treated separately. The task of establishing these different functions is ongoing, but for the current study, we will be content to explain the different interpretations and uses of fronted arguments as the result of the realisation of the information-structural primitives of contrast and focus, combined with the discourse status of givenness. The move to do away with topic should not be interpreted as a denial of the discourse functional phenomena that have been treated in terms of topic or topic feature, only that we need to find other theoretical solutions than an intuitive but vague notion to analyse these effects.

Moreover, it may well be that in specific analyses of some linguistic phenomena the term topic is being used for a consistent notion, and thus the general use of the term topic is unproblematic and has no effect on the particular analysis. However, the argument presented here is that the term is being used in different studies without any consistency, which is problematic, and it would benefit the development of a consistent analysis if other terms are being used for clearly different discourse functions.

Taken together, the unresolved issues that have surrounded the notion of topic since its conception suggest that the phenomena that are argued to relate to topic are not expressions of a single concept, but a number of different functions.

The definition in terms of pragmatic aboutness, together with the fact that not all assertions have topics, raises the question of what these topicless assertions are about.

2.6.4 Discourse development and Thematic progression

Previously, we looked at one of the more influential notions that have been proposed for modelling the information that all discourse participants share in terms of the common ground. However, although this theoretical concept gives us a coherent and systematic way to separate information that has already been established in discourse from information that is just being introduced, it is a rather crude concept that ignores many of the phenomena that relate to discourse coherence and developments, as exemplified by the criticism expressed by Kamp (1981). One aspect of this, reference tracking, i.e. which references that we have established as being part of the common ground, are at any one point in the discourse development available for being linked to by pronominal elements. The common ground does not provide any information of referents, apart from them being either in the common ground or not. Several solutions have been suggested to address this issue. A number of the proposed solutions employ the notion of salience (Arnold, 1999; Blumenthal-Dramé et al., 2017; Chiarcos et al., 2011).

A factor that has been advanced as being important for the order of elements in a sentence is the information flow, i.e. the organisation of information that will simplify the incorporation of new information to the common ground or discourse model. Placing information that provides a frame for the rest of the sentence or links the information to the common ground initially will facilitate the processing of information for the reader/listener and contribute to the coherence of the text (Biber et al., 1999, p. 897).

The analysis of topics is often framed in minimal question-answer contexts which downplays the relation between topics of consecutive sentences. Going back to Daneš's notion of thematic progression (1974), a distinction is made between topics that are carried over from the previous sentence, sometimes called *continued topic*, as illustrated in (139) from Erteschik-Shir (1997), and topics that are changed from the topic of the previous sentence, called shifted topics or "switch" topics, as in (140).

(139) $[\text{John}]_{\text{T}}$ likes to read. $[\text{He}]_{\text{T}}$ is intelligent and industrious and will go far.

(140) $[\text{John}]_{\text{T}}$ envies Mary. $[\text{She}]_{\text{T}}$ is intelligent and industrious and will go far.

It has often been assumed that new discourse referents are introduced in the manner shown in (139) where they are initially part of the predication/comment of a sentence with an already established referent as subject. This makes the referent then available to be selected as topic of a subsequent sentence. Alternatively, a new discourse referent can be introduced in thetic or stage-topic sentence, such as (140), from Erteschik-Shir (1997), her exampel (26).

(141) $[\text{Time,Place}]_{\text{T}}$ A boy came into the classroom. $[\text{He}]_{\text{T}}$ gave the teacher the book he was holding in his hand.

Erteschik-Shir bases her notion of stage-topic on similar treatments of spatial-temporal indices as topics suggested in Gundel (1977) and Reinhart (1981). According to Erteschik-Shir only stage-level predicates can have stage-topics.

There are, however, a number of exceptions to this principle of having to introduce discourse referents as part of the predication/comment before using them as topics. All referents that are considered to be part of the discourse context, such as the interlocutors and their real-world context can be used without any previous mention. In addition, entities like "the President" and "the sun", which are uniquely salient (E. F. Prince, 1981b) are constantly available. There are

also several types of inferred discourse reference that can be used, such as entities "anchored" by other referents, as illustrated below, from E. F. Prince 1981b:

(142) I got on a bus yesterday and the driver was drunk. "a bus" = unanchored, "the driver" = anchored by "a bus"

(143) A guy I work with says he knows your sister. "a guy" = anchored by "I work with"

Discourse referents can also be culturally anchored, such as "the neighbour across the hall" or "my car". The latter is anchored both by the possessive pronoun to the speaker, but the referent has to be culturally acceptable as part of the set of entities that is normally in such a possessive relation. Thus, "my rocket" would require that it is introduced in the discourse model/common ground.

It should be noted, however, that text type can also influence the ability to use new, non-inferable discourse referents. As pointed out by Lambrecht (1994, p. 196) there are genre specific possibilities. In fiction texts, characters can be used as if they were established referents without previous mention. They are accommodated in the sense discussed by Lewis (1979). Example (144) is the opening line from Jim Crace's *Arcadia*.

(144) No wonder Victor never fell in love. (JC1.1.1.s1)

By using a proper noun as subject with a discourse referent that has not been established and is not universally salient, the addresser indicates that the addressee should process this as if it were an established referent. Clark and Haviland (1977) call this "addition" and note that this is often used as a literary device.

Svenonius (2002) argues that the pre-verbal position in V2-languages such as German and the Scandinavian-languages is the default position for shifted or switch topics. If the topic is a continued topic, the position will be filled by non-topical elements that signal changes in the discourse model such as shifts in the temporal-spatial frame. Only in cases where there is no shifted topic and no changes in the discourse model will the pre-verbal position be filled with a continued topic.

The pre-verbal position according to Svenonius (2002) is the default position for "the elements in the sentence that is regarded by the speaker as most significantly redirecting the emphasis of the discourse". This includes "contrastive focus, speaker oriented adverbials, discourse connectives, scene-setting adverbials, and switch topics.

Lambrecht (1994) also includes the criterion for topics that they need to be "discourse referents" according to Karttunen (1969). This means that only referring expressions can be topics, and that the referent must be pragmatically established in the universe of discourse of the interlocutors.

What counts as information structure of a sentence is not the truth value of the proposition expressed by it but its *information value* in a particular discourse. This information value depends not only on the meaning of the sentence but also on the presuppostional situation in which the sentence is uttered.

A crucial point for Lambrecht is to make a distinction between pragmatic properties of referents, such as activation and identifiability and pragmatic relations like being topics and having focus. Lambrecht disagree with E. F. Prince (1984) analysis of fronted objects as in "History I found to be dry" as signalling the activation state of a referent. Instead the idea is to signal that the fronted object is a particular kind of topic. The reason for needing a fronted construction is to prosodically mark the object and had this been done in its canonical position, it would be interpreted as the focus of the sentence. In order to have a contrastive or referent-establishing function it needs to be fronted.

The condition of interpretability provides the best explanation for the relationship between topic function on the one hand and the activation and identifiability properties of topic referent on the other. "In selecting a topic for a sentence, a speaker makes a communicative decision as to the "point of departure" (Lambrecht, 1994, pp. 162–163) for the new information, i.e. as to the entity that she wishes to convey information about.

(a) The speaker makes hypothesis concerning the status of the referent of the topic in the mind of the addressee at the time of utterance

(b) The speaker decides on the form of the sentence in which the topic is to be coded.

(c) The speaker makes a communicative decision as to the "point of departure" for the new information, i.e. the entity she wants to convey information about.

The concept of "aboutness" which is normally referred to as "topic" is a pragmatic relation between the discourse model and a discourse referent of a sentence that is added to the discourse. The discourse model is assumed to be a structured representation of the communicative context and the information introduced in the discourse at the point the sentence is added. A topic has the properties of being both given, i.e. already in the discourse model, and among the most salient referents, i.e. being among the discourse referents in the model that is at the top of the referent stack. The default realisation of the topic is the syntactic subject. The function of the topic is to serve as the reference point relating the new information to the already established information in the discourse model. A distinction is made between "aboutness" topics which are discourse referents and "stage" topics which are temporal-spatial indices.

2.7 Prosody

An important development in the research on word order and discourse functions such as argument fronting is to understand how morphosyntax and phonology interact. An important contribution to the nascent field of information structure and its phonological realisation is Halliday (1967). Developing a theoretical model of language based on functional criteria, Halliday makes insightful contributions to the way syntax and phonology interact when discourse develops. He describes utterances as phonologically divided into information units, *tone groups*, which do not have to align with syntactic constituents, although they are often co-extensive. Thus, the language user can make different choices in how to map these tone groups onto syntactic structures. However, when tone groups align with clauses, Halliday consider this to be the unmarked option, whereas when a tone group does not constitute a syntactic clause this is considered marked. This is illustrated in (145a–145c) with double strokes indicating tone group boundaries.

(145) a. //John saw the play yesterday// (unmarked)
b. //John //saw the play yesterday// (marked)
c. //John //saw the play //yesterday// (marked)

Example (145a) where the tone group is co-extensive with the syntactic clause is unmarked, whereas (145b) and (145c) are marked. The division of syntactic units into tone groups is significant for the interpretation of the utterance as it signals the way syntactic structures are separated into information units. According to Halliday, tone group boundaries are determined primarily by information-structural considerations but are also constrained by phonological principles.

Moreover, the tone group is the domain for the placement of information focus, which indicates the locus of new information, i.e. the central part of the information unit. Each tone group will have an information focus which is realised as a tonic syllable, a phonologically prominent syllable, primarily by pitch movement, with an optional pre-tonic syllable. There is a close connection between the notion of new information and the one of focus, but they are independent concepts. All constituents are either new or given and their relation to focus is that all focus elements are new, but not all new elements are focal.

The dichotomous separation of linguistic constituents into given or new is another functional dimension that influences the form of an utterance. Halliday points out that it is the speaker who presents information as given or new, based on his or her assumption about the knowledge shared by the speech participants; it is not some objective state of different discourse referents. Given information is expressed by anaphoric elements such as proforms and ellipsis. Halliday describes new information as being textually and situationally non-derivable, which does not mean it has to be previously unknown information, or information not previously introduced into the discourse. Thus, if the answer to a question like "Would you like tea or coffee?" is "I would like tea please" the DP "tea" would be considered new as it is the instantiation of the variable in the open proposition [I would like X (to drink)], although it is a "known" alternative, previously introduced. New information also includes contrastive or unexpected information.

Each tone group will have one information focus, which means that having several foci in a single syntactic structure requires that it is divided into several tone groups. However, the position of the focus within the tone group is not decided by the process of mapping syntax with information units. We can thus have the same division into tone groups but different location of the information focus as shown in (146) with bold typface indicating focus.

(146) a. //**Mary**// always goes to **town** on Saturdays//
b. //**Mary**// always goes to town on **Saturdays**//

However, the interpretation will be affected by both the placement of tone group boundaries and the placement of the information focus. We would then have two distinct interpretations and, most likely, different contextual constraints between (147a) and (147b).

(147) a. //on **Saturdays** //Mary always goes to **town**//
b. //on **Saturdays** Mary always goes to //**town**

Halliday describes (147a) as "normal" and (147b) as "unlikely" because of another dimension of organising the sentence as a discourse unit, which Halliday describes as *thematization*. This notion is based on the ideas of Communicative Dynamism (Firbas, 1971), but unlike the original theme-rheme structure, where placing theme before rheme is regarded as a tendency which can be overridden, for Halliday, thematization is a strictly sequentially defined functional organisation of the syntactic main clause, with theme being the constituent that is placed first in the clause. Depending on the clause type, different constituents will be considered the unmarked option. For declaratives, it is the subject.

According to Halliday, placing a constituent clause initially indicates how the current utterance is linked to the previous discourse. This is regarded as a structural realisation of the function of thematization. However, this strong dependency on left-to-right ordering has faced much criticism (Gundel, 1977, among others), primarily on the grounds that it merely provides an alternative way of describing the clause-initial placement of a constituent. However, applying a thematization analysis to the sentences in (147b) Halliday reflects that although dividing information into tone groups and placing focus within them are essentially independent, the fact that placing the time adverbial clause initially suggests that this is a unit of information and should be delimited with a

tone group boundary before the subject. The fact that it is a marked theme, i.e. not the subject in a declarative sentence, makes it functionally harmonious with making the thematic unit into a separate tone group. According to Halliday (1967), unmarked themes do not have any general tendency to form their own tone group; however, with marked themes Halliday argues that this is the "predominant pattern" both with contrastive and non-contrastive tones, and provides the examples in (148), single slash notation indicates foot boundary:

(148) a. //these **houses** //my grandfather **sold**//
b. //**that** //I /don't/**know**//
c. //**tomorrow** //John's taking me to the **theater**//

Halliday's discussion about the interaction between notions of discourse status, i.e. given and new, and information structural function on focus is of particular relevance for the analysis of argument fronting. Halliday notes that the interpretation of what is given and new depends on the context in which an utterance appears, but also argues that the linguistic properties of the units involved play a decisive role. For example, anaphoric elements can only be information focus (i.e. most informative/non-retrievable information) if they are contrastive. As information focus in English is realised by a tonic nucleus, which is broadly equivalent to the nuclear pitch accent in autosegmental-metrical accounts, making an anaphoric element carry nuclear accent will cause it to be interpreted contrastively.

Halliday (1967) is foundational to the theory of information structure, establishing many of the notions that constitute its theoretical primitives and that continue to feature in more recent overviews of information-structural terminology and concepts (Krifka, 2008; Neeleman et al., 2009), although the exact definitions and their relations are part of an ongoing debate. From the perspective of IS-fronting and other discourse related word-order constructions, Halliday (1967) explains their interpretation and usage as dependent on the combination of thematization, tone group division, the encoding of discourse referents as given or new, and the position of the information focus, applied to syntactic structure.

Approaching many of the same phenomena as Halliday, Chafe (1976) discusses the way speakers accommodate their speech to the assumed cognitive state of the hearer at the time of utterance, i.e. in addition to expressing propositional content, a speaker will indicate how this content is related to the discourse, which he calls *information packaging*. Chafe's aim is to give a clearer description of terms related to the study of information packaging, such as topic, given, and contrast, among others and how they can be distinguished from each other.

2.7.1 Auto-segmental Metrical phonology

Within prosodic phonology, a simple sentence can be analysed as consisting of hierarchically organised prosodic units (Nespor & Vogel, 2007; Selkirk, 1980, 1984). Metrical theory represents prosodic prominence in terms of hierarchically organised structures. Prosodic phonology extends this notion of hierarchical organisation to a hierarchy of prosodic constituents (Nespor & Vogel, 2007; Selkirk, 1980, 1984). Autosegmental-Metrical (AM) approaches to intonation build on this prosodic and metrical structure by relating tonal events to metrically prominent positions and prosodic boundaries (Ladd, 2008; Pierrehumbert, 1980).

For the purpose of describing focal prominence, three higher-level prosodic constituents are particularly relevant: the prosodic word, the phonological phrase, and the intonational phrase. The head of each constituent is metrically prominent relative to the other elements at the same level. This hierarchical organisation can be represented using a metrical grid, in which an x indicates the head of a prosodic constituent and the height of the columns represents relative prominence, as illustrated below:

ι	(			x)
ϕ	(x)	(		x)
ω	(x)	(x)	(	x)
	John	ate	a	donut.

Although prosodic constituents often correspond to morphosyntactic constituents, the two representations are not strictly isomorphic. In particular, they differ in the extent to which they permit recursive structures. Prosodic representations have traditionally been assumed to obey a strict hierarchy, as expressed in the Strict Layering Hypothesis (Selkirk, 1984). AM approaches to intonation incorporate this prosodic hierarchy into the representation of intonation by associating tonal events, such as pitch accents, with prominent positions and tonal boundary events with prosodic boundaries (Ladd, 2008; Pierrehumbert, 1980).

There seems today to be a fairly large acceptance of the view that placement of nuclear pitch accent in a clause is a combination of communicative choice and morpho-syntactically based default principles. Thus, both the intuitions captured by Chomsky and Halle (1968)'s Nuclear placement rules and Bolinger's communication-based principles (Bolinger, 1972) have to be acknowledged.

As discussed, the information-structural notion of focus is closely related to prosodic prominence in both English and Swedish. However, a relevant question is what relation between information structure and phonology we should assume.

Typologically, a distinction is made between languages with fairly strict requirements for where in the clause the nuclear pitch accent should be placed, and those with a more flexible placement of the nuclear pitch accent. Ladd (2008) discusses this and refers to, Vallduví (1991), which describes the first category as [−plastic], which includes Italian and Catalan, and the second as [+plastic] which includes English and Greek.

However, even languages with relatively flexible placements of nuclear pitch accent may require word order changes relative to a canonical order to achieve the correct prosodic realisation. Lambrecht (1994) suggests that the process of fronting complements is driven partly by a need to avoid topical elements from receiving nuclear pitch accent. Xu (2004) argues that Mandarin Chinese is a [−plastic] language which places the focal element in the structural position which is the default position for focal constituents rather than marking the focal element in its canonical position (morphologically or phonetically).

Although both English and Swedish are languages where the placement of nuclear pitch accent is relatively free, this does not mean that we will get the same interpretation of a canonical structure with the nuclear pitch accent placed on a constituent other than the default last constituent, and a non-canonical word order that places the same constituent in final position where it receives nuclear pitch accent. Thus, in a sentence such as (149) the default placement of the nuclear pitch accent is on"brother". However, if we move the direct object to final position but continue to give it nuclear pitch accent, we do not get the same interpretation as shown in (150).

(149) John gave the book to his BROTHER.

(150) John gave his BROTHER the book.

The difference is generally described in terms of wide and narrow focus, i.e. when a constituent receives nuclear pitch accent in a non-default position, it will be interpreted as having narrow focus, and consequently non-felicitous in an out-of-the-blue context.

Moreover, the fact that it has been hard to find any systematic differences in the properties of the accent that indicate wide vs. narrow accent, means that the default placement of the nuclear pitch accent on the final constituent makes the sentence ambiguous between wide and narrow focus (However, see Sityaev and House, 2003, among others).

Restructuring the sentence can thus have two main purposes: (a) moving a constituent out of the sentence final positions to prevent a constituent from being perceived as having broad focus, and (b)moving constituents into a sentence final position to prevent a constituent from being perceived as having narrow focus.

However, it could be argued that intonation doesn't necessarily have to disambiguate between broad and narrow focus since this will be clear from the context, thus in contexts like (151) we will perceive the nuclear accent on *John* as having broad focus, whereas in (152) *John* will be interpreted as narrow.

(151) Q: How did you get that information?
A: [I called JOHN]$_{\text{F}}$

(152) Q: Who did you call?
A: I called [JOHN.]$_{\text{F}}$

2.7.2 Prosodic phrasing

In terms of phonology, an important aspect for the analysis of the interpretative potential of a particular syntactic structure is prosodic phrasing, i.e. establishing the boundaries between prosodic constituents and what consequences different prosodic organisation have on accent placement and communicative function. Of particular importance for displaced argument constituents, like argument fronting, is the distinction between phonological- and intonational phrases and how these are mapped onto syntactic constituents.

According to Selkirk (2005) a standard sign of the edge of an intonational phrase is a H-L% continuation raise combined with a H* accent, the appearance of a longer pause (size 4 boundary index, compared to size 3 for Major Phrase), also upward pitch reset.

The phrasing of prosodic units will affect the interpretation of any linguistic structure that is phonologically realised. Much of the interest has been directed at the placement and various characteristics of pitch accents on syllables in words and particularly the placement of the nuclear accent and its relation to the discourse functional notions of topic and focus. Edge tones have been regarded as indications of continuation and illocutionary force, but it may also be influenced by syntactic structure and the prosodic realisation of boundary tones caused by the morphosyntax-phonology mapping may also affect the interpretation of a structure. A relevant point in this respect is the way the left-edge of an intonational phrase is phonologically realised by edge tones.

Following Hamlaoui and Szendrői (2015, 2017) who argue for an analysis that places the left boundaries of intonation phrases based on the position of the finite verb, there is a difference between V2-languages and subject first-languages like English in that the verb in the former raises to a position in the CP-layer whereas in English the finite verb remains in Spec-IP. This affects the prosodic structure by creating a nested IP-structure in English whenever an element which is not base generated above the subject is displaced to the CP. This can be compared with the discussion in Féry (2011) which is primarily concerned with the recursive nature of prosodic phrases, but briefly touches on larger structures, suggesting the analysis in (153a).

(153) a. [Maria]ι [hat [ein Buch]$_{\text{p1}}$ gekauft]ι
b. [Maria]ι [ist [nach Berlin]$_{\text{p1}}$ gefahren]ι

Féry (2011) gives as an example of contexts that may result in recursive intonational phrases when the subject is a contrastive topic or an aboutness topic.

2.7.3 Accent placement

The distribution of pitch accents within a prosodic constituent is fundamental to signalling information structure, establishing the relative prominence between given and new information. Within Autosegmental-Metrical (AM) phonology, the placement of pitch accents is widely held to be governed by the interplay between syntactic constituency, discourse context, and pragmatic focus.

2.7.3.1 Focus-to-Accent Theory and Focus Projection

A foundational framework for understanding the relationship between prosody and focus is the *Focus-to-Accent* (FTA) approach (e.g., Ladd, 2008; Selkirk, 1984; von Stechow & Uhmann, 1986). The core thesis of FTA is that semantic or pragmatic focus requires pitch accentuation, where the focal constituent must contain the primary prosodic prominence—typically the *nuclear accent*—of its prosodic phrase.

Since the mid-1980s, the FTA paradigm has underpinned influential modelings of focus semantics (e.g., Krifka, 2006a; Rooth, 1985, 1992) as well as accounts of the syntax-prosody interface (e.g., Féry, 2006; Steedman, 2000). A central question within this framework is how a single pitch accent on a sub-constituent can signal broad focus over an entire verb phrase or clause. This is accounted for using *focus projection* rules (Selkirk, 1984, 1995), whereby focus assigned to a head or argument can project up to higher syntactic nodes, rendering the larger phrase focal without requiring multiple individual pitch accents.

2.7.3.2 Challenges to Focus-to-Accent: Clause-Type Specificity

Despite its widespread acceptance, the universal application of Focus-to-Accent has been challenged. For instance, Bocci et al. (2021) demonstrate that *wh*-questions in Italian do not conform to standard FTA predictions, as the focal *wh*-element does not consistently carry the main prosodic prominence. To reconcile such empirical counterexamples with focus theory, it has been proposed that accent placement rules may be *clause-type dependent*. Under a Questions-under-Discussion (QUD) framework (Roberts, 1996), while declaratives default to standard FTA principles in response to explicit or implicit discourse questions, non-declarative structures (such as interrogatives or imperatives) may follow distinct, illocution-driven prosodic conventions.

2.7.3.3 Prosodic Realization in Specific Systems: Swedish and Beyond

The phonetic and phonological realisation of focus placement varies substantially across languages. In languages like English or German, focus assignment primarily alters accent location and pitch range. In contrast, Scandinavian languages like Swedish integrate focus marking with a complex word-accent system (Accent 1 vs. Accent 2).

As shown by Myrberg and Riad (2015) and building on Bruce (2012), focus in Swedish is marked not only by shifting stress, but by appending a dedicated focal H tone to the lexical word accent. This results in distinct two-peaked pitch contours for focalised Accent 2 words. Consequently, accent placement in Swedish serves a dual function: maintaining lexical pitch accent contrasts while simultaneously signalling post-lexical focal prominence within the intonational phrase.

2.7.4 Syntax-phonology - contrastive differences

A central aspect in the explanation of the differences between the use and interpretation of argument fronting in English and Swedish is the differences in phonological realisation of topic-

fronted constructions in the two languages. It has been argued in various studies, based on both introspective and corpus data Féry (2006), that in a co-text like in (154) the natural pitch contour for an English fronted structure is to have two pitch accents: one on the fronted DP and the other on the predicative part of the remaining clause, as in (154).

(154) (What about Henry? How did Susan inform HIM?)
HIM she CALLed

However, in Swedish the general pattern is to have only one pitch accent, which is placed at the end of the fronted element/beginning of the predicative part, as in (155).

(155) Q: (Henry då? Hur informerade Susan HONOM?)
Henry then? How informed Susam him?

A: Honom RINGde hon.
Him called she.

Q: (What about Henry? How did Susan inform HIM?)
A: HIM she CALLed

The argument for considering these constructions as equivalent, although they have distinct prosodic patterns and partly different interpretations, is that in both constructions the nuclear pitch accent is on the verb. Thus, in Swedish both the canonical and the fronted construction place the nuclear accent on the verb. In order to understand how the two languages differ syntactically and prosodically and how this affects the interpretation and the communicative options available, we need to establish what is generally congruent in the two languages in relation to different co-texts. The questions used in (154) and (155) are intended to provide a context similar to that used by Jackendoff (1972) for what is now considered to be contrastive topics, but with a prosodic prominence on the object rather than the subject. This type of context is congruent with a T-fronted construction in English as (154) illustrates. However, despite placing prosodic prominence on the object 'him' in the question 'how did you inform HIM', there is no perceived markedness to using a canonical structure with a single nuclear accent on the verb, as in (156).

(156) (What about Henry? How did Susan inform HIM?)
She CALLed him.

This option is also, as already mentioned, available in Swedish, also with full congruence, as shown in (157).

(157) (Henry då? Hur informerade Susan HONOM?)
Hon RINGde honom.

In terms of phonology, we get two different patterns in English, where there are two pitch accents in the fronted construction of T-fronting in English, whereas the canonical structure in English has the same pattern as both the fronted and the canonical structure in Swedish. The fronted structure in English thus stands out by having a pitch accent on the fronted element. The only other possibility of placing a pitch accent on the fronted element is to make it the nuclear accent. However, this will not be felicitous in the same co-text as in (154 and 156) as shown in (158). Instead we seem to require a co-text congruent with the matrix clause is given and the fronted element contrastive (159).

(158) (What about Henry? How did Susan inform HIM?)
#HIM she called.

(159) (What about Henry? Susan neither called me nor Peter, but...)
HIM she called.

The same restrictions and conditions also appear to apply to Swedish, with (160) being the equivalent to (159).

(160) (Henry då? Susan ringde varken mig eller Peter, men...)
HONOM ringde hon.

From these examples, it is clear that the difference between English T-fronted construction and the English canonical structure, as well as the Swedish fronted and canonical structure, is phonological, primarily the placement of pitch accents. The question that we need to answer is therefore, what decides and restricts the placement of pitch accents in these structures. We will at this point tentatively propose that this is caused by the combination of syntactic structure and the way phonology is matched up with this structure.

Starting with the canonical structures in English and Swedish in (156) and (157), we have a subject and an object that are both given and pronominalised.

As discussed in Section 3.2, standard generative analyses account for the structural differences between English and Swedish in terms of finite verb movement. While the English finite verb remains within the inflectional layer (I^0, with the subject in [Spec, IP]), the Swedish finite verb raises to the C-domain (C^0, with the subject or fronted constituent in [Spec, CP]) to satisfy the Verb-Second (V2) constraint.

Myrberg and Riad (2015) discusses the prosodic realisation of various information structural constructions presented in Krifka (2008) and notes a great number of similarities in the placement of pitch accents between the two languages. The tone of contrastive topics are, however, distinct where the English L*H tone in English is a (H)LH-accent in Swedish. Myrberg and Riad observes that there is rarely any boundary tone following the contrastive topic accent. Also, any element in initial position will receive this prosodic accent regardless of its information structural function.

2.7.5 The placement of nuclear pitch accent and the interpretation of Focus

In the prosodic analysis of an utterance, one constituent carries the primary phonological prominence. Traditionally termed the *nucleus* or *sentence stress*, this prominence is standardly referred to as the *nuclear pitch accent*. The principles governing which constituent receives this primary accent have been central to generative linguistics, involving a direct mapping between syntactic structure, prosodic weight, and information status.

A foundational formulation of this mapping originates with Jackendoff (1972), who established the *Focus Prominence Rule*. Jackendoff proposed that the semantic feature [+FOCUS] must formally correspond to the main stress of the sentence, establishing that a constituent marked for focus must contain the nuclear accent. Within modern interface frameworks, this insight was developed further by Reinhart (2006) as the *Stress-Focus Correspondence Principle*. Reinhart argues that default prosodic rules (such as the Nuclear Stress Rule) automatically assign main stress to a default structural position (e.g., the rightmost constituent in English). When the intended focal domain does not coincide with this default position, a costly interface operation—prosodic shift or focus-driven accentuation—is triggered to shift the nuclear accent to the focal target.

In Autosegmental-Metrical (AM) phonology, this interaction between accent placement and syntactic structure is mediated by a hierarchical prosodic organisation (Ladd, 2008; Selkirk, 1984). Syllables and feet are grouped into prosodic words, which in turn map onto higher-level domains: the *prosodic phrase* (*p*-phrase) and the *intonational phrase* (ι-phrase). Although these prosodic

constituents correspond roughly to syntactic phrases and clauses, the mapping is non-isomorphic. Because nuclear pitch accentuation and focus projection primarily operate at the level of the *p*-phrase and *ι*-phrase, the present overview focuses specifically on these domains.

Building on these interface principles, this section provides a descriptive background for analysing the discourse-functional properties of object- and complement-fronted structures in Swedish and English. We will briefly go through, the realisation of focal elements through the placement of nuclear pitch accents; phonological phrasing and destressing given information.

Consider the canonical English declarative in (161).

(161) Robert sold his old car.

In accordance with default nuclear stress assignment (Reinhart, 2006), placing the nuclear accent on the rightmost element (*car*) yields a wide range of focus readings (broad focus on the VP or S, or narrow focus on the object). However, applying Jackendoff's focus rule via prosodic shift allows the accent to fall on non-default elements (*Robert* or *sold*), restricting the reading strictly to narrow focus on the accented item.

(162) a. ROBERT sold his old car.
b. Robert SOLD his old car.
c. Robert sold HIS old car.
d. Robert sold his OLD car.
e. Robert sold his old CAR.

Further, we can conclude that with the exception of (162e), all the placement of pitch accent will result in the type of focus normally described as contrastive or narrow focus, i.e. assuming the general interpretation of focus which evokes establishing of a value among possible alternatives, the alternative set consists of referents equivalent to only the element receiving the nuclear pitch accent, i.e. the alternative set for (162a) would be {Lisa, John, Peter..} and the alternative of (162b) would be {bought, painted, repaired..} etc.

The placement of the pitch accent on CAR can function as narrow focus but also makes it possible to 'project' to include only the head of the DP, the entire DP, the VP and the entirety of the sentence. This flexibility of interpretation means that placing the nuclear pitch accent on the final content word is the default assumption if no other indication exists in the context. This observation is the origin of the discussion surrounding the nuclear stress rule (NSR) (Chomsky & Halle, 1968) and objection to a strictly structural definition on placing nuclear pitch accent (Bolinger, 1972).

The definition of nuclear pitch accent is that it is the last pitch accent in the intonational phrase, which we can for the moment take to be more or less equivalent to the syntactic clause. This means that we cannot have any post-nuclear pitch accent. However, something that adds flexibility and complexity to the placement of nuclear pitch accents is the fact that we can manipulate phonological phrasing. Essentially, we can divide the morphosyntactic structure into intonational phrases that are smaller than a clause.

Moreover, of the higher-level phonological units: prosodic words, prosodic phrases, intonational phrase, there is no strict adherence to these being isomorphic to morphosyntactic constituents. Although there are restrictions on how to form phonological phrases across morphosyntactic boundaries (Selkirk, 2005), phonological phrasing is a resource for expressing discourse functional meaning, i.e. for communicative reasons we can make the subject into a separate intonational phrase to express surprise or incredulity, as in (163).

(163) ROBERT!? // sold his old CAR!?

Consequently, we need to have a firm grasp of what we can assume of the way a particular structure can be rendered phonologically, based on a particular written structure in a certain context. The influence that morphosyntactic structure has on the phonological rendering of an utterance can be observed in the different information-structural constructions, like clefts, and passives. By manipulating the structural position in which certain clause elements are placed, these elements will be given default pitch accent, i.e. an *it*-cleft will place a particular phonological phrasing on the propositional information, splitting up the predication, and naturally create two focal positions.

(164) It was RObert // who sold his CAR.

Finally, the flip-side of phonological prominence is attenuation or deaccenting, which occurs when an element which in similar structures and context would receive nuclear stress, would get reduced prominence, and often realised with a proform, as illustrated by the exchange in (165).

(165) Q: What did Robert do with his old car?
A: He SOLD it.

In the prototypical cases, elements that have recently been mentioned and can therefore be considered salient in the context are treated as given and do not receive nuclear accent. The pitch accent instead falls on the verb. A systematic account of when and what type of elements will be deaccented has attracted some attention since it is unclear if givenness is something that is a property of referents or denotations, or even more general information types, as was discussed in chapter 2.

A relevant aspect of deaccenting and the use of proforms in these contexts is the available proforms in a language. It is an important parameter for deciding what the discourse-functional possibilities are. The fact that Swedish third-person singular impersonal pronouns are homonyms to demonstrative pronouns and definite articles makes it possible to use these as both strong and weak pronominals, as well as with both anaphoric and deictic reference. In contrast, the pronoun *it* is weak and cannot be fronted, similar to German *es*. We will look at this in more detail in the section where we discuss fronting of pronominal elements with no prominence.

2.7.6 Characteristics of Swedish and English fronted constructions

Based on the interaction between syntax, prosody, and information structure, fronted constructions in English and Swedish can be classified into three distinct prosodic types:

(i) Fronted element **with** prosodic prominence and a final pitch accent.

(ii) Fronted element **without** prosodic prominence and a final pitch accent.

(iii) Fronted elements **with** prosodic prominence **without** a final pitch accent.

Examples of the different phonology-based categories are found in (166 – 168). An example of (i) is presented in (166); (ii) is examplified in (167), and (iii) with (168).

(166) De röda skorna med KILklack fann jag i Siiris SKÅP.
The red shoes with heels found I in Siiris cupboard
'I found the red shoes with WEDGE heels in Siiri's CUPboard.'

(167) Den skulle du ha HÖRT.
That(one) should you have heard.
'You should have heard it'

(168) Tick-SWOSH they went.

These prosodic patterns in fronted structures in English are discussed in Gundel (1977), E. F. Prince (1981b), Ward (1988) and Lambrecht (1994). E. F. Prince categorises the difference in information structural terms where (i) is refered to as *topicalization* and (ii) as *focus preposing*. She uses the following examples to illustrate the difference.

(169) MACADEMIA nuts, I can't AFFORD.

(170) MACADEMIA nuts I think they're called.

Prince observes that in topicalizations there are two prosodic units with a short break or pause between them sometimes orthographically indicated with a comma, whereas Focus fronting only consists of a single prosodic constituent. (Dryer, 2003) argues for a further subcategorization based on the fact that in topicalizations the initial pitch accent is optional. If Dryer is right, Swedish and English have the same possibilities but where the use of the English constructions is much less frequent. One of Dryer's main claims is that topicalisation is a form of a double focus structure of which only some involves a fronted element.

2.7.6.1 Placement of nuclear pitch accent in English

Given a sentence like the one in (171) there will be three syllables that carry a pitch accent: *ang* in *Angus*, *pee* in *peeled* and *spar* in *asparagus*. The standard analysis is to regard the right-most pitch accent as the one that is the nuclear accent and the one that will be perceived (although not necessarily instrumentally) as the most prominent.

(171) Angus peeled the aSPARagus.

The assumption for this analysis is that the context in which the sentence is uttered is a so-called out-of-the-blue context, i.e. a context that is not prompting the utterer for any particular piece of information and where no information in the sentence has been mentioned and activated previous to the utterance. The nuclear pitch accent is often indicated by rendering the whole word or the prominent syllable in uppercase letters.

The fact that adding consituents to the right of *asparagus* will lead to the nuclear pitch accent being moved toward the right edge even though this information is morphosyntacically subordinate, as in (172).

(172) Angus peeled the asparagus with a KNIFE/from the MARket.

However, it is clear that different contextual conditions will affect the placement of nuclear pitch accent. These conditions can be complex and involve different types of inferencing, but as a diagnostic, it is common to use various types of constituency questions, indicating what information is regarded as the key contribution to discourse, resulting in the question-answer pairs in (173).

(173) a. Q: Who peeled the asparagus?
A: ANGus peeled the asparagus.

b. Q: What did Angus do with the asparagus?
A: Angus PEEled the asparagus.

c. Q: What did Angus peel?
A: Angus peeled the aSPARagus.

Some observation can be made from these examples; any pitch accent that occurs after the nuclear pitch accent is prosodically reduced, in natural discourse this result in proforms and ellipsis, so that the answer to (a) is simply *Angus*, (b) *He peeled them* and (c) *the asparagus*. As a consequence, the rightmost pitch accent is always the nuclear pitch accent.

Secondly, there is no difference in the placement of the nuclear pitch accent between the out-of-the-blue context where the entire utterance is new information and (c) where the contribution is object. This raises the question of what exactly placing the nuclear pitch accent on a particular constituent actually serves in terms of communicative function.

Thirdly and related to the second point, the information-structural notion of Focus and the placement of nuclear pitch accent are clearly related, but it needs to be established more formally what this relation is.

2.7.6.2 Placement of nuclear pitch accent in Swedish

Despite a long tradition of research into word accents (e.g., Bruce, 1977), sentence-level intonation in Swedish has traditionally been described as under-studied (Riad, 2014, p. 255). However, recent work—most notably by Myrberg (2010), Myrberg and Riad (2015), and Riad (2014)—has significantly advanced our understanding of the Swedish syntax-prosody interface.

In a broad Germanic comparison, Riad (2014, p. 255) notes that the default distribution of the nuclear pitch accent in Swedish closely mirrors that of English: the nuclear accent is assigned to the rightmost prosodic word within the phonological phrase (p-phrase), placing broad focus default prominence near the right edge of the clause (Myrberg, 2010). Functionally, Swedish nuclear accentuation operates in a manner comparable to other Germanic languages, serving as the primary exponent of informational prominence and focus projection (cf. Féry, 1993; Gussenhoven, 1992; Selkirk, 1984).

Despite these functional similarities, important structural differences exist between Swedish and other Germanic systems. As Myrberg (2010) observes, the mapping rules governing prosodic phrasing and accent distribution reflect language-specific syntactic parameters. In terms of interface mapping, Myrberg and Riad (2015) establish the baseline assumption that the intonational phrase (ι-phrase) in Swedish interfaces directly with syntax at the clause/sentence level (CP or IP). Within this domain, post-lexical focal tones (H_f) interact with the underlying lexical word accents (Accent 1 vs. Accent 2), giving rise to distinct multi-peaked contours that are structurally absent in non-tonal Germanic languages like English or German (Bruce, 1977; Myrberg & Riad, 2015).

This shared right-edge default behaviour diverges substantially in how each language handles prosodic phrasing under argument fronting. In English, fronting an object to [Spec, CP] typically triggers structural re-phrasing: rather than simply shifting the clause's single nuclear accent, the fronted constituent forms its own independent intonational phrase (ι-phrase). This results in a two-ι-phrase structure for a single clause, where the fronted element carries a contrastive topic accent (L+H* L-H%) in the first phrase, followed by a second nuclear accent in the remaining clause (Jackendoff, 1972; Steedman, 2000). In Swedish, fronting an argument does not necessarily induce an ι-phrase boundary. Because Swedish permits non-prominent fronting, a fronted object can remain unaccented and be phonologically integrated into a single, overarching ι-phrase alongside the verb and subject. In such configurations, default nuclear prominence (H_f) is assigned exclusively to the rightmost constituent of the unified phrase (Myrberg & Riad, 2015). Consequently, while English fronting enforces a complex, multi-phrase intonational structure, Swedish allows syntactically fronted constituents to remain prosodically unmarked within a single intonational domain, facilitating their function as non-contrastive continuation topics.

2.7.7 Summary

In this chapter, some of the most central notions of information structure have been presented and discussed to arrive at the definitions that will be used in the current study to analyse the discourse semantic and pragmatic effects of argument fronting in the English and Swedish written texts. The analysis of information structure involves relating any sentence added to discourse in terms of its relation to the assumed shared knowledge among the discourse participants, the common ground. In addition, modelling the development of each new sentence, referred to by Krifka (2008) as common ground management, also requires the notion of Question under Discussion (QUD) where the contribution of each new sentence is seen as a response to an implicit question. This enables an analysis of sentences in terms of their question-answer congruence, solving some problems of the initial concept of common ground.

Moreover, the notion of information status, i.e. the distinction between information that is presented as being already part of the common ground and information that updates the common ground, traditionally termed given and new, involves establishing the relation between a taxonomy of familiarity, with a taxonomy of linguistic forms representing these information states. In English and Swedish, the formal expressions are morphosyntactically different types of nominal expressions, and so called information packaging construction (Huddleston & Pullum, 2002, p. 1365), as well as phonologically, the absence or presence of pitch accents, and prosodic phrasing.

The close connection between information status and phonological realisation in English has resulted in much research being directed toward establishing the definition of the semantic/pragmatic notion of focus and the placement of the nuclear pitch accent.

Chapter 3

Linguistic model and morphosyntactic analysis

This chapter presents the general theoretical model of linguistic knowledge that will be the basis for the analysis of the phenomenon of argument fronting in English and Swedish used in this study. It places the theoretical discussion in a particular school of though with respect to the nature of the cognitive system of language and the way it is organised, i.e., the language faculty is a distinct module of human cognition which in itself consists of separate modules with their own set of principles. See (Scheer, 2011, pp. 497–562) for an extensive discussion about modularity in generative theory and Curtiss (2013) for supporting cognitive evidence.

A short overview is given of the type of morphosyntactic analysis that forms the basis for the discussion about the way argument fronted constructions differ from canonical ones in their discourse function, and the way we can understand the existing differences in interpretation and use between argument fronting in English and Swedish written texts.

3.1 Introduction

Four main areas will be the main focus of this chapter; first, the question of how to incorporate a theory of information structure and discourse status in a general linguistic model; secondly and closely related to the first is an evaluation and subsequent rejection of incorporating information structural concepts into morphosyntactic derivation; the third area is the syntactic analysis of the so-called verb-second phenomenon in languages like Swedish and German, and how that differs from the structure of English. Fourthly, we need an account of the syntactic differences between VP-ellipsis (VPE) and VP-proform (VPP). The results of these discussions will be decisive for the analysis of why there are differences in the use and discourse function of argument fronting in Swedish in English, particularly with regard to T-fronting.

In terms of morphosyntax, this study is based on a relatively standard theoretical framework of formal or generative linguistics, but adopts a specific stance on the issue of where the phenomena of information structure and discourse status should be placed in this theory: at the morphosyntax–phonology interface rather than within morphosyntax proper. This chapter will present the main arguments against keeping information structure within morphosyntax, but the discussion of the interface analysis is done in chapter 6.

3.2 The architecture of grammar and information structure

The current study does not commit to a specific theory of syntax, but assumes some fundamental properties of such a theory, which makes it generally compatible with those described as formal or generative in the literature.

One central tenet is a modular view of language competence, which means that the theoretical model of human-language knowledge is separated into different components based on types of linguistic phenomena. The number and granularity of these modules may vary, but we will assume that morphosyntax, semantics, and phonology are distinct modules. Theories of each module are formed on principles that are module specific, and interaction between modules are conducted through interfaces, see Arka et al. (2021) for LFG, and Ramchand and Reiss (2007) for Minimalism. The exact organisation of the general model of language competence, often referred to as the *architecture of grammar*, and the function of the interfaces are, however, still contentious and the focus of intense research (Hu & Pan, 2019; Irurtzun, 2019; Sauerland & Alexiadou, 2020). Nevertheless, the assumption that linguistic representations are subject to constraints arising from distinct grammatical components is widespread in generative approaches and provides a common point of departure for theories of the syntax–phonology interface.

More specifically, according to Minimalism, the latest iteration of transformational-generative grammar (Chomsky, 1993, 1995), the modular architecture is organised such that the morphosyntactic system is separately connected to the logico-semantic (LF) and a phonological module (PF). There is no direct link between the semantic and the phonological structure as they both independently interface with the morphosyntactic representation. Thus, there is no direct transfer of representations from meaning to phonology, only via the representation of morphosyntax. Much of the work within the theory thus involves working out the principles within each linguistic system, along with the rules and restrictions that regulate how these systems interact.

Moreover, it is argued that no information will enter the morphosyntactic derivation that does not originate in the lexicon (Chomsky, 1995) and no information will be present at the interfaces that cannot be interpreted by the modules. These are all conditions being adopted based on general principles for what is required of a model that relates linguistic form with meaning to generate well-formed linguistic expressions. Although they give needed guidance on the overall limitations of theoretical notions and organisation, there is still considerable variation in how this general outline should be interpreted when devising more specific analyses. Thus, this section will try to outline what specific assumptions we will make of the general theoretical model, and be transparent about what we will attempt to answer based on the information available, and what will have to be considered areas for further research.

Although this study takes much of its theoretical machinery, especially regarding morphology and syntax, from formal syntactic theory, most of the theoretical discussion will be concerned with the way morphosyntactic structure interfaces with phonological principles and discourse through the use of information-structural functions and the discourse status of discourse referents and propositional content.

The main argumentation rests on the syntactic module passing syntactically well-formed structures to the interface between syntax and phonology, and the way these structures 'match up' with the constituents of phonology. Apart from the aspects directly affecting our analysis of information structure, the precise model for how morphosyntax generates these structures is therefore not at issue.

However, to place the discussion about the interpretation of data, in a theoretical context, we need to give a brief overview of the main outline of a formal theory of syntax. This information aligns closely to what is presented in introductory textbooks such as Adger (2003), and Radford (2004). Thus, we will follow the standard syntactic description of sentence structure by dividing

it into three domains, called the CP domain, the IP/TP domain, and the VP domain. These are broadly conceived of as representing different interpretive/functional aspects of morphosyntactic structure Platzack (2001, p. 23), and Haegeman (2012); the VP-domain concerns the predicate/argument structure and the thematic roles inherent in the properties of the lexical verb; the TP-domain enhances the thematic structure to a state-of-affairs, and specifies the grammatical relations between all syntactic elements included in the structural representation of this state-of-affairs, such as tense, person and aspectual information; finally, the CP involves the relation between the sentence and discourse. Although generative theorizing links this domain structure closely to linguistic form, similar ideas of domains can also be found in functional theories but where these dimensions are seen as purely semantic (Dik, 1997a; Hengeveld, 1989). Also, the view of left edge of the sentence as the domain of the relation between the clause content and discourse is reflected in descriptive reference grammars like Quirk et al. (1985) which describe the initial part of a clause as instrumental in establishing clause structure type, structural relation (super/subordination) and semantic type.

However, it is worth pointing out that this way of linking structure and domains of conceptual interpretation typical of generative-transformational syntax has the repercussion that there is a strict sequential constraint in that no elements claimed to belong to the linking of proposition to discourse should be found below elements that belong to the TP or VP-domains. Clearly, this requires that syntactic means of linking sentence and discourse need to be clearly defined and demarcated from other means of combining sentences into discourse.

The way the different domains are constructed is by combining lexical items into larger structures, phrases, and clauses. Following Larson (1988) and Kayne (1994) among others, only binary branching is permitted, and syntactic phrases have the general structure as in 3.1, where ZP is the specifier, SpecXP, and YP the complement of X^0 (Chomsky, 1972; Jackendoff, 1977).

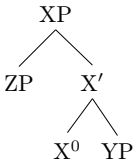


Figure 3.1: Generic phrase structure, following X-bar theory

Further, the subject is base-generated in the specifier of a head in the VP-domain. We can simplify this to Spec-VP although it is taken to be generated in the spec of VoiceP or *v*P. The resulting structure is shown in Figure 3.2, and is taken to progress in the same way in English and Swedish. At this stage of the derivation, we are concerned with the assignment of θ-roles, i.e. the thematic semantic roles (such as Agent, Theme, or Goal) assigned by a predicate to its argument constituents.

Structure building continues with the TP-layer where the standard analysis is to argue for language specific differences based on the behaviour of lexical and main verb. In English, the main verb remains in the VP receiving tense through Agree from the T^0. However, the tensed auxiliaries move to/internally merge at the head of TP. One of the standard motivations for the different behaviour of main and auxiliary verbs is the relative position of the main verb to adverbials, e.g. the fact that the finite verb remains after the adverbial *Cain always hated Abel* vs. *Cain has always hated Abel* is taken to show that the main verb remains low, whereas the auxiliary is inserted in the head of TP. In addition, the subject/external argument will move to Spec-TP because of a general Extended Projection Principle (EPP), which posits that with a

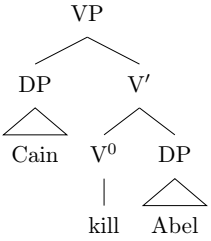


Figure 3.2: VP-layer of the sentence *Cain killed Abel*

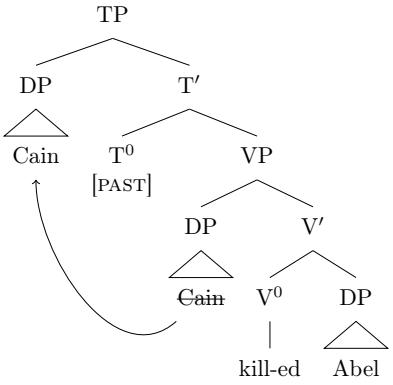


Figure 3.3: English TP-structure

finite verb there must be an overt subject in Spec-TP, even if there are no semantic argument, thus we get expletives as in *it is raining* and *there is free beer*. As a result of these processes we end up with the structure in Figure 3.3.

Essentially, transformations are ways to theoretically construe syntactic relations as movements or agree-processes by placing the same syntactic object in several structural positions of a functionally motivated phrase structure. However, the motivation and theoretical machinery for transformations radically change with the launching of the Minimalist program Chomsky (1993, 1995). From transformations being free, and only restricted by clearly defined constraints, transformations are now measures of last resort that are triggered by features on phrasal heads and lexical constituents. Thus, the reason the DP *Cain* moves from Spec-VP to Spec-TP is that it is argued that there is an unvalued EPP-feature on *Cain* that must be evaluated by pairing up with valued EPP-feature at the T^0. Something that is done by establishing a Spec-Head relation, or the valued feature C-commanding the element having an unvalued feature of the same type. The main impact of this for an investigation of fronted element is that it requires some specific feature that compels a constituent to move from its canoncial position to the left edge of the clause.

The syntactic structure of canonical declarative clauses in Swedish and English is that they differ in a systematic way along the lines of the difference between verb-second languages (such as German and Swedish) and general SVO-languages such that in the former the verb moves from

TP^0 to C^0.

It is generally assumed that the external argument of the verb is introduced in the Spec-VP in order to be assigned a θ-role and is moved/re-merged to Spec-IP to check ϕ-features. Unlike in English, where the verb remains in IP and the external arguments in Spec-IP, in V2-languages like Swedish, the finite verb moves to head of CP and the external argument moves to Spec-CP. However, it is unclear in this model where a theory of information structure will be included as it clearly has both discourse semantic and phonological effects, and will have communicative functions.

This situation has resulted in different ways of handling this state of uncertainty. Among the syntacticians there is a tendency to adopt Rizzi's (1997) incorporation of information structure into the syntactic module, which being discourse functional must be considered to be pragmatic. Although conceptually at odds with the philosophical underpinnings of minimalism, which stipulate that any context-dependent linguistic representations be kept out of syntax, it is argued that it is theoretically consistent given the need in minimalism to trigger any displacement in the syntactic structure by the system of checking grammatical features through achieving local structural relations. Another alternative is to assume that the constituent movements happen post-syntactically either in LF or PF.

However, among those phonologists and those with a greater focus on information-structural aspects and the syntax-phonology interface, including a greater emphasis on contextual constraints there is a stronger tendency to not assume independence of the semantic and phonological representation, but to argue for a model where the syntactic and phonological principles have module specific principles but that the units and structures in each module then need to align or match.

The standard analysis for the structure of the left-edge of declarative clauses in a basic SVO-language like English and a V2 VO-language such as Swedish is primarily focused on a difference in how high up the syntactic tree the external argument/subject and the verb is placed.

3.3 The role of morphosyntax

In this study, we adopt the view that discourse construction is guided by language-independent principles. Information-structural notions—such as givenness, contrast, and focus—along with the strategies used to develop coherent discourse, are taken to be universal. However, the exact linguistic realisation of these principles depends on the specific grammatical inventory of a given language; for instance, givenness may be marked intonationally in one language and morphologically in another. The theoretical and methodological tools used to analyse how individual linguistic properties contribute to discourse coherence are discussed in the present chapter and in Chapter 2.

It is generally agreed that morphosyntactic structure will play a significant part in any explanation of phenomena like argument fronting, but the way morphosyntax interacts with other parts of the linguistic faculty has yet to be established. Within formal syntactic theory, some linguists promote the view that the role in deciding the discourse functional properties of particular language structures, like TF-fronting, is solely the domain of mophosyntactic principles. However, we will instead argue that it is necessary to explain the syntactic structure of individual utterances in terms of the way syntactic and phonological principles interact relative to some discourse representation, similar to the model presented by Büring (2016) among others, where the independently generated syntactic and phonological structures are paired up or matched to produce the final utterance.

One of the theories that puts almost all explanatory force in the hands of morphosyntax, even

for such apparent pragmatic functions as discourse development, is Rizzi (1997) and the school of linguistic minimalism referred to as 'cartography'. However, the integration of pragmatic or contextual functions into syntax creates a significant problem for a theory like minimalism, the latest version of the generative transformational grammar, in that it is central to the theory that each distinct grammatical behaviour is connected to some grammatical feature which projects its own functional projection.

Rizzi (1997) main idea is that the CP-domain of the syntactic structure needs to be split into more specific functional projections such as TopP, FocP, FinP, and ForceP, and this idea has been met by a fairly broad acceptance in investigations of syntax. However, in contrast to FinP and ForceP which can be claimed to be more anchored in properties of the individual sentence, FocP and above all TopP are fundamentally within the boundary of discourse semantics or pragmatics. The problematic aspects of TopP and FocP are also that such discourse functional notions have turned out to be notoriously difficult to define with any degree of preciseness, which makes it very challenging to know if the motivation for being endowed with topic and focus features is consistent across different languages or indeed different structures. Any non-canonical structure which involves displacing constituents from lower down in the structure to a position in the left-periphery is ripe for a TopP, FocP analysis, but little effort is generally spent in trying to establish exactly what these Topic and Focus features are meant to represent. Undoubtedly there is a need for a richer structure in the area above the IP domain in languages and for that reason the added functional projections according to Rizzi (1997) is attractive. However, it is not conducive for the effort to understand the nature of these displacements to describe them in terms of topic or focus-based as if that carries any form of explanatory force, instead of the "for some discourse functional reason"-type of hypothesis that they really represent. Notably, Sigurðsson (2010) refers to Frascarelli and Hinterhölzl (2007) characterize their split up of Top and Foc into subcategories with independent properties, as convincing; however, he then pursues an analysis where these functional projections are replaced by a single Context Linking-head. In effect, adopting an analysis more in tune with a view where syntax incorporates some generic context category rather than a number of highly specific discourse functional ones.

The problem is thus both conceptual and descriptive. It is conceptual in the sense that it goes against a central tenet in a strictly modular theory like Minimalism, called the principle of full interpretation (Bošković, 2013; Chomsky, 1986). According to this principle no information should be included in the syntactic derivation that cannot be given an interpretation at the interfaces, i.e. if there is a topic feature in morphosyntax, it must be assigned both a semantic and phonological interpretation. The lack of any clear semantic effect dependent on assigning topic-status either to *Cain* or *Abel* in *Cain killed Abel* seems to violate this principle. One way around this is to stipulate that the topic and focus features are introduced into the derivation through the lexicon (Aboh, 2010), which suffers from being an ad-hoc solution in that these would then be the only contextual features that are introduced in this way.

The problem is descriptively problematic in that the function of topic seems to defy any consistent descriptive characterisation. It is generally clear, even among the proponents for there being some linguistic primitive termed topic, that it is not trivial to find some definition of this notion that is able to identify topics within a single language, let alone cross-linguistically. This is pointed out by Hartmann and Winkler (2013, p. 1) who specify the problem by asking what and how many grammatical features represent the core functions normally discussed in the literature in terms of morphosyntactic theory.

In addition, topic and focus do not behave like most other constituents that are displaced to check features in that they often have the option to remain in their canonical position. In some cases there is a clear interpretive difference between canonical and fronted position, but there are also examples where there is no discernible effect. The solution suggested by Miyagawa (2011) to

the problem of optionality, i.e. claiming that it is possible to have two propositionally equivalent structures that both are well-formed as long as there is some interpretable difference, is thus also problematic.

The possibility of movements being optional is a problem for theories advocating for the syntactic nature of information structure, since minimalism stipulates that movements of constituents should only happen as a 'last resort', triggered by the need to establish a relation between the head of the phrase and its specifier in order for some grammatical feature to be eliminated. Any movement where there is little indication of a clear grammatical effect and where movement is optional, such as heavy-NP shift, extraposition, scrambling and topicalisation, there is a need to both establish what feature triggers the movement and why this trigger is optional. Culicover and McNally (1998) discuss this and concludes, based on the phenomenon of Heavy NP-shift, that there are two plausible options to pursue in order to handle optional displacements: In the first one, you maintain the syntactic analysis where all movements are based on the existence of specific functional projections and the need to check features through a Spec-Head agreement, which pushes the issue from the mechanism of movement being optional to the existence of a functional projection being so. Moreover, in cases such as Heavy-NP shifts, this results in a complex set of movements to arrive at the final surface order, and where these individual movements need some justification independent of the final result. In the second one, syntax and information structure are separate systems which need to match up. In such a model the different options are syntactically possible but pragmatically constrained.

The latter solution is attractive for a number of reasons. Assuming a modular system, in which morphosyntax is seen as primarily a way to relate semantic meaning with morphosyntactic form independent of contextual information, introducing pragmatic notions such as topic and focus is theory-internally problematic. Moreover, given that we can develop a convincing analysis for the way morphosyntactic structure can be made to match up with a phonological representation according to the discourse functional principles of information structure, we achieve a stricter modularity and jettison the option of using a structure from the module of morphosyntax to that of discourse semantic/pragmatics, i.e. morphosyntax constraints the well-formedness of sentences, and the interaction of information structure and discourse properties will regulate what structures can be used in what contexts.

The problem of optionality is also discussed as one of the objections to regarding topic and focus as syntactic features (Szendrői, 2017) in addition to problems related to the heterogeneous behaviour of focal elements leading to a proliferation of focal heads.

Looking at the way topic and TopP is used in syntactic analyses, there are examples of the problematic nature of topic as a grammatical feature. When incorporated in syntactic textbooks like Adger (2003, p. 332), which argues for a parametric notion of features in terms of strength, the topic feature stands out as being neither strong nor weak in English but rather 'optional', whereas in a language like French it is simply listed as 'unknown'. It can be argued that this is just the effect of this phenomenon not being sufficiently investigated, but it is more likely caused by the pragmatic nature of topic resisting a formalization in terms of syntactic features.

Another suggestion that also to a large extent relies on morphosyntactic properties is Frey's analysis of fronting constructions in German Frey (2005), which distinguishes between two primary argument fronting principles: True A-Bar Movement (TAB) and Formal Movement (FM). They differ in both their syntactic behaviour and their semantic interpretation. TAB can originate from any position in the clause from which the standard principles for A-bar movement can move to Spec-C and will receive a contrastive interpretation, whereas FT is a purely formal movement of the highest available element below C^0 which has no effect on the interpretation.

Critique based on theory-internal issues has been raised by Branan and Erlewine (2021) which point out a range of theoretical issues with using a syntactic F-feature marking focus elements in

syntax, among them the lack of pied-piping in focus movement compared to the way *wh*-movement behaves. The resulting solution boils down to how to handle the semantic interpretation and the behaviour of pragmatic particles.

3.4 The Syntax-Phonology interface

When it comes to the syntax-phonology interface, there are different views on its principled design. The following section will give a brief summary of the main theoretical options, but for a more detailed overview, see Frota and Vigário (2018).

There are a couple of central issues that need to be addressed for any analysis in terms of the syntax-phonology interface which assume the basic morphosyntactic theory of minimalism. The first is not restricted to minimalism but applies to any theory that assumes a linguistic model consisting of different independent submodules, such as the separation into a semantic (LF) a phonological (PF), and a morphosyntactic module. Given such a modular system, we have to take a stand on the issue of whether the interaction between morphosyntax is unidirectional, i.e. morphosyntax affects phonology but phonology has no way to constrain syntax, or bidirectional, where the influence can go both ways. One influential formulation of the former is expressed as the principle of phonology-free syntax (PPFS) laid out in Miller et al. (1997), Zwicky (1969), and Zwicky and Pullum (1986).

There have been various objections to the PPFS, based on empirical evidence that are interpreted as requiring bidirectional influence. One analysis promoted by Culicover and Jackendoff (2005) assumes a parallel development where morphosyntax and phonology exert influences on each other at various points. Another way to allow phonology to influence morphosyntax is to view the interaction between these modules in terms of the morphosyntax generating structures according to morphosyntactic principles that are then constrained phonologically by a model that 'filters out' any structure that does not adhere to the principles of phonology (Büring, 2013). One implementation of this filtering architecture is provided by Optimality Theory (A. Prince & Smolensky, 2004) where competing morphosyntactic outputs are evaluated against ranked interface constraints (e.g. Samek-Lodovici, 2005; Selkirk, 2000; Truckenbrodt, 1995).

The second issue is more specific to a minimalist syntactic theory. One of the disruptive changes that were made in the transition from the Government-and-Binding (GB) model to Minimalism is the requirement that any displacements of constituents, reformulated as external merge, which are originally introduced at some other position in the structure, termed internal merge, must be compelled to move by having one or more grammatical features that are unvalued. These unvalued features must be paired up with a valued feature of the same type to eliminate these features. The unvalued features must generally be checked by being placed in a structural relation to the valued equivalent that results in displaced elements. This is a significant change from the GB theory, which allowed free movement, formulated as move-alpha, and the primary concern was not the motivation for movement but the principles constraining movement. Displacement has thus essentially become semanticized in having some interpretive effect (Newmeyer, 2009; Rizzi, 1997).

From the perspective of the analysis of argument fronting, this then begs the question of what feature requires the argument to be displaced to the left-edge of the sentence. This is particularly problematic given the vague notion of topic and the fact that often the canonical and fronted structures can be used in the same context with only minor interpretive differences, whereas they are incompatible in others.

Among the alternatives to a cartographic analysis is Cormack and Smith (2000) which replaces the discourse dependent and function-specific topic and focus features with gap and Φon features.

One of the most influential alternative approaches to a model that assumes a relation between morphosyntax and phonology, where the phonological component simply adopts the morphosyntactic structure with only minimal structural effects is Reinhart (2006).

Bocci et al. (2021) gives an overview of the syntactic vs interface theories of Focus. The syntactic approach is to ignore the inclusiveness condition, which stipulates that no extra features to those which are part of the numeration/lexicon can be part of the syntactic derivation, with the benefit that it is possible to maintain a parallelism between foci and *wh*-phrases by claiming that both are endowed with syntactic *wh*-features. The interface-approach dispenses with any grammatical features of discourse-semantic nature and instead the matching of syntactic structure and prosodic principles will place nuclear pitch accents on the head of the prosodic domain that is focal. The connection between focus and *wh*-movement is instead that semantically the focus element results in an alternative interpretation where the focal element is replaced by a set of possible alternatives in addition to the normal propositional representation. Similarly a *wh*-question is interpreted as an open proposition with the *wh*-constituent replaced by a set of alternatives.

3.5 VP Pronominalization and VP Ellipsis

We will end this chapter with a short presentation of one of the contrastive morphosyntactic differences between English and Swedish that has been argued to contribute to the more frequent use of fronted structures in Swedish texts: the use of special proforms in Swedish for the verb phrase, termed by Houser et al. (2007) *verb phrase pronominalisation* (VPP).

As shown in corpus studies of Swedish argument-fronted constructions (Jörgensen, 1976; Rahkonen, 2000, 2006; Westman, 1974), a fairly large proportion of the fronted elements in both written and spoken data are pronominal elements whose antecedent is a verb phrase or a clause. A possible contributing factor to the comparatively low occurrence of fronted constructions in English could therefore be related to the fact that English only has proforms for clausal antecedents and lacks VP-proforms. An analysis providing a syntactic account of these different formal possibilities is presented in Platzack (2012), who compares the use of VP Pronominalization, VP Topicalization, and VP Ellipsis in a range of Germanic languages. Platzack shows that both English and Swedish allow VP Topicalization, but that only Swedish allows VP Pronominalization, while only English allows VP Ellipsis.

Whereas VP Ellipsis (VPE) has attracted considerable attention and been investigated to the extent that van Craenenbroeck (2017) has described it as "an industry of its own", overt VP anaphora/proforms have received considerably less attention (Baltin, 2012; Mikkelsen, 2015; Ström Herold, 2009; Weir, 2024). Moreover, studies of the relation between VP Pronominalization in languages such as Swedish, which have these proforms, and the use of VP Ellipsis in languages such as English, which lack them, are rarer still, with Mikkelsen (2015) being an exception. In addition, the different grammatical properties of the proforms available in the two languages affect the possibilities for fronting in English and Swedish. Swedish third-person neuter pronouns are versatile elements in that they can refer to antecedents of different types, such as:

- **Nominal referent:** *huset* ('the house') → *det* ('it'), as in
- **Verbal phrase (vP/VP):** *åka till Spanien* ('go to Spain') → *det* ('∅/ do so') shown in (175)
- **Clausal complement (CP):** as illustrated in (176)

(174) a. Vi passerade **ett stort hus** på väg till byn.
We passed a large house on way to the village.
'We passed a large house on the way to the village.'

b. **Det** hade vi sett tidigare.
It had we seen previously.
'We had seen it before.'

(175) a. Q: Ska ni **åka till Spanien**?
will you go to Spain
'Q: Are you going to go to Spain?'

b. A: Ja, **det** ska vi.
yes it will we
'A: Yes, we will (do so).'

(176) a. Q: Tror du **att Kalle kommer**?
thinks you that Kalle comes
'Q: Do you think that Kalle is coming?'

b. A: Ja, det sa han.
yes it/that says he
'A: Yes, he said so.'

It is worth noting that in (175) and (176), it would be possible to have the VP proform in-situ, but the topicalized versions are arguably more neutral, whereas the in-situ version would be marked as implying something unexpected. According to Mikkelsen (2015), in Danish only the topicalized version is felicitous.

The differences in morphosyntactic possibilities in English and Swedish in terms of VP-Ellipsis and VP Pronominalisation, as well as the different properties of third person neutral pronouns in the two languages, are relevant for an explanation of the different possibilities in the use of fronting.

3.6 Summary

This chapter has outlined the theoretical framework underlying the analysis of argument fronting. It adopts a modular view of grammar in which morphosyntax, semantics, and phonology constitute distinct components whose representations interact at interfaces. In particular, the chapter has argued against incorporating discourse-functional notions such as topic and focus directly into morphosyntactic representations, and instead assumes that the discourse-related interpretation of syntactic structures is determined through the interaction of morphosyntactic and phonological representations at the syntax–phonology interface. The chapter has also reviewed the syntactic differences between English and Swedish relevant to argument fronting, including the structure of the left periphery in V2 and non-V2 clauses and the different grammatical possibilities associated with VP Pronominalization and VP Ellipsis. These assumptions provide the basis for the analysis of the discourse functions and cross-linguistic differences in argument fronting developed in the following chapters.

Chapter 4

Fronting in Swedish and English original texts

This chapter presents the quantitative and distributional data on the fronting constructions found in the Swedish and English original texts. They are classified according to clause element and phrasal category, and the type of canonical structure the fronted sentences are related to. Although the size of the material is too small to support statistically reliable generalizations about written language as a whole, the quantitative data give a preliminary indication of the representativeness of the sub-corpus, and the overall distribution of fronted constructions in the material.

4.1 Introduction

A general overview of the occurrences of fronted constructions in the original texts is shown in Table 4.1. It presents the absolute and relative frequency of fronted structures in the English and Swedish original texts divided into fiction and non-fiction text types. Previous quantitative studies on fronting in English and Swedish have shown a larger number of fronted structures in Swedish originals (M. Johansson, 1996), and also that there is more fronting in fiction texts in English (Biber et al., 1999). These predictions were to some extent borne out by the findings in the English and Swedish original texts in the present study.

Table 4.1: Fronted complements in English (EO) and Swedish (SO) original texts.

	EO		SO	
Text Type	*N*	Per 10,000	*N*	Per 10,000
Fict	54	8.6	157	25.3
Non-Fict	11	1.6	62	11.2
Total	65	5.0	219	18.7

The absolute number of fronted constructions in the Swedish originals is about 3 times higher than in the English originals, which is a smaller difference than might be expected by the numbers reported in previous studies. Biber et al. (1999) reported a frequency of 50–100 occurrences per million words in conversation and news and 200–300 occurrences per million words in prose and

fiction, and calculating the occurrence per million words for the data in the present study, we would end up with approximately 500 occurrences per million words in the English originals. The difference is a consequence of different definitions of what constitutes fronting, especially the inclusion of structures with inversion as part of the fronted construction in the present study. In both text types, inverted structures make up a large part of the total set of fronted structures, as seen in Table 4.2. In Biber et al. the inverted structures are given their own category. If the inversion category is separated out, the frequency is around 200 instances per one million words, which is similar to the number for the prose and fiction category.

A factor that can influence the frequency in the present study is the fact that the material consists of a fairly limited number of texts. The relatively small number of texts gives the stylistic characteristics of each individual text a fairly large impact on the general distribution. As can be seen from the texts in the present study, the number of fronted constructions varies quite substantially among the different texts, as seen in (4.2).

Table 4.2: Basic word order and inversion in fronted constructions (EO).

	EO		
Construction	Fict	Non-Fict	Total
Basic word order	22	5	27
Inversion	32	6	38
Total	54	11	65

Table 4.3: Distribution of fronting in Swedish and English original texts.

Swe Fiction		Swe Non-Fiction		Eng Fiction		Eng Non-Fiction	
Text	*n*	Text	*n*	Text	*n*	Text	*n*
AL	54	ASSI	19	AT	8	ABB	2
AP	52	BJ	13	JC	7	CS	1
GT	26	CO	9	MA	12	DM	3
KOB	8	CP	7	RD	8	JL	3
MS	19	HG	14	RR	20	LT	2

At the same time, the number of fronted constructions in the Swedish originals is smaller than the results presented in Jörgensen (1976) which showed an exceptionally large number of fronted constructions, which, in terms of occurrences per one million words, would be closer to 10,000. This can be compared with a frequency of around 2000 per one million words in the present study. Jörgensen's study was, however, based exclusively on spoken data, and included many occurrences of constructions which may be particularly frequent in spoken discourse, such as expressions with a fronted pronoun *det*, as discussed in chapter 3.

It is worth noting that the figures presented in Biber et al. (1999) show a higher proportion of fronted constructions in registers like fiction and prose than in conversation and news. The fact that Biber et al. combines a spoken register (conversation) and written (news) makes it hard to conclude anything about the differences in distribution between written and spoken English when it comes to fronting. No large study of naturally produced language has thus investigated the difference in frequency of fronted constructions between spoken and written language. Intuitively, however, the assumption would be that spoken language contains more fronted constructions

than written.

4.2 Clause function of fronted elements in the English original texts

There are few syntactic restrictions on what type of clause element can be fronted in English (Birner & Ward, 1998). The limited number of examples in the English data cannot give any indication of the general distributions of fronted structures in English, but the figures show that there is a fairly wide variation in the type of fronting that occurs in the English original texts as shown in Table 4.4.

Table 4.4: Clause elements in English original texts.

Text Type	ADVL	PC	Verb	Part	Obj	Total
Fict	37	6	4	3	3	53
Non-Fict	4	2	2	2	1	11
Total	41	8	6	5	4	64

The largest group of fronted elements in the English originals consists of different types of initially placed obligatory adverbial. The majority of these exhibit inversion of subjects and verb, but four examples have basic word order, as shown in (177) – (180). All of these examples includes the adverbs *here* and *there* with locative meaning.

(177) And **there** Joseph stayed, ...

(178) **Here** the flagstones began, ...

(179) "**There** it is," Fred said.

(180) **Here** I come, ...

Examples (177) and (178) can be argued to be compositional, i.e. not fixed phrases, in the sense that content words can be exchanged for synonymous expressions without loss of idiomaticity and communicative function. In contrast, the examples in (179) and (180) have more the characteristics of being idiomatically fixed.

Of the constructions with initial adverbial and subject-verb inversion, the great majority consists of an initial prepositional phrase or adverb expressing locative meaning, as in (181).

(181) **On the table** were three plates with food on them.

The only example of a prepositional phrase that is fronted without inversion can be seen in (182).

(182) **At the school gates** he left the car with a muttered goodbye and did not look back.

Syntactically, we analyse the *at*-phrase as a complement since it is licensed by the verb *leave* and thus it is included as an instance of fronting. Discourse functionally, however, there are many similarities between (182) and an adjunct like the one in (183).

(183) **At the local BDA office** Leith found the executives responsible for livestock to be frank and helpful.

The verbs in the clauses with inverted adverbial fronting have meanings of location or movement such as *be*, *come*, *stuffed*, *stand*, *spill out* and *lay*, but also perception verbs like *see*, as is shown in (184).

(184) **Inside** could be seen a kind of outer hall, an orangery with bay trees and lilies in tubs on the mottled white marble floor.

In (184) the full inversion is combined with a passive verb phrase which enables the omission of the agent of the "seeing" relation and makes it possible to place the subject in a position where it can get nuclear stress.

As discussed in chapter one, a construction that has a similar discourse function to locative inversion structures is the presentative *there*-construction with initial adverbials such as those in (185) and (186).

(185) **Down in the cellar** there are shelves and shelves of white rats and black mice, special ones that aren't wild. ...

(186) **In one room** there are glass cages with snakes in them, snakes bigger than any we 've ever seen before.

The prepositional phrases in these sentences can be argued to be fronted complements on syntactic grounds, but functionally they appear closer to initially placed adjuncts.

As can be seen in Table 4.4, adverbial fronting makes up the great majority of fronted structures in the English originals. There are almost twice as many fronted adverbials in the English original text as there are other types of fronting combined.

Among non-adverbial fronting, the most frequent type is predicative complements. Often these will cause a subject-verb inversion (Biber et al., 1999, p. 902), as in (187).

(187) **So old and honoured** were the patterns of the trading pitches that Rook, or so he claimed, could have walked as sure-footed as a village cat.

However, in four instances, the basic word order is kept in the matrix clause, as illustrated in (188).

(188) And **a strange sight** it was, ...

In three of the four occurrences of predicative complement fronting, the subject complement is an argument of the copular verb *be* (188). In one instance, however, the structure is slightly more complex, as the sentence structure is a passive clause with a fronted predicative complement (189). The quotations in this example should not be taken as indicating direct speech, but rather as signalling non-literal use of the phrase.

(189) **"Pitch money"**, it was called.

A clear difference between adverbial fronting and predicative complements on the one hand, and other types of constituent preposing, on the other, is that inversion does not occur in the latter group. There are two examples of fronting of direct objects in the material, as shown in (190) and (191), and one instance of fronting an oblique object, (192).

(190) **The other woman** she identified as certainly Davina Flory's daughter Naomi.

(191) **The first, the teleological criticism**, I think is wrong and I will now try to show why.

(192) **To Pemberton** he said, "Break the window.".

In (190) the object is a complement of the verb *identify* and is part of a SVOC pattern, whereas in (191) the fronted element is the embedded subject in the clausal object of the verb *think*. In terms of discourse function, we can observe that both examples are closely connected to the meaning of copular constructions without fronted constituents such as:

(193) The other woman was Davina Flory's daughter Naomi.

(194) The first, the teleological criticism is wrong.

In terms of the linear presentation of information, the two types of constructions appear very similar and appear to have closely related pragmatic functions. The oblique object is dependent on the verb *say* and part of the pattern S *say* DO IO.

In addition to the fronting of complements of the lexical verb, there are also a number of structures where it can be argued that the verb phrase itself or parts of it are fronted. In the English original many of the fronted elements are anaphoric expressions in the form of the proform *so*. The pattern for fronted *so* expressions is to accompany the fronting with partial inversion (Ref), leaving the operator in second position and replacing the rest of the verb phrase with the proform *so*, as seen in (195) and (196).

(195) "**So** did I!" the brother said

(196) **So** have costs.

The antecedent of the proform is normally expressed in the previous sentence and the fronted *so*-pattern results in additional meaning of 'too' is added to the *so*-construction. Given the sentence and context of 197, the proform *so* is analysed as an example of VP-fronting with the operator in second position. An equivalent non-fronted sentence would be as in example (197).

(197) "I heard a voice!"
   a. "So did I!" the brother said.
   b. "I heard a voice too!" the brother said.

The only instance in the English original texts of VP-fronting of a non proform is a construction with *do* insertion as shown in (196).

(198) **And get involved** he does, ...

In the constructions where verbs or verbs together with complements are fronted, we experience that it was possible to front only parts of a phrase. This phenomenon is also found with some noun phrases and subordinate clauses.

With certain noun phrases, the prepositional complement of the noun head is fronted, as a way to confine or limit the truth of the proposition. In the material, all the examples are *of*-phrases, as the one in (199) and (200).

(199) **Of all the prizes that come from surviving more than fifty years** the best is the freedom to be eccentric.

(200) **Of the customers**, one said twelve and another said eighteen.

The inclusion of these examples rests on the analysis that the prepositional phrases are raised from the complement of the noun head of the subject, i.e. (199) and (200) are derived from:

(201) The best of all the prizes that come from surviving more than fifty years is the freedom to be eccentric.

(202) One of the customers said twelve and another said eighteen.

In terms of the syntactic possibilities of raising a part of a phrase to front of the clause, this appears in both the English and Swedish original texts. As they can be given a similar analysis, the discussion of these constructions will be presented in the section 4.3 dealing with the clause elements of the Swedish original material.

Another type of raising is the placement of a constituent of a subordinate clause at the front of the superordinate clause, as in (203).

(203) The first, the teleological criticism, I think is wrong and I will now try to show why.

In this example, the subject of the subordinate object clause of *think* is moved to the front of the superordinate clause. Syntactically, it is notable that fronting is only possible when the subordinate clause does not have a subordinating conjunction, thus (4.2) is ungrammatical.

(204) * The first, the teleological criticism, I think that is wrong.

In this section, the examples of fronting in the English original texts have been presented according to the clause element of the fronted constituent. The data confirm the findings of previous studies in that there are a fairly small number of fronted constructions and that the majority of these are obligatory adverbials, generally accompanied by subject-verb inversion. In addition, predicative complements also occur a number of times, again mostly with an inversion of subject and verb. Fronting of other constituents such as objects and verbs is infrequent. We could also see that there are a number of elements that syntactically are part of phrases or subordinate phrases that occur sentence initially.

4.3 Clause function of fronted constituents in the Swedish originals

As in English, there are few syntactic restrictions on the type of clause element that can be fronted in Swedish (Teleman et al., 1999). Although the number of examples is somewhat larger in the Swedish material, the frequencies are still not sufficient to give any statistically significant indications of the distribution of fronted elements in Swedish.

Dividing the Swedish original text into fiction and non-fiction texts, we can see that there is some correlation between types of fronted clause elements and text category, as shown in Table 4.5. The fiction texts have a large proportion of fronted direct objects and predicative complements, whereas the great majority of fronted elements in the non-fiction texts are obligatory adverbials.

Table 4.5: Clause elements in Swedish original texts.

Text Type	ADVL	Obj	PC	Neg	VP	Total
Fict	43	66	34	10	4	157
Non-Fict	56	2	4	0	0	62
Total	99	68	38	10	4	219

The clause function with the largest number of occurrences in all the Swedish original text is the obligatory adverbials. This is consequently similar to the situation in the English original

texts. Indeed, some of the adverbials in the Swedish originals have a locative function similar to that of the English originals. For instance, in the category of fronted obligatory adverbials, we find constructions with verbs such as *vara* ('be'), *finnas* ('exist/occur'), (205)and *flytta* ('move')(207). Other instances involve semantically more complex verbs like *redovisa* ('show/present').

(205) **I denna kategori** fanns också den stora svenska flottbasen Karlskrona.
In this category *exist also the large Swedish naval base Karlskrona.*
'In this category there was also the large Swedish naval base of Karlskrona.'

(206) **In i ordens värld** hade han flyttat.
Into in words world *had he moved.*
'He had moved into the world of words.'

(207) **I tabell 4** redovisas en enkel sortomvandling.
In Table 4 *is presented a simple unit conversion.*
'A simple measurement unit conversion table is presented in Table 4.'

However, the characteristics of some of the adverbials in the Swedish originals differ from those in the English texts in several distinct ways. Semantically, the adverbials in the English material mainly express locative meaning, whereas in the Swedish material, the adverbials have more diverse meanings. Among them are complements of communication verbs like *påminna* ('remind') och *tala* ('speak') as in (208) and (209). Also complements of verbs like *styra* ('control') shown in (210).

(208) **Det där** ville Skalle-Per inte gärna bli påmind om.
That there *wanted Noddle-Pete not particularly become reminded about.*
'Noddle-Pete did not enjoy being reminded of this occasion.'

(209) **Skogen** hade de talat om.
Forest *had they spoken about.*
'They had talked about the forest.'

(210) **Över detta** styr centralmakten med samma myndighet i varje del av området.
Over this *rules central government.DET with same authority in every part of the territory.*
'The central government rules every part of this territory with the same authority.'

Syntactically, the Swedish obligatory adverbials also differ to some extent from the English ones which have more of an object relation to the verb. This is demonstrated by the possibility of being able to passivise the clauses with the prepositional phrase as subjects.

This ability to front elements that function as objects is the most significant difference between the Swedish and English fronting constructions in the material. In the material of Swedish originals, we can see that fronting of objects (211) is among the most common types of fronting in the Swedish original texts.

(211) **Den** kunde vi undvara.
That/it *could we do_without.*
'That we could do without.'

This is a noticeable difference compared to the distribution of fronted clause elements in the English originals as the number of fronted objects in the English original texts made up a small part of the overall occurrences.

The distinction between direct objects on the one hand, and obligatory adverbials on the other is to a large extent dependent on the analysis of the phrasal boundaries between the verb head and the following complement. Although there are syntactic and semantic arguments for making such a categorical division, the differences within the group of obligatory adverbials and the way this category relates to optional adverbials presents a situation where the data more closely resemble instances on a gradual scale of the strength of the relationship between the verb and the complement/adjunct.

The relatively frequent use of fronted object is, however, register dependent and is primarily used in fiction text, and in particular some specific fiction text in the material, as seen from Table 4.3.

In addition to direct objects, predicative complements are also more prevalent in the fiction category. The main verb is often a form of *vara* ('be'), as in (212) but other verbs like *stå* ('stand') also occur (213).

(212) **Men rädd** var hon inte.
But afraid *was she not.*
'But she was not afraid.'

(213) **Eldröd** stod solen mitt i himlen över det vita Karelen.
Fiery red *stood sun.DET middle in sky.DET over the white Karelia.*
'Fiery red, the sun stood high in the sky above the white of Karelia.'

A few examples of fronted verb phrases were also found in the Swedish original material as shown in (206). In addition to being fairly rare in the material, it was also a structure that was primarily found in one of the source texts, Astrid Lindgren's *Ronja Rövardotter* (AL1).

(214) **Dansa** lärde hon sej också.
Dance *taught she REFL. also.*
'She taught herself to dance, too.'

In the Swedish original texts there were examples of fronted sentence adverbials that are not part of the argument structure of the verb, but still obligatory in the sentence such as negations.

(215) **Inte** har väl du varit med om detta?
Not *have really you been with of this*
'You haven't really experienced that.'

In the Swedish original text, there were a number of anaphoric expressions, normally the pronoun *det* ('it/that'). Generally, it is clear from the valency structure of the main verb and the context what clause element they represent. In the majority of cases, they are direct objects as in (216b).

(216) a. "Hur vet jag vilka som är vildvittror och grådvärgar och Borkarövare",
How know I which REL are wild harpies and grey dwarfs and Borka robbers
frågade Ronja.
asked Ronia.
'How will I know which are wild harpies and gray dwarfs and Borka robbers?" asked Ronia.'

b. “**Det** märker du”, sa Mattis.
It *find out you said Matt.*
“‘You’ll find out,” Matt said.’

Similar to the English original text, the Swedish texts contained examples of fronted preposition phrases which syntactically function as a complement of a noun phrase. They are consequently raised from their canonical position to the initial position in the main clause. Many of these are prepositional phrases with the preposition *av* (’of’) as the head, as in (217).

(217) **Av den totala volymen säkringar** utgör 84% säkringar mot SEK.
Of the total volume hedges *comprises 84% hedges against SEK.*
‘Of the total hedged volume, 84% comprised hedging against the Swedish krona.’

This phenomenon is described in Swedish reference grammars as *adverbialised attribute* (Teleman et al., 1999, p. 448). The process of adverbilasing the modifier/attribute has a tendency to occur when the noun head and the modifier have a certain type of solution, one of which is that the noun head expresses a part or the collection which is referred to by the complement, e.g. in sentence like (218).

(218) En liten bit av tårtan finns kvar.
A small piece of cake.DET exists left
‘A small piece of cake is left.’

In this example, the prepositional phrase *av tårtan* is a complement in the noun phrase, the prepositional phrase can be separated from the rest of the noun phrase either to the right or left of its canonical position.

(219) a. En liten bit finns kvar av tårtan.
A small piece exists left of cake.DET.
‘A small piece is left of the cake.’
b. Av tårtan finns en liten bit kvar.
Of cake.DET exists a small piece left.
‘Of the cake a small piece is left.’

As shown in this section, the syntactic functions of the fronted elements in the Swedish texts are different from the English originals, primarily because of the many fronted objects and predicative complements in the Swedish fiction texts. The adverbials also showed some differences in being more diverse and in some cases having a more object-like relation to the licensing verb.

4.4 Phrasal category of fronted elements in the English originals

In terms of the type of phrasal categories that the fronted element can be realized as, there are few restrictions in English (Huddleston & Pullum, 2002). The distribution of phrasal types of fronted constituents is shown in Table 4.6.

In the material, the decidedly most common phrasal type is prepositional phrases and adverb phrases. Primarily because of the large proportion of obligatory adverbials in sentence initial position, which are generally realised by prepositional and adverb phrases.

Table 4.6: Phrasal categories in English original texts.

Phrasal Type	*N*	%
PP	32	48.0
AdvP	15	23.0
NP	8	14.0
Proform	5	8.0
AdjP	3	5.0
edP	1	1.5
VP	1	1.5
Total	65	101.0

The prepositional phrases are fairly short and with limited internal structure both with clauses of basic and inverted word order. Among the structurally most complex examples can be seen in (220) where the preposition still has a relatively simple noun phrase complements. The adverb phrases generally consist of a single adverb head such as *here* (221) and *there*.

(220) **From a Chinese jar on the table** spilled out the long tendrils of a trailing plant.

(221) **Here** was the refectory table he had been imagining, glowingly polished and with fruit on a polished wooden platter in the centre of it.

The fronted nominal elements in the English original text are all full noun phrases. Fronted pronouns do not occur in the English originals, which is possibly unexpected given that other proforms have been fronted and the fact that pronouns generally refer back to an antecedent, and thus in terms of information structure both known and salient. Many of the noun phrases function as predicative complements licensed by the verb *be*, as in (222).

(222) **A remarkable woman**, she was.

The only example of a fronted noun phrase functioning as a clause element other than a predicative complement is in (223) where the noun phrase has the function of direct object.

(223) **The other woman** she identified as certainly Davina Flory's daughter Naomi.

One instance of an *ed*-participle was also found as a locative complement with full inversion, as shown in (224).

(224) **Clamped to the man's ears** was a headset for one of those miniature tape recorders.

Fronted adjective phrases occur as realisations of predicative complements. In the material, all the adjective phrases are part of inverted structures, such as (225).

(225) **Equally vulnerable to human weapons** are the armoured giants of the animal world, the rhinos and the elephants.

Verb phrases are possible as fronted element but require that an operator is outside the fronted element (Huddleston & Pullum, 2002; Quirk et al., 1985). In the present material, the operators are *do* or *have*. In most of these examples, the VP is represented by the proform *so*, except for the example shown previously in (198).

From the data on the phrasal types of the fronted elements, we can see that the English originals contained a predictably large number of prepositional phrases and adverb phrases, given that the main clause function in fronted position is obligatory adverbials. It can also be noted that the only proforms that occur in the English originals are *so* with no examples of fronted pronouns.

4.5 Phrasal category of fronted elements in Swedish originals

Looking at Table 4.7 we can see that the most common phrasal category of the fronted elements in the Swedish original text are prepositional phrases. As was the case with the English originals this was as would be expected since the most frequent clause elements in the Swedish original text are obligatory adverbials.

Table 4.7: Phrasal categories in Swedish original texts.

Phrasal Type	*N*	%
PP	69	31.0
Proform	69	31.0
AdvP	33	15.0
AdjP	21	10.0
NP	20	9.0
InfP	3	1.0
Clause	2	1.0
VP	1	0.5
Total	218	98.5

The great majority of the fronted prepositional phrases have locative meaning both in a literal (226) and a more abstract sense (227).

(226) **I asken** låg flera tänder, antagligen farmors.
In box.DET *lay several teeth presumably Grandmothers.*
'There were several teeth in the box, presumably Gran's.'

(227) **I rörelseresultatet** ingick realisationsvinster från fastighetsförsäljningar med
In operating profits *was included capital gains from property sales of*
MSEK 90 (429).
MSEK 90 (429).
'Operating profit included capital gains from property sales of MSEK 90 (429).'

Example (227) is fairly typical of the use of prepositional phrases in the non-fiction text where they often have discourse relational functions expressing how information is presented or related, using verbs like *framgå* ('be clear'), *visa* ('show'), *ange* ('state') etc.

Fronted prepositional phrases also occur with a passive form of the verb *mena* ('mean'), as shown in (228). It is interesting to note that placing the subject pre-verbally, like in (229), which is ostensibly in its canonical position makes the sentence be markedly worse in acceptability than with the prepositional phrase fronted.

(228) **Med detta** menas den samlade inverkan av luftens och de omgivande
With this *means the overall effect of air.DET.GEN and the surrounding*
ytornas temperatur (medelstrålningstemperaturen).
surfaces temperature (mean radiation temperature).
'This is understood to mean the overall effect of the temperature of the air and the surrounding surfaces (mean radiation temperature)'.

(229) ?? **Den samlade inverkan av luftens och de omgivande ytornas temperatur (medelstrålningstemperaturen)** menas med detta.

In almost all instances of fronted PPs, they function as the only internal complement of the verb phrase. In only one example does the prepositional phrase function as an indirect object/obligatory adverbial with the verb *veta om* ('know about/aware of') as in (230).

(230) **Zetterbergskans lössläppthet** vet vi inget om och prästbarn och
Zetterberg's looseness *know we nothing about and minister_ children and*
prästbarn.
minister_ children.
'Zetterberg's looseness we don't know anything about, and what do you mean a minister's children?'

A number of prepositional phrases have the preposition *för* ('for') as head which semantically restricts the application of the proposition of the rest of the clause. In (231) the verb *gäller* ('apply') requires a complement, similar to English *applies + to* NP.

(231) **För svenska emittenter** gäller kravet "K1" enligt Standard &
For Swedish issuers *applies requirement.DET K1 according_ to Standard &*
Poor's.
Poor's.
'For Swedish issuers, the requirement is K1 according to Standard & Poor's.'

However, depending on the verb and constructions these prepositional phrases are part of, their status as complements can be more or less clear. In (231) the diagnostics identify the prepositional phrase as a complement, but in (234) it appears more as an adverbialised attribute.

Within prepositional phrases that appear to be obligatory according to diagnostics like coordination, and the relative placement of adverbials, can still exhibit different characteristics such as the ability to strand prepositions. According to the literature obligatory adverbials have a tendency to strand prepositions, but as seen in (232) and (233), there is variation among the obligatory adverbials. We may here have a phenomenon that could be analysed according to the arguments presented in Huddleston and Pullum (2002) for separating between obligatory PP-complements based on whether the preposition is 'fixed' or 'mobile'.

(232) **Skogen** hade de talat om.
Forest.DET *had they talked about.*
'They had talked about the forest.'

(233) **Till detta** bidrog hans ovanligt varma leende som gjorde att en iakttagare
To this *contributed his unusually warm smile which made that an observer*
tvekade i sin bedömning.
hesitated in REFL. evaluation.
'This was reinforced by his unusually warm smile, which was enough to put any observer off his stride.'

(234) **För massaved** påverkades marknadsläget av den osäkra situationen inom
For pulpwood *was affected market situation.DET of the uncertain situation within*
massa- och pappersindustrin.
pulp and paper industry.

'The market for pulpwood was affected by the uncertain situation in the pulp and paper industry.'

The verb *påverka* has passive form and has the complements the subject *marknadsläget* and the agentive adverbial *av den osäkra situationen inom massa- och pappersindustrin.* The prepositional phrase *för massaved* restricts the application of the proposition but is not licensed by the verb in the way that was the situation in (231). Instead the prepositional phrase has a semantic scope that equal to it being a complement of the noun of the subject, i.e. *marknadsläget för massaved* which is the way the translator has rendered the translation, *the market for pulpwood.*

However, in contrast to the English translation, the Swedish original has a different discourse functional property, which enables the scope of the restriction *för massved* to be more global. By separating out the preposition phrase and placing it initially, it functions as a frame which can apply to not only the current sentence but also the discourse beyond. The Swedish original makes a shift in topic that is more general and global than the English translation where the shift is more specified to the 'market of pulpwood' not pulpwood in general, and the shift therefore becomes more tied to the particular information in the sentence rather than a shift from one line of products to another. Sentence (231) and (234) are thus syntactically different but discourse functionally similar.

Semantically related to the prepositional phrases are the fronted adverb phrases, which in the material consists of single adverbs expressing position or direction, as *dit* ('there') in (235) and *här* ('here') in (236).

(235) **Dit** hör exempelvis hörselskadade, hjärt-kärlsjuka samt personer i
There *belong for example hearing impared, cardiovascular diseased and people in*
höga grader av stress eller ångest.
high degrees of stress and anxiety.

'These include, for instance, individuals with impaired hearing, sufferers from cardiovascular diseases and persons at high levels of stress or anxiety.'

(236) **Här** lämnas endast några kommentarer.
Here *leave-PASS only some comments.*

'. . . we shall make no more than a few brief comments here'.

Looking at the type of clause categories that occur in English and Swedish, the most obvious difference is the pronoun category. In the English originals the only type of proforms that was found was constructions with *so.* In the Swedish original texts, the diversity of types and frequency of pronouns are much larger.

The anaphoric expressions can be divided into two categories, pronouns and proforms where the antecedents are other phrase categories than nominals. In the pronoun category we find among other s personal, demonstrative and reflexive pronoun, as shown in (237) – (239).

(237) **Honom** dyrkade farmor.
Him *worshiped Grandmother.*

'Gran worshipped him.'

(238) **Det där** ville Skalle-Per inte gärna bli påmind om.
That *wanted Noddle-Pete not particularly be reminded of.*

'Noddle-Pete did not enjoy being reminded of this occasion.'

(239) **Själv** sade hon inget.
Herself *said she nothing.*
'She herself said nothing.'

The most frequent use of fronted pronouns is, however, the pronoun *det* which may function not only as a proform for nominal expressions but also as a proform for other phrasal categories and as an expletive element. Categorizing *det* therefore required an analysis of the context in which it was used.

In (240) we can see from the previous sentence that *det* refers back to the entire clause, and should therefore be classified as a proform for nominal expressions.

(240) Torin är väl inte fin nog. **Det** anser inte du heller.
Torin is probably not good enough. ***It/So*** *think not you either.*
'Torin isn't good enough for them. You don't think so either.'

There are also a number of other pronous such as *så*, *sådana/sådant*, as illustrated in (241) and (242).

(241) **Så** är det sagt, svarade Herwart Klammer och sträckte fram handen till ett
So *is it said answered Herwart Klammer and extended forward hand-DET to a*
bekräftande handslag.
confirming handshake.
"That's what they say," said Klammer, holding out his hand to seal their relationship with a handshake.

(242) **Sådant** sköter livet om alldeles själv.
Such take_ care life-DET about completely itself
'Life takes care of that.'

The proform *så* also functions as an antecedent other than a noun phrase as in (243) where the antecedent is the immediate discourse rather than some individual clause constituent.

(243) Så är det hemma hos oss, ...
That is EXPL. home at us ...
'That's what it's like at home ...'

Fronting of full noun phrases is not as frequent as fronting of pronouns and from the way these occur in the text material, it is also clear that fronting of noun phrases seem to have a stylistic dimension, which make it relatively common in certain texts whereas it does not occur at all in many of the texts. Primarily fronted noun phrases occur in fiction texts, as illustrated in (244) and specifically in the texts *Hundstjärnan* by Agneta Pleijel (AP1) and Astrid Lindgren's *Ronja Rövardotter* (AL1).

(244) **Fast sällsammare vandring** hade hon då aldrig prövat.
However stranger walk *had she then never tried.*
'But she had never made as strange a journey.'

The few instances of noun phrase fronting that were found in the non-fiction texts had a different character from the ones in the fiction texts. Similar to the situation with fronted prepositional phrases, the examples in the non-fiction texts relate discourse organisational content, as in (245),

(245) **En god illustration av detta** hittar man i en amerikansk studie, ...
A good illustration of this *find one in a American study...*
'A good illustration of this can be found in an American study ...'

The Swedish original text contained a number of adjective phrases which in the vast majority functioned as predicative complements. Many of the fronted adjective phrases were modified by the emphatic *så* which indicate that the adjective phrase is used to make a comparison to a level or degree of a property salient in the previous discourse, as illustrated in (246).

(246) **Så bråttom** har du väl inte
That in a hurry *have you really not.*
'Are you really in that much of a hurry?'

Fronting of verbs is rare in the Swedish original texts, and only one occurrence of fronting of the lexical verb of the main clause is found, shown in (247). The need to have a finite verb in second position requires the use of a form of *göra* ('do') in the canonical operator position.

(247) **Och kom** gjorde han.
And come *did he.*
'And he came.'

Infinitive clauses occur twice and both times the fronted element is the bare infinitive. In Swedish the infinitive marker is often omitted when fronted Teleman et al. (1999), as shown in (248).

(248) **Dansa** lärde hon sej också.
Dance *taught she REFL also.*
She taught herself to dance, too.

There are also a couple of examples of fronted subordinate clauses, in both cases they are introduced by the general subordinating conjunction *att* ('that'), illustrated in (249).

(249) **Att det var något särskilt med Siiri** begrep de inte.
That it was something special with Siiri *understand they not.*
'They didn't understand there was something special about Siiri.'

Looking at the phrasal categories of the fronted elements in the Swedish originals shows that there are many nominal elements as fronted constituents, both pronominal elements and full noun phrases.

4.6 Summary

In this chapter, the fronted complement structures found in the original material have been described in terms of the clause function and phrasal categories of the fronted constituent. The data show us that there are a larger number of fronted constructions in the Swedish originals at a ratio of approximately 3 to 1. Looking at the relative number of the types of clause elements and phrasal categories in the material, there are, however, some similarities between English and Swedish.

Looking at the clause element of the fronted element, the amount of fronted obligatory adverbials and predicative complements is large in both languages. In terms of phrasal categories,

both languages have a large number of prepositional phrases and a fair number of adverb and adjective phrases. There are also a small number of finite and non-finite verb phrases. The most striking difference between the two sets of original texts was the occurrence of fronted objects and fronted noun phrases in the Swedish material. There are occurrences of fronted noun phrases and fronted objects in the English original text, but they are fairly rare. However, the distribution of clause elements and phrasal categories in the two languages seems to indicate that there is at least some overlap between the languages in how the fronted elements are used communicatively. The question we need to answer is why we find fewer overall instances of fronting in the English material, and what restricts the possibility of using fronting in English, if syntactically the potential for fronting is the same.

Chapter 5

Fronting in Swedish and English translations

This chapter presents the occurrences of fronted constructions when Swedish and English original texts are translated into the other language, i.e. Swedish into English and vice versa.

5.1 Introduction

The use of a parallel corpus when comparing the two languages makes it possible to find out what expressions are considered to be equivalent in the same discourse context, and thus enables a comparison not only of formally equivalent structures but of a way of finding structures that are discourse functionally equivalent.

The chapter reports on the type of constructions the translators consider to be the equivalent of a fronted construction in the source language and the type of constructions in the original languages that are best rendered as fronted constructions in the translation. As already established, fronting is much more frequent in Swedish, and it is therefore often the use of fronting in Swedish originals and Swedish translations that are compared to the equivalent structures in English.

By examining the translations of original texts, the analysis seeks to establish what types of constructions are equivalent to fronted constructions in the other language. A comparison of the morphosyntactic, semantic, and pragmatic properties of the fronted constructions and their equivalent structures in the other language will reveal how these properties are related and how they interact to decide on the use of a particular linguistic form in a particular context.

5.2 English translations of fronting in Swedish originals

The previous chapter looked at the different forms of fronting constructions that were found in the Swedish and English original texts. These data show similarities between the two languages in terms of both having the syntactic possibilities of fronting most types of constituents, but also revealed differences in specific configurations and frequency, such as the low frequency of fronting objects and predicative complements and the lack of fronted non-stressed pronouns in English. Taking the differences between the original texts as an indication of language specific possibilities and tendencies in the way certain meanings are expressed in the two languages, the assumption is, as formulated by M. Johansson (2002), that the distribution in the translations will reflect these differences.

The translations are used to compare the fronted constructions in the originals with what has been considered to be equivalent in the other language, and also to see what structures in the translations are rendered as fronted structures in the originals. As we saw in chapter 5 the general frequency does, not surprisingly, confirm the assumption that the low frequency of fronted constructions in the English originals are also reflected in a low frequency of fronted structures in the translation of Swedish originals, in spite of the fact that the Swedish original texts have a comparatively high number of fronted structures. The types of alternative structures in the English translations and how they relate to particular Swedish fronted constructions is thus highly relevant for an understanding of the relation between formal properties and discourse functions.

5.2.1 Types of translation strategies

The overall largest number of fronted constructions were found in the Swedish originals (SO) with 212 tokens. In terms of the way these are rendered in the English translations (ET) they can be divided into four main categories:

1. English translation is a morphosyntactically equivalent constructions of the original fronted structure but with the fronted element in its canonical position.
2. English translation is reformulated.
3. English translation is a fronted construction.
4. English translation omits the fronted element (ellipsis).

Examples of these are found in (250) – (253) respectively.

(250)	**Detta** har pappa tagit tillbaka många gånger. (AP1.3.s51)	Dad has taken it back many a time. (AP1T.3.s53)
(251)	**Detta** sa jag till honom när jag satt med vårt nyfödda barn i mina armar och jag minns att jag fick kväljningar när jag insåg att jag låtsades även i en sådan situation. (MS1.2.s42)	That's what I told him as I held our newborn child in my arms, and I remember that I felt sick when I realized that, even in that situation, I was being hypocritical. (MS1T.2.s35)
(252)	**Vackra och galna** och grymma var vittrorna. (AL1.2.s138)	Beautiful, mad, and ferocious, the harpies were. (AL1T.2.s139)
(253)	**Det** vet jag inte. (MS1.10.s61)	'I don't know.' (MS1T.10.s65)

Example (250) illustrates a case where the English translation maintains a fairly similar construction as the original but leaves the equivalent of the pronoun, which is fronted in the original, in its canonical position. Even though the translation has changed the pronoun from a demonstrative pronoun in the original *Detta*, which by default would be given more prosodic emphasis in this context, to a prosodically weaker personal pronoun *it*, it is considered to be a non-fronted equivalent of the original example.

The re-construction group which is illustrated by (251) is the most heterogeneous category, with a number of different representations of the original's fronted structure. In this example, the fronted object in the form of the demonstrative pronoun *Detta*, is rendered as a reversed *wh*-cleft, *That's what I told him*.

In (252) the fronted subject complement in the Swedish original *Vackra och galna och grymma* has been maintained as a fronted subject complement in the English target text *Beautiful, mad and ferocious*.

The sentence in (253) is an example of a translation where the fronted element in the original has been omitted. It is not possible to determine whether the ellipsis is a null object or a topic-drop, but the fact that English does not allow weak fronted pronouns (Biber et al., 1999) suggests that this is a translation that is similar to the category one with a phonetically unrealized object in canonical position.

Table 5.1 shows the general distribution of the different types of translations of the Swedish fronting into English.

Table 5.1: English translation types of fronting in Swedish originals.

Direction	Canon.	Re-form.	Front.	Omit.	Total
SO → ET	101	87	18	6	212

The two largest categories, canonical position and re-formulation, can be regarded as representing different approaches to the way the information structure of the original is rendered in the translation. Leaving the element that is fronted in the original in its canonical position will allow the translator to stay close to the original in terms of lexical choice and verb pattern, as shown in (250) where the original "tagit tillbaka" is translated with the semantically close "taken back". However, by choosing to make the translation close to the original in these aspects, this approach results in a change in the linear order of the constituents which affects the information structure. The translation will therefore lose the pragmatic characteristics that the original fronted structure has.

Re-formulation, on the other hand, will be less concerned with finding semantically close lexical equivalents of the original expression, and instead focuses on finding a propositional equivalence which allows a closer adherence to the information structure of the original. As demonstrated in (251) the original fronted construction, glossed in English as 'That told I to him' has been translated as a reversed *wh*-cleft, which makes it possible to preserve the same referent as the initial element in the translation, and maintain a prosodic prominence on the first constituent. The fact that one non-canonical, pragmatically marked construction in Swedish has been replaced by another type of non-canonical construction in the English translation could be an indication that the pragmatic functions of the particular information-structural constructions, such as fronting and clefts, differ in the two languages.

Compared to the canonical and reformulation categories, the other two categories are relatively infrequent. The number of fronted constructions in the Swedish original that has been translated by a fronted construction in English would be expected to be low, given the low frequency of fronted constructions in English original texts reported in previous studies and as shown in the data of the current study.

It is hard to draw general conclusions based on the distribution of these examples because of the low number of tokens, and they do not belong to a consistent type of fronted construction. The example in (252) appears to be fairly marked as it fronts a focal element. This type of 'Focus movement' in English is discussed by E. F. Prince (1981a) who concludes that this construction requires that the fronted element is the value of an attribute and that the subject is salient in the discourse. However, she does not go into detail about the discourse function of focal movements. In the current material only two of the fronted translation are examples of focus movement, which is dominated by locative inversion. It is interesting to note, however, that it is one of the few types of fronting that occurs in the English originals.

The category with the smallest number of tokens are translations where the fronted element is omitted in the translation. In contrast to the category of fronted translation, the type of structures in the omitted category show a great deal of similarity. Similar to the sentence in (253), the constructions in the source text that are rendered as phonetically unrealized elements are all instances of the unstressed object pronoun *det*, where the pronoun replaces parts of the predication or clausal complements. This type of structure was reported to be the most frequent type of object fronting in Jörgensen (1976)'s study of Swedish spoken discourse. It appears that it is generally not possible in English to front an unstressed pronoun, as indicated by the corpus data in Biber et al. (1999). It is clear that in (253), it would not be possible to replace it with a fronted construction like(254), and this construction seems unacceptable regardless of context.

(254) *It I don't know.

This is interesting from the perspective that often the main criteria for fronting in English is the discourse status of being 'given' in some sense. Clearly even from a very general perspective this characterisation needs to be given a more detailed analysis.

The possibilities in Swedish seem less restrictive and in many contexts either the structure in (253) or a structure with an omitted complement, such as in (255), can be used felicitously.

(255) Jag vet inte.
I know not.
'I don't know'

However, in other examples, an omitted complement is not a possibility in Swedish, such as in (256) where the omission of the pronoun, would not be acceptable.

(256) "Har du TV på rummet, pappa?"	"Do you have a TV in your room, Dad?"
"Nej, det har jag inte/*Nej, jag har inte, sa jag och såg på snön som smälte runt skorna, "jag skulle i alla fall inte få tid att titta." (GT1.1.s3-GT1.1.s4)	**"No, I don't**," I said as I watched the snow melt around my shoes. "In any case, I wouldn't have time to watch it." (GT1T.1.s4-s5GT1T.1.)

This brief look at the different translation strategies thus suggests that there are differences between the two languages both in terms of the ability to front unstressed elements and the use of reversed *wh*-clefts in English for certain fronted structures in Swedish. Our aim is now to find general categories of fronting in the Swedish original and match these with the types of translation to discover similarities and differences in their pragmatic properties.

5.2.2 Types of English translations of fronting

To find out if there are any general trends between types of fronting and type of translation, we can break down the translation types into the clause element of the fronted element, which is shown in 5.2.

5.2.2.1 Translations of fronted objects and predicative complements realised as proforms

As can be seen, the most common way of translating fronted objects is to use a close morphosyntactic representation in the target language but keep the object in its canonical position. It is also clear that of all the translations that have used this type of translation, fronted objects make up the largest proportion (47 of 101).

Table 5.2: English translation types according to fronted element in Swedish originals (SO → ET).

Fronted Element	Canon.	Re-form.	Front.	Omit.	Total
Object	47	16	3	5	70
Predicative complement	15	17	2	0	33
Adverbial	30	52	11	0	93
Negation	6	2	1	0	9
Predication	3	0	0	1	4
Total	101	87	18	6	212

Looking at the type of fronted objects we find in the Swedish originals, we can see that we have both full noun phrases and different proforms, the most common of the latter being the pronouns *det*, *detta* and *den*. Among the translations the equivalent form is in the majority of cases the demonstrative pronoun *that* with some examples of *so*, as illustrated in (257) and (258).

(257) Därinne bakom de röda murarna finns många sådana hundar.
Det vet jag nu.
Och i huset bredvid experimenterar de med människor.
Det visste jag inte då. (AP1.1.s204-s206)

Inside those red walls, there were lots of dogs like that.
I know that now.
And in the next building they experiment on people.
I didn't know that then. (AP1T.1.s209-s211)

(258) Torin är väl inte fin nog.
Det anser inte du heller. (GT1.2.3.s44-s45)

"Torin isn't good enough for them."
"You don't think so either." (GT1T.2.3.s46-s47)

It is clear from the context of (257) that the fronted element is not intended to be contrastive, instead it is the polarity of the verb phrase (*know*, *didn't know*) and the temporal reference (*now*, *then*) which are contrastive. This would indicate that there is a difference between a Swedish fronted element and an English fronted element, which has been suggested to always have a contrastive quality, as proposed by Chafe (1976). Assuming that we can distinguish three main topic types from their principle function, as proposed by Frascarelli and Hinterhölzl (2007) – *aboutness topic*, *contrastive topic* and *familiar topic* – the realizations of these different topics would possibly be implemented in different ways in the two languages. Fronted constructions in Swedish appear to include the function of aboutness topic, as suggested by the example in (257).

Similarly, in (258) the Swedish fronted element is non-contrastively given, with the added information being the viewpoint of the addressee in the emulated dialogue. Again the proform in both the original and the translation indicates the givenness of the referent, and the context excludes a contrastive interpretation. The discourse function of *det* as aboutness topic in the Swedish original and the English translation are maintained, but they are realized by different structures.

Assuming that fronted constructions can be used with aboutness topics in Swedish, this account does not say anything about why the Swedish original uses a fronted construction rather than a canonical one. More speculatively a default prosodic realisation of the fronted and non-fronted construction in the contexts of (257) and (258) would place more stress on the pronoun in the canonical position, giving it a more contrastive quality, as indicated in (259) and (260).

(259) Därinne bakom de röda murarna finns många sådana hundar.
*there.inside behind the red.*PL *walls.*PL *exists many such dogs*
'There are many such dogs in there behind the red walls.'

a. Det vet jag NU.
*it know.*PRS *I now*
'I know that now.'

b. ? Jag vet DET nu.
*I know.*PRS *that now*
'I know that now.'

(260) Torin är väl inte fin nog.
Torin is presumably not nice enough
'Torin isn't nice enough, I suppose.'

a. Det anser inte DU heller.
*it consider.*PRS *not you either*
'You don't think so either.'

b. ? Du anser inte DET heller.
*you consider.*PRS *not that either*
'You don't think that either.'

Thus, fronting in the Swedish original in this context becomes a case of avoiding default prominence on the fronted element. In contrast, fronting the pronoun in the English translation would instead cause the pronominal element to receive more prosodic prominence.

This need to avoid default stress on the pronominal element that are fronted appears to account for examples of proforms that are left in their canonical position in the translation. However, this explanation appears problematic when we look at the other main type of translation where the fronted elements in the Swedish originals are translated with reformulations that enable the linear order of the information to be maintained in the translation. Given that we have examples in the originals of a similar type as the ones in (257) and (258), this will be a problem for the analysis presented so far.

In (261) we can see an example of where the fronted object pronoun "det" has been translated with a reformulation in the English translation instead of being rendered as construction with the object in its canonical position. In this case, the translation is a reversed *wh*-cleft or reversed pseudo-cleft (Huddleston & Pullum, 2002).

(261) Vem Adèle än var, så var hon ovälkommen, allra mest om hon talat sanning. Hon kom ur någon dunkel berättelse ur Lameks förflutna. **Det** tänkte jag. (AP1.2.s220-s222)	Whoever Adèle was, she was not welcome, least of all if she had told the truth. She came from some obscure story in Lamek's past. That's what I thought. (AP1T.2.s220-s222)

In terms of the linear distribution of information, the choice of a reversed pseudo-cleft in the translation enables the English equivalent of the Swedish fronted object construction to retain the pronominal element, referring back to the previous propositions, as the initial element in the clause. This would be unexpected if we follow the previous reasoning about the relationship between the fronted object constructions in Swedish and the canonical structures in the English translations that was presented for (257) and (258). The crucial difference, however, is that

in (261) the fronted element in the original will be given prosodic prominence, if we assume a prosodic structure congruent with the context, i.e. in the Swedish original neither *tänkte* nor *jag* can be prominent in this context. Although the number of examples with fronted objects that are reformulated in the translation is fairly small, it is interesting to note that almost half, seven out of sixteen tokens, are translated with reversed pseudo-clefts.

The fact that this correlation exists in several of the texts also suggests that it is not a type of correspondence that is specific to a particular translator but something that is perceived to be a shared discourse function between the two constructions, as shown in (262) and (263).

(262) **Detta** sa jag till honom när jag satt med vårt nyfödda barn i mina armar och jag minns att jag fick kväljningar när jag insåg att jag låtsades även i en sådan situation. (MS1.2.s42)

That's what I told him as I held our new-born child in my arms, and I remember that I felt sick when I realized that, even in that situation, I was being hypocritical. (AP1T.2.s220-s222)

(263) **Det** behöver vi alla. (KOB1.2.s108)

"That's something we could all do with." (KOB1T.2.s107)

The similarity in discourse functions between fronting and reversed pseudo-clefts is also supported by the findings of M. Johansson (2002) study which investigates clefts from a contrastive perspective using the ESPC. In M. Johansson's data English reversed *wh*-clefts are often translated as fronted objects in Swedish.

The fact that the fronted object in (261 needs to be prosodically prominent indicates that it has a specific relation to the surrounding context. The pronominal form of the fronted element shows that the referent is in some sense 'given' in the present context. As we have seen previously, the combination of prosodic prominence and givenness would generally be properties resulting in a contrastive interpretation. As one of the requirements of English fronted constructions is that the fronted element should be contrastive, it seems reasonable to assume that it would be possible to translate the Swedish constructions in (261) – (263) with an English fronted object. However, the relationship between the fronted, prominent object and the rest of the proposition is different from that of fronted constructions where the fronted element is contrasted.

As outlined in Huddleston and Pullum (2002), in terms of information packaging, the clefted element in a reversed pseudo-cleft can be described as **foregrounded**, whereas the rest of the clause is being **backgrounded**. The initial element in the reversed pseudo-cleft and the fronted element in English fronted construction have similar relations to the previous context in being both given and highly salient. They also have a similar relation to the rest of the information of the clause. The main difference between the two constructions lies in the relation between the non-clefted/non-contrastive part and the previous context.

In contrast to the reversed pseudo-cleft, the backgrounded part in the fronted construction is highly salient, and thus clearly presupposed, whereas in the cleft, the backgrounded information is not presupposed from having been introduced as common ground in the previous context, but is being made presuppositional by the cleft-construction. It is thus not presupposed that something has been *thought*, *told* and *done with* from having been previously stated in the context. Instead, this information is presupposed by being introduced as backgrounded information in a pseudo-cleft. The non-clefted information is accommodated into the common ground as presupposed information. Again, we can see that it is not syntactic restrictions that prevent the use of a fronted structure in the English translation, but the incompatible information structure, as is shown by the unacceptable use of fronting in (264).

(264) Whoever Adèle was, she was not welcome, least of all if she had told the truth.
She came from some obscure story in Lamek's past.
That's what I thought/***That** I thought. (AP1T.2.s220-s222)

It would therefore seem that both the English reversed pseudo-cleft and the fronted construction can correspond to a Swedish fronted-object construction in terms of discourse function. Although the same Swedish syntactic construction can thus serve as the translation equivalent of both English structures, the different discourse functions of the Swedish fronted construction may be reflected phonetically. In particular, the contrastive and reversed-pseudo-cleft interpretations may differ in their prosodic prominence or placement of the pitch accent.

Intuitively, there seems to be greater prominence in a contrastive context, as in (265) compared to the use of the same construction in (261).

(265) **Det** tänkte jag, men andra saker har jag inte funderat över.
that thought I but other things have I not thought about
'That I thought, but I have not thought about other things.'

Also, several studies have argued for specific patterns of nuclear pitch depending on the particular information structure of a construction, e.g. Büring (1997), and Frascarelli and Hinterhölzl (2007).

In addition to the fronted pronouns, *den*, *det* and *detta*, there were also three examples of the proform *så* that exhibit the same information-structural properties as English reversed pseudo-clefts as illustrated by (266). In terms of discourse function the examples with *så* share the relevant characteristics with the other fronted proforms with prosodic prominence.

(266) **Så** sade farmor, och hon lät inte direkt ovänlig när hon sade det. (AP1.4.s58)	That's what Gran said, and she didn't sound exactly unfriendly when she said it. (AP1T.4.s61)

One of the examples combines the proform *så* with the adverb *här* making the fronted element *så här*, which results in the clefted element being an adverb *here* in the English translation, as seen in (267).

(267) **Så här** gör man med sådan smörja.(GT1.2.5.s132)	Here's what to do with such rubbish. (GT1T.2.5.s139)

It is interesting to note that syntactically we have rather a different structure from the previous reversed pseudo-clefts as the generic pronoun in the Swedish original *man* (*one/you*) results in a subjectless construction with a *to*-infinitive phrase. What is maintained is the order of information.

As the examples presented so far illustrate, most of the fronted elements in the Swedish originals are various types of proforms. This would be expected if it is a requirement for fronting that the referent of the fronted element is salient in discourse. Following the suggested "Givenness hierarchy" in Gundel et al. (1993), the fronted objects mentioned so far have the cognitive status of "in focus" or "activate", i.e. referents that are indicated to be highly salient in the current discourse. However, the fact that a referent is activated enough to be realised as a proform high in the givenness hierarchy is clearly not sufficient to front it in Swedish, and the data from the corpus suggest that without any additional pragmatic features, referents that are "in focus" or "activated" are less likely to be fronted. We clearly need a better understanding of the relationship between activation status and other discourse-functional properties.

The fact that different information structural constructions in Swedish and English seem to have similar discourse functions, such as Swedish fronting and English reversed *wh*-clefts may indicate that combination of formal and functional properties in the two languages result in different constructions having the same discourse function.

The Swedish original texts included a sentence that demonstrated an interesting combination of different types of constructions, as shown in (268).

(268) "För nu lever Mattisätten vidare, men Borkaätten, **den** går det rakt åt pipsvängen med." (AL1.1.s63)

Now Matt's line will live on, but Borka's line will be finished and done for." (AL1T.1.s66)

Here we find a structure which includes both a right dislocation and a fronted structure in a sentence with an expletive subject. If we break down the different types of information packaging present in (268) and relate them to the canonical structure, we can get some idea of the relation between formal properties and discourse function. The canonical sentence (269) introduces an expletive subject in (270) which moves the canonical subject to the end of the sentence. However, this movement is reversed again when the complement of the PP (271) is fronted and is then subsequently left-dislocated by the introduction of a pronominal *den* in the clause (272).

(269) Borkaätten går rakt åt pipsvängen.
Borka's line goes straight to pipe bend.DET
'Borka's line will be finished and done for'

(270) Det går rakt åt pipsvängen med Borkaätten.
It goes straight to pipe bend.DET.
'Borka's line will be finished and done for'

(271) Borkaätten går det rakt åt pipsvängen med.
Borka's line goes it straight at
'Borka's line will be finished and done for'

(272) Borkaätten, den går det rakt åt pipsvängen med.
Borka's line it goes it straight to pipe bend.DET
'Borka's line will be finished and done for'

From the perspective of how these structures distribute the information throughout the sentence, it seems somewhat puzzling that an expletive subject is used together with a fronted object since the two configurations seem to cancel each other out. It is clearly too simplistic to regard this only in terms of placing particular constituents in different positions in the clause.

The translation in (268) keeps the same referent sentence initially, but simply uses the canonical structure which is equivalent to the one in (269). It seems highly plausible that the combination of different information-structural configurations will have discourse functional effects, and only a closer analysis of what these effects are and how they interact can begin to explain the usage of this particular structure in this context.

5.2.2.2 Translations of fronted objects and predicative complements realised as full NPs

In addition to the different types of proforms functioning as fronted elements in the Swedish material, there are also a number of sentences which have a full NP as the fronted element. Looking at the way these examples are translated into English, we find that a great majority of them use a translation equivalent that keeps the fronted element in it canonical position. Not even the instances where the fronted element is clearly contrastive, as in (273) does the English translation use anything but a canonical structure.

(273) Inte var han då så högväxt och vacker som Mattis, det var bra, tyckte hon.
Men han såg stark ut, det kunde inte förnekas.
Kort var han visserligen men bredaxlad och kraftig, och rödhårig dessutom med testar som spretade åt alla håll. (AL1.1.s27-s29)

But he was not as tall and as handsome as Matt; that was a good thing, she thought.
He looked strong, though; there was no denying that.
He might be short, but he was broad shouldered and powerful, with tufts of red hair sticking out in all directions. (AL1T.4.s33-s35)

In example (273), the immediately preceding context makes the fronted element highly salient in this section of the text. As the attributes of the referent of "he" in the first two sentences are focal information, we need to explain how focal information, which is prototypically new, can be regarded as salient in the context. A way to make this more explicit has been suggested by Birner and Ward (1998).

They argue that although the particular values of the attributes are new, the context establishes the open proposition of "he is X, where X is a member of the set {personal attributes}". This makes the set of values that can be attributed to the referent established as common ground, and will in turn link each new attribute to the "given" set of attributes. According to Birner and Ward, this explains the possibility of fronting in English in contexts such as (274), Birner and Ward's example (22).

(274) Customer: Can I get a bagel?
Waitress: No, sorry. We're out of bagels. *A bran muffin I can give you.*

However, in the sentence with the fronted construction, the information that the referent of *he* is not "as tall and as handsome as Matt" makes the physical traits of the referent salient. We can then argue that the use of fronting in the Swedish original is used to create a link to the attributed physical value established earlier and to create a contrast between the property of being short and the property of being "broad shouldered and powerful".

Yet, the context in (273) does not trigger a fronted construction in the English translation, which suggests that the type of contrast that is illustrated by the Swedish original does not have the same properties as a fronted structure in English. Although a fronted construction seems possible syntactically, the context in (273) appears to reduce the acceptability of a fronted construction, as in (275).

(275) ? Short he might be, but he was broad shouldered and powerful, with tufts of red hair sticking out in all directions.

A similar occurrence of a fronted adjective phrase is found in the same text (AL), with the same solution in the translation of retaining the English predicative complement in its canonical position, shown in (276).

In this example the open proposition of "The Borka robbers are X where X is a member of the set personal attributes" is less explicitly expressed in the text, but at the point where the fronted construction is used to create a link to the common ground and as a contrast to another property, the context makes it possible to infer an open proposition that can establish the necessary link from the previous descriptions of the robbers, and makes it possible to front essentially focal information. The fronting also indicates the contrast between being "bearded and unwashed and noisy and wild" and being "a dirty devil".

Clearly the translator in these two cases did not regard the contexts to be such that it was necessary or possible to use an English fronted structure, or any other pragmatically marked sentence structure. As it is possible to have fronted elements with contrastive meaning in English,

it is reasonable to assume that the aspect of the communicative situation that restricts the use of a fronted construction is the context.

(276) Borkarövarna är skitstövlar hela bunten", sa Per, och det höll alla med om.
Vilken tur att Mattisrövarna var så mycket bättre, tänkte Ronja.
Hon såg på dem där de satt vid långbordet och slurvade i sej soppa.
Skäggiga var de och lortiga och bråkiga och vilda.
Men ingen skulle komma och kalla dem för skitstövlar, så att hon hörde det. (AL1.2.s103-s107)

The Borka robbers are all dirty devils, the lot of them," said Noddle-Pete, and everyone agreed with him.
Wasn't it lucky that Matt's robbers were so much better, Ronia thought.
She looked at them as they sat at the long table slurping up their soup.
They were bearded and unwashed and noisy and wild, but no one was going to call them dirty devils in her hearing. (AL1T.4.s33-s35)

A tentative explanation of this phenomenon can be found in the data of Birner and Ward (1998). Their investigation found that fronted AP where the type of fronted clause elements with the most limited range of fronted constructions. In addition, they found that a great majority of the fronted APs occurred together with another fronted AP construction, i.e. these fronted constructions occurred in contrastive pairs, as shown in (277), Birner and Ward's example (41c).

(277) The Philadelphia Fish & Co. is grilling fresh seafood soooooo good the competition is "broiling" mad... **Casual and affordable** we are. **Expensive** we are not.

In this and most of the other instances in Birner and Ward's study, the use of a fronting requires that there is another instance of fronting that is being contrasted, it is thus not possible to leave one of them out and still have a felicitous expression, as demonstrated in (278).

(278) # The Philadelphia Fish & Co. is grilling fresh seafood soooooo good the competition is "broiling" mad... **Casual and affordable** we are.

This suggests that in English, it is not sufficient for fronted predicative complements to have a link to an open proposition that is derived from the information introduced into the common ground: there must also be a second fronted construction to contrast against. The fact that these structures place nuclear stress on the linking verb expressing the polarity of the clause leads to the conclusion that, in English, fronting APs requires not only that the fronted element be linked to the previous discourse in a contrastive relation, but also that there be a contrast in polarity in the focus position.

This can be tested by changing the context in (273) and (276) to include a second structure with a polarity contrast, and as seen in (273) and (276). It is clear that the use of APs in English is much more restricted than similar constructions in Swedish.

(279) The Borka robbers are all dirty devils, the lot of them," said Noddle-Pete, and everyone agreed with him.
Wasn't it lucky that Matt's robbers were so much better, Ronia thought.
She looked at them as they sat at the long table slurping up their soup.
Bearded and unwashed and noisy and wild they were, but dirty devils they weren't. (AL1T.4.s33-s35)

It is clear from the examples in the Swedish originals that the structures with fronted APs have different restrictions. Judging from the limited number of examples in the data, the Swedish construction requires a link between the fronted element and the previous context and a contrastive

relation to another element of the inferred set. However, in contrast to the English equivalent fronted AP structure, the non-fronted part of the clause is clearly presupposed and is made prosodically less prominent. Moreover, there does not have to be a contrast in polarity.

As can be seen in (273) the second part of the contrast only contains the element being contrasted with the fronted element, e.g. "bredaxlad och kraftig". The contrast is also between the negative and positive attributes that the referent possesses ('short' vs 'broad shouldered and powerful'), not between attributes that he does and does not possess.

Similarly, in (276) the contrast is between the set of attributes of the robbers ('bearded', 'unwashed', 'noisy', 'wild') and the attribute of being a 'dirty devil'. Although there is a contrast in what they are and what they aren't, the polarity doesn't have to be focused. The structures are also less parallel than in many of the examples by Birner and Ward since the contrast is between a fronted AP and an NP. It is thus less idiomatic with a contrast as in (280) than as in (281).

(280) #Skäggiga och lortiga var de, men skitstövlar var de inte.
Bearded and dirty were they, but dirty devils were they not
'Bearded and dirty they were but dirty devils they weren't'

(281) Skäggiga och lortiga var de, men inga skitstövlar.
Bearded and dirty were they, but no dirty devils
'Bearded and dirty they were but no dirty devils'

The phenomenon in English of combinations of fronted elements and a focus on the polarity of the proposition can be observed also in other structures than paired fronting of APs. Birner and Ward (1998) present a set of constructions that leave the finite auxiliary in end focus position. They refer to these as 'proposition assessment' and there are a number of different subtypes, such as 'proposition affirmation', 'proposition suspension' and 'proposition denial' which are exemplified in (282), (283), and (284) respectively.

(282) It's hard to imagine that this topical little island was a battleground, but *a battleground it was.*

(283) It's odd that Diane said that, *if say it she did.*

(284) The NBA's collective-bargaining agreement sounds as though it was written by the same people who put together the International Revenue Service's long form. *Simple it isn't.*

Although these constructions have distinct characteristics in the way they are used in discourse, they all have in common that they place the default nuclear stress on the finite verb carrying the polarity of the proposition. From a contrastive perspective, it can also be observed that the equivalent Swedish construction will not place the finite verb in final position, although in the proposition denial category the negation would be placed sentence finally (*Enkelt är det inte*).

As always, we have to be somewhat cautious in drawing too general conclusions on the basis of these findings, partly because of the low number of examples, but also because the majority of these structures are found in a single text, Astrid Lindgren's *Ronja Rövardotter*. However, although we can claim that this type of fronting is characteristic of Lindgren's style, which often exhibits features typical of oral discourse, and the style of the translator, it is still necessary to explain why these structures are possible in certain contexts but not others, or why they are much more restricted in their use in English.

A different structure with fronted adjective phrases is found in text other than AL. In this construction the fronted constituent is modified with the adverb "så", i.e. "så stora", och "så arg" as shown in (285), and (286).

(285)	Hur stor var slanten? Sidner måttade med fingrarna. **Så stora** är bara hästpremieringsmedaljerna. De är inte mycket värda. (GT1.2.6.s26-s28)	"How big was the coin?" Sidner measured with his fingers. "Only prize medals for horses are that big. 'They're not worth much."' (GT1T.2.6.s34-s37)
(286)	Farmor gick i en sky av svart vrede genom de gula vetefälten. Hon talade inte med någon hon mötte, men varje gång hon satte foten i landsvägsdammet gungade marken av hennes stora vrede. **Så arg** var hon att hon resten av livet vände ryggen åt det håll varifrån hon kommit. (AP1.3.s24-s27)	Gran walked in a cloud of black fury through the yellow wheat fields. She didn't speak to anyone she met, but every time she set foot in the dust of the roadway, the ground beneath her bounced from her great rage. She was so angry, for the rest of her life she turned her back on the direction from which she had come. (AP1T.3.s26-s28)

In these two examples, the fronted AP in the Swedish originals are both contextually salient and 'given', whereas the element after the verb are new or focal. Still, the fronted constituents receive primary stress. In (285) the syntactic structure is a traditional Subject-Verb-Predicative complement (SVC) where the complement is linked to the previous text by pointing to the previous sentences non-verbal answer to the question"how big".

Syntactically the verb-second requirement in Swedish results in there being no difference between a fronted construction, equivalent to the English "That big the medals are" and an inverted structure "That big are the medals". Neither of these structures are, however, possible in English even though in some instances inverted constructions are possible, i.e. with locative inversion.

From an information structural perspective, the Swedish sentence adheres to the principle of having 'given' information first and the 'old' information last. Prosodically, it is, however, marked. The English translation on the other hand, places the main stress at the end which is the unmarked option, but in terms of the order of information, it places new information before given, and heavier, more complex constituent before shorter, simpler structures. This would indicate a difference in what pragmatic function is given priority in the two languages.

In (286) the relation between the fronted element and the surrounding context is different from (285). Here the new information is not in the subject but in the complement of the adjective head. The Swedish original thus allows a structure where a element which is information-structurally given is placed initially and receives prosodic prominence, whereas the new, heavy information is placed at the end. In English, however, there are stricter limitations on placing given but stressed information, initially.

In addition to fronted APs, the data also contain some instances of fronted adverb phrases, as in (287.

(287)	Jag slöt ögonen och kände mig illamående. **Så långt** hade det alltså gått sedan sist. Det var värre än jag tänkt. (AP1.2.s130-s132)	I closed my eyes, feeling sick. That's how long it had been since the last time. It was worse than I had thought. (AP1T.2.s137-s139)

In this example, the restrictions on structuring the information are more complex interactions between syntax and pragmatic principles. By introducing a demonstrative pronoun "that" as subject of the main clause and reducing the rest of the sentence to a subordinate clause, it is

possible to maintain the general order of the information compared to the original, i.e. “så långt hade det alltså gått sedan sist” och “...how long it had been since the last timE”. The use of the demonstrative pronoun “that” will, however, mean that the main stress will be on the pronoun rather than other constituents in sentence.

This can be regarded as a way of splitting the semantic and pragmatic function of the fronted element in the Swedish original between two separate elements in the English translation. The linking function of placing a stressed constituent clause initially is being handled by the demonstrative pronoun being the subject of the main clause, whereas the element triggering the inference that the previous information demonstrated the length of time is the predicative complement “how long” which is given emphasis by being fronted by *wh*-movement in the subordinate clause. The fact that the English construction achieves this by placing these elements clause initially in the main and subordinate clause respectively suggests that the clause initial position has particular pragmatic importance in both languages.

In addition to the full NPs that work as predicative complements, the data also contains a number of fronted object that are realized as full noun phrases. These are distinct in function and information structure compared to pronominal object fronting, and predicative complements that takes the form of full noun phrases.

One property that is often claimed to be central to the possibility of fronting a constituent is that it should be in some respect “given” in the current context. This is clearly the case when the fronted element is a proform as previously discussed. What remains to be seen, however, to what extent this also applies to fronted elements is a full noun phrase.

In the Swedish original texts, 19 out of 73 fronted object (26%) were realized by full noun phrases, all but two occur in fiction texts. Looking at the way they have been translated into English, we can note that the great majority of them are rendered by structures that leaves the fronted element in its canonical position.

One observation that can be made from the Swedish data is that in some contexts a full NP can be used as an aboutness topic. Although this type of pragmatic function is fairly rare in the Swedish original material, the examples reveal some interesting aspects of the interaction between syntactic and pragmatic properties of sentences. It has previously been observed (Ref.) that topical elements often shift and re-appear repeatedly in the development of discourse. Several referents can at any given point in the discourse be salient to various degrees.

Chafe (1976) discusses this in terms of referent *activation* and referent *accessibility*, i.e. to what extent different referent are activated in the consciousness of the discourse participants. Chafe argues that morphosyntactic form and prosodic properties are indications of referents activation states. Pronominal form and weak prosodic manifestation are characteristics of highly active referents. In addition to referents that are currently being activated, there are others that are in a semi-active state. According to Chafe (1987), there are two main types of semi-active referents: previously activated referents and referents that are part of a schema that is currently being evoked in the discourse. The fact that only a small set of referents can be activated at a certain point in the discourse means that when there is a shift from one referent to another, in essence a shift of topic referent; the previous topic becomes less activated but not, according to Chafe, inactive. These referents can thus fairly easily be re-activated since they are accessible by being part of the common ground. In a similar manner, it is assumed that discourse participants' ability to access certain referents is derived from them having access to knowledge structures which are based on their experience of the world, often described as schemas, e.g. a restaurant schema. Such cognitive structures will make it possible to infer referents from our knowledge of a typical restaurant, such as a menu, a waiter, etc.

Whereas pronouns are typically associated with currently activated referents, fronted noun phrases in Swedish can be used to re-activate a semi-active referent. Looking at the example in

(288) we can see how the fronted construction is used with a more global scope than the previous uses that have been primarily concerned with the relationship between two adjacent clauses.

The background to the passage in (288) is that Matt's fort has been split in two by a lightning strike, which at the beginning of the text excerpt has enraged the character Matt. However, in the third line it is stated that his anger subsides quickly, and he reflects on the positive aspect of having a smaller fort. This leads into a section describing an event where the character Noddle-Pete previously got lost in the vast fort. At the end of the section the question of Matt's views on the split fort is re-introduced, which in the Swedish original is signalled by fronting the object noun phrase "hela borgen". The fronting is here used a text structuring device which in the English translation is performed by the initial prepositional phrase "in any case".

The possibility of using fronted constructions in Swedish without an automatic contrastive interpretation and in a context where the referent of the fronted element does not have to be activated in the immediately preceding sentence makes it possible to employ fronting as a way to create global coherence and as a discourse structuring device.

(288)

Mattis rasade som ett vilddjur.	Matt was raging like a wild animal.
Hur kunde något sådant få hända hans fäders gamla borg?	How could this have been allowed to happen to his forefathers' old fortress?
Men Mattis orkade aldrig rasa länge för en sak, och han kunde alltid hitta på tröstegrunder.	But Matt could not go on being angry about anything for long, and he could always find reasons to take comfort.
Nåja, vi får inte så mycket irrgångar och källarhålor och skräp att hålla reda på.	'Oh, well, we sha n"t have so many twists and turns and cellar pits and rubbish to keep track of.'
Och nu behöver kanske ingen mer gå vilse i Mattisborgen. Ni minns hur det var när Skalle-Per gick bort sej och inte kom tillrätta på fyra dar!"	'And perhaps no one will need to get lost in Matt"s Fort any more.'
	'Remember what it was like when Noddle-Pete went astray and did n"t turn up for four days!"'
Det där ville Skalle-Per inte gärna bli påmind om.	'Noddle-Pete did not enjoy being reminded of this occasion.'
Kunde han rå för att det gick så illa?	'Was it his fault he had gotten lost?'
Han hade ju bara försökt ta reda på hur stor och kolossal Mattisborgen egentligen var och hade som sagt funnit den stor nog att gå vilse i.	He had only been trying to find out how vast and rambling Matt's Fort really was, and had indeed found it big enough to get lost in.
Stackarn, han var så gott som halvdöd, innan han äntligen hittade tillbaka till stora stensalen.	Poor thing, he was almost half dead before he finally found his way back to the great stone hall.
Rövarna hade tack och lov skrålat och väsnats så att han hörde det på långt håll, annars hade han aldrig kommit rätt.	Thank goodness the robbers had been bawling and kicking up enough noise for him to hear them a long way off; otherwise he would never have gotten back.
Hela borgen har vi ändå aldrig använt", sa Mattis.(AL1.1.s79-s90)	In any case, we have never used the whole fort," said Matt, "and we will go on living in our hall and bedrooms and tower rooms where we have always lived. (AL1T.1.s82-s93)

A similar function can be seen in a later part of the same text, as show in (289) where again

Matt's fort is re-activated, and turns the discourse from other things Ronia has explored and learnt to the topic of the fort.

(289) Ronja aktade sej och övade sej mer än Mattis och Lovis visste om och blev till sist som ett friskt litet djur, vig och stark och inte rädd för nånting.
Inte för grådvärgar, inte för vildvittror, inte för att gå vilse i skogen och inte för att ramla i älven.
Ännu hade hon inte börjat akta sej för att trilla i Helvetesgapet, men det skulle hon snart göra hade hon tänkt.
Mattisborgen hade hon annars utforskat ända upp till murkrönet.
Hon hittade i alla de ödsliga salarna, där ingen mer än hon någonsin satte sin fot, och hon gick inte vilse i några underjordiska gångar och mörka hålor och källarvalv. (AL1.2.s85-s89)

Ronia watched out and practiced more than Matt and Lovis knew, and in the end she was like a healthy little animal, strong and agile and afraid of nothing.
Not of gray dwarfs, not of wild harpies, not of getting lost in the forest, and not of falling into the river.
So far she had not begun to watch out for Hell's Gap, but she planned to start soon.
Besides that, she had explored Matt's Fort right up to the parapets.
She found her way into all the deserted rooms, where no one ever set foot, and she did not lose her way in underground passages, dark pits, and cellar vaults.(AL1T.2.s88-s92)

In (289) it is clearer than in (288) that the referent of Matt's fort is semi-active at this point and is being re-activated. Also, as in (288) the English translation uses a discourse marker in initial position to indicate the shift in topic and leaves the semi-activated referent in its canonical position. In contrast to many of the other uses of fronted constructions in the Swedish originals a non-fronted construction does not seem completely infelicitous in this context, which suggest that semi-active referents often can be re-activated in structures similar to the ones in the English translation, i.e. with an initial discourse element to signal the topic shift and the referent in a canonical position, as suggested in (290).

(290) I övrigt hade hon utforskat Mattisborgen ända upp till murkrönet.
Besides that had she explored Matt's Fort right up to parapets.DEF
'Besides that, she had explored Matt's Fort right up to the parapets.'

When the phenomenon of fronting is discussed in the literature, one of the discourse functions that is generally assumed to be the reason for placing a non-subject argument in initial position is contrast. By displacing a constituent from its canonical position to a clause initial position, and giving it prosodic prominence, the fronted element will receive a contrastive interpretation. As the comparison between Swedish and English data in this study has shown, this is a rather simplified account of the actual functions. Still, there are examples of fronted noun phrases in the Swedish original text that have a contrastive interpretation, as seen in (291) and (292).

(291) Lite ljug och påhitt kan nog vara nöjsamt ibland.
Men **såna där etterförjordade dumheter** får du låta bli att sätta ihop. (AL1T.3.s131)

A bit of lying and pretending may be fun sometimes.
But you've got to stop this make-believe.(AL1T.3.s132-133)

(292) Utan att han själv riktigt visste hur det gått till balanserade han på en stol i köket med spik i munnen och lagade farmors trasiga tak.
Kaffe fick han för besväret, och smörgåsar, men inget mer. (AP1.3.s43-s44)

Without his really knowing how it had happened, he would be balancing on a chair in the kitchen with nails in his mouth, mending Gran's ceiling.
He was given coffee for his trouble, and bread and something, but nothing more. (AP1T.3.s44-s45)

The example in (291) conforms to the prototypical relation between contrasting elements with the introduction of concepts belonging to a set in the first sentence (lying, pretending) which is being contrasted by another member of the set (make-believe). The contrast is also reflected in the predication, where 'be fun' is contrasted with 'got to stop'. Also, the contrast is signalled by the contrastive conjunction, and in the Swedish original, the fronting of the contrasting item. As discussed in Molnar (2001) the notion of contrast is generally associated with focal information, and the distinction between the pragmatic notions of focus and contrast can be more or less clearly manifested in different languages. English is regarded as a language in which contrastiveness is not obligatorily expressed linguistically. Judging from the contrastive data in the current study, Swedish has stronger requirements to express this pragmatics function formally.

Comparing (291) with (292) we can observe that they differ in the way the contrastive set is established. In (292) the fronted element 'Kaffe' cannot be infered to belong to any previously established set, as was the case in (291) instead the contrasted set is presented in the following constituent in same sentence, i.e. the set of possible compensations for fixing the ceiling. This type of structure can in turn be compared to fronted structures in the Swedish originals where there is no explicitly mentioned contrastive set, as in (293).

(293) De skrubbade och skrapade henne för den händelse hon medförde ohyra.
Hon såg på dem med sina blekblå ögon och lät dem hållas.
Det var krig.
Mat fick hon, och en säng att sova i.
Att det var något särskilt med Siiri begrep de inte. (AP1.3.s119-s122)

They scrubbed and scraped her in case she had brought vermin with her.
She looked at them with her pale-blue eyes and let them have their way.
It was war.
She was given food, and a bed to sleep in.
They didn't understand there was something special about Siiri.(AP1T.3.s119-s122)

Here the fronted element is focal but cannot be linked to any contrastive set in the previous discourse, nor is there, like in (292) any clear indication of a contrast set following the fronted construction. However, by using a fronted construction which give emphasis to the things that the character Siiri is given, there is an implication that these items are contrasted with all the things she was not given. This aspect of the Swedish original is not captured in the English translation. The implied meaning that Siiri is being given less than she should is strengthened in the following sentence, which is also a fronted construction of an object clause.

In this sentence the fronted construction makes the fronted element appear presupposed, i.e. although it has not been established in the previous discourse that Siiri is something special, the fronting of that information indicates that it should be incorporated into the common ground as if it was already given.

A similar effect can be seen in (294) where "de röda skorna med kilklack" is new information but is being presented as presupposed. Part of that effect is the use of a definite article, but also the syntactic construction which links the referent to the previously mentioned dress, thus creating an anchor set in the terms of Birner and Ward (1998). Again, this implication is not

carried over to the same degree in the English translation, which does not evoke a set consisting of the dress and the red shoes, in the way that is done in the Swedish original.

(294) Jag letade efter Siiris klänning i träkistan, den svarta som hon haft när hon dansade med pappa till tonerna av "Au bal de l'amour" och det doftade av grönska och brännvin.
Jag fann den, den luktade lite mögel, det gjorde inget, den var som jag mindes den: urringad och med små veck, och snäv över magen.
De röda skorna med kilklack fann jag i Siiris skåp.

I was looking in the wooden chest for Siiri's dress, the black one she had been wearing when dancing with Dad to the tune of 'Au bal de l'amour' and that smelt of greenery and liquor.
I found it, though it smelt slightly of mould, which didn't matter, because it was just as I remembered it, low-necked and with small pleats, tight across the hips.
I found the red shoes with wedge heels underneath a whole mass of rubbish, newspapers and old letters in Siiri's cupboard.

As previously mentioned, only two of nineteen occurrences of full NP object fronting were found among the non-fiction text. These two examples, however, indicate that in addition to containing a limited number of fronted constructions, the non-fiction texts also contain occurrences of fronting of a somewhat different type than the fictional texts.

(295) I konsekvens med sin kritiska grundsyn förespråkade Wallgren strikta rutiner och regelbunden läkarkontroll av barn och lokaler, samt för hemmavaro vid sjukdom.
Något egentligt empiriskt underlag för sina rekommendationer hade Wallgren inte (Familjestödsutredningen 1978)(CP1.3.4.2.1.s4-s5)

In line with his basically critical view, Wallgren advocated strict routines and regular doctor's examinations for the children and the premises, and recommended sick children to stay at home.
Wallgren did not really have any empirical foundation for his recommendations (Socialdepartementet 1978).(CP1T.3.4.2.1.s6-s7)

The text excerpt in (294) illustrates the difficulty of analysing discourse in terms of rather imprecise categories such as given and new, and also to give such a classification to entire constituents with no recognition of the complexities of the combination of presupposed and newly introduced propositional information that is often found together in a single element.

The fronted noun phrase is clearly part of 'new' information as there has not been any mention previously of "empirical foundations" or the lack thereof. At the same time, the modifying "för sina rekommendationer" ("his recommendations") links the noun phrase to the previous context.

Some discussion of this problem can be found in among others in Lambrecht (1994) who suggests a nested analysis where given and new elements can be found within other constructions that as a whole are analysed as having a particular information structural function, i.e. in a sentence with thetic interpretation like "My CAR broke down", the focal NP "my car" includes the topic of the sentence which is the reference to the speaker "my".

Similarly, we could then analyse "Något egentligt empiriskt underlag för sina rekommendationer" as the focus of the main clause but within this focal elemenent we find the topical "för sina rekommendationer". By fronting the object, the nuclear stress is placed on the negation, giving greater emphasis on the lack of empirical evidence. In the translation the sentence has a

less marked information structure by keeping the topical subject of the previous sentence in clause-initial position and placing the focal information last.

The possibility to introduce new or focal material clause initially was also found in the fiction texts in which it was translated with a presentative construction in English, as seen in (295). Again, the unmarked information structure with the topical *det* as subject and the new information in default final position has been replaced with a marked structure where new information is introduced in initial position. Similar to (294) it is also a negated sentence where the negation receives the nuclear stress.

The English translation uses a less marked way of incorporating the new information by employing a presentative structure, which prototypically is said to introduce previously unknown information into the common ground. These examples would suggest a greater possibility in Swedish to place focal information clause initially, both when they include topical information as in (295) and when it is all new as in (296).

(296)

Man har rekommenderat mig nummer sextiosex, upprepade Fröij mycket tydligt men utan att höja rösten.	"But I've had number sixty-six specially recommended," repeated Frey emphatically but without raising his voice.
Tyvärr.	"I'm sorry.
Det är inte möjligt.	It's not possible.
Vi har ingen rätt med det numret på matsedeln.	We don't have any dishes on the menu with that number.
Pröva i stället sextiofemman.	Try sixty-five instead.
Ankan.	The duck.
Den är alldeles utmärkt.	It's excellent."
Något språkligt missförstånd kunde det inte vara fråga om.	There could be no question of linguistic misunderstanding.
Till och med ä:na var utpräglat stockholmska och förrådde att den unge mannen var Sverigefödd. (KOB1.1.s191-s199)	His vowels had the ring of a Stockholm dialect, and it was clear the young man had been born in Sweden. (KOB1T.1.s175-s183)

The second example of a fronted object noun phrase in the non-fiction material, does not in the same way place new information sentence initially, but it enables a shift in topic where the fronted element indicates the link between the previous context and the information introduce in the current sentence. As can be seen in (297) the Swedish original uses a construction with the generic pronoun *man* ('one/you') whereas the English translation uses a passive construction.

(297) Kanske vågar man tolka kunskapsläget idag så här: 1. Eventuella skillnader kan inte vara stora. 2. Det finns metodiska problem som är svåra att komma runt och som tenderar att överskatta sjuklighet bland människor med hög utbildning. 3. Daghemsutvecklingen kan ha bidragit till att utjämna eventuella skillnader mellan högre och lägre sociala klasser.
En god illustration av detta hittar man i en amerikansk studie, där det tydligt framgår hur daghemsvistelse modifierar klasskillnaderna i incidensen av nedre luftvägsinfektioner (Margolis et al. 1992). (CP1.3.2.3.1.s24-s26)

Perhaps we could dare to interpret the state of our knowledge today as follows: 1. Any differences that exist cannot be great. 2. There are methodological problems which are difficult to get around, and which tend to overestimate morbidity among people with high education. 3. The development of day-care centres may have contributed t'
A good illustration of this can be found in an American study which clearly showed how the use of day-care centres modified class differences in the incidence of lower respiratory tract infections (Margolis et al. 1992).(CP1T.3.2.3.1.s25-s27)

Both original and translation have, however, very similar linear order of information, and these two different constructions in Swedish and English appear to be discourse-functionally equivalent. If this is generally true, we would expect to see passives used in English translations of these types of Swedish constructions with impersonal pronoun *man*. A quick search of the corpus of the current study reveals numerous examples that suggest that this is a general discourse functional equivalence, as shown in (298) and (299), but a more systematic investigation of this will have to be made to confirm this. For the present discussion, the main point is to recognise that two syntactically distinct constructions, fronting in Swedish and passives in English, seem to be discourse functionally equivalent in certain contexts.

(298) **I Stockholmsområdet** har ***man*** ansett kvalsterförekomsten vara låg, men tecken finns på att problemet kan öka också på nordligare breddgrader. (BJ1.1.3.3.2.s8)

In the Stockholm region the incidence of mites ***has been considered*** low, although there are signs that the problem may also increase in more northern latitudes. (BJ1T.1.3.3.2.s9)

(299) **Dessa och en del andra skillnader** tog ***man*** hänsyn till i multiple logistic analyser. (CP1.3.4.1.2.s31)

These and some other differences ***were taken*** into account in multiple logistic analyses. (CP1T.3.4.1.2.s32)

More generally, the use of fronting for managing topic shifts and the progression of discourse is the most common type of discourse function for the fronted constructions in the non-fiction text in the Swedish originals. Apart from the examples of object fronting that have been discussed previously, the overwhelming majority of fronting are constituents with an adverbial function.

5.2.2.3 Translations of fronted adverbials

This category can be argued to be more weakly tied to the transitivity of the main predicate, and in many ways these examples have more in common with initially placed adverbial adjuncts than with the fronted objects and predicative complements that we have looked at so far. Still, the placement of these elements in initial position in English is in many cases clearly more restricted than adjuncts. The tendency for these constructions to occur primarily in fiction text in the Swedish originals is also something that should be given a more systematic treatment.

One of the more common forms of adverbial fronting and a prototypical way of rendering that into English can be seen in (300).

(300) **I nedanstående tabell** visas de tio största ägarna i AssiDomän per 31 december 1998. (ASSI1.3.4.s1)

The table below shows the ten largest shareholders in AssiDomän at 31 December 1998. (ASSI1T.3.4.s1)

In terms of the linear order of information the texts in the two languages are very similar. At a certain point in the discourse, a contextually salient reference point is used to link new information to the common ground. In (300) a metatextual reference to a table in the text is used to introduce new information to the shared knowledge of the discourse participants.

Syntactically the structures differ in that the Swedish original uses a passive construction where the reference point is realized as a fronted prepositional phrase that is licensed by the verb, in this case the verb *visa* ('show') in passive form. The English translation, on the other hand, uses an active sentence where the reference point is the subject. Both languages would be able to use the type of structure employed in both the original and the translation. The Swedish text would with little difference be able to use an active clause with the NP complement of the preposition as subject. However, a fronting construction in English would result in the verb being placed last and receive the nuclear stress, i.e. *In the table below the ten largest shareholders in AssiDomän at 31 December 1998 are shown.*

The information that links the new information to the common ground can also be part of the propositional content of the discourse, as illustrated in (301). In this example, the general discourse topic is "different types of hypersensitivity". The link to the common ground is *the hypersensitivity field.* Similar to (300) the Swedish original uses a fronted prepositional phrase licensed by the verb, whereas the English translation realise the same referent as the subject. In contrast to (300), however, the Swedish original is not a passive construction but the active form of "belong".

(301) **Till området "överkänslighet"**hör också några andra luftvägssjukdomar.

The "hypersensitivity" field also includes a few other diseases of the respiratory passages.

This type of fronted prepositional phrase which has been translated by the same referent as the subject constitutes approximately one third of the fronted adverbial elements in the Swedish original (18 of 52).

The strategy employed by the translators is a "re-formulation" strategy which aims to maintain the linear order of information instead of attempting to find a morpho-syntactic representation that mirrors the one in the original. The other main strategy that has been discussed previously is to opt for a closer morpho-syntactic representation but without fronting a non-subject constituent. This second strategy is less often employed for translations of the adverbial fronting in the Swedish originals, but do occur, as shown in (302) and (303).

(302) **I tabell 4** redovisas en enkel sortomvandling. (BJ1.1.2.6.s23)

A simple measurement unit conversion table is presented in Table 4. (BJ1T.1.2.6.s22)

(303) **Till dessa aktörer** hör också biståndsgivarna. (CO1.1.2.2.s5)

Assistance donors are to be counted among these actors. (CO1T.1.2.2.s4)

Typical of these types of equivalences are the relatively short length and limited complexity of the syntactic structure. The effect of placing the linking information at the end and the new information initially is comparatively small. Structurally they have similar properties to sentences that can undergo inversion in English.

Looking ahead to the types of fronting construction that were found in the English originals we can note that inverted structures where the adverbial element is placed pre-verbally and the

subject is placed after the verb are one of the most common types of fronting. In the examples in the English originals, however, the verb is generally required to be a copula verb and the fronted element is often a prepositional phrase, as shown in (304). The similarity in the two structures is, however, the way information is structured linearly.

(304)	**On each disc** is a shell or a star, an orange maple leaf, a white rose.	På varje skiva finns en snäcka eller en stjärna, ett rödgult lönnlöv, en vit ros.

In the Swedish original non-fiction texts, there are also a number of examples where the referential link is simply a deictic adverb such as *här* ('here') and *dit* ('there'). The most common way of rendering these into English is to replace the deictic adverb with a full expression of the antecedent referent that the adverb is pointing at. Often this is a noun phrase that is made into the subject, as in (305) but in one instance it is translated with a cleft (306).

(305)	Till sådana drag hör familjehushållet som grundenheten för produktion, reproduktion och konsumtion, familjejordbruket som den vanligaste produktionsformen och bondehushållens plikt att till jordägare och stat lämna en del av sitt överskott som ett dominerande drag i resursfördelningen. **Dit** hör vidare en viss marknadsintegration, framför allt förmedlad genom de från det övriga samhället på olika sätt avsöndrade städerna. (HG1.2.s3-s4)	These features include the family household as the basic unit of production, reproduction, and consumption; the family farm as the most common form of production; and the duty of the peasant household to hand over part of its surplus to landowners and the state as one of the predominant elements in the redistribution of resources. Other features include a certain degree of market integration, especially through the intermediacy of the towns, which were separated in various ways from the rest of society. (HG1T.2.s3-s4)
(306)	I bostäder är situationen en annan. **Här** vistas man längre tid och hänsyn måste tas till spädbarn, åldringar och allergiker med flera.(BJ1.1.2.6.s5-s6)	The situation is different in dwellings. It is in dwellings that we spend more time and allowance must be made for infants, elderly people, persons suffering from allergies etc.

Both examples replace the adverb with a noun phrase with the full semantic value of the referent. The difference between these two examples are found in the role of the referent in the previous and current sentence. In (305) the first sentence in the Swedish original front the prepositional phrase *till sådana drag* ('to such features') which refers a set of features, and the second sentence keeps the same syntactic structure but replaces the prepositional phrase with the adverb *dit*. In the translation the set of features is the subject and the set is also made the subject of the subsequent sentence. Both original and translation thus exhibit what Daneš (1974) has termed "continuous theme".

In (306) the relation between the initial elements of the two sentences is somewhat different in the translation. The first sentence keeps the syntactic structure of the original but do not front the adverbial, which in this context is new information. Thus the the initial element on the second sentence is linked to focal information in the previous sentence which in Daneš's terms is a "linear thematic progression". This suggest that "thematic progression" i.e. the way given and new information progresses in discourse will affect the requirements and possibilities of using a particular information structural construction.

When discussing the different types of fronting of adverbial elements in the Swedish originals, it is worth pointing out that placing elements like deictic adverbs in initial position, with very

similar discourse functions as the examples presented here, are more frequent than indicated by the numbers reported here. The reason why all of these examples are not included in this study is that they have been analysed as adjuncts rather than complements of the verb. An example like (307) will consequently not be included as the locative *där* ('there') is not licensed by the verb *utgå* ('proceed'). However, as can be seen in the translation, the strategy of rendering this in English is similar to (305) and (306).

(307) **Där** kan man i princip utgå från maximalt 8 timmars exponering och personer som är vuxna, relativt friska och inte överkänsliga. (BJ1T.1.2.6.s3-s4)

In this context, it is possible to proceed in principle on the assumption of exposure for a maximum of 8 hours and individuals who are adult, relatively healthy and not hypersensitive. (BJ1T.1.2.6.s3-s4)

The distinction between verb licensed adverbial elements and adjuncts when placing them clause initially is thus less pronounced than when we are looking at objects and predicative complement.

5.2.2.4 Swedish fronting translated as English fronting

The previous discussion has looked at the translation strategies employed in rendering Swedish fronting in English where certain discourse functional properties are maintained. Looking at the total number of fronting in the Swedish original texts, only eighteen where translated by fronting in English, and of those, eleven where different types of adverbial fronting. There are two main types of fronting of adverbials in the English translations: constructions where only the fronted element is displaced, and fronting combined with postponement, i.e. inversion. The difference between these two are related to the discourse status of the subject. In contexts where the subject is given and salient, the English translation uses fronting, as seen in (308) whereas in context where the subject is new or a re-introduced referent, the translation is an inverted structure, illustrated in (309).

(308) **Där** är den, säger han och pekar på en stor och blank bil. (MS1.8.s24)

'**There** it is,' he says and points towards a large, shiny car. (MS1T.8.s24)

(309) **I kartongen** fanns en grön marsipanklädd Fahlcrantz special. (AP1.2.s69)

Inside the carton was a Fahlcrantz special, iced with green marzipan.

Among the few examples of English fronting in the translations that were not adverbials we can find some examples of fronted predicative complements, shown in (310) and (311). These are normally examples of focus fronting as they introduce new information. These are among the types of fronting that are also found in the English original texts.

(310) **Vackra och galna och grymma** var vittrorna. (AL1.2.s138)

Beautiful, mad, and ferocious, the harpies were. (AL1T.2.s139)

(311) **'Klagosång över en död Borkarövare'** heter den. (AL1.4.s44)

'Lament for a Dead Borka Robber,'i it's called. (AL1T.4.s39)

The restrictions on fronting focal elements in English appear to be dependent on both syntactic structure and information structure, as originally observed by E. F. Prince (1981a). In standard English, structures that front focal elements expressing attributes of the subject are accepted whereas other types of semantic roles are less felicitous. However, in dialects with a West-Germanic influence such as Yiddish, fronting is allowed with a wider range of clause elements/semantic

roles. E. F. Prince thus makes a distinction between fronting which is acceptable in the standard varieties of English such as (312) and what she terms Yiddish-fronting, shown in (313), E. F. Prince's examples 37a and 42a respectively.

(312) 'Now they're coming out with a hydraulic crane. ***Cherry pickers*** *they're called.* They're so very easy to upset...'

(313) 'She works with me. ***Twenty years*** *we've been here almost.* They demand more from a hairstylist and you get more money for your work'

Judging from E. F. Prince's examples of Yiddish-fronting, the focal information fronted in these constructions is giving emphasis to express that this information is not only new, but unexpected or surprising. Comparing (313) with the examples of fronting found in the Swedish originals, we find no direct equivalent to this type of fronting. It is clear that syntactically it is unproblematic to front temporal/locative references that are licensed by the verb as in (314). However, in the Swedish original data the fronted elements are often salient to a greater degree than what is shown in the examples of Yiddish-movement, and they generally lack the implication of expressing information that is surprising or unexpected.

(314) Han fick bevittna hur hon bröt upp ting efter ting, så att de blev rika, skimrande av betydelser.	He witnessed how she forced open one thing after another so they became rich, glimmering with meaning.
In i ordens värld hade han flyttat.	He had moved into the world of words.

The lack of this type of fronting in the Swedish originals could simply be a matter of the small number of fronting constructions in the material, rather than that these types of constructions not being felicitous in Swedish. Intuitively, it seems acceptable to have fronted structures with emphasis of surprise or unexpectedness, as in (315). However, the data in the Swedish originals cannot confirm this use of fronted constructions.

(315) Det är märkligt att vi inte träffats tidigare. Tjugo år har vi bott
It is strange that we not met before. ***Twenty years*** *have we lived here,*
här, utan att ens veta om varandras existens.
without that even know about each other's existence.
'It is strange that we have not met before. We have lived here twenty years without even know of each other's existence.

However, E. F. Prince's claim that fronting focal elements that are not attributes of the subject is restricted to certain V2-influenced dialects may be too restrictive. Some observations indicate that given contexts where the focal element is of a surprising or unexpected nature, these types of fronted construction may also be possible in standard English. Birner and Ward (1998) illustrate the possibility of focus-fronting with an example like (316).

(316) A: Where can I get the reading packet?
B: In Steinberg. Six dollars it costs.

Birner and Ward argue that the possibility of fronting new information in this example is dependent on the fact that the property of cost is inferable from the context. The property of cost is part of the salient set of properties of the reading packet, i.e. the cost of the reading packet can be related to the different properties of the reading packet which has been made salient by speaker A asking for one of its properties.

However, even when there is a less obvious inferred set that can be used as a link to the fronted element a fronting construction can be used, given that the fronted element is expressing surprising or unexpected information, similar to the situation with the example of Yiddish-fronting in (315). The example in (317) is taken from the BBC radio program "Seven day Saturday" illustrates a construction similar to the one in (316) but where the ability to use a fronted construction is possible because of the emphasis that is placed on the fronted constituent signalling it being surprising or remarkable, rather than having a easily inferable set.

(317) Presenter: Also you may notice that my script comes and goes at various points because ... the printer gave me the wrong script. SIX BILLION this building cost.

The speaker here foregrounds the information about the cost of the building both by prosodic means and by fronting it, to emphasise the surprising or remarkable situation of having a costly new building that is still not working properly. This may suggest that this type of fronting is highly felicitous in certain contexts but that they are typical of spoken discourse and may be rare in written corpora.

5.2.2.5 Swedish fronted negations

In the Swedish original texts we find constructions that place negative adverbs like *inte* ('not') and *aldrig* ('never') in initial position. The fronting of these constituents differs in significant ways from the fronting of other types of adverbial elements, which motivates that they are treated as a separate category.

Firstly, the negative adverb *inte* differ from all other categories included in this study in that they cannot be fronted in English. There is no structure in English with a fronted *not* that can be used as translation, as we can see if we look at the Swedish sentence in (318).

(318) Inte behövde hon vara rädd.
Not needed she be afraid.
'She didn't need to be afraid'

Secondly, fronting the equivalent to the Swedish negative adverbial *aldrig*, i.e. *never* will syntactically require that the word order in English becomes verb-second, such as in (319).

(319) Never had she met anyone of such mildness.

Looking at the way these types of fronting are translated into English we find that six out of nine occurrences have used English structures with the negation in its canonical position. One have used a fronted construction and two reformulations.

Fronting the negative adverb *aldrig* in the Swedish original has the effect of putting heavier emphasis on the fronted constituent, often creating a sense of the uniqueness the circumstances described in the sentence, as demonstrated by (320).

(320)

Jag går från vettet av det här, tänkte hon, det blir slutet för mej!	I 'm going to lose my wits this way, she thought; it will be the end of me!
Då kom där djupt inne från dimman några stilla ljuva klagande toner, där kom en sång, och det var den underligaste sång.	Then, from deep in the thickest mist, there came soft, sweetly plaintive notes –; a song, and it was the most wonderful song.
Aldrig hade hon hört något som liknade den, å, hur vacker den var, hur den fyllde hennes skog med sin ljuvlighet!	She had never heard anything like it. Oh, how lovely it was, how it filled her forest with its beauty!

However, the relative difference in emphasis between the fronted and canonical placement seems rather small. There is no obvious text-structural effect, and conceptually the negative adverbial does not directly link up with anything in the previous context. Clearly the translators believed that using the fronted construction in English would be less faithful to the meaning expressed in the original than the canonical structure, although that would have been syntactically possible.

The only example of a fronted construction with *never* is when the adverb is modified by another temporal adverb, as in (321), giving the fronting a clearer contrastive relation to the current events.

(321)	Då såg hon något som skrämde henne. Överallt mellan träden glimmade ögon, ja, där hade kommit en ring av ögon runt stenen som vaktade på henne, och hon hade inte märkt det. **Aldrig förr** hade hon sett ögon som kunde lysa i mörkret, och hon tyckte inte om dem.	Then she saw something that frightened her. Everywhere, among the trees, eyes were gleaming. There was a ring of eyes all around the stone, watching her, and she had not noticed anything. **Never before** had she seen eyes that shone in the darkness, and she did not like them.

It is worth noting that in the complete ESPC, instances of initial *aldrig* is generally translated with fronted constructions of *never* when *aldrig* is combined with adverbs such as *förr* and *mer*, resulting in translations of *never before* and *never again*. Even when the Swedish original does not have an adverb modification, the translation will include it as in (322).

(322)	Aldrig har vi varit varandra så nära. (PCJ1.5.s36)	Never before have we been so close to each other. (PCJ1T.5.s35)

This suggests a distinction in English between fronting *never* when it is negating the truth of the proposition, and when it has the meaning of comparing or contrasting the current proposition with previous or coming actions.

The use of fronted *inte* is primarily of interest as it demonstrates that fronting a constituent may have different interpretations depending on the context in which it is used. According to Teleman et al. (1999) initially placed *inte* can have two separate functions: reservation/objection or emphatic negation. The first function requires a context where the proposition in the sentence is given or salient, as illustrated in (323).

(323)	Minsta felsteg kunde föra vilse i dimman, det visste Ronja. Men rädd var hon inte. Med händer och fötter kände hon sej fram, stenar, träd och buskar var hennes vägmärken, det gick sakta, men nog skulle hon vara hemma, innan Lovis sjöng Vargsången. **Inte** behövde hon vara rädd. (AL1.4.s173-s176)	Any wrong step could lead them astray in the fog, Ronia knew. But she was not afraid. She felt her way forward with hands and feet. Stones, trees, and bushes were her signposts. They moved slowly, but she would still be home before Lovis sang the Wolf Song. There was no need for her to be frightened. (AL1T.4.s157-s162)

In the context, Ronia is trying to convince herself that she does not have to be afraid of getting lost when trying to find her way home. The fronting of the negation *inte* is here a less certain denial of the truth than a sentence with the negation in its canonical position.

This more modal function of the negation is picked up in the translation, which is the only time a fronted *inte* is rendered with a reformulation. Instead of a sentence with the negation in canonical position, the less certain negation is formulated by *There is no need for....* In several other cases, however, this distinction between the canonical placed negation and the fronted negation is not reflected in the translation.

The emphatic negation is in contrast to the objection/reservation function of the fronted negation a complete negation similar to the canonical position, but adds the element of enforcing or emphasising the negation. Contextually it is not the proposition which is presupposed but the negation, i.e. the context assumes that something is not true, but the event or action that is untrue is not given. This is demonstrated in (324).

(324) Hur litet hon än gillade framsidan av sin karl var baksidan, ryggen och de smutsiga fotsulorna än värre att ständigt behöva beskåda.
Du har glömt mocka under Rosa, skrek hon, det är tredje dagen i rad jag fått göra det själv.
Och **inte** har du varit ute i huggeriet heller.

However little she cared for the front side of her man, his bottom, back, and the dirty soles of his feet were even worse to have to look at constantly.
"You've forgotten to muck out around Rosa," she shrieked.
"That's the third day in a row I've had to do it myself.
And you haven't been out in the masonry, either."

In this context the fronted negation is signalling the additional action that has not been done. It is established in the immediately preceding context that the man has not cleaned the cowshed, and by fronting the negation, the information structure makes the negation the link to the previous context and places the focal material after the given information.

The main point here is the way the discourse function of the fronted element differs depending on the context in which it is places together with the information in the sentence. Any explanation that attempts to base the analysis solely on the initial placement of the negation has to be able to account for the difference that the context makes for the way the clause is interpreted.

5.3 Swedish translations of fronting in English originals

The types of fronting found in the original texts, together with the way fronted constructions in the Swedish originals were translated into English, lead us to expect a high degree of correspondence between the two languages. In particular, we would expect the great majority of fronted constructions in the English originals, many of which consist of different types of verb-licensed adverbials, to be rendered as fronted constructions in the Swedish translations. As shown in 5.3, this prediction largely holds true.

Table 5.3: Swedish translation types of fronting in English originals.

Direction	Canon.	Re-form.	Front.	Omit.	Total
EO → ST	3	13	48	1	65

There are, however, a number of reformulations in the translation. Although this group is not as large as the same category in the Swedish translations, it is perhaps larger than would be expected in relative numbers, 41% in the Swedish originals versus 28% in the English originals.

The type of translation which loses out to the fronting category is canonical constructions. Only in three cases were the fronted constructions translated into Swedish structures with the fronted element in its canonical position. Only in one instance did the Swedish translation omit the fronted element in the English original.

However, when we look at the different categories of translation we must keep in mind that the English fronted constructions consist of two distinct types: constructions where only the fronted element is displaced, which we can call *basic fronting* illustrated in (325), and constructions where the fronted element is displaced at the same time as elements which are canonically placed pre-verbally are placed after the main verb, i.e. *inversion*, shown in (326).

(325)	Macon often recalled that director. **Jim**, his name was, Jim Robinson or maybe Robertson - a burly, white-whiskered man with a crew cut, wearing a suit coat, as if in respect, over a Redskins T-shirt. (AT1.2.s272-s273)	Macon kom ofta att tänka på den där föreståndaren. Jim hette han, Jim Robinson, eller kanske Robertson – en stor, kraftig karl med vita polisonger och kortsnaggat hår som, kanske för att visa sin respekt, bar kavaj utanpå sin Redskins T-shirt (AT1T.2.s267-s68)
(326)	The waiting room at the Meow-Bow smelled strongly of disinfectant. **Behind the counter** stood a thin young woman in a ruffled peasant blouse. (AT1.3.s230-s231)	I väntrummet på Vovojam kändes det en intensiv lukt av desinfektionsmedel. Bakom disken stod en ung kvinna i en utanpåskjorta med rysch. (AT1T.3.s228-s229)

As discussed in connection with similar structures in the Swedish originals (examples 308ff), the discourse functional effects of these two types of constructions are very different. In (325), the focal information, *Jim*, is preposed, placing the information that is linking back to the previous context, *his name was* at the end of the clause. In (326), on the other hand, the locative reference which is placed initially, *behind the counter* is creating a link to the spatial context established in the previous discourse, *the waiting room at the Meow-Bow* while the focal information is placed at the end of the clause. In the English original data these two types constitute the great majority of cases. It is consequently an exception in the English original texts to "topicalize" i.e. preposing given information unless this is accompanied by a movement of the subject in post-verbal position.

In terms of translations into Swedish, the equivalent structures of both of these types also occur frequently in the Swedish original texts which makes it interesting to look more closely at those instances where the translators choose a different option than the equivalent construction. Looking at the two major types in the English originals, basic fronting and inversion, we find that among the basic frontings 11 out of 26 examples (42%) are re-formulated in the translations, whereas only 4 of the 39 (10%) inversions have another translation than the equivalent Swedish construction. This suggests that there is a greater discourse functional effect in the English inversion-structure and the Swedish equivalent structure.

In some instances, formally distinct constructions seems to have fairly little discourse functional relevance. In the great majority of the cases in the English original material, there will be inversion between the locative adverbial and the subject as shown in (326) but an alternative construction will employ a combination of fronted adverbial and an existential construction like the ones in (327).

(327) There are staircases, also of wood, which creak when we climb them, and banisters we aren't supposed to slide down, and iron radiators that make banging noises and are either stone cold or blazing hot.
On the second floor there are corridors leading into other corridors, lined with shelves that contain jars full of dead lizards or pickled ox eyeballs.
In one room there are glass cages with snakes in them, snakes bigger than any we 've ever seen before. (MA1.2.5.s30-s32)

Det finns trappor, också de av trä, som knarrar när vi går uppför dem, och trappräcken som vi inte får åka ner på, och värmeelement av järn som smäller till och som är antingen iskalla eller glödheta.
På första våningen finns det korridorer som leder till andra korridorer, kantade av hyllor där det står glasburkar fyllda med döda ödlor eller konserverade oxögonglober.
I ett rum finns det glasburar med ormar i, ormar större än alla vi sett tidigare.(MA1T.2.5.s30-s32)

According to Birner and Ward (1998) the difference between an inverted structure as (328a) and the existential construction in (328b) is the discourse status required of the initial PP. With inverted structures the fronted element only has to be less new than the focal information, i.e. both can be relatively new in the discourse, whereas with a fronting and a *there*-construction, the fronted element has to be discourse-old.

(328) a. In one room are glass cages. (Inversion)
b. In one room there are glass cages. (Fronting + presentative construction)

However, this analysis requires a fairly subtle distinction between degrees of information-structural value. The locative reference in (326) seems in many aspects similar in terms of "giveness" compared to the locative PPs *On the second floor* and *In one room* in (327). In the Swedish translations very little difference can be detected between the two structures. Although there is a tendency to use the verb *finns* ('exist'/'occur') with presentative constructions, this verb is also commonly used with inverted structures, as shown in (329).

(329) The motels are always far from anything, set against a dark wall of forest, their lights glimmering in the uniform obscuring night like those of ships, or oases.
They have gas pumps outside, human-sized, with round discs on top, lit up like pale moons or haloes minus the head.
On each disc is a shell or a star, an orange maple leaf, a white rose. (MA1.2.2.s52-s54)

Motellen ligger alltid långt borta från allt, mot en mörk vägg av skog, och ljusen från dem glimmar i den oföränderliga fördunklande natten som ljusen från båtar eller oaser.
De har bensinpumpar utanför, stora som människor, med runda skivor upptill, upplysta som bleka månar eller glorior minus huvudet.
På varje skiva finns en snäcka eller en stjärna, ett rödgult lönnlöv, en vit ros. (MA1T.2.2.s52-s54)

The restriction for using fronting with *there*-construction is more likely related to the same restrictions that apply to existential constructions in general, i.e. the discourse status of the referent of the subject, i.e. often only permitting indefinite noun phrases.

(330) Rook stooped to recover one last dropped banknote from the tunnel floor.
It was moist with Joseph's blood.
Next to it was the clipping from the catalogue, covered by the photograph which Con had given Joseph. (JC1.1.4.s298-s300)

Rook böjde sig ner för att plocka upp en sista sedel från golvet i tunneln.
Den var fläckad av blod.
Bredvid den låg urklippet från katalogen, täckt av fotografiet som Con hade givit Joseph. (JC1T.1.4.s274-s276)

This context seems the most resistant to a presentative-construction although it seems at least marginally possible, as shown in (331).

(331) Rook stooped to recover one last dropped banknote from the tunnel floor.
It was moist with Joseph's blood.
?Next to it there was the clipping from the catalogue, covered by the photograph which Con had given Joseph. (JC1.1.4.s298-s300)

(332) One idea that supports this view sees the stripes as a means for the zebra to identify which species it belongs to.
There are three main species surviving today.
Up in the north of the range there is the big Grevy's zebra, with its very fine, narrow stripes. (DM1.2.s47-s49)

En idé som stöder detta synsätt ser sebrans ränder som en metod för sebran att urskilja vilken art den tillhör.
I dag finns det kvar tre stora sebraarter.
I norra delen av utbredningsområdet *lever "revysebran (kungssebran) som har mycket tunna och smala ränder.*(DM1T.2.s47-s49)

Inverted structures are generally described as having a set of formal characteristics such as having a main verb that is intransitive with a meaning that is often quite weak. In addition, the fronted element should be relatively light or at least no heavier than the postponed subject, and often have locative meaning. However, most of these formal properties can be given an explanation in terms of the different constituents information-structural properties, as argued by Birner and Ward (1998, among others). One effect of this, can be seen in the restriction on having pronominal elements as postponed subjects. Elements that are fronted in inverted structures perform the basic functions that are often characteristic of canonical subjects, i.e. they create a link to the previous context, and is often signalled as the most salient referent by being reduced to a proform. There is thus a distinction between a subject that is a salient referent and a subject which is indicated to be the most salient referent. This means that it is possible to use inverted structures when the subject referent is salient but not the most salient, such as in (333a) but inversion is not possible if the referent is the most salient one, illustrated in (333b).

(333) a. Here is Johnny!
b. #Here is he!
c. Here he is!

The fact that the subject is the most salient referent normally prevents it from being placed in focus position and requires it to be placed in its canonical place, as in (333c). This is reflected in the examples from the English originals, as shown in (334) and (335).

(334) She knocked on his door and asked if he would be kind enough to show her the famous bird.
Fred was delighted and led her up to his bedroom where a truly magnificent blue and yellow parrot sat in a tall cage.
"**There** it is," Fred said. (RD1.4.s57-s59)

Hon knackade på och frågade om han ville vara så snäll och visa henne sin omtalade fågel.
Fred blev stormförtjust och gick före henne uppför trappan till sitt rum, där en verkligt magnifik blågul papegoja satt i en hög bur.
Här är den", sa Fred.

(335) He always had to make his appearance felt immediately by creating a lot of noise and clatter.
One could almost hear him saying, "It 's me!
Here I come, the great man himself, the master of the house, the wage-earner, the one who makes it possible for all the rest of you to live so well! (RD1.6.s35-s37)

Han var alltid tvungen att demonstrera sin närvaro genom att föra en massa oväsen.
Man kunde nästan höra honom säga: "Det är jag!
Här kommer jag, den store mannen i egen hög person, husets herre och familjens försörjare, den som gör det möjligt för alla er andra att leva så gott! (RD1T.6.s35-s37)

In these examples, it can be seen that it would be possible to replace the pronoun subjects with a full NP with maintained referents and still felicitously place them in final position, as shown in (336) and (337).

(336) Fred was delighted and led her up to his bedroom where a truly magnificent blue and yellow parrot sat in a tall cage.
There is the bird, Fred said.

(337) One could almost hear him saying, "It 's me!
Here comes the great man himself, the master of the house, the wage-earner, the one who makes it possible for all the rest of you to live so well!

Swedish does not have the same restriction on placing highly salient pronominal elements in clause-final focus position and makes no syntactic difference between indications of a salient subject referent and the most salient subject referent in terms of information structure. Clearly the syntactic restriction of only allowing one element before the finite verb is more dominant in Swedish than the function of signalling the information value of the referents.

However, it cannot be assumed that all occurrences of inversion in English are discourse functionally equivalent to a syntactically similar structure in Swedish. The use of fronted deictic adverbs like *here* ('här') and *there* ('där') with inversion can have discourse functional properties in English, which differ from the syntactically equivalent structure in Swedish, resulting in reformulations in the translations. Two examples of this are found in the material, shown in (338) and (339).

(338) All right, Macon told him, "let 's plan our strategy."
Strategy?"
Where we 're going to sit."
Why would we need strategy for that?"
It 's you who asked to see this movie, Ethan. I would think you 'd take an interest in where you 're sitting. Now, **here**'s my plan.

Då gör vi upp vår strategi."
Strategi?"
Var vi ska sitta."
Varför behöver vi en strategi för det?"
Det var du som ville se den här filmen, Ethan. Då tycker man att du skulle vara intresserad av var du sitter. Nu ska du få höra min plan.

(339) And now he waited for his return in the tunnel under Link Highway Red.
He squatted on his haunches, smoking, and studying the picture Con had given him – a snapshot of a market stall.
The man amongst the vegetables and fruit was a younger Rook, smiling, scarf unknotted at his throat, his clothes all black.
Here was the man to ambush, frighten, rob.

Nu väntade han på att Rook skulle komma tillbaka genom gångtunneln under trafikleden.
Han satte sig på huk, rökte och studerade bilden som Con hade givit honom, ett amatörfoto som föreställde ett marknadsstånd.
Mannen mitt uppe bland frukten och grönsakerna var en yngre, leende upplaga av Rook, klädd helt i svart och med en oknuten halsduk runt halsen.
Det var honom han skulle lägga sig i bakhåll för, skrämma och råna.

A contributing factor in the reformulation of the translation in (338) is most likely the use of the adverb *now* as a discourse marker before the inverted structure in the English original, which cannot be used with the same functional characteristics in Swedish. The translator has therefore opted for the phrase *Nu ska du få höra* which combines the discourse structural function of *now* with the information structural function of the fronted *here* with inversion.

More importantly, however, are the particular pragmatic properties of the fronted *here* in (338), which distinguish it from other uses of fronted adverbials. From the perspective of discourse function, constructions with initial *here* and *there* with inversion can be used with two distinct pragmatic functions. In the first type, the fronted adverb has a clear locative reference, similar to fronted adverbial PPs. The meaning of the adverb is either anaphorically or deictically referring to a point in space. An example of this was shown in (334).

The second type of fronted *here* and *there* with inversion has a much vaguer locative meaning and is instead used primarily to introduce new information, similar to the use of demonstratives like *this* and *that* (Huddleston & Pullum, 2002), as demonstrated in (338), repeated here as (340).

(340) It 's you who asked to see this movie, Ethan. I would think you 'd take an interest in where you 're sitting. Now, **here**/**this**'s my plan.

The difference is clear when we compare the fronted construction with a structure with the adverb in its canonical position, shown in (341).

(341) a. Now, here is my plan.
b. Now, my plan is here.

This equivalent fronted adverb in Swedish, *här*, seems to have a more limited function as a way to present new information, and tend to use demonstratives like *så här* and *det här* to a greater extent. An indication of this is that these Swedish demonstratives when fronted are translated with initial *here* in English, which can be seen in (342) and (343).

(342) Ge hit boken,fräste magistern.
Sidner räckte den stum till honom.
Så här gör man med sådan smörja.
(GT1T.2.5.s137-s139)

"Give me the book," hissed the teacher.
Sidner dumbly handed it to him. "**Here** 's what to do with such rubbish."
(GT1.2.5.s130-s132)

(343)	Han såg elak ut.	He looked evil.
	– Gåvor kräver gengåvor, sa han retsamt.	"Presents demand presents in return," he said teasingly.
	Det här är ministerns invigningstal. (KOB1.2.s192-s194)	"**Here** 's the minister's inaugural speech. (KOB1T.2.s189-s191)

The difference between the two languages when it comes to the discourse functions of fronted *here/här* does not reveal a great deal in terms of general differences in the way information structure interacts with syntactic constraints, but it highlights the fact that the mapping between a particular structure and its discourse function is often language specific.

This is also demonstrated in (339) where the Swedish translation is also restructured. In this example, the fronted adverb with inversion is translated by a cleft-construction in Swedish. In the discourse preceding the excerpt shown in the example, it has been established that the referent of the *he* in the initial sentence has been asked to rob a man, consequently the final noun phrase in the English original *the man to ambush, frighten, rob* is given. The initial adverb links the referent mentioned in the immediately preceding sentence with the description of a possible individual, 'the man who was going to be robbed', established earlier. The main focus is placed on the head of the noun phrase, the *man*.

Comparing it to the cleft-construction in the Swedish translation, we can see information structural similarities. The part after the cleft-constituent, often referred to as the the cleft-clause, *han skulle lägga sig i bakhåll för, skrämma och råna* is generally assumed to be presupposed information, and the cleft constituent, *honom*, is in focus.

In terms of discourse function, a cleft construction links a referent to a previously introduced nominal expression; the structure of the expression has to signal how the expression and the referent are paired up. From this perspective, the inverted construction with initial "here" and a cleft have similar properties.

The constructions in English thus have similar prosodic properties, as demonstrated in (344) and (345). They consist of two tone units where the first one establishes the referent in the discourse that will be linked to an expression, i.e. *Here was the man* and *It was him*. The focus in this tone unit is the final noun phrase. The second tone unit presents the previously mentioned expression that will be linked with this referent, i.e. *to ambush, frighten, rob* and *that he was going to ambush, frighten, rob*.

(344) Here was the man to ambush frighten, rob.

(345) ? It was him that he was going to ambush, frighten, rob.

However, although the two constructions have similar structural and prosodical properties in English and equivalent structures in Swedish, the pragmatic properties of the English and Swedish constructions differ, which makes using the equivalent structures in translations unidiomatic. As previously discussed, the use of initial adverb *här* in Swedish is not used to introduce focal information which makes the Swedish translation of (339) infelicitous. At the same time, the structure in (345) is rare in English, as indirect objects are rarely used as cleft constituents (Quirk et al., 1985, p. 1050).

As discussed in chapter 4, the great majority of fronted constituents in the English original texts have the syntactic function of adverbial, often expressing locative meaning. They thus make up a subsection of the large group of adverbials in the English text that are placed in clause initial position. The main differentiator between the two types of initial adverbials is that fronted adverbials, like all fronted constituents are licensed by the verb and thus obligatory, whereas most fronted adverbials are linked to the whole proposition. Being licensed by the verb formally restricts the possibility of where to place the adverbials, but to a great extent the discourse

functions of the different categories of initial adverbials are in many ways similar regardless of their internal relationship with other clause constituents. For example, placing a prepositional phrase clause initially is often done to link a referent in the previous sentence with the current proposition. However, the communicative difference between adverbials that are licensed by the verb and those that are not are often rather small, as seen in (346). The original sequence (346, 346b) taken from Margaret Atwood's (MA) *Cat's Eye*.

(346) Our sleeping bags are khaki and thick and lumpy, and always feel a little damp.
   a. Underneath them we put groundsheets, and inflatable mattresses.
   b. Underneath them we noticed that the ground was rain-soaked and muddy.

The fact that the verb *put* licenses the prepositional phrase *underneath them* whereas with the verb *notice* the prepositional phrase is not part of the argument structure seem to have little pragmatic effect. There is thus a difference between licensed adverbials and other licensed clause elements in that the restriction on fronting licenced adverbial elements are less strict than the restrictions on other types of valency constituents. The behaviour of licensed adverbials is thus closer to the behaviour of non-licensed adverbials than to other licensed constituents.

The type of fronting that is most most common in the English original texts apart from fronted adverbials is subject complements marked as focus. This is consequently a matter of 'focalization' rather than 'topicalization'

The fronted subject complement provides new information to the common ground and is prosodically marked as the focal constituent. The rest of the sentence is assumed to be presupposed. However, the presupposed part of the sentence is normally not explicitly mentioned in the previous context, but it must be possible to link the fronted construction to information in the previous context. A more specific account of this has been suggested by Birner and Ward (1998), which was discussed in section 1.5.1.1.

(347) The drops on the windshield grew closer together.
Macon switched his wipers on.
Tick-swoosh, they went - a lulling sound; and there was a gentle patter on the roof. (AT1.1.s20–22)

Dropparna på vindrutan föll allt tätare.
Macon kopplade på vindrutetorkarna.
Tick-svisch, lät de – det var ett sövande ljud, och regnet smattrade lätt mot biltaket.(AT1T.1.s20–s22)

The specific sound of the windshield wipers is new information but the idea that wipers that have been switched on make a sound, is inferable from the context. The saliency of the information is strong enough for it to be marginally possible to allow only the new information to be expressed, as example (348) shows.

(348) Macon switched his wipers on.
   a. Tick-swoosh, they went - a lulling sound; and there was a gentle patter on the roof.
   b. ?Tick-swoosh - a lulling sound; and there was a gentle patter on the roof.

The part of the clause that is left in it canonical position contains information that has not been mentioned before in the context, and is thus new, but it adds information that is generally inferable from the context.

In the material there are two examples of structures that demonstrate two distinct types of fronting in English, and how they relate to the surrounding context, which motivates the use of fronting. They also serve as examples of the discourse functional differences between English and Swedish fronting.

In the first example, shown in (349), the fronted constituent is the deictic adverb *there* which anaphorically refers to the previously mentioned prepositional phrase *in the corner of the goods car which carried Victor's name*. As previously discussed, fronting of anaphoric elements in English are possible when they are deictic elements that are prosodically prominent.

(349) When the klaxon blew to mark the train's departure, Joseph lifted his final load – a plastic travel-tank marked URGENT: LIVE FISH – and stowed it in the corner of the goods car which carried Victor's name.
And **there** Joseph stayed, as quietly as a slug in fruit, until the Salad Bowl Express set off for town.

När tågvisslan tjöt för att markera att det var tid för avgång lyfte Joseph upp sin sista last – en stor plastdunk märkt BRÅDSKANDE: LEVANDE FISK – och stuvade in den i ett hörn av godsvagnen som bar Victors namn.
Och **där** blev han kvar själv också, tyst som en snigel bland frukten, tills Salladsexpressen rullade i väg mot stan.

In terms of discourse status, the fronted element is clearly 'given' whereas the rest of the sentence consists of new information. At this point in the text, the information that the character *Joseph* will remain in the goods car is unexpected and new. However, by fronting the adverb, it is possible to mark the location as highly salient while simultaneously giving it a prosodic prominence. If the adverb has been placed in its canonical position, it would been given end focus and appeared to be part of the focal part of the clause.

In the Swedish translation, the adverb is also fronted, but as fronting in Swedish does not carry default prosodic prominence, the fronting construction in itself does not constitute the same communicative significance, which prompts the translator to add the reflexive *själv* and the adverb *också* as glossed in (350).

(350) Och där blev han kvar själv också
And there was he left himself also
'And there Joseph himself stayed as well'

Notably, the English original uses the full noun phrase *Joseph* for the subject referent of the previous sentence, which in most context the default form would be a personal pronouns, similar to what has been used in the translation, which render the subject as *han* ('he') in the Swedish version. One reason for using the full NP in the original may be that this emphasizes the adverb as the most salient referent and the main topic, which is less clear if the subject is reduced to a pronominal element. In this example a highly salient and given referent is fronted to indicate the link to the previous sentence and to give it prosodic prominence without making it interpretable as a focal element.

In the second example, illustrated in (351), the fronted element is an indefinite noun phrase, which according to Gundel et al's givenness hierarchy (Gundel et al., 1993) signals a low degree of accessibility, what they call "type identifiability". However, in this context, the indefinite noun phrase is a predicative complement in a copular clause, and thus attributes a quality to the referent of the subject *it*. Although the relation is different from the fronted adverb in (349) where the linguistic form did not add any new information but simply linked to the previous referent, it shares the characteristic of referring back to previous information, i.e. the description of the girl reading.

(351) Over the next few afternoons Mrs Phelps could hardly take her eyes from the small girl sitting for hour after hour in the big armchair at the far end of the room with the book on her lap.
It was necessary to rest it on the lap because it was too heavy for her to hold up, which meant she had to sit leaning forward in order to read.
And **a strange sight** it was, this tiny dark-haired person sitting there with her feet nowhere near touching the floor, totally absorbed in the wonderful adventures of Pip and old Miss Havisham and her cobwebbed house and by the spell of magic that Dickens

De följande eftermiddagarna kunde fru Gudomlig knappt slita blicken från den lilla flickan, där hon satt i den stora fåtöljen längst in i biblioteket med boken i knäet.
Hon var tvungen att stödja den i knäet, för den var för tung för henne, och det i sin tur innebar att hon var tvungen att sitta framåtlutad för att kunna läsa.
Och hon utgjorde verkligen en märklig anblick, denna lilla mörkhåriga person som satt där med fötterna dinglande långt ovanför golvet, fullständigt uppslukad av Pips underbara äventyr och av änkan Havisham och hennes hus som var fullt av spindelväv och av

Again, fronting the constituent places extra emphasis on the predicative complement and indicates that it is linking back to the previous context, and makes it possible to make a connection to the statement that *Mrs Phelps could hardly take her eyes from the small girl....* The fronted construction thus becomes a confirmation of the implied meaning that the previous description was remarkable. A non-fronted construction would indicate that the predicative complement is brand new which would have been infelicitous. We can see similarities with the other examples of fronted predicative complements, in that the fronted constituent introduces new information that is still highly salient.

The Swedish translation has opted for a non-fronted construction and using a pronominal reference to the girl, *hon*, as subject. The confirmation of the implied strangeness of the scene that is expressed by the fronted construction must instead be expressed by the adverb *verkligen* ('really'/'indeed'), as glossed in (352).

(352) Och hon utgjorde verkligen en märklig anblick
And she made really a strange sight
'And she really was a strange sight'

When it comes to the fronting of objects and indirect objects, English seems to require not only that the fronted element is prosodically emphasized but also that the reference of the fronted element is in a relation of contrast with some other member of an explicit or implicit mentioned set. Although this is generally true for the few examples of object fronting found in the current material, the contrastive relation is somewhat different from what is normally assumed in intuitively constructed contexts.

The example in (353) demonstrates a context where there is a clear contrastive relation which enables the fronting of the object.

(353)	Before running outside to boil a kettle and lay a tray – she seemed a hyperactive restless woman – she had confirmed his own identification. The dead man on the stairs was Harvey Copeland, the elder of the dead woman at the table Davina Flory. **The other woman** she identified as certainly Davina Flory's daughter Naomi.	Innan hon störtade ut för att sätta på tevattnet och duka brickan – hon verkade vara en hyperaktiv, rastlös kvinna – hade hon bekräftat hans egna identifieringar. Den döde mannen i trappan var Harvey Copeland, den äldre av de döda kvinnorna vid bordet Davina Flory. Den andra kvinnan var helt säkert Davina Florys dotter, Naomi.

Previous to this part of the text, the set of murdered people has been introduced, which consists of one man and two women. In (353), the different members of the group are being identified. The fronted construction is used to contrast the second woman from the first. A clear contrast is created between the 'other woman' and the others in the set by the fact that the other members of the set have been explicitly listed, and have been given attributes of the same type, i.e. names.

The use of fronting enables the referent of the noun phrase *the other woman* to be linked as contrastive topic in relation to the previous context without having to be the subject of the sentence. In contrast, the translation renders the topic referent as the subject of the sentence, which by default endows the referent with the properties of a topic. This arguably changes the meaning of the sentence both in terms of evidentiality and epistemic modality: the original expresses that the body was identified by a particular person with certainty, whereas the translation presents the identification as having been made with a high degree of certainty. However, in terms of the choice of topic, both original and translation have the same type of information structure, despite realising the propositional content with different sentence structures.

The second example has a relationship that is less clearly contrastive, as shown in (354).

(354)	He tried to look through the knife-blade sliver of bright light that was the gap between the edges of those curtains. He could see nothing but a dazzlement. The others were behind him, silent but close behind him. **To Pemberton** he said, "Break the window."	Han försökte kika in genom den rakbladstunna strimma av starkt ljus som utgjorde springan mellan gardinerna. Han kunde inte se något annat än ett bländande ljus. De andra stod bakom honom, tysta men tätt intill honom. Han sa till Pemberton: "Slå in fönstret."

In the context, it has been established what individuals are present, and at this point in the narrative the perspective is that of the character referred to with the pronoun *he*. The salience of the rest of the group is raised by them being made the subject of the third sentence of the example. However, the fronting in (354) differs from (353) in that the relation between the referent of *he* and the other members of the set is not part of the contrastive interpretation. The contrast is only in terms of picking out one member of the set to the exclusion of the others, but there is no parallelism in terms of the predicates of the members, i.e. the fronted construction in (354) would have had a different contrastive quality if the preceding context had a communicative relation between *he* and the rest of the members of the set, for example with a previous contexts of "He motioned to Sue and Tom that they should stand back. To Pemberton he said, 'Break the window."' As it stands, the fronting of the prepositional phrase primarily provides a 'frame' that restricts the application of the predication to a particular domain, in this case to a particular referent. The fronted construction in (353) would thus be close to what Chafe (1976) describes as a "Chinese topic", in contrast to a "real topic" in (354). The fact that the Swedish translation has

opted for a canonical rather than a fronted construction can be interpreted as an indication of the framing function being less dependent on fronting than topics with a more distinct contrastive interpretation.

5.4 Fronting in the Swedish translations of non-fronted constructions in the English originals

In the previous sections, we have looked at contexts in the English and Swedish original texts where fronted constructions have been used and studied what type of structures they have been translated into. In this way it is possible to investigate the interactions between basic properties of information structure, and discourse contexts. However, the use of parallel corpora also makes it possible to take the translations as the starting point, and investigate what original structures triggered the use of fronted constructions in the translations. In the cases where the fronted constructions in the translation resulted from a fronted construction in the original, these have already been dealt with in the previous sections. In some cases, however, the fronted constructions in the translations correspond to non-fronted constructions in the original.

In Table 5.4 the types of structures in the English originals that were translated into fronted structures in the Swedish originals are presented.

Table 5.4: Fronting in Swedish translations and their origins in English originals.

Direction	Canon.	Other Structures	Front.	Empty	Total
ST → EO	29	27	11	27	94

The different categories are similar to those used in the presentation of the fronted constructions in the original texts and their corresponding versions. The exception is the category "Reformulation", which has been replaced by "Other structures". The reason is that, when the Swedish version is taken as the point of departure, the English original cannot be considered a reformulation of it. Rather, the source of the fronted construction in the Swedish version is a structure that differs from it in a way that prevents it from being classified as either "canonical", i.e. a structure corresponding to the Swedish version but without fronting, or "omission", in which the fronted element in the Swedish version has no phonological realisation in the original.

As can be seen from Table 5.4, only a small proportion of the fronted elements in the Swedish versions have their source in fronted constructions in the English originals. The great majority of fronted constructions in the Swedish versions correspond to non-fronted constructions in the original texts. The methodological assumption is that the relationship between fronted constructions in the Swedish versions and their source structures in the English originals will reflect the patterns found when fronted constructions in the Swedish originals are rendered in English. This assumption is borne out by the distribution of the anaphoric proform *det*, which is frequently found as the fronted element in the Swedish versions. It has two main types of source structure: canonically placed demonstrative pronouns and elliptical structures in which the material corresponding to the fronted element is not phonologically realised in the original. These two types are illustrated in examples (355) and (356), respectively.

(355)	Sarah telephoned Macon and asked if she could come get the navy blue rug from the dining room. "Navy blue rug," Macon repeated. (He was stalling for time.) "I wouldn't mention it except you never liked it," Sarah told him. "You said it was a mistake to have a rug where people were eating." Yes, he had said that. (AT1.3.s1-s6)	Sarah ringde och frågade om hon kunde komma och hämta den mörkblå mattan i matsalen. Den mörkblå mattan", upprepade Macon. (Han försökte vinna tid.) Jag skulle inte ha frågat annars, men du har ju aldrig tyckt om den", sade Sarah. Du sa att det var fel att ha en matta där man äter." Ja, **det** hade han sagt. (AT1T.3.s1-s6)
(356)	My mother goes to Aylesbury every afternoon to play bingo, Matilda had said. "She doesn't know I come here." "But that's surely not right," Mrs Phelps said. "I think you'd better ask her." "I'd rather not," Matilda said. (RD1.1.s94-s98)	"Min mamma åker till Aylesbury varje eftermiddag för att spela bingo", hade Matilda sagt. "Hon vet inte att jag går hit." "Men det är väl inte rätt", sa fru Gudomlig." "Jag tycker att det vore bättre att du bad henne om lov." "**Det** vill jag helst inte", sa Matilda." (RD1T.1.s95-s99)

In the English original in (355) the demonstrative pronoun *that* which refers to the indirect speech clause in the preceding sentences *it was a mistake to have a rug where people were eating* is in its canonical position where its receives nuclear stress. In the context it confirms the truth of the assertion in the previous sentences and therefore receives prominence on the auxiliary verb *had*. The Swedish translation fronts the pro-element and the sentence stress is placed on the auxiliary *hade* and nuclear stress on the main verb *sagt*.

In (356) the main verb and the object have been omitted in the English original, leaving the adverb and negation to receive nuclear stress, whereas in the Swedish translation there is a structure similar to (355) with a prosodically de-accented fronted proform *det* and a nuclear focus on the negation. In terms of information structure the English text only retains the focal information whereas the Swedish translation the link back to the previous context is placed clause initially.

As we have seen previously, the same structures with fronted proform *det* are also found in the Swedish original text, but not the same number of occurrences as in the Swedish translation where more than half of the examples are of this type (52 of 95). Not all of them, however, have the type of source constructions shown in (355) and (356).

However, the general principles are fairly consistent in that references to newly introduced, salient information in the English original texts are realised as proforms placed in their canonical position, further down in the sentence structure. This often means that topical information is placed at the end of the sentences. On the other hand, in the Swedish texts, both original and translations, the proform is fronted to appear before focal information, and in terms of the ordering of information, thus more closely follows the principle of placing constituents from left to right based on the degree to which they contribute with new information to the discourse as suggested in the theory of 'communicative dynamism' (CD) (Firbas, 1971).

In some relatively few cases, the non-subject proform is realised as a personal pronoun rather than a demonstrative one, as shown in (357) and (358).

(357) I sit down and say to myself, how can I convert a mileage reading of one hundred and fifty thousand into only ten thousand without taking the speedometer to pieces?
Well, if I were to run the car backwards for long enough then obviously that would do it.
The numbers would click backwards, wouldn't they?
But who's going to drive a flaming car in reverse for thousands and thousands of miles?
"You couldn't do it!"

Jag slår mig ner och frågar mig själv hur jag ska kunna förvandla tjugotusen mil till tvåtusen mil utan att plocka ner hela mätaren.
Tja, om jag skulle ta och backa bilen tillräckligt långt skulle det uppenbarligen greja biffen.
Räkneverket skulle gå bakåt, eller hur?
Men vem fan vill sitta och backa en bil i hundratals mil?
Det kan man inte göra!"

(358) Macon wondered if even this moment would become, one day, something he looked back upon wistfully.
He couldn't imagine it; he could n't think of any period bleaker than this in all his life, but he 'd noticed how time had a way of coloring things.(AT1.2.s171-s175)

Macon undrade om även detta ögonblick en dag skulle bli någonting som han såg tillbaka på med längtan i sinnet.
Det trodde han inte; han kunde inte minnas att han någonsin hade genomlevt en lika dyster period, men han hade ju lagt märke till att tiden sätter färg på saker och ting. (AT1T.2.s169-s172)

In both examples the translations have fronted a proform referring back to a verb phrase or entire clause. The purpose of the underlined sentences are to reject the propositions introduced in the preceding context and it is thus the negation of the propositions that is the focal information. The main link to the previous context is the proforms *it*/*det* but in contrast to the English original in which the proform can remain in its canonical position, the anaphoric function of the proform leads to them being placed sentence initially in the Swedish translations.

Comparing (355) where the proform was a demonstrative proform *that* with (357) and (358), we can see that it would be possible to front a demonstrative pronoun with a greater prosodic emphasis and a contrastive interpretation, but fronting the proform *it* is ungrammatical, as shown in

(359) Yes, THAT he had said.

(360) *IT you couldn't do!

It is clear that objects that are topical has a much greater tendency to be fronted in the Swedish translation than in the English original. and fronting in Swedish doesn't necessarily have to express a clear contrast in the same way as is the case in English. This is illustrated in example (361) where the pronominal object *him* in the English original remains in its canonical position, whereas in the translation the object *honom* is fronted.

(361)

"Whoa", the girl said.	"Hoppsan", sade flickan.
She frowned at the card she'd pulled.	Med rynkad panna betraktade hon kortet
"Edward?" she said.	som hon hade plockat fram.
"On Rayford Road?"	"Edward?" sade hon.
"That's right."	"På Rayford Road?"
" We can't accept him." (AT1.3.s171-s176)	"Ja, det stämmer."
	"Honom kan vi inte ta emot." (AT1T.3.s170-s175)

In this excerpt from Anne Tyler's *The Accidental Tourist* the dog called Edward is highly salient at the point where the underlined sentence is introduced and clearly topical. In the English original it is sufficient to realize it as a personal pronoun in object position to make the link to the previous context clear, at the same time the use of *we* as subject is motivated by the importance of the action of "accepting/not accepting" the dog. Also placing the object in initial position would require prosodic prominence which would lead to a contrastive reading, implying a membership of a delimited group from which the referent of him is selected.

This contrastive interpretation is not present in the Swedish translation where the object has been fronted. Instead the fronting indicates the link to the previous context and the status of the object referent as the most salient one. Fronting of given referent can be a way to highlight a referent in Swedish, and isn't connected to contrastive interpretation as exclusively as in English. Thus the interpretation of the fronting of topical objects in the two languages can be paraphrased as in (362-363).

(362) Him we can't accept. (X to the exclusion of all other possible candidates cannot be accepted)

(363) Honom kan vi inte acceptera. (X, the current centre of interest cannot be accepted)

However, not in all context will it be possible to let the topical information remain in its canonical position and let the anaphoricity of the proform link the referent to the previous context. The fact that fronting of topical information in English will result in a contrastive interpretation results in the use of other constructions to indicate to topicality of a constituent. The different possibilities of indicating the topic in English and Swedish can manifest itself in the use of fronted constructions in the translations for other types of structures in the original, such as the use of fronting to translate passives, as illustrated in (364).

(364) The big dyeing job was done twice a year at the hairdresser's, but every month or so in between, Mrs Wormwood used to freshen it up by giving it a rinse in the washbasin with something called PLATINUM BLONDE HAIR-DYE EXTRA STRONG.
This also served to dye the nasty brown hairs that kept growing from the roots underneath.
The bottle of PLATINUM BLONDE HAIR-DYE EXTRA STRONG was kept in the cupboard in the bathroom, and underneath the title on the label were written the words Caution, this is peroxide. (RD1.6.s5-s7)

Två gånger om året lät hon färga det hos hårfrisörskan, men ungefär en gång varannan månad sköljde hon igenom det i handfatet för att det inte skulle förlora färgen, och då använde hon sig av något som på etiketten kallades PLATINABLONT HÅRFÄRGNINGSMEDEL EXTRA KRAFTIGT.
Detta medel hjälpte dessutom till att dölja de fula råttbruna partier som växte ut efter hand.
Flaskan med PLATINABLONT HÅRFÄRGNINGSMEDEL EXTRA KRAFTIGT förvarade hon i badrumsskåpet och på etiketten stod det också Varning! Innehåller vätesuperoxid.

Here the bottle of hair-dye is introduced in the first sentence and placed clause initially to signal it topical function in the underlined clause. In the translation, the same basic word order is maintained but the clause is active with the topical object fronted and with the subject expressed. In this context there would be little difference in the interpretation of the Swedish translation between using a passive or active construction.

Another source for fronted topical objects in the Swedish translations are non-agent subjects in active clauses, as shown in sentence (2) in (365).

(365) This was because he had developed a system that enabled him to sleep in clean sheets every night without the trouble of bed changing.
He 'd been proposing the system to Sarah for years, but she was so set in her ways.
What he did was strip the mattress of all linens, replacing them with a giant sort of envelope made from one of the seven sheets he had folded and stitched together on the sewing machine.
(1) He thought of this invention as a Macon Leary Body Bag.
(2) A body bag required no tucking in, was unmussable, easily changeable, and the perfect weight for summer nights. (AT1.2.s6-s65)

Det berodde på att han hade tänkt ut en metod som gjorde att han kunde sova i rena lakan varenda natt utan att behöva bädda om.
I flera år hade han föreslagit Sarah att göra på det sättet, men hon satt fast i sina gamla vanor.
Han gjorde så att han befriade madrassen från allt sänglinne och ersatte det med ett slags jättelikt fodral, gjort av ett av de sju lakan han hade vikt och sytt ihop på symaskin.
Macon Learys Kroppsfodral kallade han sin uppfinning.
Ett kroppsfodral behövde man inte stoppa om sig, det trasslade inte ihop sig, var lätt att byta och lagom tunt om sommarnätterna. (AT1T.2.s61-s65)

In the English original the concept of 'a body bag' is introduced in sentence (1) and therefore topical in the following sentence where it is realized as the non-agent subject of "required". In the Swedish translation, the original structure has been rendered as the verb *behövde* ('need/require') but with the generic pronoun *man*('you/one') as subject and the Swedish equivalent of 'body bag' *ett kroppsfodral* as a fronted object. A possible reason for the using a construction with a

generic pronoun as subject rather than the original structure is the use of a transitive verb *stoppa om* that requires an agent subject and a reflexive pronoun, as can be seen in the gloss in (366).

(366) Ett kroppsfodral behövde man inte stoppa om sig
A body bag need one not wrap around oneself
'You don't need to warp a body bag around you'

The translation alters the meaning of the original somewhat, but the examples in (364) and (365) demonstrate the discourse functional equivalence of on the one hand subjects in passive sentences and non-agent subjects in English and fronted objects in Swedish. In most contexts it would be possible to use Swedish equivalents to the English passives and non-agent subjects but as is shown in (overstran72) there can be other aspects of the sentence structure that makes fronting a more idiomatic choice. Fronting in Swedish is to a much greater extent a way to 'topicalize'.

However, sentence (1) in (365) also illustrates an example of where a focal element is fronted in the Swedish translation where the original has the focal element in its canonical position. An important factor in the structural differences between original and translation is the type of verb phrase construction used in the two languages. In the English original it is the structure 'X thought of Y as Z' whereas the Swedish translation 'Z kallade Y X', i.e. 'X called Y Z'. Fronting from that structure in English would be ungrammatical as in (367).

(367) Macon Leary Body Bag he thought of this invention as

I contrast, it would be marginally possible using a verb equivalent to the Swedish *kalla* ('call'), as in (368). The felicity of this construction would further improves if the object is encoded as more salient with the use of a pronoun, as in (369).

(368) ? Macon Leary Body Bag he called this invention.

(369) Macon Leary Body Bag he called it.

The context makes available the idea of an invention, and a salient property of that invention is its name. This is similar to the type of fronted construction that was illustrated in 325. In contexts where a referent have been made salient, inferable properties of that referent can be linked to that referent by fronted noun phrases that often functions as predicative complement. However, in English the fronted element needs to be the most focal element which restricts the type of information that can be included in the non-fronted part of the clause. In Swedish this tolerance is somewhat greater which is why the translation of sentence (1) in (365) is felicitous whereas the equivalent in (368) is only marginally possible in this context.

Another non-fronted source of fronted constructions in the translation are initial adverbials with presentational constructions like in (370).

5.5 Fronting in the English translations of non-fronted constructions in the Swedish originals

When it comes to the translation of the fronted constructions in the English translation the great majority had their origin in fronted constructions in Swedish fronted constructions. However, there were also a small number of examples where the English translations used a fronted construction for a non-fronted construction in the Swedish original. One type of fronting construction in the English translation that does not originate in a fronted construction is with the fronted proform *so* as illustrated in (370 - 371).

(370) Du är inte så bortkommen", sa Birk, och genast hoppade han efter.
Men Ronja väntade inte på honom.
Med ett nytt språng flög hon tillbaka över klyftan.
Så kunde han stå där och glo efter henne bäst han ville!
Du skulle ju ge mej en på trynet, varför gör du inte det då", sa Birk.
"Nu kommer jag."
"Jag ser det", sa Ronja.(AL1.3.s56-s58)

You 're not so bad," said Birk, and a second later he had jumped after her.
But Ronia had not waited for him.
With another leap she flew back across the chasm.
"You were going to punch me – why didn't you?" Birk said.
"I 'm coming!"
"**So** I see," said Ronia. (AL1T.3.s52-s58)

(371) Han kan vara mycket påfrestande, svarade pressekreteraren.
Men det är alla ministrar.
Å andra sidan fortsätter han att visa mig stort förtroende.
Du vet kanske att både han och jag har varit ordförande i samma SSU-klubb.
Visserligen med femton års skillnad men ändå.
Det är sådant som förenar, svarade Gerd Angerbo leende.
Det påstås så. (KOB1.2.s171-s176)

"He can be very trying," admitted the press secretary, "but then, all ministers are the same.
On the other hand, he still takes me very much into his confidence.
Maybe you know we were both chairman of the same Young Socialists group.
Admittedly, there were fifteen years between us, but even so."
"That's the kind of thing that binds people together," remarked Gerd with a smile.
"**So** they say."

The proform *so* are fronted in the English translations in both examples although the original texts have the objects in canonical position. It is questionable to what extent this demonstrates some more principled difference between the discourse functions in the two languages, since the constructions with the proform *so* in English have some specific properties. For many verbs, fronting is obligatory and placing the proform in canonical position would not be possible, as is the case with (370) where **I see so* would be ungrammatical. Other verbs like *believe* or *think* can be constructed with the proform in both fronted and canonical position. It is interesting to note that we found the reverse situation between fronted and canonical object placement with the use of *so* in the material, as previously shown in (258) where ***Det*** *anser inte du heller.* is rendered as *You don't think so either.*

One of the examples of fronted English translations of non-fronted structures in the Swedish originals is the result of particular idiomatic expressions that are difficult to consider to be formally equivalent structures, as shown in

(372) Birk skrattade.
Du är hårdare än sten, rövardotter!
Men du hittar bättre än jag till Mattisborgen.
Kan jag få hålla fast vid en flik av din kolt, tills vi har kommit ut ur skogen?"
Du håller så lagom", sa Ronja. (AL1.4.s161 - s165)

Birk laughed.
You 're harder than stone, robber's daughter!
But you can find your way to Matt's Fort better than I can.
May I hold on to the edge of your tunic until we 're out of the woods?"
That you can't," said Ronia, but she untied her leather rope, the one that had saved him once before, and handed him one end. (AL1T.4.s146 - s150)

5.6 Summary

The analysis of the different types of fronted construction that were found in the Swedish and English original texts and the way they have been translated English and Swedish respectively, give some insight into the syntactic structure, distribution and discourse function of fronted constructions.

Looking at the fronted construction in Swedish originals, we can see that the great majority of these are fronting of different types of anaphoric noun phrases, often in the form of types of proforms, such as *so* and personal, as well as demonstrative pronouns. However, when these are compared with the English translation, we can see that the congruent structures in English do not exhibit any movement to the beginning of the clause by the anaphoric elements. Moreover, there is a clear correspondence between, on the one hand, unstressed anaphoric pronouns in Swedish and English translations where the proform is left phonetically unrealised, and on the other, prosodically prominent proforms in Swedish and canonical placement in the English translations. The examples in the material tentatively support the hypothesis that the difference in structures in the two languages has to do with the mapping between presentational focus and syntactic structure.

This difference between the languages is supported by the findings when comparing fronted constructions in the Swedish translations with the type of structures they correspond to in the English originals. Although there are overall fewer fronted constructions in the Swedish translations than in the Swedish originals, we find that the Swedish translation in some cases fronts anaphoric proform when the English originals place the proform in its canonical position. Although the number of instances is fairly low, there is some tendency that fronted pro-nouns in Swedish have the same discourse functional properties as reversed *wh*-clefts in English.

In the Swedish original texts, there are also a number of instances of fronted constituents with full lexical form, primarily as adjectives and nouns. Discourse functionally, there is a more diverse set of applications of the fronting of full lexical constituents than with fronting of proforms, which to a large extent linked back to a previous mentioning of the same referent or propositional content. Among the fronted adjective phrases, we can distinguish two types of discourse functions corresponding to two different types of morphosyntactic form. Non-modified adjectives are fronted with contrastive function. By fronting the adjective phrase, it is possible to indicate the anaphoric link to previous mention of the property described by the fronted constituent, at the same time as it contrasts with some property that it introduces in the following context. The English translations opt for placing the adjectives in a canonical position. However, looking at previous studies it is clear that there are contexts in which contrastive adjectives are fronted in English, although there seems to be a preference for both properties in the comparison to be fronted.

The other function of the fronted adjective phrases is to indicate a comparison to some previously mentioned properties. In these instances, the adjectives are modified with *så/so, that*. Again, the fronting construction enables the sentence to create a link to the previous context and also to introduce the comparison to a property which is introduced in the discourse and should therefore be placed in focal position. The English translations tend to place the adjective phrase in canonical position, but looking at data from English monolingual corpora, we can establish that this type of fronting is clearly possible in English, possibly with the difference that in English there is a stronger requirement for the subject to be less salient than is necessary in Swedish. The relative restrictive functions of fronted adjectives make them fairly infrequent, and no examples were found in the Swedish translations.

Fronted noun phrases exhibit great variation in their discourse functions in the Swedish original texts, and one distinct difference between Swedish and English fronting is the use of fronted noun phrases as discourse structural constructions, by re-introducing previously mentioned referents or propositional material. The fronted noun phrase is both discourse-old and hearer-old; however, there can often be a great deal of information between the fronted anaphoric link and the introduction of the relevant antecedent material. In terms of saliency, it is thus less prominent in the mind of the interlocutors and requires a greater effort to draw the correct inference. The English translations make use of connectives clause initially to indicate the structural break. Generally, the English texts often make use of initial adverbial phrases and clauses to relate the latest addition to a discourse.

Noun phrases are also fronted to express a contrast which can both relate to a previously mentioned concept or point forward to some information that will be introduced. The fronted element is normally focal information, as is often the entire predication, whereas the subject is topical. This type of fronting in Swedish is distinct from the general principles of fronting in English, as described by Birner and Ward (1998), as their requirement for fronting predicts that they cannot be fronted unless they have an antecedent in the previous context. In the Swedish originals there where also some examples of fairly heavy constituents which contains pronominal expressions linking back to the previous discourse, but which themselves are indefinite noun phrases. Looking at the English translations, there is no consistent congruent structure for the Swedish fronting but one of them is passive construction. Comparing the relation between this type of fronting in Swedish and the English translation, we can see that this relation is mirrored in the way Swedish fronted structures in translations have their origins in English originals.

A large group of fronting constituents in both Swedish and English consists of verb-licensed adverbials, to a large extent their discourse function is similar to more loosely attached adjunct adverbial in that they establish the sentence spatial or temporal reference point, or how the proposition of the sentence relates to some external spatio-temporal reference point. As many of the English examples are locative inversions, the surface word order of the two languages is often very similar. The Swedish data do, however, result in a greater number of fronted negations and negative polarity items.

In English original texts, one of the more frequent examples of fronting of full-lexical phrases is different types of subject complement. This is supported by the occurrence of similar structures in the English translation. This type of fronting is generally focal information that is fairly easily inferred from the context.

The English originals also includes examples of fronting for emphatic purposes, similar to the constructions reported by (Birner & Ward, 1998) in terms of "proposition assessment", see chapter 3. The fact that the Swedish translation opt for non-fronted constructions, although congruent structures are generally possible, shows the restricting influence of lexical choices for fronting.

Chapter 6

Differences of fronted structures in English and Swedish

This chapter argues that the way morphosyntactic and prosodic properties interact in English and Swedish results in different discourse functional possibilities for argument fronted constructions in the two languages. We also claim that this analysis goes some way towards giving a more systematic account of how argument fronting is used in the two languages, with an emphasis on IS-fronting.

6.1 Introduction

In the earlier parts of this thesis, we have presented previous research on the syntactic structure of IS-fronted structures in English and Swedish, as well as studies looking at the discourse functions of these constructions. We have also presented the empirical findings from a small corpus study of argument fronted structures in English and Swedish written text and the way these have been rendered in the translations into the other respective language.

Moreover, we have provided a brief overview of the theoretical framework within which we will analyse and try to explicate the way argument-fronted structures work in English and Swedish, both how these structures differ from the canonical structures in each language and how they differ between the two languages.

The central proposal is that the way the phonological representation is mapped onto the morphosyntactic structure in the two languages results in the different discourse functional properties of fronted structures in the two languages. In addition, the information status of referents (Given-New), as well as the information-structural function of sentence elements (Topic-Focus) will also play a decisive role in deciding the pragmatic constraints of a fronted construction in English and Swedish. Moreover, we will refer to basic properties of the pronominal elements in the two languages, to explain the limitations in English fronting, which combines with the central analysis of the interface differences between the two languages to further reduce the frequency of argument fronted structures in English. Thus, the claim is that constraints in usage are based on differences in general principles and not on properties of individual constructions, as has been claimed by among others Michaelis and Lambrecht (1996).

As a first step, we employ the different theoretical notions we have already established in an analysis of the data of our empirical investigation.

6.2 Main patterns in the corpus data

It is worth repeating at this point that previous research and reference works (Biber et al., 1999; Birner & Ward, 1998; Huddleston & Pullum, 2002) have demonstrated that syntactically there are fairly small differences in the possibility of fronting particular clause elements in the two languages, i.e., morphosyntactically most types of clause element can be fronted in both languages. In terms of the position from which the arguments originate, data indicate that Swedish can more readily front from syntactic positions that have traditionally been regarded as islands, like complex NP Ross (1967) as argued by, among others, L.-G. Andersson (1982), Engdahl (1982, 1997) and Löwenadler (2015).

Moreover, the discourse-contextual conditions that are required for fronted constructions to be felicitous are generally fairly restricted in both languages. Also, in many instances where the Swedish texts have fronted constructions, it would often be, at least marginally, possible to use the same structure in English, but with the result of making the expression stylistically marked or somewhat degraded idiomatically. The differences in acceptability of fronted constructions in a particular context in the two languages is therefore in some instances more subtle than the categorical distinction between grammatical or ungrammatical, and more a question of different degrees of acceptability.

There are, however, examples of fronting that are possible in Swedish but ill-formed or infelicitous in English. The current analysis argues that both the higher acceptance of fronting in some contexts in Swedish and the absolute blocking in others in English have their origin in language-dependent implementations of the mapping between morphosyntactic structure and phonological structure. Also, the greater possibilities in Swedish reflect the discourse functional properties of argument fronting that are dependent on the morphosyntactic and phonological difference in the two languages. As discussed earlier, this argumentation is based on the view that information structural constructions like argument fronting includes the representation of the implicit phonology of these structures.

6.2.1 Fronted structures in the corpus data

As was shown in chapter five, a notable difference between the Swedish and English original texts is that most of the fronted elements in the English texts are obligatory adverbials and that the number of fronted objects and predicative complements is limited. Also, the great majority of instances are found in fiction texts.

In Swedish there is a larger number of fronted objects and predicative complements, and the fronted obligatory adverbials are more evenly distributed between fiction and non-fiction texts. The type of fronting structure that is the most central for our argumentation is object/predicative complement fronting of full NPs and pronominal elements. We will therefore start by analysing various variants of fronting of objects and predicative complements, and move on to look at the fronting of verb-licensed adverbials, both with normal and inverted subject-verb word order, in the later parts of this section.

6.2.1.1 Fronted objects and predicative complements

Looking at the fronting data from the Swedish original text, it is clear that there are two main types of fronting in Swedish and neither of them correspond in any systematic way to a fronted construction in the English translation. The two types are:

(a) Fronted object and complement *without* focus/contrast interpretation

(b) Fronted object and complements *with* focus/contrast interpretation

Example (373) illustrates (a) and (374) (b).

(373) Den skulle du ha hört. (AL1.2.s160)
That/It should you have heard
'You should have heard it'

(374) Mat fick hon och en säng att sova i.
Food got she and a bed to sleep in
'She was given food and a bed to sleep in'

In languages like English and Swedish, Focus is expressed primarily through placement of nuclear pitch accents, and given the fairly limited restrictions that morphosyntactic structure places on where the pitch accent can be placed in these languages (Vallduví, 1991; Vallduví & Engdahl, 1996), the same morphosyntactic structure will have different contextual constraints.

In addition, word-order changes which have no logico-semantic effect may also affect the discourse function of the structure with the way phonological structures maps onto a particular morphosyntactic structure. The claim that is made is that a particular sentence will only be felicitous in some specific context if it is possible to apply a phonological structure to that structure that gives it an information-structural representation appropriate for the discourse context.

Because English and Swedish have different morphosyntax and different ways to map phonology onto its syntax, the same type of structure, IS-fronting, in the two languages will have different discourse functions, something which will be apparent in different distributions of the structures and different frequency in the corpus material. Swedish fronting of nominal referents functioning as objects and subject complements have a different discourse-functional profile, which makes it possible for them to appear in more contexts than the English equivalent.

The corpus study reports a higher frequency of object and complement-fronted structures in the Swedish text, and also indicates that the default phonological rendering of the examples demonstrates a phonological difference of these structures in English and Swedish. In English, the fronted elements in their respective contexts must receive prosodic prominence to be felicitous, whereas in Swedish, fronted elements can have reduced prominence. Note, however, that intuitively and in the literature (E. F. Prince, 1981b), there are examples of fronted objects or predicative complements that do not receive pitch accent, which needs to be recognised, and although we will claim that the main differences between Swedish and English fronted structures, especially fronting of object-NPs will be explained as mainly an interface phenomenon, these data require a more nuanced discussion.

Looking at the different categories of fronted objects and complements in terms of their phonological rendering in their context, we can identify prosodic correlates to the information structural categorisation listed above. We can separate the categories based on, on the one hand, whether the fronted element receives added prominence, i.e. nuclear pitch accent at the right edge with prominence on the fronted element, and on the other, whether there exists a nuclear pitch accent at the right edge of the clause. The focus-based categories above would have the equivalent phonology-based categories:

(i) Fronted element **with** prosodic prominence and a final pitch accent.

(ii) Fronted element **without** prosodic prominence and a final pitch accent.

(iii) Fronted elements **with** prosodic prominence **without** a final pitch accent.

We will primarily focus on the first two categories as they represent the types of structures that can help to explain the differences in the properties of the argument structures in English and Swedish.

The main task in the next sections is to show how the combinations of morphosyntactic and phonological structures can account for the different discourse functions and contextual interpretations in the Swedish and English data, both in terms of original and translation.

6.3 Analysis of Swedish and English argument fronting

The analysis promoted in this study is based on the claim that the discourse functional constraints of a sentence in discourse are dependent on the contextual interpretation of its phonological structure, more specifically the position of the nuclear pitch accent and the phonological phrasing, in both written and spoken data. This means that although the main empirical data in this study are written texts which do not contain any phonological information, a central part of the argumentation is based on sentences' implicit phonological structure, an approach that was argued for in detail in section 1.8.3.

Moreover, we will also discuss the properties of pronominal elements in Swedish and English, more particularly the differences between weak and strong pronominal elements, and what this entails for the possibility to front different pronominal elements in English and Swedish. Initially we will give the general outline of the line of argumentation, followed by a more detail discussion developing the argumentation.

1. **Syntax-phonology interface determines the discourse functional constraints**

   The phonological structure of an utterance is dependent on its morphosyntactic structure and the principles for how phonological structures are mapped onto the syntactic one. This includes the possible placement of pitch accent and nuclear pitch accent, and prosodic phrasing.

   The constraints on fronting argument-licensed constituents in English and Swedish are primarily based on the possibility to express a particular information-structural construction with the prosodic characteristics allowed by the principles for syntax-phonology mapping in the two languages.

2. **Intonational phrase (ιP) boundaries are determined by the highest verbal projection**

   Following the work of Hamlaoui and Szendrői (2015, 2017) and Szendrői (2017) regarding a more flexible mapping between morphosyntax and phonology, the intonational phrase is no longer isomorphic of the syntactic CP-layer, instead the left edge of the intonational phrase will be based on the highest position of the verb or verb elements, the higest verbal projection (HVP). English which is a straight SVO language and Swedish which is a verb-second (V2) language will have different positions of the verb in declarative clauses. The standard analysis for these types of languages is that in languages like English the verb stays lower in the syntactic structure, in the head of IP/TP whereas in V2 languages the verb rises to the higher CP-domain. As a consequence, fronted elements in English will be outside the left edge of the intonational phrase containing the main part of the clause, whereas in Swedish the fronted elements stays inside the same ιP.

3. **English and Swedish have different inventories of pronominal types** Another important aspect in this analysis is the differences in pronominal types in English and Swedish. As is well known in the literature, the third-person singular inanimate pronoun *it* is weak in the type of classification developed by Cardinaletti and Starke (1999). This prevents *it* from fronting for morphosyntactic reasons. The Swedish non-animate pronouns *den* and *det* have both strong and weak forms (Hellan & Platzack, 1999).

6.3.1 Fronting of NPs functioning as object/predicate complements in Swedish and English

The main question we are faced with is how to account for the differences in English and Swedish fronted constructions. Our primary focus will be on the more restricted possibility to front NPs, both pronouns and full NPs, functioning as objects and subject complements in English, and in a second step we will expand our analysis to the other types of argument fronting.

Based on the corpus data, the two languages seem to have similar possibilities when both the fronted element and the clause final constituent are perceived as having prosodic prominence. This represents the prototypical Topic-fronting structure as reported in English, where the purpose is generally to introduce a new Topic (switch topic) or indicate a contrast to some previously evoked information.

(375) **The other woman** she identified as certainly Davina Flory's daughter Naomi. (RR1.4.s270)

Den andra kvinnan var helt säkert Davina Florys dotter, Naomi.(RR1T.1.4.s271)

(376) **Zetterbergskans lössläppthet** vet vi inget om och prästbarn och prästbarn. (GT1.2.3.s62)

"Zetterberg's looseness we do n't know anything about, and what do you mean a minister's children?"(GTl1T.1.2.3.s63)

The possibility becomes more restricted when only the fronted element receive prominence, which represents the Focus-fronted structure in the literature.

(377) **A remarkable woman**, she was. (RR1.4.s138)

Man måste säga att hon var en märklig kvinna.(RR1T.1.4.s138)

(378) **Vackra och galna och grymma** var vittrorna. (AL1.2.s138)

Beautiful, mad, and ferocious, the harpies were.(AL1T.1.4.s139)

These structures are found in both the Swedish and the English corpus material, showing that although these structures may not completely correspond in terms of discourse functions, there is nothing in the mapping from morphosyntax to phonology that blocks their usage. However, it is in the third type that we find a significant difference: in Swedish it is possible to have fronted NPs, often pronominal elements, with reduced prosodic prominence, similar to in-situ objects, as has been observed by Lindahl and Engdahl (2022). This reduction can in some cases result in the elision of the fronted element. Although there are no examples in the corpus material, the construction is well attested in the literature, so called topic-drop, which has been studied in depth in Mörnsjö (2002).

The central analysis is based on a difference in the phonological structures between Swedish and English argument fronted structures resulting from the different morphosyntactic make up of the verb-second structure in Swedish and the straight SVO-structure in English. This difference makes it possible for Swedish to use fronted structures, not only to indicate a contrast or a link to some previously evoked entity, as described by Birner and Ward (1998) in English, but also as a way to use argument fronting in a way similar to the use of de-stressed subjects indicating continuing topics. Looking at the translation data, this is where we see the clearest difference: fronted elements in Swedish with no prominence is a structure that is generally not possible in English, or only in restricted contexts, resulting in translations that render the Swedish fronted structure as an English in-situ structure.

(379) "Den där vildoxen", sa han, "den som ni skulle ta vid hornen och vräka i Helvetesgapet, det blev väl en duns, kan jag tro, så det dånade i hela Mattisborgen?"
"Åt gröt, om du kan tugga den, och lämna vildoxarna åt mej", sa Mattis.
"**Dem** ska jag nog bli färdig med när tiden är inne." (AL1.4.s90-s92)

"That wild bull," he said, "the one you were going to take by the horns and heave into Hell's Gap – I suppose that was a crash that rumbled all around Matt's Fort?"
"Eat your porridge, if you can chew it, and leave wild bulls to me," said Matt.
"I'll deal with them when the time comes.". (AL1T.4.s90-s92)

6.3.2 The discourse functional effects of a flexible approach to intonational phrase boundaries

Assuming a theory of prosodic-phonology where morphosyntax interact with phonology through a prosodic hierarchy, the established view (Selkirk, 2011) have been to assume that morphosyntactic categories of word, phrase and clause would isometrically map onto the prosodic equivalents of prosodic word (ω), prosodic phrase (ϕP) and intonational phrase (ιP). The left- and right boundaries of the intonational phrase would thus coincide with the edges of the CP.

However, working on such diverse languages as Hungarian and Bàsàá (a Bantu language), Hamlaoui and Szendrői (2015) propose an analysis of the correspondence between syntax and phonological structure in which the limits of the intonational phrase are not based on the delimitation of the syntactic clause but on the position of the finite verb, or more specifically "the highest projection in the root clause (see Downing 1970), to which the verbal material (i.e. the verb itself, the inflection, an auxiliary or a question particle) is overtly moved or inserted, together with the material in its specifier" (Hamlaoui & Szendrői, 2015, pp. 80–81). They abbreviate this projection as the highest overt verbal projection (HVP). Thus, anything that sits above or outside the HVP cannot be integrated into that core ι-phrase and must be spelled out as a separate prosodic constituent.

In V2-languages such as Swedish, the asymmetrical approach to verb-second phenomena (Holmberg, 2015) states that the verb in non-embedded clauses will move to the CP-layer, as in (380).

(380) $(_{CP}\ \text{Object}_i\ (_{C}\ \text{Verb}_j\ )\ (_{TP}\ \ldots\ \text{t}_j\ \ldots\ \text{t}_i\ )\ )$

The HVP theory of the extent of the *iota*P will thus align its left edge with the edge of the CP resulting in the fronted elements remaining inside the intonational phrase.

In English, on the other hand, there is no indication that the verb moves, and it is argued that it stays lower in the structure, in the TP.

(381) $(_{CP}\ \text{Object}_i\ (_{TP}\ \text{Subject}\ (_{T}\ \text{Verb}_j\ )\ (_{vP}\ \ldots\ \text{t}_j\ \ldots\ \text{t}_i\ )\ )\ )$

Because the limits of the intonational phrase is not dependent on the extent of the CP-structure but on the structure that holds the verb, the result is that the fronted element falls outside the prosodic unit including the rest of the clause, and must therefore form its own prosodic constituent. Thus, the flexibility that exists in +plastic languages (Vallduví, 1991) like English and Swedish within an intonational phrase cannot prevent the fronted element from receiving pitch prominence as each prosodic constituent must have a head that is given prominence.

An effect of this is that English and Swedish fronted constructions will receive different phonological structures. The syntactic structure in (382) shows the effect of verb and fronting being in the same phrasal projections whereas (383) demonstrates the effect when the fronted element move out of the phrase harbouring the finite verb.

(382) $(\text{XP}_j\ \text{V}_i\ ...\text{t}_i\ ...\text{t}_j)_{\iota P}$

(383) $(\text{XP}_i\ ...)_{\iota P}(\text{V}\ ...\text{t}_i)_{\iota P}$

If we implement these structures in Swedish and English we would get (384) and (385).

(384) $(\text{Det löste han})_{\iota P}$

(385) $(\text{That }\ )_{\iota P}(\text{he solved})_{\iota P}$

This is based on a traditional approach where prosodic constituents do not nest. However, the exact phonological representation of these structures in the two languages is ultimately an empirical question. Nested structures has been avoided according to the Strict Layer Hypothesis (Nespor & Vogel, 2007; Selkirk, 1981), but this hypothesis has become increasingly challenged. In Ladd (2008), it was shown that coordinated clauses in English requires a recursive analysis, and data from Japanese have supported the need for abandoning or softening the SLH (Ishihara, 2003, 2004, 2007; Kubozono, 1989), as suggested in the overview of the Syntax-Prosody interface by Bennett and Elfner (2019) where they conclude that the SLH is too strong. Moreover, Féry (2010) presents data in English that support a recursive phonological structure. However, for the main analysis the difference between two consecutive ιP and one ιP nested inside another is not the main issue in this thesis, as the key point for the analysis is that in English the fronted element will be placed outside the intonational phrase requiring it to be given prosodic prominence.

It is worth briefly exploring some of the consequences of this analysis. The possibility to use fronting to a greater extent in Swedish than in English is not just an effect of the English being restricted to fronted elements that have prosodic prominence. It is also the greater flexibility in Swedish fronted structures. As Swedish fronted structure can manipulate the prosodic phrasing to assign a separate ιP to the fronted element, it is possible to two prominent elements as in English or restrict it to a single one. It is also worth considering the dual function of fronted elements as pointed out by Lindahl and Engdahl (2022) that it has both an internal function as signalling the relation within the clause, and an external function relating the role of the fronted element to the previous discourse.

Moreover, as will be further discussed in the section on fronted adverbials, the flexible approach to intonational phrase boundaries also aligns with the discourse functional characteristics of locative inversions as they will normally have the main prominence on the final subject element rather than the fronted verb-licensed adverbial. Something that is predicted from the fact that in these structures the verb has been moved to a higher position in the clause, enabling the nuclear accent to fall only on the final element.

The higher frequency of object fronting in Swedish compared to English is driven by a difference in the discourse functional possibilities of the fronted structure. In Swedish (V2), argument fronting can target structurally weak, non-contrastive elements. In English (SVO), fronting is strictly reserved for pragmatically marked configurations since the fronted element must be given prosodic prominence, drastically reducing the construction's overall utility and frequency in the language.

6.3.2.1 German data as a counterargument

The advantage of a principled account as presented in the previous section is that the difference in discourse functional possibilities between English and Swedish fronted structures is that it is based on general principles of morphosyntax and phonology, and does not have to revert to properties that are attributed to particular structures based on conventionalisation. However, such an analysis also makes a stronger claim on the way all V2-language should work. Thus, if

Swedish argument fronted structures have greater flexibility because it is a V2-language, all V2 languages should exhibit similar flexibility.

Looking at some German data that have been discussed in the literature, they seem to challenge the current analysis since the weak pronoun *es*, the equivalent of the third person non-animate pronouns *it* in English and *den/det* in Swedish appear to defy the predicted possibility of being able to front without the need to carry prosodic prominence, as the one in (386) demonstrates.

(386) * Es habe ich gestern im Park gesehen.
It have I yesterday in Park seen.
'I saw it yesterday in the Park'

However, as was discussed in section 3.5 this does not mean that fronting elements in German have to have prominence, but rather that some pronouns such as *es* in German and *it* in English are weak pronouns (Cardinaletti & Starke, 1999). The example in (386) can be rendered felicitously, without prominence on the fronted element, if the fronted element is the demonstrative *das* as in (387).

(387) Das habe ich gestern im Park gesehen
That have I yesterday in Park seen.
'I saw that/it yesterday in the Park'

6.3.2.2 Fronted verb-licensed adverbials

Generally adverbials that are verb-licensed appear to be more readily moved than objects, but if the verb is semantically weak fronting will result in a sentence that is infelicitous without some additional adjunct at the end. Thus, "The cat sat in the window" but not "In the window the cat sat" unless "In the window the cat sat every afternoon" Instead the significance of fronted verb-licensed adverbial lies in the discourse linked effect of fronting, especially with deictic adverbs like *here* and *there*.

A closer look at the category of fronted verb-licensed adverbials reveals a range of different types of constructions, but their acceptability is generally more dependent on the position of the main verb and the placement of nuclear pitch accent than the prosodic properties of the fronted adverbial, which sets this category slightly apart from the fronting of objects and predicative complements.

When fronting may result in the verb being placed in the position where is receives nuclear accent by default, other structures can be used in English, such as a *there*-existential construction. As shown in (388) – (390).

(388) I denna kategori fanns också den stora svenska flottbasen

(389) In this category the large Swedish naval base of Karlskrona also was.

(390) In this category there was also the large Swedish naval base of Karlskrona.

By inserting the expletive subject *there*, it is possible to avoid placing the verb at the right periphery and avoiding nuclear accent on the verb. Although *there*-constructions are limited in their ability to combine with definite noun phrases, contextual conditions can facilitate this, as detailed in Ward and Birner (1995).

In a number of examples with fronted verb-licensed adverbial, nuclear stress on the finite verb is the desired outcome in combination with an increased prominence on the fronted adverbial. These constructions generally consist of a deictic locative adverb like the medial *there* or proximal *here* in fronted position and verbs of motion or being at the right edge, as in (391)–(392).

(391) And there Joseph stayed ...

(392) Here the flagstone began ...

The initial adverb links back to a salient position established in the context and at the same time the verb needs to receive nuclear accent. Discourse functionally, it would not be possible with a canonical word order with focus on the adverb, and narrow focus on the verb. Narrow focus on the verb is generally interpreted as verum focus (Höhle 1992) which is not normally the intended meaning, as show in (393) and (394).

(393) And Joseph STAYed there ...

(394) The flagstones beGAN here

The analysis of these fronted structures will also have bearing on a particular type of adverbial fronting which we have yet to discuss, i.e. the full inversion of external and internal arguments.

As M. Johansson (1996) noted, the great majority of Swedish fronted constructions in original text are rendered as canonical SVO-clauses in English. This means that the difference must be more fundamental than the differences observed in the difference in the use of fronting of verb-licensed adverbials.

Chapter 7

Discussion

This chapter discusses how the current study contributes to the description and theoretical understanding of argument fronting in Swedish and English, as well as argument fronting in general. As with all investigations of specific linguistic phenomena, the goal has been to arrive at solutions that can be generalised across languages and contribute to a better understanding of the way morphosyntax, phonology and discourse interact to establish the communicative function of a particular expression.

The discussion will be organised according to the main areas in which the study is argued to have developed the knowledge of the phenomena being studied: (a) methodology, (b) empirical finding, and (c) theoretical model.

7.1 Introduction

Different languages allow for different configurations to express the same propositional content, and one of the tasks of the current thesis is to explicate why similar syntactic constructions in two languages, in this case argument fronting, have different pragmatic constraints. Argument-fronted constructions have garnered attention both in terms of the way they should be analysed syntactically and in whether the different properties should be explained as being morphosyntactic or effects of the way morphosyntax and phonology interact.

A key question for investigations looking at the pragmatic aspects of argument fronting is in what way a canonically structured declarative structure differs from argument fronting in terms of communicative function. A common observation is that full noun phrases that are displaced from their canonical position post-verbally and placed in pre-verbal position will be interpreted contrastively, i.e. compared to (395a), the structure in (395b) gives the object added prominence implying that it is contrasted with the "things" that were not investigated.

(395) a. The police investigated the crime.
b. The crime the police investigated.

Some of the explanation has been argued to have its origin in the default phonological representation that is given to a fronted construction, i.e. added prominence to the fronted constituent and the nuclear accent on the main verb. This prevents the fronted construction from being interpreted in the same way as the canonical construction when the entire sentence is in

focus. Generally, it has not been seen as a simple topicalisation process, which just changes the topic element in the two sentences, with no other changes in the interpretation of the structures. The fact that the predication following the topic is not a syntactic constituent but a NP V combination that is not a well-formed constituent, in contrast to a left-dislocation "The crime, the police investigated it".

Still, the standard description of argument fronted constructions is that it topicalises the internal argument for reasons of establishing it as the referent about which something is being predicated. Although more accurate descriptions can be found in the literature, systematic comparisons of information-structuring constructions (Huddleston & Pullum, 2002, p. 1365) in order to develop a more language independent account of the discourse functions of these constructions are still relatively rare. Moreover, such investigations that attempt to find theoretically-based explanations rather than descriptive generalisation are even less common. Thus, by combining insights from the past 30 years of research on morphosyntax, prosodic phonology, and information structure, the current study contributes to the development of analytical models for establishing and comparing different structures, discourse functions, and constraints.

7.2 Methodology

The use of corpus data in investigating specific linguistic expressions is well established in linguistics, to the extent that this is now regularly used also in studies based on formal morphosyntactic theory, which have traditionally mainly relied on intuitive data. However, cross-linguistic investigations with corpora are normally limited to form-based searches which requires that it is possible to define equivalent formal structures in two or more languages, i.e. search for a structurally defined construction in language A and the equivalent structure in language B, and compare the meaning and use of the constructions in the respective language.

Following the ideas and work of S. Johansson (2007, 2012), the current study has made use of a parallel corpus, the English-Swedish Parallel Corpus. The corpus makes it possible to compare instances of a particular structure with their translations in the target language. This, in turn, allows communicatively equivalent expressions to be compared, such as an argument-fronted construction and its most natural rendering in the same context. This can reveal aspects of communicative usage that are not possible when comparing formally equivalent constructions, or a particular construction, with an expression that is most commonly associated as the equivalent structure.

A growing body of work has used parallel corpora for its unique possibilities to both compare original texts in two or more languages, and compare original texts with their translations. Many of these studies have investigated constructions with information-structural properties, or information-structuring constructions such as Ebeling (2000), M. Johansson (2002), Fredriksson (2016), among others. The current study belongs to the same methodological tradition, but deviates from the great majority by not relying as heavily on having a large number of examples in order to do descriptive generalisation, but instead focused more on finding theoretically grounded analyses of the possibilities and constraints of the contextualised examples of argument fronting found in the data.

7.3 Empirical findings

An aspect of the current study that is somewhat unorthodox is the strong emphasis on phonology in explaining phenomena in written texts. Although prosodic phonology and Auto-Segmental analysis of spoken language are firmly established as being central to the explanation of pragmatic

phenomena such as information-structural functions like givenness and focus, they are rarely part of the analysis of naturally produced written data. However, this thesis shows that adopting the developments in prosodic phonology made by Féry (2006), Féry and Ishihara (2009), and Hamlaoui and Szendrői (2015, 2017) enables ways to explain the differences in discourse functions that are demonstrated in the corpus data. Thus, providing support for the sentiment expressed by Féry (2006) in the introduction to her paper as "This paper shows that the syntactic structure of German sentences is shaped by the formation of prosodic phrases and that many puzzles are solved if prosodic phrasing and accents are systematically considered in formal grammar."

The comparison of argument fronting in Swedish original texts and the way they are rendered in the English translation demonstrates that in the majority of the cases, the English translation will render the argument fronted constructions with canonical SVO structures.

7.4 Theoretical model

The approach taken in the current study is that we cannot explain instances of argument fronting in natural discourse by analysing them as being purely syntactic derivations caused by the existence of some syntactised pragmatic feature of [+topic] or [+focus] which requires that these constituents are placed in phrasal projections with corresponding unvalued feature at the left edge of the clause, resulting in the fronting of the object or predicate complement.

As has been previously argued in this thesis, this simply shifts the explanatory burden from the need for constituent to be displaced to the nature of being a carrier of a topic or focus feature. Moreover, given that these information-structural notions still defy clear semantic or pragmatic definitions, little is gained by this way of formalising argument fronting. In addition, the optionality of many of the topicalized structures raises questions that have yet to be resolved. Instead, the analysis promoted in the current study is based on a view of grammar that constrains the derivation of well-formed sentences based on the principles within different grammatical subsystems: morphosyntax, phonology, information structure, and the principles regulating the way the different representations interact at the interfaces.

These two approaches: syntax internal vs interface-based are well known in the literature, but either has yet to be fully worked out, especially the latter. Thus, the current study contributes to the development of an interfaced-based approach by suggesting how such an analysis can avoid some of the weaknesses of a syntax-internal analysis for argument fronted construction in English and Swedish, and the way such an approach can account for differences in discourse function that are hard to explain with a purely syntactic model. Moreover, an interface-based theory can more readily combine theoretical notions regardless of whether they originate in functional and formal linguistic theories.

7.5 Combining formal and functional linguistic theories

The theoretical notion of information structure has been developed primarily within functional theories of language (e.g. Firbas, 1971, 1992; Hajičová et al., 1998; Halliday, 1967; Mathesius, 1964) , where the focus has been on figuring out what pragmatic/communicative properties different information structural configurations have. This has resulted in less attention to the formal properties of these structures and the way models of information structure interact with other subsystems of grammar, i.e. phonology and morphosyntax. However, given that the attention to phenomena that involve the combination of morphosyntax and discourse has gradually increased within formal theories of linguistics, combining insights from both functional and formal linguistic research has become increasingly possible.

The model of grammar used in the current study is able to straddle the traditional separation between formal and functional theories of language, and in doing so achieves a greater degree of exactness, at the same time as it is able to account for a wider range of data. One of the strengths of formal theories of morphosyntax is their well-developed analytical framework, which allows a detailed analysis of structural constraints, but its theoretical domain is limited to that of morphosyntax. Thus, the approach is generally either to first establish whether something can be considered to be a phenomenon analysable as morphosyntactic, and if not to consider it to be outside the remit of core syntax and in need of a different type of analytical model. However, for those scholars who are sympathetic to a formal model of morphosyntax, but interested in phenomena that encompass aspects that have to do with the externalisation of abstract linguistic structures, e.g. phonology, and how they are used in a co-text, have developed analytical tools that deal with the way morphosyntax interacts with other subsystems of grammar such as phonology, and how that can account for the discourse functional properties of utterances.

Chapter 8

Conclusions

This chapter presents the main conclusions of the corpus study and discusses the implications of the proposed account of the differences in the use of fronted structures in Swedish and English written texts. It also suggests some areas for further investigation based on the findings of the current study.

8.1 Introduction

Producing discourse requires more than expressing syntactically well-formed sentences that present relevant propositions, it also involves indicating linguistically how the propositional content relates to the participants common ground and the information state of discourse referents. The present study contributes to the understanding of how these two primary functions interact and how language specific properties result in syntactically similar structures in English and Swedish having partly different discourse functions.

8.2 Corpus study

Using the English-Swedish Parallel Corpus (ESPC) various types of argument fronting in Swedish and English written texts are investigated together with the translation of the original texts into the other language. The corpus material includes a set of examples of argument fronting in the two languages and what the translator considered to be the equivalent structure in the same co-text in the target language. Previous corpus studies and anecdotal observations suggest that these constructions are infrequent in both languages, but that the Swedish constructions can occur in a wider range of co-texts and are more discourse-functionally diverse than the corresponding English construction. The corpus material is consistent with these previous observations.

The Swedish text, both originals and translation contained more argument fronted construction than the English material. However, the argument fronted examples in the English texts are partly of different types than the Swedish text. The two dominant types in the Swedish material are, on the one hand, constructions where the fronted constituents consist of noun and adjective headed phrases functioning as direct objects or predicative complements, as in (396).

(396) **Kaffe** fick han för besväret, och smörgåsar, men inget mer. (AP1.3.s44)
Coffee got he for the trouble and sandwiches but nothing more.
He was given coffee for his trouble, and bread and something, but nothing more. (AP1T.3.s44-s45)

On the other hand, there are proforms representing VPs or clausal complements, as shown in (397).

(397) "**Det** ska jag komma ihåg", sa Ronja. (AL1.2.s71)
That will I remember said Ronia
"I'll remember that," said Ronia. (AL1.2.s71)

In the English material, the most frequent type of argument fronting is structures with fronted obligatory adverbials accompanied with full inversion of the subject and verb.

(398) **Behind the counter** stood a thin young woman (...). (AT1.3.s231)
Bakom disken stod en smal ung kvinna (...).
Bakom disken stod en ung kvinna (...). (AT1T.3.s228)

In terms of the corresponding structure for fronted sentences, we find that only 8% of the Swedish fronted constructions were rendered as argument fronted construction in English, whereas 73% of the English structure translated into argument fronting in Swedish. This gives some support to the more limited use and distribution of the fronted construction in English.

The corpus study gives us some indication of the type and frequency of fronted constructions in the two languages, but one of the main purposes of using material from a parallel corpus was to find patterns in the correspondence between argument fronting and its equivalent expression in the other language, whether it be a translation or original. The main aim was to arrive at an analysis of fronted structures in Swedish and English that will explain and predict their interpretation and use.

The most common English equivalent of Swedish argument fronting of a DP, AdjP or Pronoun constituent is a canonical structure. This is consistent with the results gained by M. Johansson (2002) which is expected as his study used data from the same corpus, although a different subset. The most frequent correspondence to Swedish VP-preposing of a VP-proform or a clausal complement is a pronominal form, often *that* but most often in canonical position, as was illustrated in (398). Looking at the English types of fronting, we find that the English argument inversion is to a large extent also found in the Swedish corresponding text.

8.3 Data analysis

Having established these correspondences between Swedish and English fronted constructions, we observe that the prototypical argument fronting in Swedish is more frequent and less restrictive than the English argument fronted construction which is occurs very rarely both in the originals and as translations of Swedish original structures including Swedish argument fronting. VP-preposition with a VP-proform, which is relatively frequent in the Swedish texts is rare in the English text and the main equivalent structure in the English text is pronouns in canonical position. The use of argument inversion is common in both Swedish and English.

8.4 Discourse functions of argument fronting

These difference in the use of English and Swedish fronted construction is argued to be based on language specific differences in the morphological, syntactic and phonological features, but that the pragmatic and discourse functional principles are essentially language independent, at least when it comes to the formation of discourse in English and Swedish.

The initial question is what the discourse functional differences are between a canonical structure and a fronted construction, on the one hand and how the English and Swedish construction differ in order to cause the distributional distinction shown in the data.

To answer the first question, we rely on the theories of the interaction of common ground and informational structural functions, as well as the discourse status of propositional information, primarily discourse referents. In both English and Swedish, the restricted word order flexibility leads to a tendency to use various means to structure propositions according to a SVO pattern or in the case of Swedish AVSO/SVO pattern. However, so-called information packaging constructions which formally consist of SVO patterns such as *there*-existential and clefts are used to adapt the presentation of propositional information and discourse referent to align with the common ground.

One aspect of discourse development is to signal the givenness of referents (which is an indication that the listener should assume this to be given, not necessarily that it is given in terms of being activated in the current discourse model). In the same way as subject DPs can represent Discourse-new but Hearer-old E. F. Prince (1981b), in the sense that the referent is retrievable, argument fronting indicates that the fronted DP is a salient referent, in contrast to left-dislocation which is a salience promoting structure, i.e. in the left dislocated example in (399) the dislocated NP is not salient in the preceding discourse but acquires this status as an effect of the structure.

(399) Those piranha, I halved my best time for the 400 meter freestyle when I saw them. (Rodman, 1974, p. 35)

The way propositional information and referents are made accessible to speech participants follows the ideas presented in Gundel et al. (1993), and Ariel (1990), and as pointed out particularly for argument fronting and left-dislocation by among others Rodman (1974). However, the argument fronted construction is restricted by the syntactic structure and the way the structured is mapped to prosodic constituents and nuclear focus to indicate the Focal elements of the structure.

According to auto-segmental phonology, within which a central area of research has been mapping syntactic and phonological representation to achieve contextual interpretations and communicative functions. A general theory of the interaction between syntax and phonology, often referred to as the syntax-phonology interface, has been developed by Schwarzschild (1999), Selkirk (2005), Büring (1997, 2007, 2009), Féry (2006), establishing the structure of phonological constituents and the placement of nuclear accent to indicate focal elements. Working within auto-segmental phonology, Hamlaoui and Szendrői (2015, 2017) argue for a syntax mapping that is more flexible than traditionally assumed, and basing the left-edge of the intonational phrase not on any syntactic notion of clause but on the placement of the verb in the structure. As a consequence, the left edge of the intonation phrase will encompass the entire clause, including the fronted element, if the verb is moved to the same phrasal projection as the fronted element, which is the case in Swedish and other verb-second languages, but will exclude the fronted element of the fronted element if the verb stays in the IP/TP projection, which is the case with English.

The effect of the difference in syntactic structure between English and Swedish, in combination of the general mapping between syntax and prosodic constituents, will result in a greater phonological prominence on the fronted element and the interpretative effect of such prominence. Several observations are in accordance with this analysis.

- Stronger contrastive interpretation of the fronted element in English
- Restrictions in fronting weak proforms, such as the third person inanimate pronoun *it* in English

- The ability in Swedish to have both nuclear and non-nuclear accents on the fronted element
- Greater range of felicitously fronted constituents in Swedish (AdjP, Neg)

In addition, we can find some arguments for the generally prosodic neutral prominence on argument inversion that is similar to verb-second languages as the verb moves to the CP layer together with the fronted element. Argument inversion is a different syntactic structure and Birner and Ward (1998) has shown that it differs also in its contextual constraints in that it only requires that the front element has no requirement to have any particular level of givenness, only to be more given than the subject. Moreover, as shown by the small corpus study, the argument inverted construction in English exhibits similar discourse functions as the Swedish typically verb-second pattern, and the question is thus a matter of what syntactic and information structural properties that allow for these verb-second remnant structure to remain in English.

The analysis of VP-proform fronting in Swedish and the equivalent structures in English which are VP-ellipsis and strong pronouns like *that* and *so*. Both of the latter have the possibility to front, but *so* may under certain conditions front as in (400).

(400) So did they.

The analysis of VP-proforms are less theoretically grounded and somewhat speculative, but there are indications that prosodic mapping may influence the limitations in English VP-proform, i.e. the lack of a fronting without having prosodic prominence will rule that out as in contexts where we have VP-ellipsis such as in (401) only nuclear focus on the subject or auxiliary is contextually congruent.

(401) a. Q: Are you going to John's party.
b. A: Yes, I am.

(402) a. Q: Ska du gå på Johans fest.
b. A: Ja, det ska jag.

By making use of a parallel corpus it has been possible to investigate fronted structures which are fairly limited in which co-texts they can be used, and by comparing English and Swedish structures using original texts and translations, the study was able to compare structures in terms of their discourse functions which enable analyses that would not have been possible had the focus been only on a syntactically defined structure.

The study's methodology has its origin in corpus linguistic and contrastive language studies, but it is also committed to a formal linguistic theory with a modular architecture of human language competence, which requires an analysis in terms of interfaces and the formulation of falsifiable predictions. Combining generative syntax, autosegmental-metrical phonology and information structure has made it possible to give an analysis of the general properties of fronted constructions, as well as show how differences in specific linguistic features results in different discourse functions for Swedish and English fronted constructions.

8.5 Looking forward

However, as always, in any focused effort to answer specific questions, there is always a need to leave others unanswered. Also, any advantages that are made, however small, will result in the discovery of areas that will benefit from further research. Some of these unexplored areas are empirical, others conceptual.

As mentioned in the methodology section, using written data when looking at fronted structure risks leaving out relevant phenomena, and restricting the study to two typologically close languages may result in results that are too restricted to be a firm basis for any broader generalizations. More data, both spoken and using other languages, are consequently areas that are ripe for exploration.

Theoretically, there are still many issues within information structure that require both new ways to come to grips with, and a general effort to battle their inherent vagueness, such as the rather ethereal quality of topicality and givenness. Moreover, as influential as Stalnaker (1978)'s notion of Common Ground is, it has been clear over time that some more overreaching take on how to devise a more structured theory of post-linguistic organization of propositional information is needed, although Reinhart (1981) did valiant attempts fairly early.

The area of prosodic phonology as a central area for information structure in languages such as English and Swedish is incontestable, but the strict layer hypothesis (Selkirk, 1981) is under attack, as is indeed phonology-free syntax (Zwicky & Pullum, 1986). Both may have significant consequences for information structure and the syntax-phonology interface.

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Appendix A

Primary Corpus Data

Code	Author/Source	Original Title & Publication Data	Translation & Publication Data
English Fiction			
MA1	Atwood, Margaret (1988)	*Cat's Eye*. Toronto: McClelland and Stewart.	Tr. M. Ekman: *Kattöga*. Stockholm: Prisma, 1989.
JC1	Crace, Jim (1992)	*Arcadia*. London: Jonathan Cape.	Tr. M. Hörmark: *Arkadien*. Stockholm: Forum, 1994.
RD1	Dahl, Roald (1988)	*Matilda*. London: Puffin Books.	Tr. M. Ottosson: *Matilda*. Stockholm: Tidens Förlag, 1989.
AT1	Tyler, Anne (1985)	*The Accidental Tourist*. New York: Knopf.	Tr. S. Bergvall: *Den tillfällige turisten*. Stockholm: Trevi, 1986.
RR1	Rendell, Ruth (1992)	*Kissing the Gunner's Daughter*. London: Hutchinson.	Tr. K. G. & L. Fredriksson: *Dödligt sällskap*. Stockholm: Legenda, 1992.
Swedish Fiction			
KOB1	Bornemark, K-O. (1986)	*Handgången man*. Stockholm: Askelin & Hägglund.	Tr. L. Thompson: *The Henchman*. New York: Dembner, 1990.
AL1	Lindgren, Astrid (1981)	*Ronja rövardotter*. Stockholm: Rabén & Sjögren.	Tr. P. Crampton: *Ronja, the Robber's Daughter*. NY: Puffin, 1983.
AP1	Pleijel, Agneta (1989)	*Hundstjärnan*. Stockholm: Norstedts.	Tr. J. Tate: *The Dog Star*. London: Peter Owen, 1991.
MS1	Scherer, Maria (1980)	*Kejsarvalsen*. Stockholm: Trevi.	Tr. K. Talbot & P. Ford: *The Emperor Waltz*. London: Quartet, 1989.
GT1	Tunström, G. (1983)	*Juloratoriet*. Stockholm: Bonnier.	Tr. P. Hoover: *The Christmas Oratorio*. Boston: Godine, 1995.
English Non-Fiction			
ABB1	ABB Ltd. (2000)	*ABB Group Annual Report 1999*. Zurich: ABB.	*ABB-koncernens årsredovisning 1999*. Zurich: ABB, 2000.
JL1	Lovelock, James (1988)	*The Ages of Gaia*. Oxford: Oxford University Press.	Tr. F. Günther: *Gaia*. Stockholm: Natur och Kultur, 1991.
DM1	Morris, Desmond (1990)	*Animalwatching*. London: Jonathan Cape.	Tr. K. Malmsjö: *Se på djuren...* Stockholm: Norstedts, 1990.
CS1	Sanger, Clyde (1982)	*Safe and Sound: Disarmament...* London: Zed Press.	Tr. U. Herz et al.: *Nedrustning...* Stockholm: Svenska FN-förbundet, 1982.
LT1	Timberlake, L. (1987)	*Only One Earth*. London: Earthscan.	Tr. H. Gezelius & D. Millwood: *En enda jord...* Stockholm: UR, 1987.
Swedish Non-Fiction			
ASSI1	AssiDomän (1999)	*Årsredovisning 1998*. Stockholm: AssiDomän.	*Annual Report 1998*. Stockholm: AssiDomän, 1999.
CO1	Cedergren & Odén (1991)	*I krisens spår...* Stockholm: SIDA.	Tr. S. Robertson: *In the Wake of Crisis...* Stockholm: SIDA, 1991.
HG1	Gustafsson, H. (1994)	*Politisk interaktion...* Author's MS.	Tr. A. Crozier: *Political Interaction...* Lund: Studentlitteratur, 1994.
BJ1	Johnson et al. (1990)	*Hus och hälsa*. Stockholm: BFR.	Tr. D. Knight: *Buildings and Health*. Stockholm: BFR, 1992.
CP1	Petersson, C. (1994)	*Förskolebarn...* Author's MS.	Tr. A. Crozier: *Preschool Children...* Lund: Lund University, 1994.

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FiF-avhandling No. 141, 2026

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This study explores argument fronting in English and Swedish, examining how syntactic structure, information structure, and prosody interact in written texts. Parallel corpus data from the ESPC shows that while non-subject fronting remains relatively infrequent overall, Swedish permits it with greater frequency and flexibility across a wider range of discourse contexts without requiring contrast. Integrating generative syntax, autosegmental-metrical phonology, and information structure, the thesis demonstrates how Swedish V2 syntax and prosodic phrasing allow unaccented fronting, whereas English fronting imposes prosodic prominence, restricting its felicitous use.

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